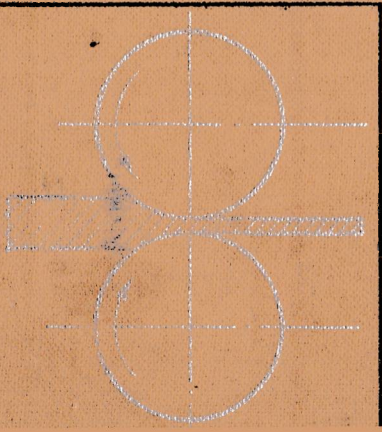
K. BURTSEV

ROLLING PRACTICE



MIR PUBLISHERS

К. И. БУРЦЕВ

ПРОКАТНОЕ ПРОИЗВОДСТВО

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K. BURTSEV

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The Russian Alphabet and Transliteration

А	а	a	К	к	k	Х	х	kh
Б	б	b	Л	л	l	Ц	ц	ts
В	в	v	М	м	m	Ч	ч	ch
Г	г	g	Н	н	n	Ш	ш	sh
Д	д	d	О	о	o	Щ	щ	shch
Е	е	e	П	п	p	Ъ		″
Ё	ё	ë	Р	р	r	Ы		y
Ж	ж	zh	С	с	s	Ь		′
З	з	z	Т	т	t	Э	э	e
И	и	i	У	у	u	Ю	ю	yu
Й	й	y	Ф	ф	f	Я	я	ya

На английском языке

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Training of skilled personnel capable of coping with the demands of the present-day technology and possessing appropriate combination of technical knowledge and practical experience has always been and remains a very important problem.

In studying the applied section of the course of metal rolling, which concerns the rolled metal production proper, one faces the complex characteristic features of this branch of the iron and steel industry: the wide, differentiated, constantly changing range of rolled products and the extreme diversity of the production process elements, equipment designs and process layouts.

The material presented in the book takes into account that allied disciplines on the theory of metal rolling, roll pass design, rolling mill equipment and reheating furnaces are studied separately, so that these sections are only touched on so far as is necessary for understanding the main subject, the rolled metal production techniques.

The rolled metal production problems are discussed at the contemporary level with due regard for the future development trends. The book reflects many years of experience of the author, who has been directly engaged in the establishment of the modern rolled metal production.

INTRODUCTION

The technological progress in all branches of industry, construction and agriculture depends to a large extent upon the development of ferrous metallurgy.

Metal users, especially in progressively developing branches, such as energetics, chemistry, aerospace industries and electronics, impose ever more stringent requirements upon the shapes, sizes, dimensional tolerances and physical, chemical and mechanical properties of the rolled metal products. High temperatures, pressures, speeds, load rates and adverse environmental conditions, which are involved in the operation of modern machines, make new demands as to the quality of materials for the machine components. The enhanced requirements for the rolled products have necessitated definite changes in the metal rolling techniques.

The range of the rolled products is largely diversified. It includes many types of sectional shapes, plates, sheets, tubes, wheel tyres and other special shapes. The rolled products are made of carbon and alloy steel of various grades including structural steel, transformer steel, corrosion- and oxidation-resistant steel, heat-resistant steel, steel with various kinds of plating, etc.

At present, the production of thin sheet, tubes and pipes, clad metal, special and high-grade metal is being extensively developed. Iron and steel production has lately undergone striking changes. While in the past one could draw a sharp boundary-line between the base-metal and high-grade-metal metallurgical plants, this distinction now is gradually disappearing as large metallurgical plants begin producing high-grade metal only.

At present, particular emphasis is placed on improving the quality and range structure of rolled products and on using metal more efficiently.

PART ONE

FUNDAMENTALS OF METAL ROLLING PROCESS

CHAPTER I. PROCESSING PROPERTIES OF METALS AND METHODS OF METALWORKING

1.1. Methods of Working Metals and Alloys

Metals, compared to other materials, possess high strength, toughness and ductility both in hot and cold states. Metal alloys incorporating additions of various chemical elements may acquire special physical and chemical properties, such as high strength, thermal and electrical conductivity, heat and acid resistance, magnetic and electrical properties, etc.

The use of metals by man dates back to great antiquity. First he made use of easy-to-work non-ferrous metals, such as gold, copper, tin and lead and then began employing iron and all its derivatives—steel, cast iron and ferrous alloys.

The main methods of making articles of various shapes from metals are casting, machining (including anodic machining) and plastic working, the latter being the most economical and efficient one. Almost all of these methods permit working of metal to be done either in hot or in cold state. Plastic working includes methods, such as forging, stamping, extruding, drawing and rolling. The most widely used method of steel working is rolling. Rolled products are used as components of various structures. The majority of machine components are made of rolled metal. Rolling can also yield finished components, which is of great advantage. Plastic working improves the structure and mechanical properties of metal.

Depending on the direction of motion of the rolls with respect to the worked metal and the direction of deformation, the following three methods of rolling are distinguished: longitudinal rolling, cross rolling and helical rolling.

In longitudinal rolling the metal is rolled between two rolls turning in opposite directions (Fig. 1.1).

In cross rolling the rolls turn in the same direction. The billet is held in place by means of a special device (force Q)

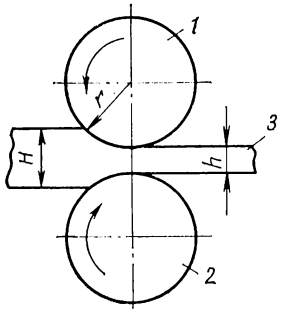


Fig. 1.1. Schematic representation of longitudinal rolling
1—top roll; *2*—bottom roll; *3*—rolled piece

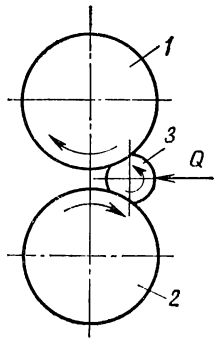


Fig. 1.2. Schematic representation of cross rolling
1—top roll; *2*—bottom roll; *3*—rolled piece

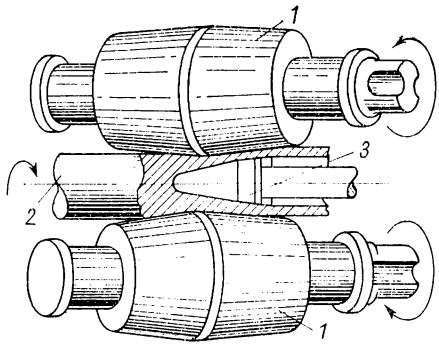


Fig. 1.3. Schematic representation of helical rolling
1—rolls; *2*—billet; *3*—mandrel

and turns around its own longitudinal axis. This method is used for making shapes which represent bodies of revolution (Fig. 1.2).

In helical rolling the rolls also turn in the same direction, but, in contrast to cross rolling, they are arranged at an angle with respect to each other. The rolls set the

round billet being worked in rotary and translatory motion. This method is used for piercing shells (roughly shaped pipes) in pipe manufacture (Fig. 1.3).

1.2. Ductility

The main processing property of metals and alloys that is associated with plastic deformation is ductility, i.e. the ability of a metal to be deformed considerably without breaking. Permanent deformation is a change in the shape of a metal which remains after the external forces causing deformation have been removed, as opposed to elastic deformation, i.e. a change in the shape which is not permanent.

The theory of plastic deformation, being a part of solid-state physics and physics of metals, has lately been extensively developed thanks to the application of modern research methods.

All metals in the solid state have a crystalline structure. There are methods of obtaining fairly large individual metal crystals, so-called single crystals. Under ordinary conditions the solidification or crystallization of metals and alloys starts from a plurality of crystal nuclei. While growing each crystal orients at random. The grain (crystal) boundaries are the surfaces of contact between adjacent crystals. Such a structure is called polycrystalline.

The grain boundaries may serve as sites of concentration of insoluble impurities, excess phases and the so-called intercrystalline substance which frequently contains low-melting compounds, such as oxides, sulphides and silicates. The strength of polycrystalline solids largely depends on the presence and distribution of the intercrystalline substance and excess phases.

When crystalline solids undergo permanent (plastic) deformation the distances between atoms in the crystal lattice change as the result of which the stable equilibrium of the atoms becomes disturbed. Usually, at a high degree of plastic deformation, elements of elastic deformation are also observed, this being associated with a definite coincidence of the direction of deformation with the orientation of crystals (grains).

The deformation spreads along the grains of polycrystalline solids in such a way that the grains, which are most favourably oriented with respect to the direction of the deforming forces, will be deformed first. Then, the deformation gradually involves other grains, which under the action of adjacent ones change their orientation in the direction of the slip planes.

Under definite conditions an intercrystalline deformation occurs in addition to the deformation of the grains themselves. At higher temperatures more favourable conditions are created for the intercrystalline deformation, due to the presence of low-melting impurities in the intercrystalline substance.

In the crystallographic plane atoms are arranged in the form of an atomic lattice which recurs again and again in parallel planes, thus forming a space lattice composed of regularly-shaped three-dimensional crystal unit cells with fixed distances between the adjacent atoms. It is known that iron can have crystal structures of two types, *viz.* the face-centred cubic structure (F.C.C.) in gamma-iron (atoms are situated at the corners and the centre of all the six faces of the cubic cell) and body-centred cubic structure (B.C.C.) in alpha-iron (in addition to the eight atoms at the corners of the cubic cell there is one atom at the centre of the cell).

Such a structure exists only within the boundaries of a single crystal. In a real polycrystalline solid the constituent single crystals differ in their spatial orientation.

When plastically deforming a single crystal or a polycrystalline specimen internal shear lines, the so-called flow lines, can be observed on the polished surface of the specimen.

In the deformed single crystal the slip occurs along a number of the most closely packed crystallographic planes, *viz.* along the plane of octahedron in the F.C.C. lattice and along the plane of rhombic dodecahedron in the B.C.C. lattice.

Plastic deformation is sometimes manifested by parallel lines which are not associated with shear or slip. This phenomenon is called twinning.

A. Bochvar distinguishes the following types of plastic deformation: intergranular displacement with recrystallization, interphase displacement with dissolution and precipitation and amorphous-diffusion displacement.

The plastic deformation of polycrystalline solids characterized by an irregular arrangement of grains and crystallographic planes occurs as a result of shears along the planes of the maximum tangential stresses. These shears occur in the first place along the slip planes located at an angle of 45° with respect to the direction of the deforming forces.

With further deformation the number of slip planes increases owing to the formation of new ones as a result of changed orientation of other grains in the direction favourable for the development of new shears. New slip planes intersect those formed earlier, strain hardening occurs along the planes of shears, the crystal lattice becomes distorted, the atoms lose their stable equilibrium and the atomic distance (lattice parameter) changes.

Thus, in the course of further plastic deformation the plastic properties of a metal deteriorate and the latter hardens (its resistance to deformation increases), i.e. strain hardening of the metal takes place.

The plastic deformation of a polycrystalline solid also occurs as a result of twinning and displacement of grains with respect to one another (intercrystalline deformation). High temperatures and the presence of low-melting impurities at the grain boundaries create favourable conditions for the intercrystalline deformation.

While changing their shape the grains become more extended in the direction of elongation of the deformed solid; a typical lineage structure, the so-called texture, is formed. The formation of one or another oriented grain structure—texture—depends on the degree and rate of deformation and also on the temperature.

The yield strength of polycrystalline metals is usually considered to be constant since they have extremely diverse grain orientation.

In plastic deformation of real metals the shears occur first and foremost in the crystal lattice regions where the atoms are displaced from their position of stable equilibrium.

Such a disturbance of the lattice is called dislocation and it is believed that the shears along the slip planes typical of the plastic deformation take place in the dislocation regions since here smaller forces (stresses) are required.

The increase of the resistance to deformation in case of strain hardening is considered to be associated with the formation of new slip planes which intersect previously formed ones.

In plastic deformation turning of some of the crystals through a certain angle (the so-called process of twinning) is also observed besides the slip. In the first place subject to twinning are grains which are not involved in the formation of the slip planes at a low degree and high rate of deformation.

Numerous tensile tests of metal specimens carried out at high temperatures and under various deformation conditions have revealed the following:

1. Within a high temperature range (above 800 °C) the values of the ultimate strength (σ_{ul}), yield strength (σ_y) and proportional limit (σ_p) virtually coincide when the deforming forces (stresses) are sharply reduced.
2. The resistance to deformation increases with the rise of the deformation rate (in case of cold deformation the effect of the deformation rate is comparatively small).
3. With the rise of the deformation rate the resistance to deformation increases only by 20-30% and then, during further deformation, remains almost unvaried.

1.3. Principles of Plastic Deformation

The following geometric regularities are inherent in plastic working.

Law of Constancy of Volumes

The volume of a metal prior to deformation V_0 and after deformation V_1 remains virtually constant:

$$V_0 = V_1 = \text{const}$$

$$F_0 \cdot L_0 = F_1 \cdot L_1 = \text{const}$$

where F_0 and L_0 = cross-sectional area and length of the metal billet prior to deformation

F_1 and L_1 = cross-sectional area and length of the billet after deformation

Some reduction of the volume of an ingot due to welding up of blow holes and cavities in hot deformation, as well as an increase of the ingot volume due to some loosening of the metal in cold deformation, can be neglected.

Law of Least Perimeter or Least Resistance

In case of unrestricted deformation metal particles flow in three directions, but mainly in the direction of the least resistance which coincides with the directions of perpendiculars drawn to the free metal surfaces. Such a flow of the metal particles in various types of plastic working is associated with the tendency of the metal cross section to have the least perimeter (a square cross section tends to become circular, a rectangular cross section tends to become oval, etc.).

Law of Similarity

According to this law, the work or forces required to cause geometrically similar deformations of geometrically similar bodies of the same material and under the same conditions are directly proportional to the ratios of the volumes or masses of these bodies:

$$\frac{A_0}{A_1} = \frac{V_0}{V_1} = \frac{m_0}{m_1}$$

where A_0 and A_1 = work of deformation of two geometrically similar bodies
V_0 and V_1 = volumes of these bodies
m_0 and m_1 = masses of these bodies

From the law of similarity it follows that forces P_0 and P_1 required to produce geometrically similar deformations of geometrically similar bodies are proportional to cross-sectional areas F_0 and F_1 of these bodies:

$$\frac{P_0}{P_1} = \frac{F_0}{F_1}$$

For geometrically similar bodies the ratio of their volumes is equal to the third power of the ratio of their linear

dimensions:

$$\frac{V_0}{V_1} = \frac{H}{h} \cdot \frac{B}{b} \cdot \frac{L}{l} = a \cdot a \cdot a = a^3,$$

while the ratio of the cross-sectional areas is equal to the second power of the ratio of the linear dimensions:

$$\frac{F_0}{F_1} = \frac{H}{h} \cdot \frac{B}{b} = a \cdot a = a^2$$

where H, B, L, and h, b, l = linear dimensions of the similar bodies

a = ratio of the linear dimensions

Law of Nonuniformity of Deformation

As a rule, all types of actual plastic working processes involve nonuniform deformation in all of the three directions. The nonuniformity of deformation is associated with the effect of factors, such as the shape of both the metal being worked and the tool, dimensions of the product, degree of deformation, effect of undeformed external parts (stiff ends) of the billet, etc. Friction at the surface of contact of the tool with the workpiece and heterogeneity of the physical, mechanical and chemical properties of the material being worked, which makes its plastic properties nonuniform (also frequently caused by nonuniform heating or cooling in hot working) are factors of special importance.

The nonuniformity of deformation may result in the disturbed continuity of a low-ductility metal or in the occurrence of residual stresses.

Apart from the above mentioned geometrical regularities of plastic deformation, the state of stress, contact friction and resistance to deformation are of great practical importance.

1.4. State of Stress

When investigating actual plastic working processes, use is made of the method of analyzing the state of stress adopted in the modern theory of plasticity. This method takes

into account the mode of application and nature of the acting forces. Stress σ is defined as the intensity of internal forces ΔP acting on an 'elemental' plane of area ΔF in the body being worked:

$$\sigma = \frac{\Delta P}{\Delta F}$$

The stress state diagram makes it possible to find out not only the conditions of the tool acting upon the deformed

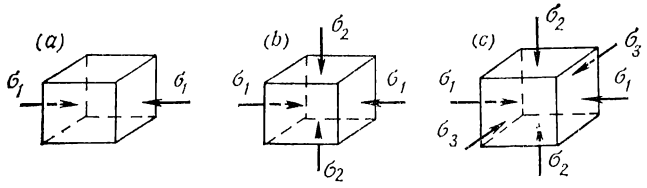


Fig. 1.4. Stress state diagrams
(*a*) linear; (*b*) two-dimensional; (c) three-dimensional

body, but also the interrelation between the external forces and stresses produced by them. The states of stress (Fig. 1.4.) may be *linear* (uniaxial tension or compression), *two-dimensional* (biaxial tension or compression) and *three-dimensional* (triaxial tension or compression). All in all there are nine states of stress, but most of them are not met with in the plastic working processes.

The linear state of stress is seldom encountered in plastic working. The two-dimensional stress system may be present in some processes of sheet metal stamping and bending while the three-dimensional one occurs in all types of rolling, forging and closed impression die forging.

A tensile stress is usually considered to be positive whereas compression is negative.

A nonuniform triaxial compression $-\sigma_1$, $-\sigma_2$, $-\sigma_3$ is the most frequently occurring type of the three-dimensional state of stress (rolling, extruding, stamping, forging).

In rolling with tension or in drawing the state of stress is $-\sigma_1$, $-\sigma_2$, $+\sigma_3$.

The state of stress $-\sigma_1$, $+\sigma_2$, $+\sigma_3$ is typical of cross and helical rolling.

When the principal stresses are equal, i. e. $\sigma_1 = \sigma_2 = \sigma_3$, shears and twinning are impossible and therefore there is no plastic deformation in this case. Under the conditions of a uniform triaxial compression only elastic deformation is possible while a uniform triaxial tension may result in brittle failure. A nonuniform triaxial compression creates most favourable conditions for plastic deformation.

For instance, inequation $\sigma_1 > \sigma_2 > \sigma_3$, where σ_1, σ_2, and σ_3 are the maximum, medium and minimum principal stresses respectively, corresponds to the conditions of longitudinal rolling.

When the principal stresses are of the same sign, the stress state diagram corresponding to this instance is called *the stress state diagram of the same sign*, but when they are of different signs, the diagram is called *the stress state diagram of opposite sign.*

The instant of transition of a loaded body (metal) from elastic to plastic deformation region is called *limiting state.* The limiting state occurs when a definite relation between the principal and yield stresses of the deformed body is reached.

There are four theories of limiting state. The first of them is based on an assumption that plastic deformation begins when one of the principal stresses reaches the yield stress. According to the second theory, plastic deformation begins when one of the largest principal strains complies with the conditions of the shear along the slip planes. Both of these theories are not corroborated by experience.

According to the third theory, plastic deformation begins at the moment when the difference between two (maximum and minimum) principal stresses reaches the yield stress of the material. However, this theory does not take into account the third principal stress.

The fourth theory complies to the highest degree with the conditions of actual plastic deformation. In accordance with this theory, the transition to plastic deformation occurs when a definite amount of energy required to cause a permanent change in the shape of a body is accumulated in the volume of the body undergoing elastic deformation.

It has been established (S. Gubkin) that the specific energy required to cause a permanent change in the shape

of a body does not depend on the type of the stress state and is constant with the same deformation factors, such as the temperature, degree and rate of deformation.

The total strain energy of the deformed body (A_t) is equal to the sum of the strain energy due to the change in the volume of the body (A_v) and that due to the change in the shape of the body (A_s):

$$A_t = A_v + A_s$$

$$A_s = A_t - A_v$$

According to Hooke's law, the principal strains ε_1, ε_2, and ε_3 are expressed as

$$\varepsilon_1 = \frac{1}{E}[\sigma_1 - \mu(\sigma_2 + \sigma_3)]$$

$$\varepsilon_2 = \frac{1}{E}[\sigma_2 - \mu(\sigma_3 + \sigma_1)]$$

$$\varepsilon_3 = \frac{1}{E}[\sigma_3 - \mu(\sigma_1 + \sigma_2)]$$

where E = modulus of elasticity of the material, kgf/mm^2
μ = Poisson's ratio, the ratio between the lateral and longitudinal elastic strains

The total strain energy is determined by the following equation:

$$A_t = \frac{1}{2}(\sigma_1 \cdot \varepsilon_1 + \sigma_2 \cdot \varepsilon_2 + \sigma_3 \cdot \varepsilon_3)$$

or

$$A_t = \frac{1}{2E}[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1 \cdot \sigma_2 + \sigma_2 \cdot \sigma_3 + \sigma_3 \cdot \sigma_1)]$$

The strain energy due to the change in the volume of the body is equal to one half the product of volume increment $\left(\frac{\Delta V}{V}\right)$ by the average stress:

$$A_v = \frac{1}{2} \cdot \frac{\Delta V}{V} \cdot \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$$

$$\frac{\Delta V}{V} = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 = \frac{1-2\mu}{E}(\sigma_1 + \sigma_2 + \sigma_3)$$

then $A_v = \frac{1-2\mu}{6E}(\sigma_1 + \sigma_2 + \sigma_3)^2$

The strain energy due to the change in the shape of the body is

$$A_s = A_t - A_v = \frac{1}{2E}[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1 \cdot \sigma_2 + \sigma_2 \cdot \sigma_3 + \sigma_3 \cdot \sigma_1)] -$$

$$- \frac{1}{6E}(\sigma_1 + \sigma_2 + \sigma_3)^2 + \frac{2\mu}{6E}(\sigma_1 + \sigma_2 + \sigma_3)^2$$

After simplification we have

$$A_s = \frac{1+\mu}{6E}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]$$

Let us assume that the state of stress is linear and $\sigma_2 = \sigma_3 = 0$ and $\sigma_1 = \sigma_s$. Then the specific strain energy due to the change in the shape of the body $A_{s.lin}$ will be

$$A_{s.\,lin} = \frac{1+\mu}{6E} \cdot 2\sigma_s^2 = \frac{1+\mu}{3E} \cdot \sigma_s^2 \text{ (equation of plasticity)}$$

where σ_s = actual resistance to the change of the shape in simple tension or compression with due regard for the effect of the temperature, deformation rate and strain hardening. σ_s is sometimes determined approximately in terms of yield stress σ_y:

$$\sigma_s = n_t \cdot n_h \cdot n_r \cdot \sigma_y$$

where n_t, n_h, n_r = coefficients taking into account the effect of the temperature, strain hardening and deformation rate on the resistance to deformation (when determining σ_y in approximate conditions of deformation it is frequently assumed that $\sigma_s = \sigma_y$).

If the expanded equation of plasticity is simplified by eliminating medium stress σ_2, i.e. if $\sigma_2 = \sigma_3$ (or $\sigma_2 = \sigma_1$), then it may be written simply as

$$\sigma_1 - \sigma_3 = \sigma_s, \text{ or } \sigma_1 - \sigma_3 \approx \sigma_y$$

when $\sigma_2 = \frac{\sigma_1 + \sigma_3}{2}$,

$$\sigma_1 - \sigma_3 = \frac{2}{\sqrt{3}} \cdot \sigma_y = 1.155\sigma_y$$

1.5. Contact Friction

Friction at the surface of contact of the tool with the metal being worked results from the engagement of surface irregularities and adhesion of clean (unoxidized and uncontaminated) metals in contact.

In practical calculations the molecular forces of adhesion are disregarded and an elementary law of mechanics (Amonton's law) established as far back as 1699 is used:

$$T = fP$$

where T = friction force
f = coefficient of friction
P = normal force

Contact friction changes the stress state conditions and thus substantially affects the process of plastic deformation. In compression, if f is close to zero, conditions are created for the development of a linear state of stress; with the rise in the value of f a three-dimensional state of stress occurs.

As the friction forces gain in strength, the force required to cause deformation and the power consumed in deformation are increased and the nonuniformity of metal flow along the height of the deformed specimen is aggravated because the metal layers at the point of contact are restricted by the friction forces.

Friction force T at the point of contact of two bodies is perpendicular to the normal force. The tangent of the friction angle (the angle between the resultant of the normal and friction forces and the normal force) is known to be equal to the coefficient of friction.

Friction forces in rolling provide, within certain limits, the possibility for the process itself to be carried out, but none the less in many instances, especially in cold rolling, one tries to reduce friction by using lubricants.

To determine accurately all the rolling process parameters, it is very important to know the value of the coefficient of friction. Empirical formulas are commonly used for this purpose. Besides the condition of the rubbing surfaces (surface finish, presence of oxides and lubricants), the rate and temperature of deformation also exert influence upon

the value of the coefficient of friction. It is reduced with the rise of the deformation rate and temperature.

The reduction of the coefficient of friction with the rise of the temperature of deformation is believed to be associated with the effect of factors, such as the reduction of the resistance to deformation, the thickness and hardness of the scale layer. Thus, the improved ductility of the metal being rolled reduces the resistance to shear of tiny projections on the metal surface which fill microcavities in the surface of the rolls.

The reduction of the coefficient of friction with the rise of the deformation rate (rolling speed) can be explained by shortening of the time of contact of the rubbing bodies, this resulting in an incomplete engagement of the irregularities of the surfaces in contact and hence, in a reduced friction force.

The coefficient of friction is also influenced by other factors, such as the chemical composition of the rolled metal, the value of the specific rolling pressure and draught, the metal thickness and the shape of roll passes. Also, the value of the coefficient of friction may, in metal biting by the rolls, differ from that in the steady-state rolling process: it is higher when biting metal and this stems from the reduced specific rolling pressure and contact surface slipping.

1.6. Resistance to Deformation

The resistance to deformation is determined proceeding from the stress state conditions and the yield stress of the deformed body. Resistance to deformation p is usually taken to be equal to maximum principal stress σ_1.

According to the general equation of plasticity

$$\sigma_1 - \sigma_3 = m\sigma_y$$

whence

$$\sigma_1 = m\sigma_y + \sigma_3$$

where σ_y = yield stress of the metal in a linear state of stress (inherent rigidity)

m = coefficient taking into account the effect of medium stress σ_2 upon the state of stress (Lo-

de's factor), changes within the range from 1.00 to 1.155

σ_3 = minimum principal stress which is designated as q in the formula for the resistance to deformation

The resistance to deformation is determined by the following formula:

$$p = m\sigma_y + q$$

Consequently, the resistance to deformation is the maximum stress under which the worked metal transits from elastic to plastic deformation region.

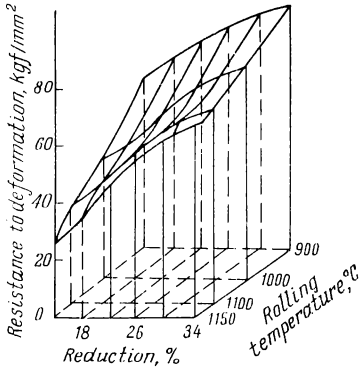


Fig. 1.5. Effect of degree and temperature of deformation on resistance to deformation in hot rolling

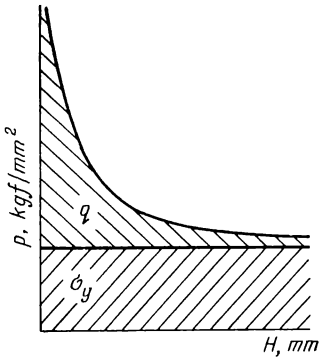


Fig. 1.6. Effect of thickness of deformed body on resistance to deformation

The resistance to deformation depends on the inherent rigidity of the metal, temperature (Fig. 1.5), strain hardening (degree of deformation) and also on the deformation rate.

The following circumstance of great importance for the rolling practice is worth noting. Resistance to deformaton p and minimum principal stress q decrease with the increase in the thickness of the deformed body (Fig. 1.6).

In case of a considerable increase of the thickness of a strip the ratio of the contact area to the volume of the de-

formed body becomes very low, the effect of contact friction is substantially reduced and the resistance to deformation approaches very closely the yield stress (linear state of stress). The diagram of resistance to deformation p consists of two regions, *viz.* σ_y region and q region. The q region limited by a hyperbolic curve characterizes the effect of contact friction on the resistance to deformation (Fig. 1.6).

The resistance to deformation decreases with the rise of the temperature and increases with the rise of the deformation rate, both in cold and hot metal working. In case of cold metal working with high deformation rates the metal ductility is improved due to generation of heat which causes diffusion and softening of the metal.

The deformation rate is defined as strain $\left(\frac{\Delta h}{H} = \delta\right)$ occurring per unit time (t), i.e.

$$\frac{\frac{\Delta h}{H}}{t} \quad \text{or} \quad \frac{d\delta}{dt} \ \text{s}^{-1}$$

The stress on the contact surface is called the specific pressure. In practical calculations the specific pressure is usually taken to be equal to the resistance to deformation and is considered to be distributed uniformly all over the surface of contact of the metal with the deforming tool.

Some investigators equate σ_1 to the vertical specific pressure, substituting term k for $m\sigma_y$:

$$p - q = k$$

In a strictly theoretical aspect the term 'resistance to deformation' is considered as the yield stress in tension of a metal of a given chemical composition under definite deformation conditions as to the degree, rate and temperature of deformation.

The specific pressure on the contact surface is determined not only by the resistance to deformation, but also by the contact friction and deformation conditions.

The specific rolling pressure is expressed as

$$p_r \approx m \cdot n_\sigma \cdot \sigma_s$$

where m Lode's factor varying within the range from 1.00 to 1.15

n_σ = stress state factor which varies within a wide range, depending on the rolling conditions and contact friction

n_σ for biaxial deformation may be expressed as

$$n_\sigma = n'_\sigma \cdot n''_\sigma \cdot n'''$$

where n'_σ, n''_σ, n'''_σ are coefficients taking into account the effect of contact friction, external zones and tension (or compression) in rolling, respectively; n'_σ and n''_σ are greater than unity; n'''_σ equals 0.7-0.8 when rolling is carried out with a considerable tension.

Assuming that $p_r \approx \sigma_1$ and $\sigma_3 = 0$ (two-dimensional state of stress), the general equation of plasticity can be written as

$$p_r = m \cdot \sigma_s$$

if $\sigma_2 = \sigma_3 = 0$ (linear state of stress), then

$$p_r = \sigma_s$$

where σ_s = stress due to the change in shape or resistance to deformation.

1.7. Strain Hardening and Recrystallization

In the course of cold plastic deformation during metal working the properties of the metal undergo changes: its strength and hardness increase while the ductility decreases, the electrical and magnetic properties alter along with thermal conductivity, and grains assume a definite orientation. Microcavities appear in the metal and its density lowers from 7.86 to 7.78 kg/m^3. Surplus energy is accumulated in the crystal lattice being distorted in cold working.

Such changes of the metal properties resulting from cold plastic deformation are referred to as *strain hardening.*

When a cold-worked metal is heated its atoms return into the lowest energy state, i.e. the so-called recrystallization takes place. Consequently, recrystallization is a process in

which the crystal structure of a substance distorted due to plastic deformation is replaced by a new, unstressed structure. In recrystallization the grains recover their initial undeformed condition.

The quality of a metal largely depends on the grain size and therefore the interrelation between the degree of deformation, recrystallization temperature and grain size is of great practical interest.

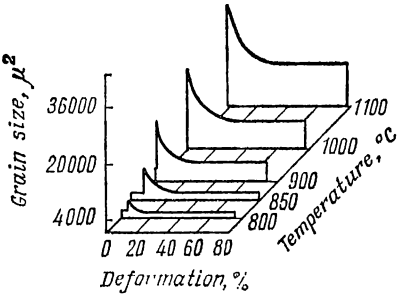


Fig. 1.7. Effect of degree of deformation and temperature on grain size of low-carbon steel

Recrystallization occurs not instantly, but takes some time. The rate of recrystallization depends on the degree of cold deformation and temperature.

According to A. Bochvar, recrystallization temperature T_r of pure metals can be approximately determined as a function of melting temperature T_m:

$$T_r \approx 0.4T_m$$

At temperatures lower than 0.25 to $0.35T_m$ a phenomenon of recovery is observed when residual stresses decrease without any changes in the crystal lattice.

With an increase in the temperature of metal heating the grain size enlarges and tiny fragments of the deformed grains become the nuclei of intensive recrystallization. In a structure with grains of uneven size recrystallization proceeds at a higher rate. The character of the metal structure after recrystallization depends on the heating temperature, time of heat and degree of plastic deformation prior to heating (Fig. 1.7).

The maximum growth of the grains is observed with a comparatively low degree of deformation (6-8%); such a deformation is called *critical.*

As the temperature augments, the position of the point corresponding to the critical degree of deformation on the diagram is gradually shifted towards the origin of coordinates. When slightly deforming an initially annealed undeformed metal only a comparatively small number of favo-

urably oriented grains undergo deformation and with such a heterogeneous structure the rate of recrystallization is increased (the growth of coarse grains results from the absorption of fine ones). In hot deformation two contrary processes, *viz.* strain hardening and recrystallization, take place simultaneously, whereas cold deformation is accompanied by strain hardening alone.

Working of a metal at a temperature below its recrystallization temperature is called cold working. Hot working is performed within a temperature range above that of recrystallization.

There are also distinguished incomplete hot working and incomplete cold working. Two types of structures, *viz.* recrystallized and unrecrystallized, turn out in the first instance. Such phenomena should be avoided in hot working since these cause nonuniform deformation and the occurrence of residual stresses resulting in a low-quality metal. There is no recrystallization in the case of incomplete cold working, but the plastic properties of a metal are somewhat improved due to the high deformation rate and heat effect.

1.8. Temperature Ranges of Hot Working

The upper limit of hot-working temperatures must be selected so as to make the most of the plastic properties of metal with minimum expenditure of power in working. The formation temperature of homogeneous solid solution may be somewhat exceeded, but the lower limit of the temperatures of working should never be as low as the eutectoid point. Metals possess the highest ductility within the range between the recrystallization and melting temperatures. On the other hand, hot working should be carried out when the metal is in a single-phase state, i.e., at temperatures higher than those of phase transformations.

An ideal condition would be, if the temperature were maintained at a constant level slightly above the temperature of iron change to the gamma state when the ductility of steel gets higher and the grain size is at its minimum. However, such a condition is not always feasible because of specific properties of various grades of steel and limited capabilities of the production equipment. The temperature

range limits of hot working of carbon steel are shown in Fig. 1.8.

In 1868 Russian scientist and metallurgist D. Chernov demonstrated that in order to obtain metal of the finest grain structure the temperature at the end of hot forming of steel must correspond to point A_3 (line *GOS*) of the iron—iron carbide diagram (Fig. 1.8).

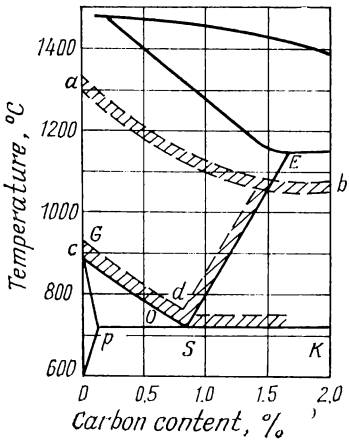


Fig. 1.8. Temperature range boundaries for hot working of carbon steel

If rolling is finished at a temperature above point A_3, coarser grains will be obtained, whereas finishing of the rolling at a temperature well below point A_3 will result in banding.

The temperatures at the beginning and the end of rolling are 1260 and 950 °C respectively for most of mild carbon steel grades, 1150 and 850 °C for carbon tool steel grades, and 1200 and 900 °C for medium-carbon steel grades.

When rolling is performed in modern high-speed mills, the ranges between the temperatures at the beginning and the end of rolling are narrowed.

1.9. Factors Influencing the Malleability of Metals and Alloys

The malleability (workability, rollability) of metals and alloys is interpreted as their ductility, i.e. the ability to be deformed considerably without breaking, under particular processing conditions, such as temperature ranges of working, degree of deformation (overall and single reductions), deformation rate, stress state character (method of working, contact friction, character of forming) and requirements as to the quality of metal after working. Being linked with one another, these conditions may either vary or be constant.

The following are the constant, initial processing factors which influence the malleability of metals and alloys: the chemical composition which defines the inherent rigidity and resistance to deformation, methods of melting and casting, ingot structure, character of preliminary deformation of cast metal and macroscopic and microscopic structures of the deformed metal. Reheating and heat treatment are also of great importance.

1.10. Methods of Determining the Malleability (Rollability) of Metals and Alloys

Under certain conditions (nonuniform triaxial compression) naturally brittle materials (for example, marble) become ductile, but under other conditions (nonuniform triaxial tension) ductile metals become brittle. Therefore, to determine the malleability (workability, rollability) of metals and alloys they are tested for ductility under the conditions most typical of each method of working. The tests performed under actual production conditions are most reliable, but such tests are not always feasible.

There is a number of indirect methods of determining plastic properties:

1. Destructive tensile, torsion and impact (impact toughness) tests.

Sometimes these tests are carried out at different temperatures. On the basis of the results of the tests and the analysis of stress state diagrams one can obtain a criterion of ductility necessary for the determination of the permissible degree of deformation.

2. Technological tests:

(a) Upsetting test: the ductility of a metal is determined by the amount of upset (reduction) of the metal specimen prior to the disturbance of continuity of the free side faces of the specimen.

(b) Bending test, expanding test (of pipes).

(c) Erichsen cupping test for determining the cold ductility of sheet metal: a round specimen cut out from the sheet is cupped in a hand screw press until the metal cracks; the ductility is determined by the depth of the obtained cup.

3. Taper rolling method. V. Grume-Grzhimailo was the first who obtained maximum draught in rolling of a pointed specimen under production conditions.

According to the method devised by Yu. Chizhikov, the malleability (rollability) of a metal is determined by taper-rolling a square specimen 40 × 40 mm in section in eccentric rolls, the reduction being varied in a range of 0-75%.

The malleability is characterized by the percent reduction under which cracks appear at the edges of the rolled wedge.

1.11. Effect of the Initial Properties of Steel and Alloys on the Ductility

The chemical composition, melting and casting conditions, character of preliminary deformation of cast metal and also the macroscopic and microscopic structures of the deformed metal are referred to as the initial properties of steel and alloys.

Effect of Chemical Composition on the Ductility of Steel and Alloys

The preponderant element in the chemical composition of steel and alloys is called the base element. So, the base element of steel is iron; in alloys with predominant nickel content the base element is nickel, etc.

The ductility of solid solutions is determined by the plastic properties of the base element. The ductility of ferro-nickel alloys is sufficiently high since both iron and nickel possess good plastic properties. This is also true for ternary alloys. Thus, if a ferro-chromium-nickel alloy forms a solid solution with a chromium content of not higher than 8%, its ductility will be as high as that of pure iron. The plasticity decreases when the elements are not completely dissolved in the solid solution, but form the so-called excess phases which occur in the metal as inclusions within the grains and at the grain boundaries (for example, carbide formers, ferritizers, sulphides and sulphide eutectics, aluminium oxides).

It should be noted that the use of extra-pure materials and microscopic additions of various elements (especially of boron) has an effect on the ductility. The conformity of the contents of individual elements in a particular grade of steel or alloy to the specified composition and the uniformity of the element distribution within the metal are also of great importance.

The effects of various elements on the plastic properties of steel are considered below.

Iron. There is a great difference between the ductility of chemically pure iron and that of commercially pure iron. At a temperature of about 1000 °C commercially pure iron (armco iron) has a reduced ductility, whereas at such a temperature the ductility of chemically pure iron is high.

Carbon. The ductility of steel with different carbon content is considered rather than that of carbon itself. As is generally known, when the carbon content of steel does not exceed 1.7%, carbon in the solid solution with iron forms austenite at temperatures above 1050 °C; in this case, with a carbon content of up to 1.4% the ductility of steel corresponds to that of iron and it is reduced when the carbon content comes to 1.4-1.7%.

When the carbon content of steel is in excess of 1.7%, ledeburite and cementite appear in addition to austenite, this resulting in a sharp reduction of the ductility.

Manganese. Manganese steels are susceptible to overheating and require strict adherence to the specified temperature conditions of heating.

Manganese has a favourable effect on the plastic properties of steel because unlike ferrous sulphides, manganous sulphides occur not at the grain boundaries.

Silicon. When silicon in steel is in the solid solution, it has no effect on the ductility. The ductility is reduced when the silicon content of steel is over 3%. The so-called vitreous phase formed by silicon oxide exerts especially harmful effect upon the ductility.

Sulphur. It is a deleterious impurity. Being low-melting, sulphur compounds turn liquid at high temperatures and, by flowing between the grain boundaries, destroy all cohesion of the grains, thus causing the so-called hot-shortness.

The effect of sulphur is somewhat allayed if manganese is present and is aggravated with the increasing oxygen content.

The increase in the sulphur content of steel favours the formation of ferrous sulphides and an eutectic with a melting temperature below 950 °C. At high temperatures of heating cohesion of the grains becomes destroyed which results in the appearance of fissures and cracks in the metal during its hot plastic deformation.

Sulphur also combines with some other elements (nickel, molybdenum), thus forming sulphides. To avoid the formation of intergranular network of sulphides in nickel and molybdenum steels, their sulphur content should be kept at minimum.

Tin, lead, arsenic, bismuth and antimony also weaken cohesion of the grains.

Oxygen. Oxygen like sulphur also causes hot-shortness because the oxides formed have low melting temperatures and concentrate at the grain boundaries. Free gases occluded in steel reduce its ductility.

Hydrogen is also deleterious since it causes internal fractures—flakes. Oxygen and nitrogen form oxides and nitrides respectively.

Phosphorus. If phosphorus is dissolved in the solid solution of iron, it does not deteriorate the ductility of carbon steel.

The increase in the phosphorus content of steel results in its cold-shortness during cold plastic deformation.

Copper. In spite of the fact that pure copper possesses high ductility, its presence in steel (or alloy) deteriorates the plastic properties of the latter, if it is not in the solid solution and, especially, if it precipitates at the grain boundaries. Structurally free copper causes hot-shortness.

Some elements which render steel hot- or cold-short are specially introduced into individual grades of steel in order to improve their machinability. For instance, sulphur and phosphorus are introduced into free-cutting steel grades. Lead is also added for the same purpose. Copper steels are used as corrosion-resistant ones.

Nickel. The nickel content in iron-nickel alloys has no effect on their ductility.

Chromium. It is conducive to the formation of an excess ferrite alpha-phase which reduces the ductility and it also has a tendency to increase the grain size during recrystallization.

Molybdenum. It may form sulphides at the grain boundaries and concentrate under the layer of scale. The workability of nickel-molybdenum alloys deteriorates when the molybdenum content is in excess of 20%.

Tungsten. Tungsten favours the formation of carbides which hamper the grain growth in recrystallization. It sharply improves the resistance to deformation.

Vanadium. It is added to steel in small quantities in order to reduce the grain size (it hampers the grain growth at high temperatures).

Boron. Boron additions greatly improve the rollability of hard-to-work steel and alloys.

Effect of Melting and Casting Conditions and Also of the Character of Preliminary Deformation on the Ductility

Many processing factors, starting from the quality of charging materials and ending with the conditions of ingot formation, influence the ductility of metal. Among these factors are the density of the metal (the presence of cavities, blisters and blowholes) and its saturation with gas, which depend on the method of production, purity of the raw materials and the kind of metallurgical fuel, and also the methods and degree of deoxidation, homogenization during heating or special homogenizing operation (levelling out of the composition at the expense of diffusion processes in the hot metal). The amount of non-metallic inclusions and their size are determined by the conditions of melting, deoxidizing and casting. Special kinds of treatment (synthetic slag, vacuum degassing) also influence the formation of non-metallic inclusions.

The initial heterogeneity of the properties and structure of a metal affects the conditions of its plastic working. Therefore the initial working of cast metal (roll cogging or hammering) is a very important part of the rolled metal production. As a rule, the plastic properties of initially deformed metal are better than those of cast metal.

Effect of Macrostructure and Microstructure on the Ductility

The structure of a metal (its macrostructure), as it is seen by the naked eye on the etched surface of specimens, gives an idea about the metal crystallization. During solidification of ingots the following three zones of crystallization are distinguished:

(1) The outer skin zone consisting of small crystals oriented at random;

(2) Zone of columnar crystals located at right angles to the ingot faces;

(3) Inner zone formed by crystals of various sizes which are also oriented at random.

Under certain conditions some grades of steel and alloys crystallize in a way different from the one described above. Some crystal zones may be lacking and the ingot may have a columnar or transcrystalline structure (the crystals grow from the periphery towards the centre of the ingot). Mixed types of structures may also occur.

Depending on the conditions of casting and cooling of ingots, the same grades of steel crystallize in different ways. Crystallization is also influenced by alloying elements. Some investigators have found that the transcrystalline and coarse-grained structures not always reduce the plastic properties of steel under the conditions of hot rolling.

In most cases of rolling ingots with transcrystalline and coarse-grained structures smaller reductions and an increased number of tilts are used. Moreover, ingots with strongly pronounced transcrystalline structure are preferably hammered prior to rolling because even with a considerable draught the cast structure is eliminated incompletely and the rolled metal will not have homogeneous (isotropic) structure.

The effect of microstructure on the ductility has already been considered earlier. A metal in the form of a homogeneous solid solution which has no excess phases at hot working temperatures is characterized by good ductility. The excess phases, e.g. alpha-phase, have an adverse effect on the plastic properties. The possibility of dissolution of the excess phases during heating should be taken into consideration. The tendency of steel to form a solid solution depends on the alpha-phase content.

With a rise in the temperature the alpha-phase content is sometimes not reduced, but, on the contrary, is even increased. The reduced ductility of a metal with an excess phase is due to the differences between the processing properties of the base solid solution (austenite) and those of the excess phase (alpha-phase—ferrite).

When heated, iron and chromium carbides dissolve almost completely in the solid solution and therefore they do not affect the ductility.

The plastic properties of steel are also influenced by the recrystallization conditions, overheating and burning in the event of a prolonged high-temperature heating.

The overheating (heating of metal to a temperature at which rapid grain growth occurs and large grains are produced) and burning (heating of metal to too high a temperature causing local fusion or oxidation of grains) result in the metal losing its ductility due to the destruction of cohesion of its grains. In a number of cases the overheated metal can be remedied by a further heat treatment, but the burnt metal is rejected.

1.12. Supplementary Processing Operations

Not long ago the vast majority of rolled products were delivered to the customers in the as-rolled condition. But the ever growing requirements for the high quality of rolled products have necessitated supplementary finishing operations.

All types of working intended to impart the rolled products more accurate dimensions and also strength, corrosion resistance and other special physical and chemical properties can be referred to as supplementary operations. Machining, drawing, etching, heat treatment (including hardening after rolling which is carried out by using the rolling heat), tinning, galvanizing and coating with plastics and insulating materials are sometimes bear the name of the fourth conversion operations.

CHAPTER II. ROLLED PRODUCT AND ROLLING MILL CLASSIFICATION

2.1. Classification of Rolled Products

All rolled products can be divided into the following principal groups: sectional shapes, sheets and plates, pipes and tubes and special rolling-mill sections.

The group of sectional shapes includes a wide range of simple and structural shapes, the most popular of which are illustrated in Fig. 2.1.

Squares, rounds, hexagons, strips and some other shapes are called simple shapes, while rolled angle, rails of all types, I-beams, channels, Z-sections, piling bars, etc. are termed structural shapes.

Automotive rim, side rings, piles, sash frames, wedges, axles, balls, etc. bear the name of special rolling-mill sections.

Various methods of production of light-weight shapes have been adopted during the past 10 years.

Efficient sectional shapes, the so-called roll-formed shapes, are produced in a fairly simple way from plates and bands of various thickness by continuous cold or hot roll-bending. Many of special rolling-mill sections for machine parts are manufactured by die rolling.

Bar mill products, depending on the dimensions of round bar sections, fall into the following classes: wire, the so-called wire rod (5-9 mm), light sections (8-25 mm), medium sections (20-100 mm) and heavy sections (80-300 mm).

As to its purpose the sheet and plate steel may be classified as follows: boiler steel, ship plate, firebox steel, automobile body sheet, transformer sheet, black and tinned plate, etc. The production of all types of cold-rolled sheet steel has been substantially increased in the last few years.

Pipes and tubes may be seamless and welded.

Great progress in energetics, chemistry, petroleum and gas industries has brought about an immense increase in the production of pipes and tubes as compared to the overall output of rolled products. According to their purpose pipes and tubes are classed into gas pipes, boiler tubes, pipes for oil pipe lines, structural tubes, etc.

The output of welded and cold-rolled thin-walled pipes made of high-alloy and stainless steel is on the increase.

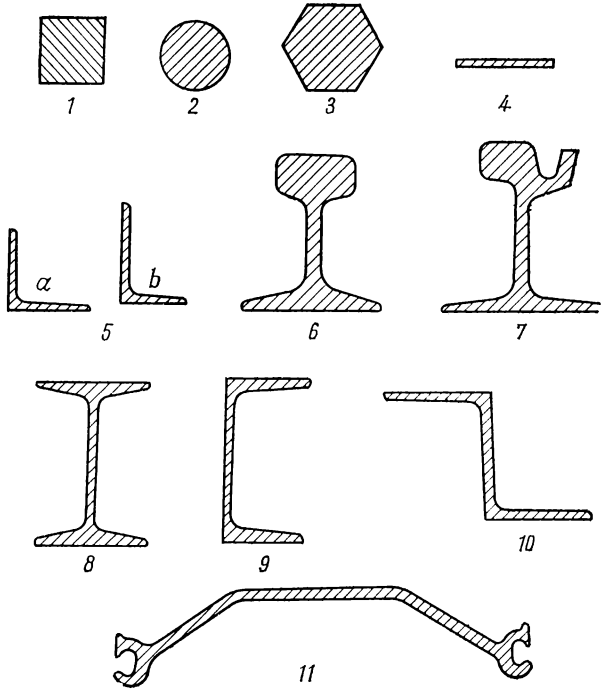


Fig. 2.1. Some types of rolled sections

1—square; *2*—round; *3*—hexagon; *4*—band; *5*—angle (*a*—equal angle; *b*—unequal angle); *6*—railroad rail; *7*—street-car rail; *8*—I-beam; *9*—channel; *10*—Z-bar; *11*—pile

Shaped pipes and pipes of variable cross section have gained wide application.

Rolled products are made of different grades of steel which are classified according to their chemical composition, methods of production, purpose and special properties.

The following types of steel are known: ordinary, high-grade and special-purpose carbon and alloyed steels. On the

basis of carbon content carbon steel may be classified as low-carbon (less than or equal to 0.3%C), medium-carbon (from 0.3 to 0.65% C) and high-carbon (more than 0.65% C). There are distinguished low-alloy steel with a total content of alloying elements of up to 4%, medium-alloy steel with the content of alloying elements in a range from 4 to 8% and high-alloy steel where the content of alloying elements is in excess of 8%.

2.2. Classification of Rolling Mills

A combination of processing machines and auxiliaries intended to produce a definite range of rolled products is called a rolling mill.

The complete rolling-mill set includes the main units which actually effect plastic deformation of a metal in the rolls (rolling proper) and accessory mechanisms which service the rolling process and also provide for transferring, shearing and finishing of the products.

Rolling mills are equipped with heating and heat-treating furnaces, intermediate heating installations and units for hardening (heat-treating) by using the rolling heat.

The line, in which the main equipment is arranged, is called the rolling-mill roll line. The principal units of the rolling-mill roll line are the working stand, transfer mechanisms and motor.

The mill rolls which actually effect plastic deformation are installed in the working stand.

The working stands may be provided with individual or group drives. The rolls of each working stand may also have a common or individual drive.

Rolling mills are characterized by a great variety of designs and processing features.

Depending on the character of plastic deformation, three types of rolling mills are distinguished, *viz.* longitudinal, cross and helical rolling mills. The longitudinal rolling has gained the widest application. In accordance with the working stand designs (the arrangement and number of the mill rolls), rolling mills are classified into the following types.

Two-high rolling mills. The mills have their working stands equipped with two rolls (Fig. 2.2). As to the character of

rotation of the mill rolls, one distinguishes the two-high reversing mills (rotation of the rolls is reversible) and two-high non-reversing mills (rotation of the rolls is non-reversible).

Three-high rolling mills. They have three rolls in the working stands and rotation of their rolls is non-reversible (Fig. 2.3).

Double two-high rolling mills (double duo mills). These mills differ from the three-high rolling mills in that their working stands have four rolls arranged in pairs at different levels (Fig. 2.4).

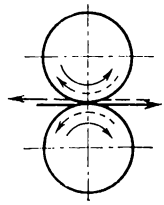


Fig. 2.2. Roll arrangement in two-high mill stand

Alternate two-high rolling mills. As distinct from the three-high rolling mills, in these mills each stand in a single line has one pass. In adjacent stands the top or bottom roll is alternately lacking, for it is not required, and coupling spindles are installed in place of the lacking rolls so as to transmit rotation to the rolls of the adjacent stand.

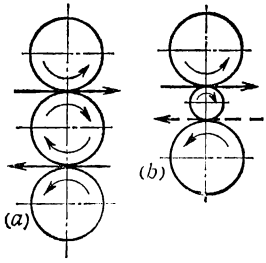


Fig. 2.3. Roll arrangements in three-high mill stand
(*a*) section mill; (*b*) sheet mill (Lauth mill)

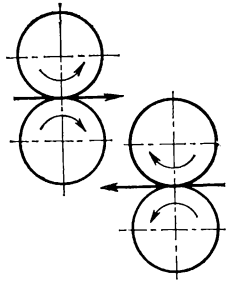


Fig. 2.4. Roll arrangement in double two-high mill stand

Universal rolling mills. The mills comprise a combination of horizontal and vertical rolls (Fig. 2.5). The vertical rolls may either be mounted together with the horizontal rolls in the same stand or installed in an individual vertical roll unit. The universal rolling mills are used for rolling flat blanks (slabs), wide strips with rolled edges (universal shape) and H-beams (Grey mills).

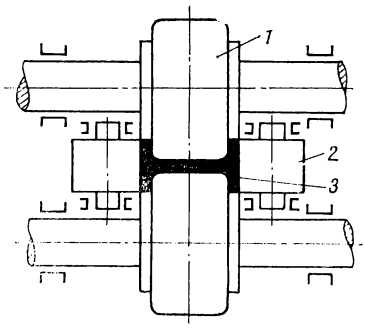


Fig. 2.5. Roll arrangement in universal mill stand
1—horizontal rolls; *2*—vertical (edging) rolls; *3*—workpiece

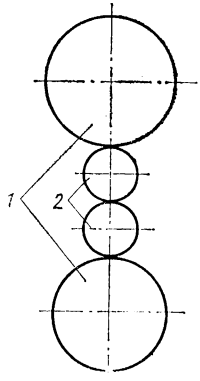


Fig. 2.6. Roll arrangement in four-high mill stand
1—backup rolls; *2*—work rolls

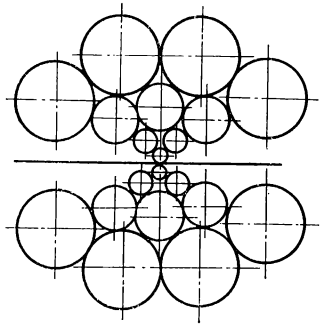


Fig. 2.7. Roll arrangement in twenty-high reversing cluster mill stand

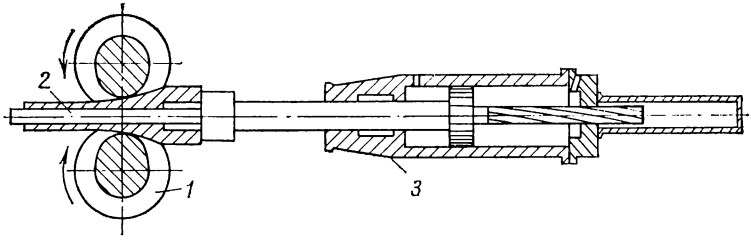


Fig. 2.8. Schematic representation of pilger rolling process
1—pilger rolls; *2*—mandrel; *3*—feeding mechanism

Vertical roll units (edgers) are widely used in modern billet, section, sheet and plate rolling mills.

Multiroll mills. There are four-high or quarto mills (Fig. 2.6), six-high cluster mills, twelve-high cluster mills, twenty-high cluster mills (Fig. 2.7) and others.

The emergence of multiroll mills is consequent to the use of work rolls of a small diameter. The minimum diameter of the rolls is limited by the permissible flexural stresses. Therefore in all designs of multiroll mills the small-diameter work rolls are supported by large-diameter rolls (back-up rolls).

Mills with backward movement of metal (pilger mills). The rolls of these mills have the working pass groove only along a portion of their periphery (Fig. 2.8). During each revolution of the rolls a metal billet (shell) is forced backward, i.e. towards the entrance of the working stand, the forward feed of the shell being effected by a special feeding device when the roll grooves form an idle pass.

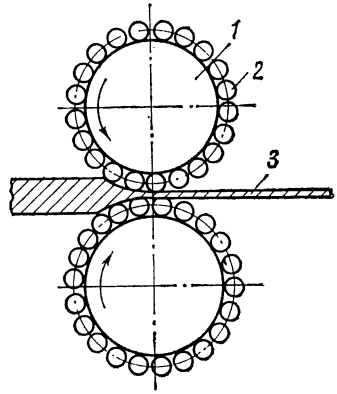


Fig. 2.9. Roll arrangement in planetary mill stand
1—backup rolls; *2*—work rolls; *3*—workpiece

Planetary mills (Fig. 2.9.) In these rolling mills a steel slab is reduced by a large number of idle planetary work rolls. The small work rolls installed in cages are friction driven by two driving back-up rolls.

Three fields, specialized according to the type of the manufactured products, have taken shape in the rolled metal production, *viz.* bar and shape production, plate and sheet production and pipe and tube production. The rolling mills are accordingly called bar and shape mills, plate and sheet mills and tubular mills. Besides, there are mills which are specially designed for rolling wheels, tyres, axles, balls, etc.

The following classes of rolling mills are known: cogging mills (blooming and slabbing mills), billet mills for rolling

semi-finished products (blooms, slabs) and mills for manufacturing finished rolled products, *viz.* heavy, medium and light section mills, beam and rail mills, skelp mills, wire and wire-rod mills, hot- and cold-rolling heavy-plate, medium-plate and sheet mills, piercing mills, pilger mills, reeling mills, plug mills, sizing mills, reducing mills, mills for welded pipe production and cold-rolling tubular mills.

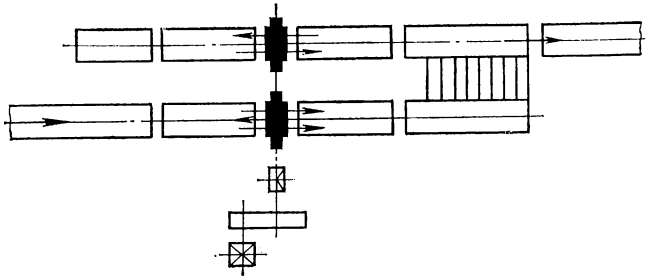


Fig. 2.10. Layout of single-line three-high mill train

The bar and shape mills are classified according to the diameter of rolls, the plate and sheet mills, according to the width of the rolled flat products or according to the length of the roll body and the tubular mills, according to the diameters of tubular products.

As to the arrangement of the working stands, rolling mill trains may be classified into the following types.

Single-line mill train (Fig. 2.10). Such an arrangement is obsolete. The main disadvantage of the single-line mill train lies in the fact that the rolling speed in all the stands is the same. In modern rolling practice the rolling speed must be increased as the metal proceeds from the first passes to the next ones and its section is reduced and the length, increased.

Two-line mill train (Fig. 2.11). This arrangement appears as the first stage of the rolling speed differentiation. Setting up the cogging stand as an individual line was done in the first half of the last century.

Multistage mill train (Fig. 2.12). To obtain the highest possible rolling speed in the last stands and thus to increase the rolling mill output, the final line is divided into seve-

ral stages (up to four). In this case, in the last (finishing) stands of old hand wire mills the rolling speed reaches 9 m/s.

Continuous mill train (Fig. 2.13). In the continuous mill train each stand represents an individual line, but the rol-

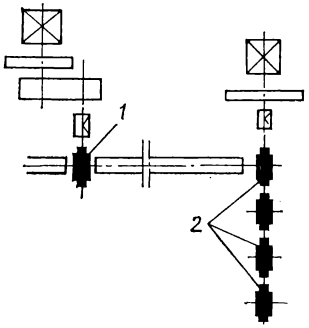


Fig. 2.11. Layout of two-line three-high mill train
1—cogging line; *2*—finishing line

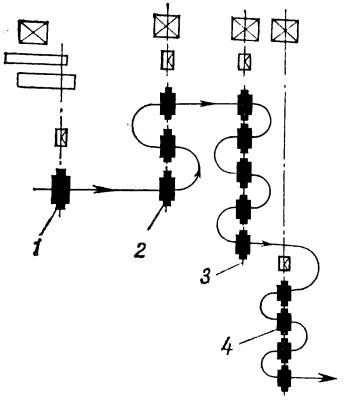


Fig. 2.12. Layout of multistage mill train
1—cogging stand; *2*—roughing line; *3*—first finishing line; *4*—second finishing line

ling direction remains constant. The rolled piece passes through the entire continuous group of stands successively and is deformed simultaneously in several stands.

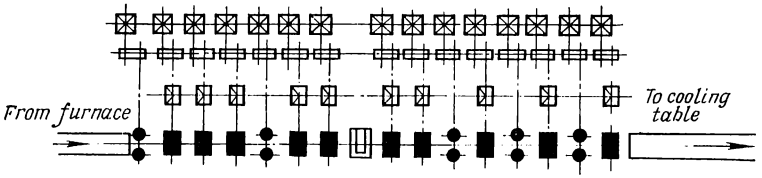


Fig. 2.13. Layout of continuous mill train

Increasing the rolling speed in subsequent stands is indispensable for the continuous mill train. Besides, a strict ratio between the rolling speeds in the consecutive stands

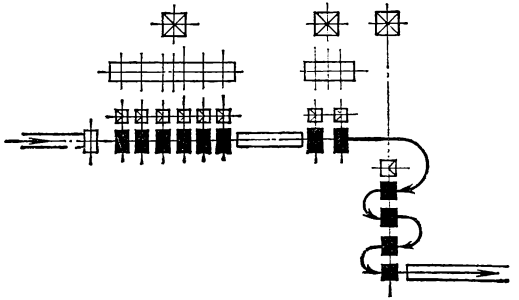


Fig. 2.14. Layout of semi-continuous mill train

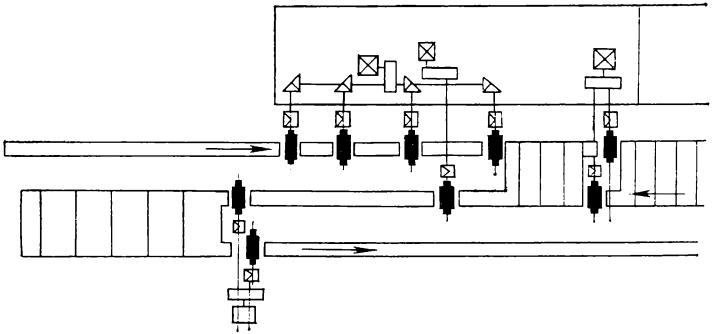


Fig. 2.15. Layout of cross-country mill train

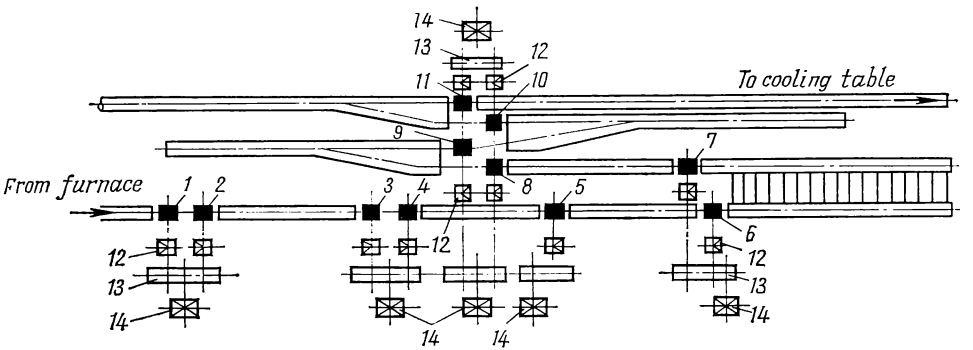


Fig. 2.16. Schematic layout for 350 mm staggered mill

1-4—working stands with rolls 450 mm in diameter; *5-7*—working stands with rolls 400 mm in diameter; *8-11*—working stands with rolls 350 mm in diameter; *12*—pinion stand; *13*—speed reducer; *14*—motor

is required, which should be proportional to the change in the cross section of the rolled piece in each stand.

If in addition to a continuous mill train there are other trains (two-high, four-high reversing or three-high) in a mill, such an arrangement is called a *combination* (*semi-continuous*) *mill train* (Fig. 2.14).

When individual stands are arranged in tandem but the rolled piece is not deformed simultaneously in two adjacent stands, i.e. the head end of the piece enters the subsequent stand only after the tail end leaves the preceding one, such an arrangement is called a *tandem mill train.*

To reduce the length of the rolling-mill shop building and to use the floor space more efficiently, tandem mill trains with a zigzag arrangement of the stands (the so-called *cross-country mill trains*) are used (Fig. 2.15). *Staggered mill trains* (Fig. 2.16) have also found application.

The continuous mill train is the most advanced contemporary arrangement.

A new method of continuous rolling, *viz.* endless rolling, is being introduced. In this case, the rolled pieces are butt-welded and the rolling is conducted without interruptions. The advantage of such a method consists not only in higher output but also in eliminating possible shortcomings associated with the rolls of each stand biting the head end of the piece.

CHAPTER III. MAIN PATTERNS OF THE ROLLED METAL PRODUCTION

3.1. Structure of Iron and Steel Production

Rolling is the final stage of iron and steel production. Inasmuch as the range of rolled products is greatly diversified, iron and steel works may have several rolling-mill shops equipped with various rolling mills in accordance with the rolled product range and output.

The iron and steel works includes three principal plants whose production processes make up the so-called metallurgical cycle, *viz.* ironmaking or blast-furnace plant, steelmaking plant and rolling plant.

Depending on the adopted structure, among the principal production departments of an industrial metallurgical complex may be the coke-oven department together with chemical by-product shops and the mining unit which in addition to the mining of ore and non-metal minerals also covers the preparation of raw materials for smelting (ore beneficiation and sintering).

The iron and steel works also includes a power-supply unit as well as a repair department and auxiliary services (heat, oxygen, compressed air and water supplies).

The structure of the iron and steel works may be integrated (ironmaking, steelmaking and rolling) or semi-integrated (steelmaking and rolling).

There are two ways of making the metal to be rolled.

The first, most popular one provides for the ingots to be fed mostly in a hot state into the soaking pits of blooming or slabbing mills. On heating, the ingots are rolled into semi-finished product, i.e. the final rolling mill feed.

The second differs from the first one in that it envisages the use of continuous casting plants without cogging mills but in some cases the cast semi-finished product is rolled in cogging or billet mills.

Some plants continue to use the old production process, *viz.* steel is cast in the shape of rather small ingots which are directly rolled into finished products.

When rolling a number of steel grades, the ingots are cooled and surface conditioned and then reheated and rolled into feeds (blooms, slabs). Surface conditioning of hot ingots is also used.

The notion of a particular structure common to plants producing high-grade steel as the one characterized by the use of insufficiently mechanized, slow-speed, labour consuming rolling mills has become obsolete. The output of high-grade steel and its field of application have expanded to such a degree, that completely integrated high-grade steel works equipped with up-to-date rolling mills are not an exception.

3.2. Ingot

All types of rolled products are made either from ingots or from continuously cast billets supplied by continuous casting plants.

The quality of ingots or continuously cast billets determines to a great extent the quality of the rolled products. The ingots and continuously cast billets have much in common.

A steel ingot is heterogeneous as to its density, chemical composition and distribution of non-metallic inclusions, this being due to the effect of various processes which accompany the ingot formation, when the metal transforms from liquid to solid state.

The mass and dimensions of the ingot are determined by the following factors: the capabilities of cogging mills and heating facilities, the requirements of rolling operations and the characteristics of the given grade of steel.

With further development of steelmaking and rolling, a tendency to increase the ingot mass is observed. The weight of each carbon steel ingot for making sectional shapes reaches ten and more tons, and that of ingots for making plates—forty and more tons. The weight of continuously cast billets is determined by the casting speed and optimum draught in rolling.

Ingots have different shapes of their cross sections, *viz.* square, rectangle, round and polygon.

The parameters of ingots (the ratio of the cross-sectional dimensions, height, taper, head size, shape of bottom and configuration of side faces) are chosen on the basis of the conditions of casting, heating and rolling and also on that of the requirements imposed upon the quality of products.

According to the character of deoxidation steel can be rimmed, killed and semi-killed.

Rimmed steel is an incompletely deoxidized steel whose crystallization in ingot moulds is accompanied by gas evolution (rimming action). Fig. 3.1 illustrates rimming ingots.

Killed steel is a steel containing small amounts of oxygen after deoxidation (Fig. 3.2). Such an oxygen content makes it impossible for the oxygen to interact with the carbon of metal during teeming, and solidification (crystallization) of steel occurs without rimming.

Semi-killed steel holds an intermediate position between rimmed and killed steels with regard to the degree of deoxi-

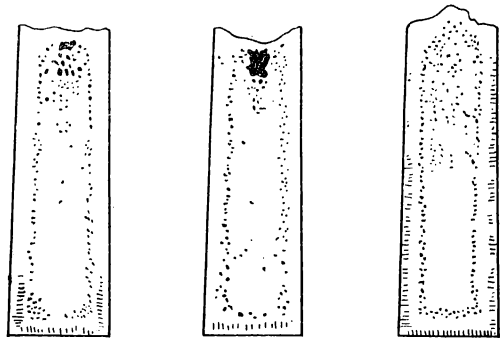


Fig. 3.1. Rimming steel ingots: (*a*) normal rimming, (*b*) violent rimming, (*c*) inert rimming

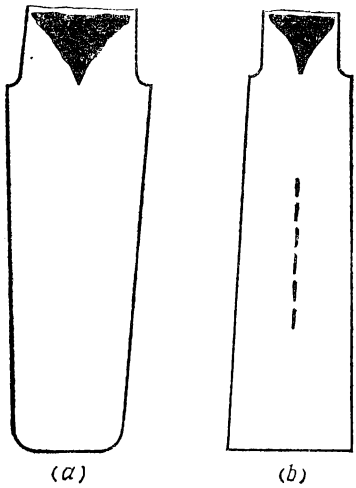


Fig. 3.2. Killed steel ingots: (*a*) big-end-up hot-topped; (*b*) big-end-down hot-topped

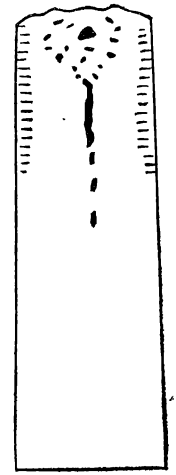


Fig. 3.3. Semi-killed steel ingot

dation. The absence of a violent rimming of steel in ingot moulds makes for a better homogeneity of the chemical composition of metal along the height of the ingot (Fig. 3.3).

In the course of solidification of killed steel a shrinkage cavity or pipe is formed in the upper portion of the ingot, which is cut off during rolling.

The killed ingots are most commonly of the big-end-up type which provides better conditions for their crystallization. A hot top with refractory lining is installed on the ingot mould. It has a taper which is opposite to that of the ingot. In the course of cooling of the bulk of the ingot conditions are created in the hot top for the metal to remain liquid for a longer period of time.

Shrinkage defects appear as the result of the decrease in the volume of the metal during solidification. Among the shrinkage defects are the zonal cavity, scattered and axial porosity and also fissures and cracks.

The formation of porosity can be minimized or even prevented by appropriately selecting the volume of the hot top, thus providing for the solidifying ingot to be fed with molten metal.

The characteristic features of the process of crystallization of a steel ingot are described below. A fine-crystalline skin zone is formed at the moment when the first portion of molten metal comes into contact with comparatively cold mould walls. Columnar crystals then begin to form. As this process proceeds the heat transfer to the mould walls sharply decreases. With decreasing temperature of steel a third zone, that of randomly oriented crystals, develops. With poor heat removal, these crystals cohere unstably with the already frozen metal, detach easily and drop down, thus forming the so-called precipitation cone.

Favourable conditions for the formation of axial porosity are created during the next period of a rapid crystallization, when the crystals precipitate, the temperature drops and the feeding of liquid metal by the hot top is already impaired. The formation of axial porosity is difficult to prevent. Porosity in the core zone of a killed ingot spreads almost along the entire height of the latter.

Welding up of axial porosity in rolling depends on the degree of reduction, the temperature and the presence of non-metallic inclusions.

The formation of scattered porosity is linked with the development of the primary zonal shrinkage cavity and

that of the columnar crystals. The greater the zonal cavity, the lesser the extent of scattered porosity, and the larger the columnar crystals, the greater the spread of this type of porosity.

Scattered porosity can be reduced by increasing the degree of deformation (draught in rolling).

The internal fissures can be of two types, *viz.* large-size single bursts (pigeon holes) and small axial fissures.

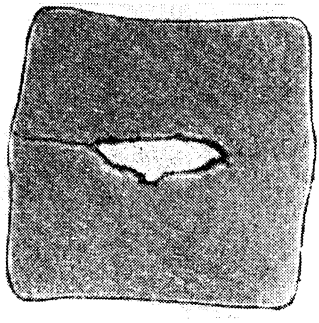


Fig. 3.4. Internal axial burst ('pigeon hole')

Large-size single bursts (Fig. 3.4) may appear in the ingot as the result of the action of thermal stresses occurring in the course of a rapid cooling or heating.

Axial fissures injure the core zone of large ingots. The fissures are also called intercrystalline shrinkage cracks. They also appear during the crystallization of some alloyed steels containing about 4 per cent of nickel. On cross templates these fissures have a specific appearance of the so-called 'spiders'. Their origin is not always explainable.

In practice, the formation of axial fissures is prevented by placing the ingots in soaking pits at higher temperatures and by increasing the degree of deformation.

The ingot surface cracks may be hot or cold. The former appear during solidification and the latter, during further cooling down to normal temperature.

Hot cracks may be transverse or longitudinal. The transverse cracks appear as a result of the hanging of the ingot

in the mould while cooling. This defect may be associated with the flowing of the liquid metal into the gap between the hot top and the mould or into erosion fractures or local recesses in the mould walls.

Longitudinal cracks appear (mainly in the ingot corners) in the course of solidification of the ingot skin. When cooling, the ingot skin contracts and forms an air gap between itself and the mould wall. Being thin the skin sometimes may not withstand the ferrostatic pressure and may rupture. In this case molten steel will flow into the formed fracture and solidify at the mould wall. This portion of steel does not weld to the skin formed earlier due to oxidation of the latter.

Cold longitudinal cracks appear (mainly in the ingot faces) when after casting the ingot is cooled insufficiently slowly. Tensile stresses arising during rapid cooling may accumulate and produce cracks, if metal lacks cold ductility. This may occasionally occur even during storage. Placing hot stripped ingots in soaking pits and increasing the time of metal cooling in ingot moulds, warm pits or furnaces may be used as measures to prevent the appearance of this defect.

Cold thermal cracks appear in the ingots of steels the structure transformations of which are accompanied by the development of high internal stresses (silicon, chromium-manganese and silicon-manganese steels).

The heterogeneity of the chemical composition of the killed ingot (segregation of impurities) results from selective crystallization the mechanism of which is such that crystals of purer metal, having a melting point higher than that of the remaining liquid phase enriched with impurities, solidify first.

Segregation is less pronounced in ingots of a smaller weight. Zonal and axial segregation are distinguished.

Checking of the chemical composition of an ingot along the entire height reveals its segregation heterogeneity. Thus in a big-end-up ingot fluctuations of the content of a number of elements are detected within the following limits: 0.5-1.1% C, 0.01-0.05% S and 0.02-0.04% P. The enrichment of metal by impurities proceeds more intensively in middle sections of the ingot and augments towards the ingot head.

Therefore in practice zones commonly designated by letters *A*, *B*, etc., are conventionally distinguished along the height of an ingot (along the length of a rolled strip).

When employing the method of bottom pouring, an increased content of impurities is observed at the extreme bottom of the ingot. After heating the ingot the zonal and axial segregation are almost completely eliminated, thanks to diffusion. To provide better conditions for the progress of diffusion appropriate heating cycles with an increased time of holding (soaking) of the ingots are selected.

Dendritic segregation is concentrated in the zone of columnar crystallization. The dendritic segregation of elements, such as manganese, silicon and nickel is commonly observed.

The billets produced by continuous casting are characterized by the homogeneity of composition along their entire length. Besides, such billets have no shrinkage defects, which is a significant advantage of the continuous casting method.

As a rule, the rimmed ingot is of the big-end-down type. The ingot is practically free from pipe. This markedly decreases cropping. The croppage from the killed ingot reaches 21%, while that from the rimmed ingot is only 3-5%.

Rimmed ingots are characterized by the presence of blowholes. If the skin blowholes do not extend to the surface of the ingot during heating and rolling, they, as a rule, weld up easily.

The carbon content of rimmed steel is not higher than 0.3%. The weldability of blowholes is restricted by this limit of the carbon content. Ductile steel ingots can be heated to higher temperatures which ensure good weldability of blowholes. Fusion of the ingot skin during heating should be avoided.

Among the disadvantages of rimmed steel is high heterogeneity of its chemical composition, structure and mechanical properties. This stems from profuse evolution of gases during ingot cooling.

A rimmed ingot has a solid surface skin, a zone of subsurface honeycomb blowholes of roughly cylindrical shape, a solid zone between the subsurface and secondary deep blowholes, a zone of secondary blowholes and a central zone or the ingot core.

Semi-killed steel poured into big-end-down moulds features small losses with crops and has a better homogeneity of chemical composition along the height of an ingot. The use of the big-end-down moulds without hot tops greatly decreases labour consumption in preparing ingot mould trains.

Rimmed ingots with higher homogeneity of chemical composition can be obtained by limiting the duration of the metal rimming in the mould. This can be accomplished by cooling intensively the mould top with water and by using heavy caps, bottle-top moulds and chemical capping of the ingot (when rimming ceases because of complete deoxidation of the ingot head by aluminium).

Three methods of teeming are known: top teeming, tundish or basket teeming and bottom teeming. The latter two methods provide a higher quality of the ingot surfaces, but they are more laborious than the former one.

Continuous steel-casting plants find an ever increasing application in the practice of iron and steel production. While continuous steel-casting plants were initially used for producing small lots of steel, lately large up-to-date iron and steel works have been built where the whole bulk of steel to be rolled is produced by continuous casting without the use of cogging mills. Continuous casting of flat steel billets (slabs) of a large weight for high-output plate mills now finds the widest application. As far as the quality of microstructure and surface condition are concerned, good characteristics are achieved by continuous casting of killed carbon and alloy steels.

Continuous steel-casting machines (machines for continuous founding) are made vertical (Fig. 3.5*a*) or curved (Fig. 3.5*b*, *c*). The bow-type machines (Fig. 3.5*c*) are preferable to organize up-to-date iron and steel production.

The principle of operation of continuous steel-casting machines is as follows. Molten metal is poured from a casting laddle into a tundish and then, into a mould. From the mould, where a constant level of liquid metal is maintained, the continuous casting being formed passes through a secondary cooling zone and enters the withdrawal rolls. The continuous casting is cut to a specified length by means of a flame-cutting device.

On passing through the mould, the continuous casting must have its skin sufficiently thick and strong to withstand the ferrostatic pressure. Crystallization must end completely during the secondary cooling. As regards possible distortion in the shape under the action of the ferrostatic pressure and the tendency to develop internal fissures and shrinkage defects, a rectangular (flat) shape of a continuous casting is the most favourable one. A higher rate of crystallization

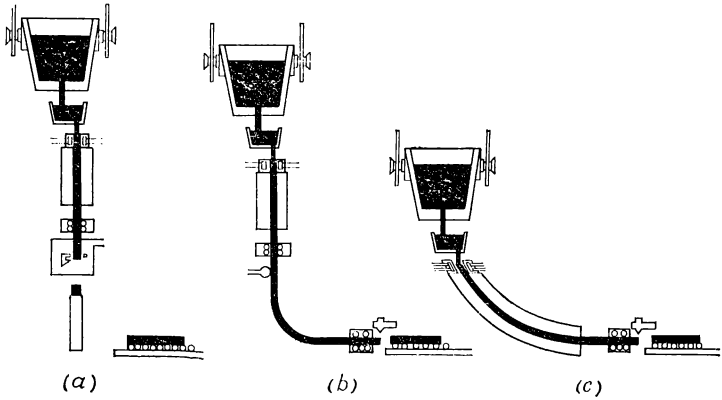


Fig. 3.5. Schematic representation of continuous casting of steel (*a*) vertical-type machine; (*b*) vertical-type machine with bending; (*c*) bow-type machine

and smaller sections of continuous castings as compared with conventional ingot helps obtain fine-grained structure and reduce the degree of segregation of impurities.

Homogeneity and fineness of grain are particularly characteristic of killed carbon and alloy steels.

An increased rate of crystallization of rimmed-steel continuous castings mitigates the development of segregation, but does not help obtain a sound outer skin.

In conventional and continuous casting steel rimming proceeds differently. The intensity of rimming increases parallel with filling of the ingot mould. In continuous casting plants, the intensity of rimming is controlled by selecting optimum heat conditions and composition of steel, blowing oxygen on the stream of molten steel, etc.

The economic efficiency of the continuous castings (the yield, outlay and maintenance cost) is rather high under any conditions.

The stability of production processes at all stages of the metallurgical conversion is decisive in ensuring high quality of finished rolled products.

The quality of metal is greatly influenced by the time of its holding in the ingot moulds, the conditions of the metal transportation, the time of removing the hot tops, the temperature of the ingots when placing them into soaking pits, etc. Of great importance are also the condition of the ingot moulds and hot tops, the quality of the mould coating, the condition of the hot-top lining, the temperature of the ingot moulds during coating, the conditions of the ingot mould train circulation in the section steelmaking shop — stripper yard—soaking pits, and the organization of the quality control.

3.3. Preparation of Metal for Rolling Process

Preparation of ingots and billets for rolling consists mainly in the removal of surface defects. The surface defects are most commonly removed from semi-finished products, *viz.* square and flat billets (blooms, slabs). In most cases the ingots are delivered hot to the heating installations of cogging mills and therefore in common they are not subjected to surface conditioning practice. In a number of cases the ingots of some particular grades of steel are conditioned after cooling and sometimes, when hot.

The removal of surface defects is a very laborious process. A great number of workers are engaged in the metal conditioning and hence decreasing the labour consumption in the conditioning operation is a very important factor for increasing the operating efficiency in iron and steel industry.

The most characteristic surface defects, which are subjected to mending, are cracks, scabs and seams. Pickling and various flaw detection methods are used for detecting concealed defects.

The depth of defects is determined in the course of surface conditioning. When conditioning a plain carbon metal, shallow flaws which will not spoil the final products during

further processing are commonly not removed. The classes of metal conditioning are determined by the specifications of iron and steel works, depending on the grade of steel and on the dimensions, shapes and purpose of the rolled products.

Surface conditioning is carried out with pneumatic chipping hammers (chippers), milling heads, and emery wheels. Scarfing with oxyacetylene cutting torches or with torches of other types is also used.

Not so long ago chipping of defects with hand pneumatic chipping hammers was the most popular method of surface conditioning. This made the iron and steel works have a large number of chippers on the staff and large areas of benches equipped with air manifolds and hoisting facilities.

Concurrently with the introduction of milling and planer-type machines for cold or hot surface conditioning, hand scarfing by means of oxygen cutting torches of various designs has found wide application.

New iron and steel works use scarfing machines which continuously scarf the piece surfaces after rolling. In this case the surface layer of metal is removed by fusion.

Continuous hot scarfing is also introduced as a means to control decarburization of the billet surfaces. Scarfing of this type is used after reheating the semi-finished products of spring and ball-bearing steels prior to rolling.

It is recommended to scarf the continuously cast rimmed-steel slabs with blowholes located near the surface after preliminary cogging in which the blowholes are welded up.

When carrying out metal conditioning, precautions are taken to exclude the possibility of formation of new defects as the result of the shape irregularity of the conditioned surface areas.

A depth limit is set for the removal of local defects in surface conditioning (commonly the depth of surface conditioning should not exceed 10% of the nominal thickness of the billet). The ratio of width to depth of chipping must be not more than 6-8 with the grooves having smooth roundings in the corners.

Hand scarfing is done by herringboning or by making parallel grooves without sharp ridges and recesses. The fused metal and slag are removed after the defects have been

scarfed. To improve the quality of scarfing the degree of oxygen purity is of vital importance.

When scarfing cold steel the surface layer of metal up to 4-5 mm deep is heated and then rapidly cooled with local stresses arising therein. In the billets of carbon and alloy steels the development of cracks is possible. Scarfing of alloy and high-carbon steels is done with preheating, whose

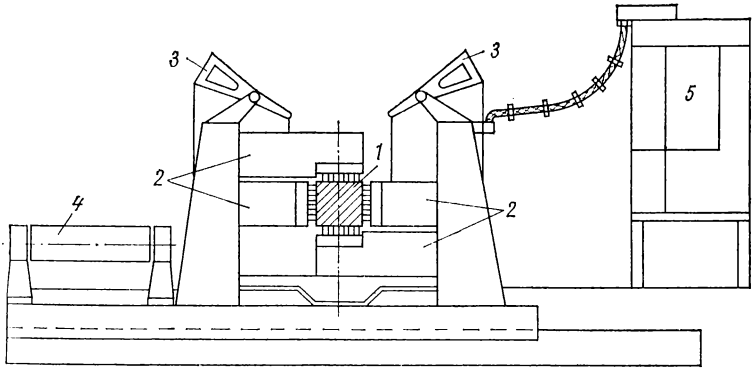


Fig. 3.6. Flame scarfing machine

1—bloom; *2*—gas-cutting blocks with torches; *3*—block balance arms; *4*—roll table; *5*—operator's pulpit

temperature depends on the content of carbon and alloying elements. The preheating temperature is usually in the range of 200-400 °C.

At present scarfing machines (Fig. 3.6) gain an ever increasing popularity. They are installed between cogging stands and shears or between continuous billet mill trains.

Depending upon the required surface finish, different depths of scarfing, economically justifiable as to the iron losses, are set for various grades of steel and billet sections. Iron losses in scarfing are most often allowed to reach 3%, but in individual cases they may be as high as 5%. The scarfing depth which is dependent on the speed of the metal movement and oxygen pressure can be as great as 4 mm for the largest sections.

When scarfing alloy steels, iron powder is occasionally fed into the flame of the cutting torches to maintain and

control the thermal parameters of the process (exothermal reaction.)

Among the present-day methods of surface conditioning is the milling of hot ingots and billets.

Into use comes machine conditioning of billets on fully mechanized and automated flow lines (Fig. 3.7) installed in billet bays. In this case all kinds of surface conditioning facilities such as milling, planing, turning, high-speed

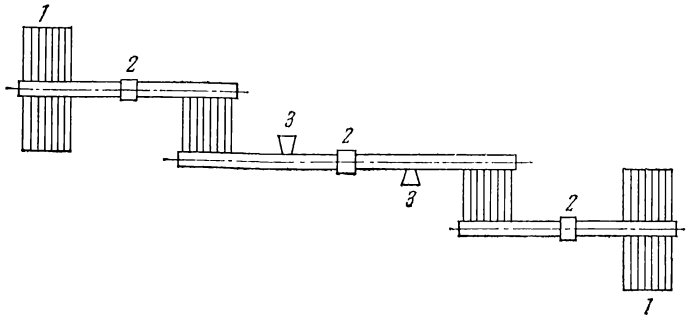


Fig. 3.7. Layout of conditioning flow line
1—billet racks; *6*—tilters; *3*—conditioning facilities

grinding and scarfing machines can be used. Automated magnetic or electromagnetic flaw detection is carried out simultaneously.

Shot-blast units are used along with pickling for descaling the surfaces of semi-finished products.

Surface conditioning by grinding is used on a wide scale, particularly at works producing high-grade metal. Alongside swing-frame grinders of simple designs, inefficient and labour-consuming, stationary highly mechanized grinders find an ever increasing application. At the same time improvements are introduced into the designs of centerless turning and grinding machines. Abroad, to raise the efficiency grinding of high-grade metal heated to 700-800 °C is used. The quality of grinding wheels specially made to handle different grades of steel and alloys is of great importance.

Central shops are set up at iron and steel works to properly manage the grinding tool service. Their task is to prepare, check and store grinding wheels, to study the durability

and service conditions of the wheels and also to collect and utilize the grinding waste.

The dynamic strength of the wheels must be ensured to provide for the safety of labour. This is checked both by the manufacturer and by the user prior to setting the wheel for operation. High-speed operation (up to 60-70 m/s) necessitates applying high-strength abrasives (flexural strength over 30 kgf/mm^2, reliable water- and heat-resistant binder with reinforcement). Of fundamental importance are also precise shape and centring of the grinding wheel.

In contrast to chipping grinding is carried out in transverse and not in longitudinal direction. Mild steels are ground with harder wheels.

The introduction of enclosed-type highly mechanized and automated machines operating at high speeds has sharply stepped up the efficiency of grinding.

The use of scarfing machines has made it possible to avoid surface conditioning of over 30% of ingots going to the adjustage.

In spite of additional iron losses (up to 2%), the use of scarfers in case of four-sided scarfing is economically justifiable.

The preparatory operations also include stacking of semi-finished products and their controlled cooling.

Stacking of semi-finished products should be organized so as to ensure uninterrupted delivery of metal to the rolling mills in accordance with the preset rolling schedule.

The areas for storing the semi-finished and finished products are selected taking into account the product flows in the bays and operations of cranes which serve the metal conditioning.

In some instances hot metal which does not require any conditioning is directly passed from cogging to rolling mills by means of crab cranes, roll tables and other transport means.

Organization of rational metal storage also envisages the keeping of a continuous stock account, marking and assorting of the metal according to the heats and, in some instances, even according to the letters assigned to each billet which identify the zones along the ingot height. The complicacy of such an accounting necessitates the application of

modern computation means and requires highly skilled personnel.

Retarded cooling of semi-finished products made of medium-carbon, high-carbon and alloy steels is carried out in piles, slow-cooling boxes, pits and special furnaces, in order to prevent the formation of small internal ruptures, called flakes.

According to the accident prevention regulations, billet and ingot yards are located in enclosed buildings having sufficient floor space and adequate aeration to provide for the cooling of the metal. The piles of metal should be carefully stacked. If the piles are stacked on racks, the latter should be checked for strength and horizontality. Depending on the available means of mechanization storing is permitted in piles of the following maximum height: 2-3 m for large ingots intended for rolling in blooming mills; 1.5-2.5 m for short billets (with a length of not more than 5 m); 4 m for longer billets; 2-3 m for slabs. The width of the passages between the piles must measure not less than 1 m for cold metal and 1.5 m for hot metal. The passages between trains and piles must be not less than 1 m. The main passages must measure 1.5-2 m in width.

To eliminate surface defects accident prevention regulations in force stipulate the use of automated and remotely controlled machines and units with suction removal of gases and dust.

When inspecting and conditioning metal with a temperature above 60 °C, the metal on the racks should be handled by means of mechanical tilters.

When the metal conditioning is done by grinding, special attention is paid to the preliminary checking and testing of grinding wheels. The grinders must be equipped with sturdy protective shrouds. The personnel should wear special goggles to protect the eyes.

3.4. Heating Metal Prior to Rolling

The heating of metal (ingots, billets) prior to hot rolling is one of the main processing operations. It must ensure the best possible plastic properties of metal and high quality of final rolled products as well.

Properly chosen conditions and temperature ranges of heating make it possible to obtain the required metal structure. In heating, the metal structure becomes single-phased due to diffusion which causes redistribution of impurities and levelling out of the metal composition (homogenization).

Overheating, burning, increased iron loss, fusing, decarburization or carburization and unsound structure are inevitable if the heating conditions are wrong.

Improper heating may result in the disturbance of the metal continuity, i.e. in the formation of cracks, fissures and seams (due to the opening of subsurface blowholes).

The heating process is determined by the following two principal parameters: the temperature range and the rate (duration) of heating.

The upper temperature of heating is limited proceeding from considerations as to the grain growth and the possibility of fusion or burning. In case of high-carbon and alloy steels special care is exercised in maintaining optimum heating temperature.

The heterogeneity of distribution of impurities lowers the melting temperature. Sharp corners and ribs of billets are usually heated more rapidly and to higher temperatures. The maximum temperature of heating is set at 100-150 °C below the solidus curve in the equilibrium diagram for the particular grade of steel.

In heating as much as 90% of the total amount of heat is transferred by radiation. The amount of heat taken by a billet depends on the temperature drop and on the characteristics of the furnace gases, masonry and billet surface.

The emissivity of the products of combustion (flame) is increased by adding coal tar, fuel oil and pulverized coal to the fuel. These additions are converted into incandescent soot carbon.

The heat transfer by convection tends to increase with an increase in the velocity of the furnace gases. In flash-heat furnaces the amount of heat transferred by convection reaches 25%.

Thermal conductivity $Q_{t.c}$ depends on the difference between the temperatures of individual layers of metal $t_1 - t_2$,

surface area F, time z, heat conduction coefficient λ and height of the billet cross section h:

$$Q_{l.c} = (t_1 - t_2)\frac{\lambda \cdot F}{h} z \text{ cal}$$

The process of heat conduction is highly complicated. The heat conduction coefficient is a variable which depends not only on the properties of materials and their density but also on the temperature.

The heat conductivity of cast metal is about one half that of worked metal and the heat conductivity of high-alloy steel is twice as low as that of low-carbon steel.

The heating conditions (rate and time) are determined by the heat conductivity and ductility of metal. In most cases, the rate and time of heating are determined by the design features and thermal capabilities of the heating facilities and not by thermal and physical properties of metal.

The possibility of instantaneously raising the surface temperature and of making use of a practically unlimited heating rate in the range of temperatures above 500 °C has so far not been utilized (for low-carbon steel there exist no limitations for the heating rate even within the temperature range below 500 °C).

Thin metal sections present no limitations as to the increase in the heating rate. For ingots, especially those with an increased C, Si and Cr content, segregated spots and subjected to non-uniform heating, the limitation of the heating rate is in force.

Alloy steel ingots with a large transcrystallization zone are prone to develop internal fissures (pigeon holes) during heating and cooling in the range of temperatures below 700 °C. When heating cold high-carbon and alloy steel ingots, these are charged into cooled cells of soaking pits.

Any unnecessary increase in the time of holding metal in a furnace causes greater iron losses and decarburization, leads to additional fuel consumption and creates conditions for welding and fusing of billets in the furnace.

Oxidation depends on the heating time and temperature and also on the nature of the furnace atmosphere and chemical composition of metal. Oxidation is explained by the two-way diffusion of oxidizing gases and metal ions. The

most typical furnace scale consists of three layers, *viz.* inner (FeO), intermediate (Fe_3O_4) and outer (Fe_2O_3). The outer layer is the thinnest (10%). The scale layer on alloy steel ingots, apart from base metal oxides, also includes oxides of some alloying additions.

Most active scaling takes place at 800-900 °C and reaches its maximum at 1250-1350 °C. The main constituent gases of the furnace atmosphere are: oxidizing—O_2, CO_2, H_2O and SO_2; reducing—CO, H_2 and CH_4, and the neutral N_2. Scale is also formed in a reducing atmosphere, but in this case it is more tough.

The scale toughness depends on the chemical composition of metal and on the heating conditions. The scale layer on chromium-nickel and copper steel ingots firmly adheres to the metal.

The oxidation of heated metal is accompanied by decarburization of its surface (in a reducing atmosphere carburization sometimes takes place).

Decarburized carbon tool and spring steels may become unfit for use. These steels are heated to lower temperatures and held in the furnace during a shorter period of time.

An elevated sulphur content in the fuel causes an increased iron loss. Saturation of the surface of heated nickeliferous steels with sulphur may lead to the formation of a low-melting sulphide eutectic which ruptures and cracks in rolling.

Flash as well as scale-free and slightly oxidizing heating (fuel-air ratio is equal to 0.5-0.8) reduce oxidation and iron loss. Sometimes coating of billets is used (sodium silicate, ammonium chloride), but this method is expensive and inefficient.

According to the heating techniques and the design of heating facilities, metal may be heated in one, two, three and more steps.

Single-step heating is done at a constant furnace temperature or at a constant heat flow. This method is used for heating plates, sheet bars (billets for piecemeal rolling of sheets), tubes and single hot ingots.

Two-step heating comprises the heating proper and holding at a constant temperature. It is used for heating hot charges of all steel grades in two-zone continuous furnaces and cold charges of carbon steels in soaking pits. Two-step

heating at a constant heat flow in the first period and at a constant furnace temperature in the second period is used for sheet packs, tubes and coils.

Three-step heating is carried out in three periods. In the first period a low heating rate is maintained with an increasing heat flow. The second period is that of a flash heat with a constant heat flow. The third one is the period of soaking at a constant temperature. This cycle is adopted in three-zone furnaces, in soaking pits for cold carbon and alloy metal, as well as in flash heat furnaces.

Multiple-step heating which consists of successive periods of heating, holding and cooling is used for heat treatment.

To ensure good workability of hard-to-work steels the holding period is prolonged, this corresponding to homogenizing.

According to the weight, shape and size of heated billets, as well as to the methods of their charging, travel, heating and discharging, heating facilities are classed as follows.

Soaking pits for heating ingots may be of single- or multiple-ingot type, centre- or side-fired, regenerative or recuperative. There are also single-ingot electric soaking pits for heating special and alloy steels.

Box batch-type and tunnel-type furnaces (including car-bottom furnaces) are used for heating large ingots and billets, plates, packs, sheets, tubes and pipes as well as coils.

Continuous reheating counter-current furnaces (two-zone, three-zone and multiple-zone furnaces) with waste-heat utilization in recuperators.

In continuous furnaces the metal charge is brought forward by means of pushers. Ingots and billets are also heated in roll-down furnaces with a sloping hearth where the metal is tilted through side doors.

Rotary-hearth furnaces (*revolving furnaces*) are widely used with modern tubular mills. They are also used for heating billets in case of piecemeal rolling of thin sheets. These furnaces have a circular rotary hearth, provide good heating conditions, are compact and easy to mechanize and automate.

Walking-beam furnaces present a number of key advantages even when compared to the present-day continuous fur-

nace designs. They offer good conditions for uniform controlled heating of metal, and billets of any cross sections and lengths can be heated therein. As regards the discharge of heated metal these furnaces can be well arranged with respect to the rolling mill.

Fast-heating furnaces (*barrel-type furnaces in tandem*). Fast heating is a progressive technological trend. A sharp decrease in the duration of metal holding in the furnace makes it possible not only to obtain high output and good quality of metal but also to solve a number of key problems of rational arrangement of the processing equipment. The thermal and physical properties of the majority of steels afford a great reserve for fast heating of rolled billets, especially at temperatures exceeding 700 °C.

The main objectives of fast heating are: rapid elevation of the metal surface temperature, uniform distribution of heat flows and firing of fuel in such a way as to have proper direction of the flame and high heat power of heating facilities. These objectives can be attained best with barrel-type continuous fast-heating furnaces. The fast-heating furnace consists of a number of small sections (often detachable) arranged in tandem. Heated billets, tubes or rods travel on rollers longitudinally. Thanks to premixing of fuel gas with the air, complete fuel combustion with a slight air excess and also to an increased heat transfer by convection, high temperatures can be obtained in the furnace sections. The design and arrangement of the burners provide for symmetric heating. Use is also made of other types of fast-heating furnaces, such as electric and high-frequency induction ones.

Through-type tandem furnaces together with walking-beam furnaces operate successfully with tubular rolling mills and modern section mills.

In high quality rolled-metal production, the required properties are obtained not only by rolling itself but also through additional heat treatment (quenching, tempering, annealing, normalizing, carburizing). Depending upon the production rate and peculiarities of processing, heat treatment is done in batch-type or in continuous furnaces.

The output of reheating furnaces is directly proportional to the furnace capacity and inversely proportional to the

heating time. It also depends on the furnace capacity factor. This factor is essential for determining the output of batch-type heating furnaces (box- and cover-type furnaces and soaking pits).

The heating time is determined by the conditions of heat transfer, which in its turn depends on the design and the furnace operating conditions. When estimating the furnace outputs correction coefficients are sometimes included to take into account the characteristic properties of heated steels.

The time of heating hot ingots in soaking pits is approximately equal to:

$$S = [7 - 0.006\,(t - 200)]\,m \text{ h}$$

where t = temperature of ingot charge, °C
m constant coefficient ($m = 1$ for carbon steel)

The output of continuous furnaces is expressed as follows:

$$O_{c.f} = \frac{24C}{S} \text{ t/h}$$

where C = capacity of furnace, t
S = heating time, h

The output of soaking pits is expressed as

$$O_p = k\frac{24C}{S + \tau} \text{ t/h}$$

where $k = 0.9$ = pit capacity factor
S = heating time, h
τ = optimum time of charging and discharging ingot, h

In accordance with the accident prevention regulations heating facilities should be located so as not to impede the inflow of fresh air to the shop. Storage of metal in bays accommodating pits or furnaces is prohibited. The furnaces should be provided with reliable heat and gas insulation. Protective shields should be installed at heating facilities and work places and also in crane cabs. Remote control of mechanisms is recommended.

The crane cabs and control stations should be air-conditioned. The use should be made of television sets, sound and light signalling systems and interlocking devices. Pits,

hollows and servicing platforms should be fenced off and provided with reliable grills.

Special attention is paid to the management of gas supply systems according to the general safety regulations for gas systems and also to the routine regulations in force at the particular works or shops. This refers to the arrangement of gas valves, regulators and other fittings. The depth of the separating water layer in the water seals of gas mains and cross-over valves should be 150 mm greater than the height of the water column corresponding to the maximum gas pressure.

The gas and air valves should be operated in such a sequence as to exclude gas blasts. The premises accommodating cross-over devices should be provided with good ventilation and safe lighting.

Letting the gas in the heating facilities after their stoppages is carried out in accordance with the approved instructions. The repairs of reheating furnaces are regulated by special instructions and orders. When heating facilities are stopped for long periods of time, it is necessary to positively disconnect them from the gas supply mains by means of gates and plate chokes. In this case the gas manifolds and stand pipes should be purged with steam until the gas is expelled completely.

3.5. Main Parameters of Metal Rolling Process

The following are the main parameters of rolling: the temperature range of rolling, the rolling speed schedule and the strain distribution among passings. Being linked with one another, these parameters determine the power conditions of the process, the roll bite conditions and wear of the rolls. They also influence the dimensional accuracy of finished rolled products and their physical and mechanical properties.

The initial temperature of rolling is selected to be as high as possible for the particular composition of steel. Taking into account that the presence of segregated spots and low-melting intercrystalline substance is possible, the temperature is set at 100-200 °C below the solidus curve. Alloy steels are rolled in a narrow temperature range which

corresponds to the optimum metal ductility. Scale properties and decarburization conditions are also taken into consideration.

The final temperature of rolling for mild carbon steels is determined by the duration of rolling and the conditions of cooling. In a number of cases the final temperature of rolling is selected in accordance with the specific properties of each grade of steel and requirements imposed upon the properties of the finished rolled products. For hypo-eutectoid steels this temperature is set at 50-100 °C above point Ar_3. These steels can be rolled at a lower temperature if further heat treatment is provided for. Rolling of hyper-eutectoid steels is finished at a temperature slightly above point Ac_1, particular attention being paid to the conditions of cementite precipitation.

The temperature drop in passings and the difference between the temperatures of the head and tail ends of the rolled strip are also of great importance.

During rolling the metal temperature drops due to heat losses resulting from the transfer of heat from the rolled strip to the surrounding air, cooling water and rolls by radiation, convection and conduction. On the other hand, heat is generated in rolling, its amount being directly proportional to the amount of work done to deform the metal. Theoretical calculations of all these components do not always give reliable results, but reasonably accurate ones can be obtained by using empirical formulas.

In practical calculations the speed of the rolled strip is determined proceeding from the peripheral speed of the rolls at their working (effective) diameters. In case of more precise calculations the line speed of the metal leaving the rolls is determined, taking into account its forward slip. The choice of optimum rolling speed schedule depends on the output and power of the rolling mill, the temperature ranges of rolling, the strain distribution among passings and, lastly, on the general rhythm of the whole rolling process.

Experience has shown that a steady rhythm of operation of all the rolling mill sections (heating, rolling, handling, shearing, straightening, etc.) joined into a single conveyer system ensures the most stable operating conditions and

high quantitative and qualitative performance characteristics as well.

In some instances, when the organization of a single common process flow is technologically impossible or inexpedient, individual operations are performed separately in independent but continuous flow lines (installations for conditioning, shearing, metal plating, heat treating, etc.).

In reversing mills the roll speed in each passing first

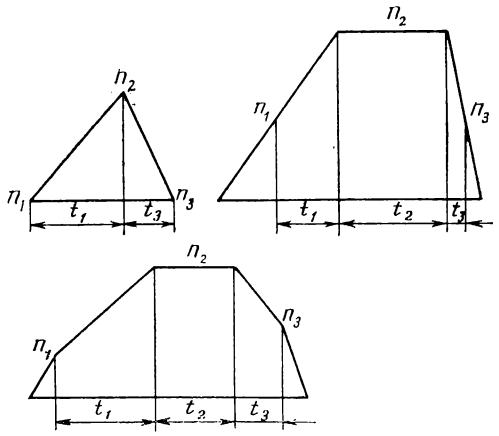


Fig. 3.8. Rolling speed diagrams

increases during acceleration from zero to the maximum value (at times constant for some period) and then decreases to zero during deceleration.

Speed diagrams (Fig. 3.8) depicting time variations of the roll speed in one passing may have the form of triangles, trapezia or polygons, depending on the character and magnitude of acceleration and on the presence or absence of the constant speed region. In rolling mills with a constant direction of rotation of the rolls, the roll speed may also vary during one passing.

The rolling speed is controlled by an operator or by means of automatic devices.

In rolling-mill stands equipped with d.c. motors transient phenomena during one passing depend on the speed-torque characteristics of the motors and on the methods of

controlling the motor speed. Especially great importance is attached to the selection of the characteristics and speed-control systems (self-regulation, compounding) of the drives for the finishing trains of continuous rolling mills. In the operation of continuous rolling mills and flow lines of continuously operating units of prime importance is the velocity pressure of the free end of the rolled strip. This pressure, which depends upon the speed, cross section and ductility (rigidity) of the strip, is of interest from the standpoint of overcoming frictional forces in the leading-in accessories of the rolling, cleaning and finishing equipment.

To achieve (reestablish) the necessary velocity pressure one has to reduce the distances between the rolling-mill stands or between other pieces of equipment, to increase the strip speed, or else to install carryover pinch rolls in the way of the free end of the strip.

3.6. Rolling-Mill Rolls, Management of the Roll and Roll Fitting Services

Since the rolls act as a working tool in the rolled metal production they determine the reliability of operation of the rolling mills and their technical and economical characteristics.

The quality of rolls is contingent upon their dimensional accuracy, surface finish, strength and wear resistance.

The rise of the output of the present-day rolling mills is also linked with the sharp increase in the roll durability which is determined by the tonnage of rolled products per roll, pass or operating area of the roll body. New rolling mills are designed to operate at a higher performance level.

The design of the working rolls should provide for their sufficient bending and torsional strength.

In multiroll mills, the transmission of torque is limited by the small diameter of the work rolls and therefore in some instances rotation is transmitted via the back-up rolls.

The rolls are made of iron or steel. Iron rolls are made of grey or partially chilled iron (soft, semihard and chilled rolls) and sometimes of iron with alloy additions. Steel rolls are cast or forged from carbon, high-carbon (hypereutectoid) and alloy steels.

Soft iron rolls made of grey and, occasionally, of alloy iron are used in low-loaded roughing stands. Hard chilled iron rolls cast from iron of a special composition, which is poured into chills providing for fast cooling of the surface layer of the metal, are used for rolling merchant and bar shapes (in case of a small depth of groove), sheet and plates.

Semihard iron rolls cast by using metal chills with a clay coating are employed for rolling heavy sections in finishing stands.

The surface hardness, the depth of the chilled layer and the distribution of hardness along the roll radius depend on the chemical composition of iron and on the melting and casting techniques.

Rolls are also made of iron inoculated with magnesium or cerium. High-strength magnesium cast iron containing up to 2.5% Ni and up to 0.04% Mg is used for increasing the wear resistance of the rolls. Sometimes a more expensive method of casting rolls is employed. With this method the metal is poured into assembled chills which provide for uniform distribution of the chilled layer over the surface of the roll grooves.

The rolls of the cogging and roughing stands of section mills, as well as the back-up rolls of hot-rolling plate mills and the rolls of cold-rolling mills, are made of steel.

Cast steel is used for manufacturing the work rolls of the roughing stands of section mills and the back-up rolls of plate mills. The section mill rolls made of high-carbon hyper-eutectoid steel, sometimes alloyed with chromium, exhibit good service characteristics. The rolls of heavy cogging mills are forged from alloy steel.

Heat-treated all-forged rolls made of steel with a carbon content of up to 0.95% and alloyed with chromium or, sometimes, with vanadium, molybdenum or tungsten, are used in cold-rolling mills which, as a rule, operate under severe conditions with high static and dynamic loads. The hardness of the quenched bodies of the back-up rolls is as high as 95 (Shore hardness) and that of the work rolls is up to 102. Stringent requirements as to the uniformity of hardness distribution are imposed upon these rolls.

Built-up rolls with steeves of high hardness are used as well. The work and back-up rolls of over 200 mm in diame-

ter for cold-rolling mills are made with axial holes to facilitate their heating before rolling and cooling during operation. The bodies and necks of the rolls are thoroughly ground (no worse than the 7th class of surface finish). This ensures high quality of the rolled plate surface and increases the wear resistance of the rolls.

Building-up and hardening of the working surface are used for the steel rolls of hot-rolling mills. The management of the roll service includes keeping a record on the operation of each roll and pass in terms of the tonnage of the rolled products; selecting roll sets for stand trains and for each stand individually, and keeping a record on them; organization of the roll storage; handling of the rolls and making preparations for roll changes; regrinding the rolls with keeping a record on the changes in the roll diameters, on their hardness and on the condition of their necks and coupling parts (wobblers or spade half couplings).

Cooling, lubrication and preparation of bearing assemblies are provided for to ensure reliable operation of the rolls. Strict adherence to the prescribed conditions of warming the rolls up prior to operation is of great importance.

The necessity for changing a pass or rolls should be determined not on the basis of an inspection of the condition of a pass or groove, but in accordance with the set schedule.

Some new methods of the roll renewal have found application. These are the electric-arc surfacing with the use of carbon or alloy electrodes or powdered carbide wire, surface hardening and hardening knurling. Knurling is performed with special rollers. The work-hardened surface layer prevents to a certain extent the formation of cracks on the working surface of the rolls caused by alternating stresses. Building-up or ragging of the rolls are used to improve the biting. Special methods of profile turning are sometimes also employed for this purpose.

Special accessories called the roll or guide fittings serve for the rolled piece to enter and leave the roll pass correctly. As to their purpose, the guide fittings may be divided into the following types.

Entry guides ensure the correct direction of the head end of the piece entering the rolls and hold the piece in the required position. Among the guide fittings of this type are

box guides, side guides, solid or roller inlet guides and inlet troughs.

Delivery guides serve to control the direction of the piece leaving the rolls after rolling.

Delivery twist guides (spreaded pipe guides, helical guides, twist rollers) are used for tilting the workpieces.

Special accessories, *viz.* troughs and repeaters, are used to pass the head end of the rolled piece from one stand to another (sometimes, from one pass of a given stand to another pass).

Among the basic requirements for the roll fittings are: sufficiently rugged design, reliable fastening allowing for the effect of high temperatures, accuracy and simplicity of setting with due regard for the changes in roll diameters and groove arrangements, absence of scratches, marks and sticking of the metal during the workpiece contacting the roll fittings.

The roll fittings are made of iron, steel, alloy and nonferrous metals. The operation of fittings is markedly improved by using roller-type devices (rolling instead of sliding friction).

Management of the roll fitting service includes replenishment and storage of a sufficient number of sets of checked components of all the fittings; replacement of old fittings by new ones; repairing and marking the fittings. Good management of the roll fitting service in the shop, correct setting of the fittings when making roll changes and going over to other shapes ensure successful operation and high quality characteristics.

3.7. Roll-Neck Bearings, Tooling and Repair Equipment

The organization of the roll-neck bearing service holds a substantial position in metal rolling. The use of assembled or all-in-one-piece chock-type composition bearings made of textolite (in place of previously employed bronze), which operate with abundant water cooling and practically without any greasy lubricants, has decreased the consumption of nonferrous metals and lubricants and diminished friction losses as well. Besides, the application of textolite has made it

possible to obtain more stable rolling conditions, increased the accuracy of rolling, simplified the adjustment and cut down the idling periods associated with changing and scraping the bearings.

Textolite (sometimes laminated birchwood or lignum vitae) is produced by pressing fillers (cotton cloth, birchwood veneer sheets, planks) impregnated with a mixture of various resins. As a result, a material is obtained with an ultimate strength of up to 10-24 kgf/mm^2 which makes it possible to have a coefficient of friction as low as 0.004-0.006.

The present-day high-speed rolling mills use enclosed film lubrication bearings, instead of roller ones. These bearings are made up of two bushings, one of them being conical, and the other, cylindrical. The conical bushing is pressed on the tapered roll neck and is located inside the cylindrical bushing. The inner surface of the cylindrical bushing is lined with a thin layer of high-tin babbit. The axial loads are taken by a thrust bearing (usually of the ball type). A high class of surface finish, centralized oil lubrication and good sealing provide for the conditions of liquid friction to be created in the gap between the two bushings (coefficient of friction is 0.003) and ensure low elastic deformations and wear under high loads. The roller bearings continue to be competitive with the film lubrication ones. They present high performance characteristics, provided their assemblies are fitted carefully, changed in due time (after expiration of the average service life set for them) and mounted correctly during the change of the rolls.

A set of rollerman hand tools usually includes sledges, hammers, wrenches, tongs, small crowbars, hand shears, loop-spliced wire ropes (slings) of suitable size, lamps for setting the passes and guide fittings, and measuring instruments (egg callipers, vernier callipers, micrometers, templates for checking the gauges of wobbler necks and guide fittings, templates for checking sections). The whole set of tools is checked regularly, stored and maintained in proper order. In addition to centralized repair shops of the rolling-mill plant or of the entire works (depending upon the scale of production), as well as to a roll shop, each rolling mill is provided with appropriate forging and machining equipment (pneumatic hammers, planers and turning lathes)

according to the specific character of the mill. Accommodation of roll-turning (roll-grinding) lathes in the rolling plant buildings has justified itself.

3.8. Cooling, Cutting and Straightening of Rolled Products

There are four types of rolled product cooling cycles.

Ordinary cooling cycle. Cooling is performed on cooling tables. The area of the tables is selected according to the maximum throughput and average time of natural cooling.

Sometimes, controlled-cooling conditions are created on the cooling tables (for example, in case of spring steel) to obtain a definite metal hardness. In these cases, a special method of placing the cooled workpieces (edgewise and without spaces) is used.

In case of usual piecemeal placing of the workpieces onto cooling tables the cooling time can be determined from the formula by M. Brovman:

$$t = 0.537 \frac{Q \cdot c}{F \cdot C}$$

where t = cooling time, h
Q = mass of one workpiece, kg
c = specific heat of the metal, cal/kg·°C
F = area of the heat emitting surface, m²
C = radiation constant (for steel $C = 4.4$ cal/m²·h·°C)

Metal cooling proceeds mainly through radiation. The amount of heat (Q) transferred from the metal to the atmosphere by radiation is determined by the Stefan—Boltzmann law as follows:

$$Q = F \cdot t \cdot C \left[\left(\frac{T}{100} \right)^4 - \left(\frac{T'}{100} \right)^4 \right]$$

(i.e., it is proportional to the fourth power of temperature).

In this formula:

F = area of the heat emitting surface, m²
t = cooling time, h
T and T' = absolute temperatures of the cooled body and environment, °K
C = radiation constant of the cooled body, cal·100⁴/m²·h·(°K)⁴

$$C = \varepsilon C_s$$

where C_s = radiation constant of black body of about (4.88-5.2) cal·100⁴/m²·h (°K)⁴

ε = coefficient characterizing the emitting surface (for steel ε is approximately equal to 0.8)

A. Tselikov, when determining cooling time t, neglects the second member of the difference in the Stefan—Boltzmann formula since T' is well in excess of T. Temperature drop ΔT during cooling of steel may be expressed in a simplified form as follows:

$$\Delta T = 0.0067 \frac{F}{g} \left(\frac{T_{av}}{100}\right)^4 \Delta t_c$$

where F and g = surface area and weight of a linear metre of the rolled section

T_{av} = average temperature at the given interval of time

Δt_c = cooling time, s

Retarded cooling cycle. This cycle is adopted for cooling high-carbon and alloy steels susceptible to flakes. Cooling is carried out in through-type fired furnaces, fired and unfired pits and boxes.

Starting from 800-900 °C, slow cooling provides for the levelling out of the temperature across the rolled product section and eliminates residual stresses. An isothermal transformation cycle, i.e. holding at 600-750 °C followed by air cooling, is used for the majority of steels subject to retarded cooling.

Accelerated cooling cycle. This cycle is adopted for cooling rods and sheets prior to coiling in order to obtain a specific structure of the metal and to decrease scale formation. Use is also made of water cooling in pipes or on roll tables.

Sheets and bands from various steels are made to undergo controlled accelerated water and air cooling down to 700-500 °C prior to coiling to obtain the most favourable and uniform structure of the metal.

High-carbon and alloy steels are water cooled to prevent the formation of carbide network.

Fast (thermal-hardening) cooling cycle provides for quenching followed by self-tempering to be done by using the rolling heat. Controlled systems of fast water cooling are used for this purpose.

Straightening and shearing of the rolled products are performed by two methods: in the main process flow (in hot and cold state) and in separate straightening and shearing units. The selection of the method is determined by the labour consumption in finishing operations, efficiency, etc. When operating the shearing and straightening units, much attention is paid to the preparation of replacement blades and straightening rolls.

3.9. Operation Management in Rolling-Mill Shops

The following are the principal elements of the operation management in a rolling shop:

1. Keeping a steady eye on the stability of the rolling process parameters: optimum metal heating cycle, rolling temperature range, rolling rhythm and speed, standard conditions of deformation (reductions and tensions in rolling), working surface condition of the rolls and roll fittings, roll cooling (or lubrication) conditions and conditions of descaling the metal surface.

The stability of the rolling process is controlled by means of pyrometers, ammeters (load indicators), dynamometers and tachometers. Hourly rolling schedules and continuous short-time testing of final and intermediate samples are used to regulate the rolling process including pass and roll changes.

2. Preparatory operations for the production, *viz.* preparing billets for rolling in accordance with monthly, weekly, daily and hourly schedules, and getting the mill ready for the operation (preparing sets of rolls for roll changes, as well as roll fittings, straightening rolls, blades and special equipment).

3. Repairs of furnace installations and of processing and auxiliary equipment of rolling mills (regular schedule, duration and characteristics of repairs and methods of their organization). Long interval repairs financed from the sinking fund of the works are called general overhauls. The periodicity of overhauls may be varied. Lengthy shut-downs of rolling mills once a year, habitual at individual plants not long ago, cannot be considered a good practice. A method of fractional overhauls with large-scale restoration

including modernization of the equipment carried out during running repairs and preventive maintenance should be considered more expedient.

Running repairs are made one to five times every month, depending on the characteristic features of rolling mills and conditions of the repair service management at the works. A method of repairs involving the replacement of whole units or assemblies is a progressive one.

Preventive maintenance between shifts and during roll changes includes, in addition to minor repairs, tightening of screwed joints, lubrication and checking of the equipment.

It is expedient that the rolling-mill repair crews be integrated so as to form the so-called specialized and centralized repair groups, especially at large works.

4. Introduction of the system of efficiency engineering in rolling-mill shops, which provides for proper allocation of jobs, multi-specialization of workers, no duplication of functions, gradual elimination of labour-consuming operations, introduction of industrial closed-circuit television systems, automation of production processes, automation of the production management and control including job rating, wage accounting, etc.

5. Planning and recording of expenditures on basic and auxiliary materials, recording of fuel, power and water consumption, of replacement equipment, rolls, repair services, of semi-finished products, workpieces in production, waste and finished products, along with the day-to-day calculation of the actual rolling and final costs.

In well-managed rolling-mill shops the routine production conditions are determined not only by the technological instructions, but also by flow charts drawn up on the basis of optimum process parameters for each type of rolled product. With the improvement of the production techniques these charts are subject to revision. The preparatory stage for the actual production, as well as primary records of the process parameters are regulated by the programmed sequence of rolling operations, hourly rolling schedules and timetables for roll changes, production change-overs and replacements of roll fittings, bearings and shearing blades.

3.10. Accident Prevention Regulations in the Rolled Metal Production

Potential hazards of rolling come from heat radiation, high travel speeds of the worked metal, concentration of moving and rotating mechanisms in the operating area, scattering of scales and shavings when shearing, sawing and conditioning the metal, formation of metal dust, release of vapours of oil and pickling products, etc.

Safety regulations for the operation of cranes, railway transport, for repairs, servicing of gas, electrical and oxygen facilities are the same as for the entire iron and steel industry. Great importance is attached to protective garments for workers of different trades, including means of individual protection (protective goggles, deflectors, nets, helmets, asbestos cover plates, special gloves, shoes and so on). Heat radiation is controlled with the help of water-net screens, curtains, safety glasses. Much attention is paid to different ventilating, blowing-off and air conditioning systems.

Moving and rotating mechanisms are guarded with special shrouds.

Finishing and high-speed working stands are reliably protected on the sides and at the top by means of removable wire guards.

In rolling-mill shops the access to standing mechanisms, rolling trains and working stands should be made absolutely safe. To this end the so-called tag system is mainly used, i.e., handing to the electricians of a division a tag taken off a corresponding starting device and serving as a pass, this confirming that the electrical circuit of the given mechanism or of the entire division has been disconnected. Special interlocks, signalling devices and mechanical shutters serve as duplicating safety arrangements. Various safety diveces (guard posts, ledges, deflectors) are used to protect the tandem rolling-mill operators.

In rolling-mill shops safe, clearly definable passages are made in the form of overhead bridges with railings or of tunnels.

Care should be taken to ensure good lighting, proper metal stacking and timely removal of rejects.

Varying operating conditions of rolling mills require working out of particular additional safety regulations in the form of production instructions which should be strictly adhered to.

3.11. Rolled Product Quality Control

The quality of rolled products is determined primarily by that of the steel melting and casting and by appropriate use of each heat conformable to the data on the actual conditions of melting and casting, as well as to the chemical analysis for each specific order.

With modern mass production the pre-requisites for obtaining high quality are steady conditions of processing and well-organized control at all stages of the metallurgical cycle.

All the processing data from melting to shipping are documented by a through-certificate of a heat or a ladle, if the metal of the given heat is tapped into more than one ladle. The heat certificate allows one to follow up the routing of each ingot mould buggy, stool or even of each metal zone along the height of an ingot.

The heat follow-up makes it possible to precisely determine the actual metal consumption factors for a given steel and to analyze all losses in processing.

One of the main functions of the procedure control is checking on the proper observance of the existing regulations for the routing of heats within the process flow and for separating reliably the heats and, sometimes, the ingots, as well as billets and bars with markings denoting the metal zones along the height of the ingot. At works producing high-grade steels the metal is checked by employing the spark test, but this should be done only as an additional safeguard against misidentification of the metal.

Lastly, the conformity of the rolled products to the order specifications is determined on the basis of the procedure control information and on that of the data subsequent to the control over the dimensions and surface condition of the products, carried out in the course of production, as well as on the ground of the final data on finishing, sorting, packing and marking, and on that of the findings obtained as a result of the final mechanical tests. For some types

of the rolled products the control involves an analysis of the macro- and microstructure.

Stepped turning, as well as X-ray, ultrasonic and other modern nondestructive tests are used for detection of internal flaws.

Works producing high-grade steels practise the so-called heat control as the basis for consigning a heat to fill important orders. This, a sort of a tentative method of control, consists in that the entire set of the rolled product tests is carried out electively on products made of individual ingots, this being followed by deciding as to the consignment of the entire heat or outlining a special type of processing.

Nowadays high quality of the rolled products is attained by carefully investigating the properties of a given steel, selecting most favourable conditions for its production and optimum ingot structure, as well as by ensuring standard processing conditions and a rational production control pattern.

Of fundamental importance is the properly selected system of the procedure control. It is closely linked with the filling in of the heat certificate. In the course of rolling quick upsetting, torsion and pickling tests are made concurrently with the intermittent and continuous control over the regularity of the shape and dimensions of the products. These tests make it possible to quickly eliminate rolling defects, such as backfins, folds, seams, surface tears, scratches and rolled-in scale.

The final control of finished rolled products is accomplished in accordance with the requirements of corresponding standards or specifications, by using specimens taken after rolling. This is often carried out in the course of production, the samples being cooled under conditions similar to those prevalent in the production. A check on flakes is done after expiration of a predetermined metal storage period (incubation period). In addition to the chemical analysis, mechanical and processing tests, internal flaws, if required, are detected by viewing the metal fracture and macrostructure.

The proportion of non-metallic inclusions, the degree of decarburization, the presence of a carbide network, etc. are determined by analysing the microstructure of the metal. A point system is employed to assess the microstructure.

The processing tests include the cupping (Erichsen), stamping and bending ones, as well as flanging and expansion tests (for tubular products) and pendulum impact tests (for rails). Repeated tests are envisaged with an increased number of specimens. Should these prove unsatisfactory, the rolled products of a given heat or batch are rejected or their grade changed to suit less important consignments.

3.12. Technical and Economical Characteristics of the Rolled Metal Production

Each ton of the annual output of rolled products, including that of the cogging division, requires capital investments 3-4 times greater than those for the blast-furnace (excluding ore mining and coke production) and steelmaking processes. The unit capital investment in rolling increase parallel with the improvement of the rolled product quality resulting from the use of additional cold rolling, heat treatment and various types of metal plating.

Taking as an example an ordinary, most popular structure of a hot-rolling shop and comparing an approximate correlation of percent costs among the main stages of rolling, we shall find that the cost of the rolling mill, including shearing and straightening facilities, will be as high as 70%, that of the heating facilities—up to 16% and of the finishing installations and store rooms—up to 14%.

It is customary to hold the rolling mill proper to be the main unit heading all other divisions of a rolling shop. Such an approach is also justified by the average upkeep expenditures of individual rolling divisions. These amount to 20-30% for heating, 50-60% for rolling and 15-30% for finishing.

The following general formula may be used for determining the actual hourly output of a rolling mill:

$$P = Q \cdot q \frac{3600}{t} \cdot k$$

where P = net yield of the mill, t/h
Q = mass of billet or ingot
q = net yield factor

t = rolling rhythm
k = rolling-mill use factor

The rolling rhythm is determined in a number of ways, depending on the rolling-mill arrangement and rolling scheme. For a single-stand blooming mill this is the total time of rolling in the stand, including intervals between passings and between successive ingots. For multistand rolling mills this is the time of passing in the finishing stand, including intervals between successive billets.

In general terms the rolling rhythm for a stand in which several passings are performed is determined from the formula:

$$t = \Sigma t_p + \Sigma t_i$$

where Σt_p = rolling time for several passings or total productive time
Σt_i = time of intervals between passings

If rolling speed v is constant (disregarding speed gain) and the length of the workpiece after passing is L, then the productive time is determined from the formula

$$t_p = \frac{L}{v} = \frac{L \cdot 60}{\pi D n} = \frac{19.1 \cdot L}{Dn}$$

where D = effective diameter of rolls
n = roll revolutions per minute

For reversing rolling mills, where the peripheral roll speed is variable, the productive time is expressed as

$$t_p = t_1 + t_2 + t_3$$

where t_1 = speeding-up period
t_2 = period of rolling at a constant speed
t_3 = slowing-down period

The duration of the speeding-up and slowing-down periods is determined proceeding from the difference between rotational speeds n_1, n_2 and n_3 and from acceleration a or deceleration b rates (rev·sec/min):

$$\text{speeding-up time } t_{s.u} = \frac{n_2 - n_1}{a}$$

$$\text{slowing-down time } t_{sl.d} = \frac{n_2 - n_3}{b}$$

It should be borne in mind that in most cases the beginning and the end of a passing do not coincide with the zero speed values in the speed diagram of a reversible motor and they are determined best with reference to the load diagrams of the motor or to oscillograms.

The duration of intervals t_i between passings of a single workpiece or between passings of successively rolled workpieces depends on the capabilities of the roll tables, trans-

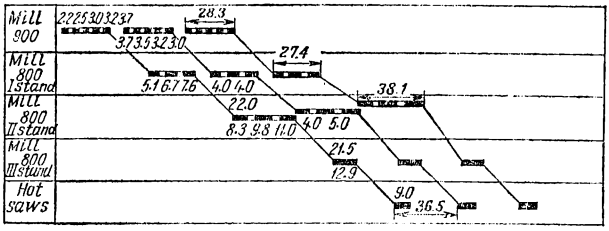


Fig. 3.9. Diagram of rolling P-50 type rails in 800 mm rail-and-structural mill (Adametsky's diagram)

fers, elevating platforms and screwdowns or, in case of old, non-mechanized rolling mills, on the time of catching and feeding the workpiece by hand.

To determine the rolling-mill output and to analyse the rolling process from the standpoint of better utilization of the equipment and possible measures to improve the output, combined diagrams (Adametsky's diagrams) of rolling operations, including shearing, straightening, etc., based on the design or time-keeping data, are used.

These diagrams (Fig. 3.9) are drawn as follows: time (in seconds) is plotted to scale along the horizontal axis, while the numbers of stands and of other units are plotted along the vertical axis in a sequence corresponding to the flow sheet. Thick lines, the length of which is proportional to the duration of the working periods or cycles or to the productive time of a stand or unit, are drawn along horizontal axes that correspond to each individual operation, while thin or dash lines show idling periods (intervals). In case of continuous or flow rolling processes passings of not one, but of a number of successive workpieces are plotted on the

diagrams. Therefore, by using the diagrams one can determine not only the rolling time in every stand or in all the stands along with the time of finishing and handling operations and the rolling rhythm, but also the general condition of the rolling process. Adametsky's diagrams are coordinated with optimum motor load diagrams, the latter making it possible to judge about the degree of the motor loading in case of a group drive of the stands and also in the event of simultaneous rolling of several workpieces in three-high stands.

The rolling-mill performance is assessed by using the so-called 'rolling-difficulty factors'. Then the output of a certain type of the rolled product or shape, more often than not, the one that is most common and simple, is taken as a reference standard. The rolling-difficulty factor for any other shape, type of the rolled product or for a certain group of strip plate will de represented by the ratio between the output of the corresponding product and that of the rolled product type taken as the reference standard.

The average output of a rolling mill is determined when designing the mill and at the time of production planning.

If rest days (holidays), time spent for overhauls, running repairs and maintenance, as well as intervals between shifts are subtracted from the overall (calendar) time, this will then represent the working time of the rolling-mill.

All the downtimes provided for by the adopted routine schedule are called running downtimes (roll changes, failures, absence of heated metal and so on). Short delays in the rolling mill operation are not taken into consideration in current accounting. They are called inapparent downtimes. These are determined through time-keeping and by analyzing the diagrams and oscillograms taken by instruments recording the rolling-mill operation.

The ratio of the net time to the actual time is called the rolling-mill use factor.

The actual time of the rolling-mill operation is the working time minus running downtimes. The net time of rolling is the actual time minus inapparent downtimes. The net time includes the productive time (the time when the metal remains in rolling-mill rolls) and shutdowns (the time spent

in doing the necessary auxiliary operations, such as catching, manipulating and so on).

The shutdowns are technically justified intervals in rolling. The running downtimes may include indispensable operations such as current roll cahnges upon wear, pass changes, replacements of blades, shears and of other rapidly wearing parts, lubrication, removing scale and so on. Thus, proceeding from the characteristic features of the process and equipment, the following rates for rolling-mill downtimes are adopted: 8-10% for cogging mills; 16-18% for rail-and-structural steel, heavy-section and medium-section mills; 10-12% for light-section mills; 6-8% for wire mills; 11-13% for continuous sheet mills and 8-10% for tandem sheet mills.

The net yield is an important technical and economical index determining the quality, output and, in the final analysis, the production efficiency.

The net yield depends upon the metal losses in the form of crops, iron loss and rejects. Standard rates are set for the ingot head croppage at blooming mills and for the workpiece end croppage after rolling. The iron loss in heating is dependent on the degree of technical perfection of heating facilities as well as on the conditions and duration of heating. The iron loss in rolling is a function of the rolling temperature and time. Rejects may be due to the failure of the products to have the required chemical composition, macrostructure and mechanical properties, as well as because of the presence of surface defects which cannot be removed.

The reciprocal of the net yield factor (less than unity) is the metal consumption factor. The metal consumption factor which takes account of the net yield from ingots is termed overall consumption factor. It equals the product of the consumption factors at individual stages of rolling:

$$k_{ov} = k_{bl} \cdot k_{bil} \cdot k_{fin},$$

i.e. the product of particular metal consumption factors at the blooming mill k_{bl}, billet mill k_{bil} and at the finishing mill k_{fin}.

For example, in case of high-grade killed carbon steel the metal consumption factors will be $k_{bl} = 1.19$, $k_{bil} = 1.04$, $k_{fin} = 1.05$ and the overall consumption factor, $k_{ov} = 1.19 \cdot 1.04 \cdot 1.05 = 1.30$.

PART TWO

FUNDAMENTALS OF ROLL PASS DESIGN

By the roll pass design or sizing of the rolled products is meant the distribution of deformations among the metal passings in rolling and also the design of the shapes and calculation of dimensions of the roll passes, as well as the design of the rolled workpiece sections.

An annular slot turned in the body of a rolling-mill roll is called the *roll groove*. A combination of the grooves in two mating rolls is termed the *roll pass*.

The design of the roll passes is based on the knowledge of the theory of plastic deformation of metal. Sizing of each rolled shape should be in keeping with the principles of the given metal rolling method and with the design features of the given rolling mill. The proper pass design provides for steady conditions of processing and ensures high quality of the rolled products, as well as the most favourable technical and economical characteristics of the production.

Consecutive changes of the shapes and dimensions of the rolled metal sections in passings are contingent upon the conditions of filling of the roll passes with the metal and is governed by the laws of plastic deformation.

The sizing of the rolled shapes consists in distribution of the shapes and dimensions of the roll passes, as well as of the workpiece sections among the metal passings, while the roll pass design involves the arrangement of the passes on the rolls and this is determined by the draught schedule, the method of rolling and the effective use of the rolls.

CHAPTER IV. SOME PROPOSITIONS OF THE METAL ROLLING THEORY

4.1. Quantities Characterizing Deformation in Rolling

Let us denote the dimensions of a rectangular workpiece prior to rolling as: H—height, B—width and L—length. These after rolling will be h, b and l, respectively. Then the law of constancy of volumes of the workpiece prior to and after rolling may be written as

$$HBL = hbl$$

$$\frac{HB}{hb} = \frac{l}{L}$$

where $HB = F_0$ is the cross-sectional area of the workpiece prior to rolling

$hb = F_1$ is the cross-sectional area of the workpiece after rolling

Hence

$$\frac{F_0}{F_1} = \frac{l}{L} = \mu$$

where μ = coefficient of draught in rolling.

Absolute reduction

$$H - h = \Delta h$$

Relative reduction is equal to the ratio $\Delta h/H$. Sometimes, it is characterized by value $\Delta h/h$ or $\ln H/h$.

Absolute lateral deformation or spread is

$$b - B = \Delta b$$

In n passings the cross-sectional area of the workpiece varies as follows:

$$F_0;\ F_1 \ldots F_{n-1};\ F_n$$

The coefficients of draught will be respectively:

$$\mu_1 = \frac{F_0}{F_1}; \quad \mu_2 = \frac{F_1}{F_2} \ldots \mu_n = \frac{F_{n-1}}{F_n}$$

Hence

$$F_0 = \mu_1 F_1; \quad F_1 = \mu_2 F_2 \ldots F_{n-1} = \mu_n F_n$$

Then

$$F_0 = \mu_1 \mu_2 \ldots \mu_n F_n$$

or

$$\frac{F_0}{F_n} = \mu_1 \mu_2 \ldots \mu_n$$

i.e., the total coefficient of draught for a number of passings in rolling is equal to the product of particular coefficients of draught for all the passings:

$$\mu_\Sigma = \mu_1 \mu_2 \ldots \mu_{n-1} \mu_n$$

The concept of mean draught is often used in practice. If

$$\mu_1 = \mu_2 = \ldots = \mu_{n-1} = \mu_n = \mu_m$$

then

$$\mu_\Sigma = \mu_m^n \text{ and } \mu_m = \sqrt[n]{\mu_\Sigma}$$

If mean draught is preset, then the number of passings may be obtained proceeding from the total draught:

$$n = \frac{\log \mu_\Sigma}{\log \mu_m} = \frac{\log F_0 - \log F_n}{\log \mu_m}$$

If the cross section of the workpiece is not rectangular, the reduction in height is determined by the formula:

$$\Delta h = \frac{F_0}{B_0} - \frac{F_1}{B_1}$$

4.2. Roll Bite Conditions

Below is considered the process of rolling metal between two plain cylindrical rolls of the same diameter which revolve at the same speed (Fig. 4.1).

Region *ABCD* is called the zone of deformation. Angles *AOB* and *DOC* are the angles of bite and arcs *AB* and *DC*—the contact arcs.

The biting of the workpiece by the rolls and filling of the entire zone of deformation with metal take place at the initial period of rolling. After the deformed head end of the workpiece leaves the rolls, steady rolling conditions set in and then the tail end of the workpiece leaves the zone of deformation.

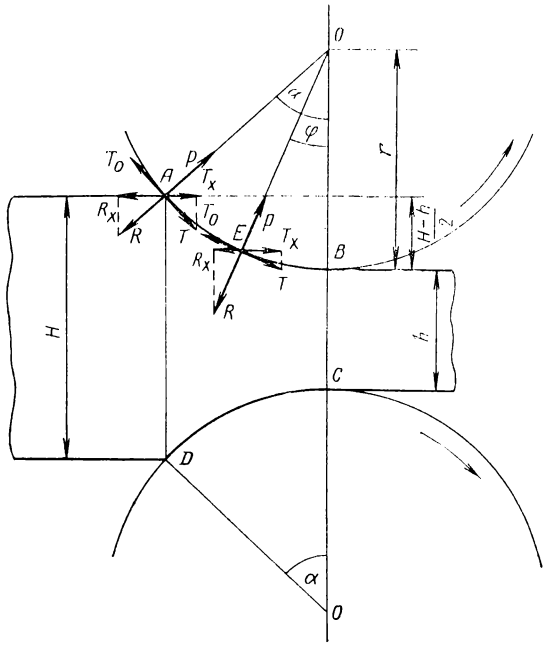


Fig. 4.1. Diagram of forces in metal biting by rolls and in steady rolling conditions

At the initial period of rolling, when the workpiece comes into contact with the rolls, normal pressure P of the workpiece on the rolls and reaction force R of the rolls appear at points A and D.

At the same points arise another two forces which act along the tangent to the roll periphery. These are retarding force of the rolls T_0 and frictional force T which pulls the workpiece into the opening between the rolls. To determine

the possibilities for the metal biting by the rolls let us consider the horizontal projections of forces T and R:

$$T_x = T \cos \alpha$$

$$R_x = R \sin \alpha$$

The biting will occur if $T_x > R_x$, or $T \cdot \cos \alpha > R \cdot \sin \alpha$, or $\frac{T}{R} > \tan \alpha$. Since the frictional force

$$T = fR$$

where f = coefficient of friction,
then

$$\frac{fR}{R} > \tan \alpha \quad \text{and} \quad f > \tan \alpha$$

By substituting the tangent of angle of friction β for coefficient of friction ($f = \tan \beta$) we will obtain the following condition of the biting: $\alpha < \beta$, i.e., the biting will be ensured when the angle of bite is less than angle of friction β. Hence, at the initial period the maximum possible angle of bite is $\alpha \leqslant \beta$.

Let us assume that under steady conditions of rolling the resultant of the pressure exerted by the metal on the rolls is applied to the centre of the contact arc at point E, or $\varphi = \frac{\alpha}{2}$.

Then

$$T_x = T \cos \frac{\alpha}{2} = fR \cos \frac{\alpha}{2}$$

and

$$R_x = R \sin \frac{\alpha}{2}$$

Hence

$$fR \cos \frac{\alpha}{2} > R \sin \frac{\alpha}{2} \quad \text{and} \quad f > \tan \frac{\alpha}{2}$$

or

$$\tan \beta > \tan \frac{\alpha}{2}$$

which means that

$$\frac{\alpha}{2} < \beta \quad \text{and} \quad \alpha < 2\beta$$

When $\varphi \quad \frac{\alpha}{n}$ the limiting conditions of the steady rolling process are determined by the following general expression:

$$\alpha = n\beta$$

i.e., if the initial biting of the metal by the rolls is ensured, the subsequent steady rolling process provides great possibilities for increasing the metal reduction in rolling. Indeed, in accordance with Fig. 4.1:

$$r - \frac{H - h}{2} = r \cos \alpha$$

or

$$H - h = D(1 - \cos \alpha)$$

By using equation

$$1 - \cos \alpha = 2 \sin^2 \frac{\alpha}{2}$$

the following equation may be deduced:

$$H - h = 2D \sin^2 \frac{\alpha}{2}$$

Assuming that

$$\sin \frac{\alpha}{2} \approx \frac{\alpha}{2}$$

we have

$$H - h = 2D \left(\frac{\alpha}{2} \right)^2 = r\alpha^2$$

The latter equation shows the direct dependence of the reduction upon the angle of bite, i.e.

$$\alpha \approx \sqrt{\frac{\Delta h}{r}}$$

Notching, building-up and special knurling of the roll surfaces, as well as wedging of the head end of the workpiece, are used to increase the frictional forces and thus to increase possible reductions in rolling. In continuous rolling mills the preceding stands create conditions of forced biting of the head end of the workpiece by the rolls of the following stands. In this case of great importance are the velocity pressure and flexural rigidity of the workpiece head end

which depend on the temperature and cross-sectional area of the workpiece, as well as on the speed and length of its free end. To obtain the required velocity pressure when rolling workpieces of comparatively small sections the tandem stands are installed as close as possible to one another or special hold-down rollers (Treibapparaten) are used.

In theoretical and practical calculations pertaining to the metal rolling process great importance is attached to the coefficient of friction.

According to the present-day notions it is believed that the coefficient of friction in rolling depends not only on the character of the rolled metal, its temperature, presence of oxides, type of lubrication, power and speed conditions of rolling, but also on the nature of slip.

Therefore, the following three coefficients of friction are distinguished: (1) coefficient of friction at the moment of biting of the workpiece by the rolls. In this case it is determined proceeding from the tangent of the maximum angle of bite; (2) coefficient of friction in slipping. In this case it is usually considered to be a factor of 1.25-2.0 lower than in biting; (3) coefficient of friction in steady motion, when the metal may slip over the surface of the rolls in opposite directions from the neutral plane. In this case the coefficient of friction is determined proceeding from forward slip or by directly measuring the tangent contact forces.

For this purpose, A. Grishkov has measured the normal pressure and frictional forces by means of a special single-pin dynamometer.

The following empirical formula deduced by S. Ekelund, which determines the relationship between the coefficient of friction, type of the roll material and rolling temperature, has found wide application:

for steel rolls

$$f = 1.05 - 0.0005t$$

for iron rolls

$$f = 0.8\,(1.05 - 0.0005t)$$

where t = rolling temperature, °C.

Diagrammatically this relationship is represented by a straight line. Coefficients which take into account the effect of the rolling speed, chemical composition of the

rolled metal and the type of the roll material are often introduced into the Ekelund formula:

$$f = k_1 k_2 k_3 (1.05 - 0.0005t)$$

where k_1, k_2 and k_3 = coefficients taking into account the effect of the type of the roll material, rolling speed and chemical composition of the rolled metal.

4.3. Neutral Plane and Angle. Forward Slip

It is well known that the head end of the workpiece leaves the rolls with a speed higher than the peripheral speed of the rolls, while the tail end of the workpiece has a speed slightly lower than the peripheral speed of the rolls. A vertical plane within the zone of deformation, in which the speed of the workpiece is equal to the peripheral speed of the rolls, is called the neutral plane and the angle corresponding to this plane is known as the neutral angle γ.

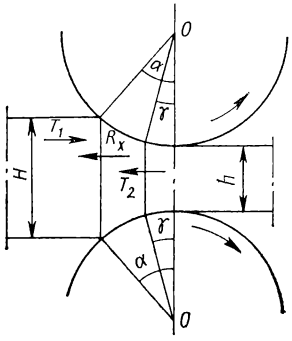


Fig. 4.2. Diagram of horizontal forces acting in the zone of deformation

According to the equilibrium conditions of the horizontal projections of forces acting in the zone of deformation (Fig. 4.2), we may write

$$T_1 = R_x + T_2$$

where T_1 and T_2 = horizontal projections of the tangent frictional forces

R_x = horizontal component of the normal pressure

The horizontal projections of the frictional forces may be expressed in terms of the normal pressure:

$$T_1 = pr(\alpha - \gamma)\beta$$

$$T_2 = pr\gamma\beta$$

$$R_x = pr\alpha\frac{\alpha}{2} = \frac{1}{2}pr\alpha^2$$

hence

$$pr(\alpha-\gamma)\beta=\frac{1}{2}pr\alpha^2+pr\gamma\beta$$

and the neutral angle

$$\gamma=\frac{\alpha}{2}\left(1-\frac{\alpha}{2\beta}\right)$$

Here we do not take into consideration the nonuniform distribution of the contact stresses along the contact arc.

When the resultant force acts at angle $\varphi=\frac{\alpha}{n}$ the neutral angle may be expressed as

$$\gamma=\frac{\alpha}{2}\left(1-\frac{\alpha}{n\beta}\right)$$

Excess frictional forces are specific for the steady rolling process (when $\alpha<\beta$). This is associated with the phenomenon of forward slip. Forward and backward slip are usually expressed in percent:

$$S_f=\frac{v_1-v_0}{v_0}\cdot 100\%$$

$$S_b=\frac{v_0\cos\alpha-v_2}{v_0\cos\alpha}\cdot 100\%$$

where v_1 = speed of the rolled workpiece leaving the rolls
v_0 = peripheral speed of the rolls
$v_0\cos\alpha$ = horizontal component of the peripheral speed of the rolls in a plane corresponding to the onset of biting
v_2 = speed of the rolled workpiece entering the rolls.

Forward slip may be determined proceeding from the equality of the second volumes of metal in different planes within the zone of deformation including the neutral plane.

There are the following second volumes:

In the plane of the workpiece entering the rolls	HB_1v_2
In the neutral plane	$H_nB_nv_n$
In the plane of the workpiece leaving the rolls	hB_2v_1

Then

$$HB_1v_2=H_nB_nv_n=hB_2v_1$$

$$v_1=\frac{H_nB_nv_n}{hB_2}$$

Since

$$H_n - h = D(1 - \cos\gamma)$$

then

$$H_n = h + D(1 - \cos\gamma)$$

consequently

$$v_1 = \frac{[h + D(1 - \cos\gamma)]\, B_n v_n}{hB_2}$$

Substituting the expression for the speed in the neutral plane $v_n = v_0 \cos\gamma$ in the above equation yields:

$$v_1 = \frac{B_n}{B_2} \frac{[h + D(1 - \cos\gamma)]\, v_0 \cos\gamma}{h}$$

then, forward slip is

$$S_f = \frac{B_n}{B_2} \frac{[D(1 - \cos\gamma) + h]}{h} - 1$$

The latter formula takes into consideration the effect of the workpiece width. Inasmuch as $\frac{B_n}{B_2} < 1$, forward slip decreases with an increase in spread. When there is no spread:

$$B_1 = B_n = B_2$$

and

$$v_1 = \frac{H_n}{h} \cdot v_n$$

Substituting yields

$$v_1 = \frac{h + D(1 - \cos\gamma)}{h} \cdot v_0 \cos\gamma$$

$$\frac{v_1}{v_0} = \frac{h + D(1 - \cos\gamma)}{h} \cdot \cos\gamma$$

Forward slip is

$$S_f = \frac{v_f}{v_0} - 1$$

or

$$S_f = \frac{[h + D(1 - \cos\gamma)]\cos\gamma}{h} - 1$$

If we assume that $v_0 \cos \gamma = v_0$, then

$$v_0 = \frac{v_1 h}{h + D(1 - \cos \gamma)}$$

$$\frac{v_1}{v_0} = \frac{D(1 - \cos \gamma)}{h} + 1$$

and

$$S_f = \frac{v_1}{v_0} - 1 = \frac{D(1 - \cos \gamma)}{h} = \frac{2D \sin^2 \frac{\gamma}{2}}{h} \approx \frac{2D \frac{\gamma^2}{4}}{h} = \frac{r}{h} \gamma^2$$

In practical calculations this simplified formula gives sufficiently accurate results.

The following, more precise expression for forward slip derived from the above equation should be used when the ratio $\frac{r}{h}$ ranges from 2 to 3:

$$S_f = \left(\frac{r}{h} - 0.5\right) \gamma^2$$

The so-called zone of metal sticking corresponds to the contact surface area where the horizontal velocities of the rolls and metal are equal. The nonuniformity of distribution of deformations and speeds of the metal moving across the section of the rolled workpiece is dependent upon the extent of the metal sticking zone. The larger the metal sticking zone, the greater this nonuniformity and the longer the time during which the metal particles at the contact surface move with the speed of the rolls. Therefore, the difference between the draughts of the contact and central layers is greater towards the end of the metal sticking zone and the greater part of forward slip will take place within the zone of non-contact deformation.

Forward slip in rolling depends on the diameter of the rolls and the thickness of the workpiece. Increasing the diameter of the rolls leads to a rise in forward slip. Decreasing the workpiece thickness results in an increase in forward slip. Forward slip decreases parallel with the increase of the lateral deformation (spread). Thus, forward slip depends on the relationship between elongation and spread. When elongation is increased by decreasing spread, the speed of the workpiece leaving the rolls is also increased.

The physical meaning of the changes of forward slip depending upon a number of rolling factors is explained by the changes of the excess of friction forces during rolling. For example, with increasing the diameter of the rolls and decreasing angle of bite α with constant angle of friction β, the excess frictional forces increase, as well as forward slip.

Interstand tension of the workpiece in continuous rolling mills has a great effect on the amount of forward slip. If such a tension is present, forward slip may be calculated by using the general simplified formula with due regard for the change of angle γ. Forward slip increases with the rise of the tension. When front and back tensions T_1 and T_0 are present, their difference is taken into account in accordance with the investigation by N. Druzhinin

$$S_f = S_0 + \alpha \frac{T_1 - T_0}{Q_1}$$

where S_0 = forward slip when there is no tension
α = angular coefficient of inclination of the forward slip line in case of tension (the tangent of the inclination angle of the line to the axis of abscissae)
Q_1 = cross section of the workpiece leaving the rolls

4.4. Spread During Rolling in Plain Rolls

Spread in rolling is the most important and complex deformation factor which should be taken into account when designing the roll passes. When metal is forged or pressed between two parallel flat surfaces the change (increase) of the cross section of the workpiece is governed only by geometrical laws.

In the process of rolling there is a complex interrelation between the acting (deforming) forces, therefore the nature and parameters of deformation are linked together by complex regularities which are not always exactly determinable by analytical means. In rolling, the workpiece widens (spreads), the spread being expressed as

$$\Delta b = B_2 - B_1$$

where B_1 and B_2 = width of the workpiece prior to and after rolling, respectively

When rolling rectangular workpieces in plain rolls, spread will be free. When grooved rolls are used, the side walls of the roll pass prevent lateral movement of the metal and in this case spread is called unfree (restricted).

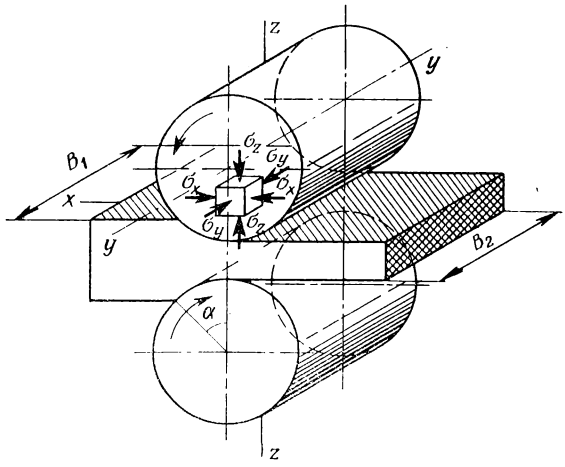


Fig. 4.3. Diagram of state of stress in rolling

In case of rolling in intricate shaped passes, where reductions and hence draughts and lateral deformation are different for different parts of the roll pass section, spread will be constrained.

Spread in rolling depends on a number of simultaneously acting factors, such as the diameter of the rolls, the rolling speed and temperature, the geometrical dimensions of the workpiece, etc.

The effect of each particular factor of rolling upon spread is considered separately, assuming that all the other factors remain constant. The character of the change of the state of stress in the zone of deformation is usually considered to reveal the qualitative pattern of lateral deformation.

In the zone of deformation the metal is under conditions of a triaxial compression (Fig. 4.3): σ_z—vertical stress

(maximum), σ_y — lateral stress, σ_x — longitudinal stress (minimum).

The dependence of the spread upon a number of factors is considered below.

The extent of the zone of deformation (the length of the contact arc) increases with a rise in the diameter of the rolls and is expressed as

$$l = \sqrt{(H-h) \cdot r}$$

With the increase of the contact arc length the sum of frictional forces acting in the longitudinal direction increases, stress σ_x rises and elongation decreases. This in its turn gives rise to an increase in lateral deformation (spread), i.e. an old rule that 'large rolls squeeze, but small ones pull' is confirmed.

As the coefficient of friction rises, the excess longitudinal frictional forces increase, this leading to an increase in the forward slip zone. In this case stress σ_x grows to a greater extent than σ_z. This reduces the longitudinal deformation and increases lateral deformation. Therefore, spread grows with increasing the coefficient of friction, all other conditions being equal.

It is well known that, as a rule, the coefficient of friction decreases as the metal temperature increases, therefore spread becomes reduced. Spread is also reduced when the coefficient of friction is decreased due to an increased rolling speed.

It is also known from experience that spread increases parallel with an increase in the carbon content of steel.

According to the data obtained by Yu. Chizhikov, a special correction factor (Table 4.1) should be introduced for alloy steels to take into account the effect of the chemical composition of steel upon the coefficient of friction. For mild steel (for example, the Ст. 1 grade steel) this factor is equal to unity.

Spread depends on the geometrical dimensions of the rolled metal. The lateral frictional forces and stress σ_y grow as the width of the workpiece increases. On the face of it, this would have resulted in a decrease in spread, but in fact, as the width of the workpiece increases, spread decreases only to some limit and then remains constant. The incre-

Table 4.1

Correction Factors for Various Types of Steel

Type of steel	Steel grade	Correction factor	
		exact	average
Pearlite-martensitic (pearlitic, pearlite-martensitic, martensitic)	У7А ШХ15 38ХМЮА Х10С2М (ЭИ107)	1.24 1.29 1.35 1.35	1.25-1.30
Austenitic	4Х14Н14В2М Х13Н4Г9	1.36 1.42	1.35-1.40
Austenitic with excess phase (ferrite, ledeburite)	1Х18Н9Т Х18Н25С2 Х23Н13	1.44 1.44 1.53	1.4-1.5
Ferritic	1Х17Ю5 Х15Н60	1.55 1.62	1.55-1.60

ase of spread upon decreasing the width of the workpiece is associated with a reduction in lateral supporting stresses while longitudinal stresses remain constant. The statement that the spread does not depend on the width holds true only for wide sheets of a fairly small thickness.

The height of the workpiece has an effect not only upon the amount of spread, but also upon its pattern. When rolling high pieces, deformation concentrates in layers near the contact surfaces, which leads to local spreading of the metal in these areas. When rolling thin pieces, their edges usually become convex in case of unrestricted spread. This is explained by the effect of contact frictional forces (Fig. 4.4).

When rolling pieces of different heights (for example, $H_1 > H_2$) with the same total reduction Δh, spreads Δb_1 and Δb_2 will be inversely proportional to heights h_1 and h_2, other conditions being equal:

$$\frac{\Delta b_1}{\Delta b_2} = \frac{H_2 - \Delta h}{H_1 - \Delta h} = \frac{h_2}{h_1}$$

This formula holds true for the case when the distribution of the reduced metal between elongation and spread is the same for workpieces of different heights.

On the other hand, the forward slip zone is decreased as the height of the workpiece is increased. This facilitates flowing of the metal in the longitudinal direction and reduces the lateral metal flow.

Reduction is the main factor that determines the amount of the spread in rolling. It has been experimentally established that the direct relation between spread and reduction

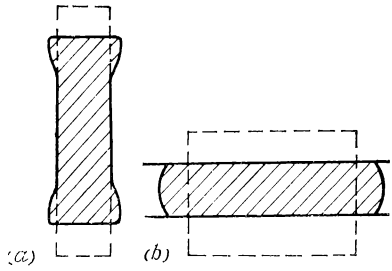


Fig. 4.4. Spread during rolling
(*a*) thick workpiece; (*b*) thin workpiece

is retained only when geometrical parameters Δh, H and h are constant. This stems from the fact that spread depends not only on absolute $(H - h)$, but also on relative (H/h) deformation. Besides, with an increase in reduction the contact arc and longitudinal supporting stresses rise and spread augments.

When rolling is done by passing metal through the rolls several times, the total spread is always less than spread when rolling is done in one passing with the same total reduction. This is explained by the differences in the state of stress within the zone of deformation in case of small and large reductions. The extent of the zone of deformation (the length of the contact arc) is decreased when reductions are small and therefore the effect of longitudinal stresses σ_x is reduced, the magnitudes of lateral stresses σ_y being the same.

In continuous rolling the interstand tension changes the relationship between longitudinal and lateral stresses

acting in the zone of deformation. Such a change in the state of stress results in a decrease in lateral deformation (spread). When designing the roll passes for continuous rolling mills, some decrease in the calculated value of spread has to be taken into account.

The need for calculating the spread in rolling created a large number of formulas.

The following, most simple formula for calculating spread was proposed by Geuze:

$$\Delta b = k\,(H - h)$$

where $k = 0.35\text{-}0.48$ is the coefficient of spread, which takes into account all rolling process factors with the exception of total reduction.

S. Petrov, a Russian engineer, and Zibel, a German scientist, established that spread is mainly a function of the length of the zone of deformation, the amount of reduction and the height of the workpiece prior to rolling:

$$\Delta b = k l_{z.d} \frac{H - h}{H}$$

By substituting in the Petrov—Zibel formula $l_{z.d} = \sqrt{r\,(H - h)}$, the length of the chord bridging the contact arc, for $l_{z.d}$, the length of the contact arc (zone of deformation), we have:

$$\Delta b = k \sqrt{r\,(H - h)} \cdot \frac{H - h}{H}$$

In this formula coefficient k accounts for the effect of a number of rolling process factors upon spread. For hot rolling this coefficient is equal to 0.35-0.45.

By making a number of assumptions, B. Bakhtinov, a Soviet scientist, has analytically deduced the following formula for calculating the spread when rolling:

$$\Delta b = 0.57 \frac{H - h}{H} \sqrt{r\,(H - h)} - \frac{H - h}{2f}$$

where f = coefficient of contact friction.

The Bakhtinov formula takes into account the coefficient of friction.

The calculation results obtained with both formulas agree and therefore the Petrov — Zibel formula may be recommen-

ded for making practical calculations with a definite degree of approximation, the value of coefficient k being preset. It should be borne in mind that the values of the coefficient of friction are also approximate.

Nevertheless, it is very important to introduce into the formulas the values of the coefficient of friction, which, in rolling, is such an important factor. Therefore, in investigations and also for complete calculations, the Bakhtinov formula is widely used.

Below is given the formula for calculating spread, as proposed by S. Gubkin:

$$\Delta b = (1-U)\left(f\sqrt{\Delta h r} - \frac{\Delta h}{2}\right)U$$

where $U = \frac{H-h}{2}$ is the relative reduction.

This formula accounts for the effect of the coefficient of friction, reduction, height of the workpiece and diameter of the rolls upon the spread.

The following formula deduced by Ekelund takes into account the effect of the width of the workpiece:

$$\Delta b =$$

$$= \sqrt{4n^2(H+h)^2\left(\frac{x_0}{B_1}\right) + B_1^2 + 4nx_0(3H-h) - 2n(H+h)\frac{x_0}{B_1} - B_1}$$

where $n = \frac{1.6fx_0 - 1.2\Delta h}{H+h}$

$x_0 = \sqrt{\Delta h r}$

The formula derived by A. Chekmarev on the basis of the relation between elongation and spread is given below in a transformed form:

$$\Delta b = \frac{2KB(H-h)}{(H+h)\left[1+(1+\alpha)\left(\frac{B_m}{R_\alpha}\right)^n\right]}$$

where $1+\alpha$ (α = angle of bite) accounts for the inclination of the roll surface

B_m = mean width of the workpiece in the zone of deformation

n = shape factor of the zone of deformation: when $B \leqslant R_\alpha$, $n \approx 1$; when $B > R_\alpha$, $n \approx 2$

K = coefficient of spread restriction when rolling in grooved rolls ($K = 1$ when rolling in plain rolls)

R_α = length of the contact arc

A. Golovin has considered that according to the law of least resistance the movement of metal particles in a horizontal plane occurs along perpendiculars drawn to the lateral surfaces of a conditionally upset parallelepiped (the horizontal projection of the zone of deformation). Such a scheme of a conditional division of the zone of deformation into four zones is used for theoretical investigations (B. Bakhtinov, A. Tselikov, D. Starchenko).

The zones adjoining the side edges of the rolled workpiece are termed the conditional zones of the tendency of deformation to spreading, great importance being attached to the effect of the outer zones. Some part of the workpiece metal in the lateral zones is deformed in the longitudinal direction and, on the contrary, a part of the metal next to the side edges shifts laterally.

Exerting influence upon the levelling out of the workpiece draught, the outer zones cause a nonuniform stress distribution along the width of the workpiece and substantially decrease its spread. This may be observed by the enlarged spread of the head and tail ends of the workpiece when one of the outer zones is absent.

The reasonability of dividing the horizontal projection of the zone of deformation into four zones for the purpose of theoretical calculations of spread is proved by the investigations carried out by B. Bakhtinov, A. Tselikov and A. Grishkov. According to A. Tselikov and A. Grishkov, spread is a function of the extent of the zone of forward slip, the logarithm of deformation, and the width and tension of the workpiece:

$$\Delta b = C_\sigma \cdot C_w \cdot l_{f.s} \cdot \ln \frac{h_0}{h_1}$$

where C_σ and C_w = coefficients which take into account the tension and width of the workpiece

$l_{f.s}$ = extent of the forward slip zone

As it has been indicated by the authors of this unique formula (which accounts for the effect of the tension and width

of the workpiece upon its spread) the calculation of the effect of the width of the workpiece, coefficient of friction and workpiece tension presents the most difficult problem in the theory of spread.

4.5. Spread During Rolling in Grooved Rolls

Above we have considered the simplest version of spread, i.e. the spread during rolling between plain rolls. In case of an unfree spread (restricted or constrained) the side walls of the roll pass prevent the lateral metal flow and the deformation of different portions of the workpiece is carried out with different effective diameters of the rolls which provide different reductions. Hence, the draughts for these portions could be different, if the workpiece shape were not of the same whole profile.

Even in comparatively simple box passes (Fig. 4.5) the distribution of metal between elongation and spread is complicated. The pressure exerted by the side walls of the pass decreases the spread and increases the elongation. In this case the spread decreases as the taper of the side faces of the box pass increases (Fig. 4.6). In diagonal passes spread sometimes may not occur or may even have a negative value.

Constrained spread is used in designing roll passes for rolling rails. In this case (Fig. 4.7) the whole mass of the section obtains a shallow draught and the base which is being vigorously reduced, naturally, cannot ensure lengthening of the whole piece. Therefore, the entire reduced metal can only go for spreading the base.

The system of the so-called slitting (swabbing) passes (Fig. 4.8) is applied for obtaining a large width of a strip when that of the workpiece is insufficient. In this case a flat or square billet is first rolled in a slitting pass and then in a flat pass. In the latter pass the sides of the section are vigorously reduced and thus obtain a large spread. However, this method should be used cautiously since the nonuniform draught along the width of the workpiece may cause its tearing.

The deformation pattern in rolling flanged sections (beams, channels, rails) is even more complex. While investigating

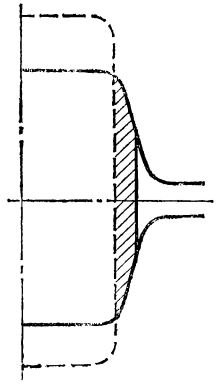


Fig. 4.5. Spread of workpiece in box pass

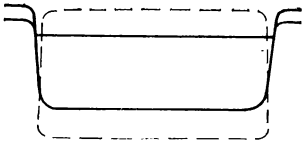


Fig. 4.6. Spread of workpiece in closed pass

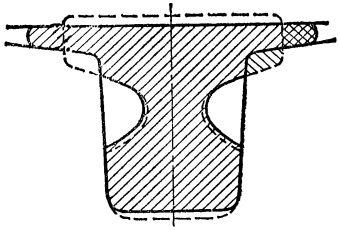


Fig. 4.7. Forced spread of rail flange

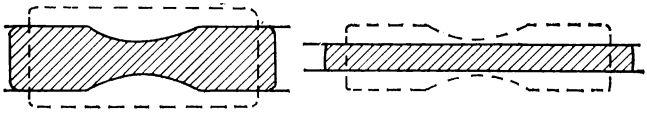


Fig. 4.8. Swabbing passes for increasing width of workpiece

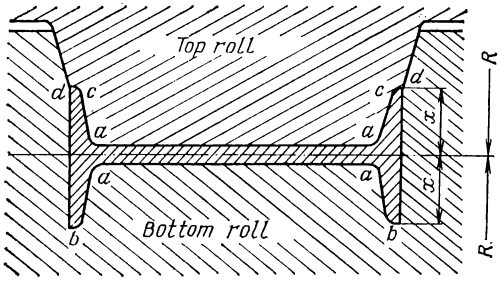


Fig. 4.9. Metal deformation pattern in flange passes according to Grume-Grzhimailo

the deformation of No. 25 beam in 800 mm dia rolls, V. Grume-Grzhimailo has given a general characteristic of the conditions of deformation of metal in flange passes (Fig. 4.9).

The web thickness is 9 mm, pass width—250 mm and flange height—110 mm. The radius of the roll part forming the web $R_w = 400-4.5 = 395.5$ mm. This radius is the same for top and bottom rolls. Taking the roll radius corresponding to the centre of gravity of the shape as 100%, then $R_w = 395.5$ mm amounts to 99%. The radius of the top roll for point c will be $400-\frac{110}{2} = 345$ mm, or $\frac{345}{400}\times$ $\times 100\% = 86\%$. The radius of the bottom roll for point d will be equal to $400+\frac{110}{2} = 455$ mm, or $\frac{455}{400}\cdot 100\% =$ $= 114\%$. Consequently, the live (top) flanges at their extreme points are worked by the roll surfaces with a relative difference between their radii (and therefore speeds) equal to $114-86 = 28\%$. The bottom dead flanges are worked only by the bottom roll at the points which have a radius equal to $400-55 = 345$ mm, or 86%. The actual speed of rolling the beam will approach that corresponding to the effective roll diameter for the beam web (i.e. 99%). From the inside the live flanges are subjected to the action of the top roll revolving at a speed which is less than the web speed. From the outside the same flanges are subjected to the action of the bottom roll revolving at a speed which exceeds the web speed. The difference between the speeds increases from the web to the ends of the live flanges. Favourable conditions are created for working intensively the flanges to increase their height as required.

The bottom flanges are under quite opposite conditions. The bottom flanges are pulled, i.e. unfavourable conditions for their working are created and a decrease of their height (pulling down) takes place. Therefore the roll passes are usually designed in such a manner that each flange is worked alternately by the live and dead parts of the pass in the top and bottom rolls. In addition to the increment and pulling down of the flanges, the spread of the entire section is also taken into account. This spread depends mainly on the web reduction.

4.6. Rolling Pressure

The rolling process design involves calculation of loads imposed on the units and parts of the rolling-mill equipment.

The rolling pressure serves as a basic initial value for these calculations.

In case of a simplified rolling scheme (plain rolls of the same diameter, rectangular workpiece) the total rolling pressure P is determined from the following formula:

$$P = pB\sqrt{(H-h)r}$$

where p is the mean specific rolling pressure (normal stress)

$B = \frac{B_1+B_2}{2}$ is the mean width of the workpiece in the zone of deformation

In a general form

$$P = p \cdot F$$

where F = area of the metal-roll contact surface.

When the diameters of the top and bottom rolls are different

$$F = \frac{B_1+B_2}{2}\sqrt{\frac{2r_1r_2}{r_1+r_2}(H-h)}$$

where r_1 and r_2 = radii of the top and bottom rolls respectively.

When rolling pressures are high, e.g. in cold rolling of a metal having a high resistance to deformation, the rolls undergo elastic deformation and the length of the contact arc increases. In this case the area of metal-roll contact surface is determined from the formula:

$$F = \frac{B_1+B_2}{2}\left(x_0 + \sqrt{(H-h)r + x_0^2}\right)$$

where $x_0 = \frac{8(1-\mu^2)}{\pi E} rp$ is the increment of the contact arc

μ = Poisson's ratio for the roll material

E = Young's modulus for the roll material

r = roll radius

p = specific rolling pressure

The increment of the contact arc for steel rolls may be approximately determined from the formula:

$$x_0 = \frac{r}{9500} p$$

When rolling intricate sections in grooved rolls the area of the metal-roll contact surface is determined either graphically or by means of special nomograms.

To calculate the contact surface area in oval-square cogging-down passes one may use the formulas as proposed by V. Drozd:

$$F_{sq} = 0.75 b_1 \sqrt{r_m (h_0 - h_1)}$$

for square-oval passes:

$$F_{ov} = 0.54 (b_0 + b_1) \sqrt{r_m (h_0 - h_1)}$$

for diamond-square or diamond-diamond passes:

$$F_d = 0.67 b_1 \sqrt{r_m (h_0 - h_1)}$$

where h_0 = height of the workpiece section prior to rolling
h_1 = height of the workpiece section after rolling
b_0 = maximum width of the workpiece section prior to rolling
b_1 = maximum width of the workpiece section after rolling
r_m = radius of a roll in the middle of the pass

In the zone of deformation the specific pressure on the contact surface (along the contact arc) is distributed nonuniformly. Many investigators tried to establish the principles of the distribution of specific pressure along the contact surface in the zone of deformation. As far back as 1913, A. Rodzevich-Belevich, one of the pioneers in investigating the rolling process in Russia, proposed a method of an experimental research into the distribution of the specific pressure by means of a unique device (single-pin hydraulic dynamometer) built in the body of the roll. It was not until the thirties that this idea was realized by Lueg, V. Severdenko, E. Frolov, N. Svede-Shvets, I. Astakhov, A. Korolev and others in the USSR and abroad.

Fig. 4.10 shows the curve of the distribution of specific pressure along the contact arc during hot rolling which was obtained by V. Severdenko.

The specific pressure is distributed nonuniformly along the contact arc: at the beginning of the arc, it corresponds to the yield strength of the metal in the initial state, then it grows, reaches a certain maximum, decreases and at the end of the arc, where the workpiece leaves the rolls, it approximately corresponds to the yield strength of the rolled metal.

The effect of various factors on the character of the distribution of the specific rolling pressure has been investigated. The specific rolling pressure increases as the absolute or relative reductions rise. When rolling carbon steel, with increasing the steel temperature up to about 400 °C the specific rolling pressure first decreases, then, in the temperature range from 400 to 600 °C, it somewhat increases and, finally, at temperatures above 600 °C, decreases intensively.

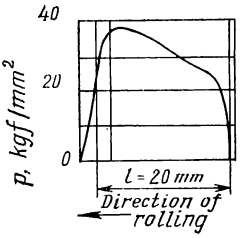


Fig. 4.10. Distribution of specific pressure along contact arc when rolling 10 mm thick piece with 54.6% reduction at 900° C

In hot rolling the specific rolling pressure increases with an increase in the rolling speed. It decreases with increasing the speed of cold rolling of very thin pieces. When rolling with front and back tension, the specific rolling pressure decreases and this is used in cold rolling. When strain hardening of the workpiece takes place, the specific rolling pressure increases. With increasing the coefficient of friction and surface roughness, the specific rolling pressure increases.

The method of measuring the total rolling pressure in laboratory and production conditions with special hydraulic dynamometers arranged under the housing screws of working stands has found wide application. Transducers used in the dynamometers send electrical signals which are proportional to the rolling pressure. These signals are directly fed to reading or recording instruments.

When rolling pressure P and area F of the metal-roll contact surface are known, the specific rolling pressure may be determined from the following formula:

$$p = \frac{P}{F}$$

4.7. Methods of Calculating the Specific Rolling Pressure

The most precise theoretical solution of the problem of determining the specific rolling pressure has been proposed by S. Gubkin, A. Tselikov and A. Korolev.

The specific rolling pressure or, as it is now usually termed, the normal contact stress (often called simply pressure) — P is dependent upon the inherent mechanical properties of the worked metal and also upon the conditions of rolling.

According to A. Tselikov, the main force acting on the rolls or the vertical component of resultant P may be expressed as

$$P = b_m \int_0^{l_a} p_r \, dx$$

where b_m = mean width of the zone of deformation
l_a = length of the contact arc
p_r = normal stress on the contact surface
dx = $r \cos 0 \, d0$
0 = running angle

The horizontal component

$$Q = P \tan \delta$$

where δ = angle between the resultant of the contact forces and the vertical axis:

$$\delta = \arctan \frac{Q}{P}$$

Force P is dependent upon factors of two classes. The first embraces factors influencing the mechanical properties of the metal worked, i.e. its resistance to deformation (inherent properties of the metal, temperature, rate and degree of deformation) and the second, factors influencing the character of the state of stress of the worked metal, i.e. contact frictional forces, outer zones, tension or compression of the workpiece, its width, etc.

The resistance to deformation is a characteristic of a material which determines its mechanical properties under particular conditions of the given working operation.

According to S. Gubkin the resistance to deformation determines the intensity of stresses sufficient to ensure plastic deformation of the given body or of its element under predetermined conditions of deformation.

In accordance with the energy theory of plasticity, there is a certain relation between the plastic deformation stress and the strain energy accumulated in the course of elastic deformation involving only a change in the shape of a body without any changes in its volume. The amount of this part of the strain energy of the body does not depend on the character of the state of stress. This proposition makes it possible to determine the mechanical properties of the worked metal by its resistance to deformation.

According to A. Tselikov, the following two characteristics are used in practical calculations:

(1) resistance to deformation σ_s in case of a linear state of stress which is called the actual resistance to the change in the shape in simple compression or tension with due regard for temperature and the rate and degree of deformation;

(2) resistance to biaxial deformation which is also called the resistance to pure shear:

$$k = \frac{\sigma_s}{\sqrt{3}} \approx 0.57 n_t n_h n_r \sigma_y$$

where coefficients n_t, n_h and n_r take into account the effects of temperature, strain hardening and deformation rate respectively, and σ_y is the yield strength of the worked metal determined under static conditions with standard rupture test machines.

Force P is determined by averaging the normal stresses on the contact surface:

$$P = F \cdot p_{av}$$

where p_{av} = average normal contact stress. (contact pressure)

F = projection of the metal-roll contact area onto a plane normal to the direction of force P

As it has been mentioned above, the average rolling pressure (specific rolling pressure) is dependent upon many factors:

$$p_{av} = m \cdot n_\sigma \sigma_s$$

where m = Lode's factor characterizing the effect of medium principal stress σ_2 on the difference between maximum σ_1 and minimum σ_3 principal stresses (according to the equation of plasticity for triaxial deformation).

When spread is comparatively large, $m = 1$ and when it is negligible (slabs, sheets, etc.), $\sigma_2 = \frac{\sigma_1 + \sigma_3}{2}$ and $m = \frac{2}{\sqrt{3}} \approx 1.15$.

n_σ is the stress state factor. As it is known, this factor may be expressed as

$$n_\sigma = n'_\sigma n''_\sigma n'''_\sigma$$

where n'_σ and n''_σ = factors taking into account the effect of contact friction and outer zones respectively. They are usually greater than unity

n'''_σ = factor accounting for the effect of tension or compression ($n'''_\sigma < 1$ for tension and $n'''_\sigma > 1$ for compression)

σ_s is a function of temperature and the rate and degree of deformation. It is usually determined on the basis of experimental data characterizing the relation between the yield strength and each particular parameter taken separately (for example, the temperature or the other two parameters, the third one being constant).

When specimens are tested at high temperatures their yield strength is usually considered to be equal to the ultimate strength.

From experimental diagrams it can be seen that:

(a) at temperatures above 1000 °C both the yield and ultimate strength of ordinary carbon steels do not depend upon the carbon content;

(b) when increasing the deformation rate, the ultimate strength (or the yield strength) increases;

(c) when increasing the degree of deformation, the ultimate (or yield) strength first rises and then (above 0.30) remains constant or decreases. While examining a steady symmetrical rolling process, A. Tselikov derived the equation of equilibrium of forces acting on finite element dx sepa-

rated from the zone of deformation in a plane parallel to the axis passing through the roll centres and located at any distance x from the plane of the metal leaving the rolls.

Pressure p_x and frictional force t_x act at every point of the contact surface and in a number of cases front or back tension may also act.

If there is no spread, the force exerted by the metal on the separated element from the entry side will be

$$(\sigma_x + d\sigma_x)(h_x + dh_x)$$

and from the delivery side, $\sigma_x h_x$.

Assuming that the direction which coincides with that of rolling is positive, the forces exerted by the rolls on the separated element will be:

in the zone of backward slip

$$2\left(-p_x \frac{dx}{\cos\varphi}\sin\varphi + t_x \frac{\cos\varphi}{\cos\varphi}dx\right)$$

and in the zone of forward slip

$$-2\left(p_x \frac{dx}{\cos\varphi}\sin\varphi + t_x \frac{\cos\varphi}{\cos\varphi}dx\right)$$

According to the laws of mechanics, when the process of rolling is steady, the sum of the projections of all the acting forces onto direction of rolling $X - X$ is equal to zero.

Then for the zone of backward slip we have:

$$\Sigma X = (\sigma_x + d\sigma_x)(h_x + dh_x) - \sigma_x h_x - \\ - 2p_x \tan\varphi\, dx + 2t_x\, dx = 0$$

If the height of the element separated from the zone of deformation is $h_x = 2y$ and the derivative $\frac{dy}{dx} = \tan\varphi$, then the equilibrium equation may be written as

$$(\sigma_x + d\sigma_x)(y + dy) - \sigma_x y - p_x \frac{dy}{dx}dx + t_x\, dx = 0$$

or

$$y\, d\sigma_x + \sigma_x\, dy - p_x \frac{dy}{dx}dx + t_x\, dx = 0$$

Thus, we have the differential equation of rolling for the zone of backward slip:

$$\frac{d\sigma_x}{dx} - \frac{p_x - \sigma_x}{y}\frac{dy}{dx} + \frac{t_x}{y} = 0$$

Respectively, for the zone of forward slip:

$$\frac{d\sigma_x}{dx} - \frac{p_x - \sigma_x}{y}\frac{dy}{dx} - \frac{t_x}{y} = 0$$

The general form of the differential equation of a steady process of rolling a rectangular piece between plain rolls will be:

$$\frac{d\sigma_x}{dx} - \frac{p_x - \sigma_x}{y}\frac{dy}{dx} \pm \frac{t_x}{y} = 0$$

where x and y are coordinates of the contact arc.

According to the law of dry friction

$$t_x = f \cdot p_x$$

Assuming that $\sigma_1 - \sigma_3 = m\sigma_y$ (according to the equation of plasticity), i.e. the vertical and horizontal stresses are principal ones for the analyzed element of the body deformed during rolling,

$$\sigma_1 = p_x \quad \text{and} \quad \sigma_3 = \sigma_x$$

$$p_x - \sigma_x = m\sigma_y = k, \quad \text{when} \quad \sigma_y = \text{const}$$

By substituting $t_x = f \cdot p_x$ in the differential equation we obtain

$$\frac{d\ (p_x - k)}{dx} - \frac{k}{y}\frac{dy}{dx} \pm \frac{f}{y}p_x = 0$$

Solving this differential equation:

$$p_x = e^{\mp\int \frac{f}{y}dx}\left(C + \int \frac{k}{y}e^{\pm\int \frac{f}{y}dx}dy\right)$$

This equation is cumbersome. Hence, for the convenience of calculations, A. Tselikov has replaced the contact arc by the chord bridging this arc. This has the following equation:

$$y = \frac{\Delta h}{2l}x + \frac{h}{2} \quad \text{and} \quad dx = \frac{2l}{\Delta h}dy$$

By making appropriate substitutions we have:

$$p_x = e^{\mp \int \frac{\delta}{y} dy} + \frac{h}{2}; \quad dx = \frac{2l}{\Delta h} dy$$

where $\delta = \frac{2lf}{\Delta h}$.

Integration constant C according to the initial conditions is:
when entering the rolls

$$p_x = k - \sigma_H \qquad \begin{matrix} x = l \\ y = \frac{H}{2} \end{matrix}$$

where σ_H = stress due to tension of the tail end of the workpiece caused by an external force
when leaving rolls

$$p_x = k - \sigma_h \qquad \begin{matrix} x = 0 \\ y = \frac{h}{2} \end{matrix}$$

where σ_h = stress due to tension of the front end of the workpiece.

Integrating and determining the integration constant yield equations which characterize the distribution of pressure during rolling:
for the zone of backward slip

$$p_x = \frac{k}{\delta}\left[\left(\frac{H}{h_x}\right)^{\delta} \cdot (\xi_H \delta - 1) + 1\right]$$

and for the zone of forward slip

$$p_x = \frac{k}{\delta}\left[\left(\frac{h_x}{h}\right)^{\delta} \cdot (\xi_h \delta + 1) - 1\right]$$

where ξ_H and ξ_h = coefficients which account for the effect of back and front tension

$$\xi_H = 1 - \frac{\sigma_H}{k}$$

$$\xi_h = 1 - \frac{\sigma_h}{k}$$

When there is no tension

$$\xi_H = \xi_h = 1, \text{ then}$$

for the zone of backward slip

$$p_x = \frac{k}{\delta}\left[\left(\frac{H}{h_x}\right)^{\delta}\cdot(\delta-1)+1\right]$$

for the zone of forward slip

$$p_x = \frac{k}{\delta}\left[\left(\frac{h_x}{h}\right)^{\delta}\cdot(\delta+1)-1\right]$$

To determine the total rolling pressure it is necessary to take the sum of pressures over the entire contact surface

$$P = \frac{B_1+B_2}{2}\int_0^l p_x\,dx$$

By using the Tselikov equations, which determine the law of distribution of the rolling pressure, we find the average pressure along the contact arc during hot and cold rolling of a wide strip (no spread being taken into account):

$$p_{av} = \frac{k}{U}\cdot\frac{\alpha}{f}\left[\left(\frac{1}{1-U}\right)^{\frac{f}{\alpha}-0.5} - \left(1-\frac{U}{2}\right)\right]\times$$
$$\times\left[1-\frac{\sigma_H-\sigma_h}{2k}\right]\ \text{kgf/mm}^2$$

where $U = \frac{H-h}{H}$ is the relative reduction

α = angle of bite, radian

σ_H and σ_h = stresses due to the back and front tension

An empirical formula deduced by Ekelund determines the relation between the specific rolling pressure on the one hand and the chemical composition of steel, temperature, the deformation rate and coefficient of contact friction on the other hand:

$$p_m = k_f(\sigma+\eta u)$$

where $k_f = 1+\frac{1.6f\sqrt{R(H_0-h_1)}-1.2(H_0-h_1)}{H_0+h_1}$ is the coefficient of external resistance

f = coefficient of friction

σ = (14-0.1t) (1.4 + C% + Mn% + 0.3Cr%) is the static resistance to deformation (kgf/mm^2) of

steel with corresponding carbon, manganese and chromium content (%)

t = rolling temperature, °C

$\eta = 0.01\,(14-0.01t)\,C$ is the coefficient of viscosity, kgf·s/mm²

C = coefficient taking into account the effect of the rolling speed:

v, m/s	C
Up to 6	1.0
6-10	0.8
10-15	0.65
20-25	0.60

$u = \dfrac{2v\sqrt{\dfrac{H_0-h}{R}}}{H_0+h_1}$ s^{-1} is the relative deformation rate

v = peripheral speed of the rolls, mm/s

H_0 and h_1 = initial and final height of the workpiece, mm

Values of the specific rolling pressure calculated from the Ekelund formula agree well with experimental data obtained during hot rolling of carbon and chromium steels in cogging, heavy section and sheet mills. For thin strips this formula gives underrated results.

The Golovin—Sims equation is used in the theory and practice of automatic control of the workpiece thickness as a function of the rolling pressure:

$$P = E_0\,(h-s)$$

or

$$h = s + \frac{P}{E_0}$$

where h = roll gap during rolling (thickness of the rolled strip)

s = roll gap prior to rolling

P = rolling pressure

E_0 = Young's modulus of the roll mill stand

$\dfrac{P}{E_0}$ = deformation of the stand

4.8. Rolling Torque and Power

The most important power parameter of the rolling process is the torque required for plastic working of metal under the given conditions of rolling. For two driven rolls the torque required to deform the metal may be presented in the following simplified form:

$$M_d = 2a \cdot P$$

where P = total rolling pressure
$a = \psi r\alpha$ is the rolling torque arm (Fig. 4.11)
r = radius of the rolls
α = angle of bite, radian
ψ = torque arm coefficient which is equal to the ratio between the length of the torque arm and that of the contact arc

According to experimental data $\psi = 0.45\text{-}0.5$ for hot rolling and $\psi = 0.35\text{-}0.45$ for cold rolling. When the rolls undergo a considerable elastic deformation (flattening) the torque arm is increased (see Fig. 4.11).

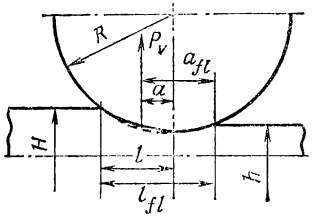


Fig. 4.11. Determining rolling torque

The front and back tension Q_f and Q_b also influence the deformation torque:

$$M_d = 2Pa + (Q_f - Q_b)r$$

The rolling torque is determined as the sum of the deformation and friction torques:

$$M_r = M_d + M_f$$

The friction torque takes account of friction in the roll-neck bearings:

$$M_f = f \cdot Pd_n$$

where f = coefficient of friction in the roll-neck bearings
d_n = diameter of the roll neck

The coefficient of friction amounts to 0.07-0.1 for bronze bearings and to 0.005-0.01 for fabric bearings.

An extra friction torque is produced in the back-up roll bearings of four-high mills. This torque referred to the axis

of rotation of the work rolls is

$$M_f = Pfd_n \left(\frac{D_w}{D_b}\right)$$

where D_w and D_b are the diameters of the work and back-up rolls.

The rolling power may be determined from the following formula:

$$N = M_r \frac{v}{r} = \frac{M_r \cdot n}{0.716} \text{ hp}$$

where v = rolling speed
n = roll revolutions per minute

The power delivered by a motor at a constant speed (static power) is expressed as

$$N = 1.025 \frac{M_s n}{\eta_{ov}} \text{ kW}$$

where η_{ov} = 0.82-0.93 is the overall efficiency of the drive train (spindles, pinion stand, speed reducer, couplings, etc.)

$M_s = M_d + M_f + M_i$ is the static torque of the motor

$M_i = K\Sigma \frac{Q_n f_n d_n}{2i_n}$ is the idling torque

Q_n = mass of each rotating part

f_n and d_n coefficient of friction and diameter of the neck respectively in the bearings of each component of the drive train

i_n = gear ratio from the motor to each component of the drive train

K = bearing preload factor (amounts to 1.1-1.2 for blooming and plate mills, 1.2-1.45 for heavy-section mills and 1.50-1.80 for light-section mills)

In reversing mills or in mills equipped with flywheels and running at a variable speed the motor torque varies during one rolling cycle or during one passing, and in a general form, is expressed as

$$M_m = M_s \pm M_d$$

where M_d = dynamic torque of the motor.

The plus sign corresponds to acceleration, while the minus one, to deceleration.

The dynamic torque is

$$M_d = J \frac{\pi}{30} \varepsilon$$

where $J = \frac{GD^2_{m.sh}}{4g}$ is the moment of inertia of rotating parts referred to the motor shaft

D = diameter of gyration

G = mass of rotating parts

g = acceleration due to gravity

ε = angular acceleration (for electric motors of rolling mills ε = 40-60 rpm/s)

$GD^2_{m.sh}$ = moment of gyration of all the rotating parts of the drive train (including the motor) referred to the motor shaft. This moment is calculated by summing up the moments of gyration of each rotating part of the drive train with due regard for gear ratio i_n from the motor to each rotating part:

$$GD^2_{m.sh} = \sum \frac{G_n D^2_n}{i^2_n}$$

When selecting the motor characteristics, the power parameters are determined by using experimental curves. The overload capacity of the motor is also taken into consideration and, finally, the motor is checked for heating.

CHAPTER V. PRINCIPLES OF ROLL PASS DESIGN

5.1. Types and Shapes of Passes

The following types of roll passes are known:

(1) cogging-down passes (used for initial cogging and reducing the cross-sectional area of ingots or billets);

(2) preleader or roughing passes (used to reduce the cross-sectional area of the rolled section and impart it the first shape);

(3) leader passes (these are next to the last passes which impart the final shape and size to the rolled section);

(4) finishing passes (used to produce the final section).

There are box, gothic, diamond, oval and square cogging-down passes (Fig. 5.1) for rolling simple sections (squares, rounds and flats). Forming passes for rolling structural sections are more complicated.

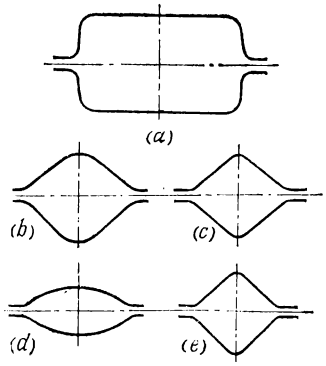


Fig. 5.1. Types of cogging-down passes (*a*) flat (box) pass; (*b*) gothic pass; (*c*) diamond pass; (*d*) oval pass; (*e*) square pass

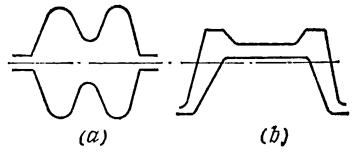


Fig. 5.2. Types of passes: (*a*) open; (*b*) closed

A roll pass is called closed if the roll parting line is outside the pass. In an open pass this line is within the boundaries of the pass outline (Fig. 5.2).

5.2. Roll Gap and Taper of Sides

Under the pressure exerted by the rolled metal the gap between the rolls increases, the units and parts of the working stand undergo elastic deformation and the clearances in their joints, which depend upon the accuracy of machining and assembling the parts, are taken up. The amount of the deformation depends upon the design and material of the working stand components.

When designing the final roll passes the clearances, elastic deformation of the stand (mill springing) and permissible amount of wear on the rolls are taken into account without fail.

The side surfaces of the roll passes are usually made not exactly upright, but inclined at some angle to the longitudinal axis of the rolls. The taper of the side walls (taper of

sides) is expressed either in percent or as the tangent of the angle of inclination:

$$\tan \psi = \frac{B_g - b_g}{2h_g}$$

where B_g and b_g — maximum and minimum width of the groove

h_g — height of the groove

The taper of sides insures the correct biting of the workpiece with respect to the centre line of the pass and prevents

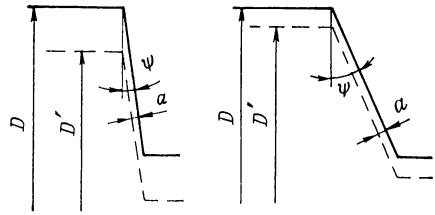


Fig. 5.3. Effect of side taper on roll regrinding

jamming of the workpiece by the side walls, as well as the danger of roll collaring. If the side walls of the roll pass are tapered, the width of the pass is retained after the roll diameter has been reduced (after redressing) and hence, the service life of the pass is increased. Therefore, one always tries to increase the side taper wherever possible (Fig. 5.3). In case of rolling with a large spread an adequate side taper and properly selected radii of curvature prevent the formation of fins (seams, laps, collar cuts) on the side faces of the rolled metal.

Grooves cut into the roll barrel are separated from one another by collars. The width of the collars is selected so as to make the best use of the barrel length, but their minimum width is limited by their strength and also by the design features associated with the location of the roll fittings.

5.3. Pass Arrangement

A line which divides in half the distance between the axes of a pair of rolls is called the roll parting line. The neutral line of a pass is such a line which, being matched

with the roll parting line, ensures straight delivery of the workpiece. In this case the moments of forces applied to the workpiece section are considered to be equal with respect to the top and bottom rolls. A line along which the roll passes are arranged is called the pass line (Fig. 5.4).

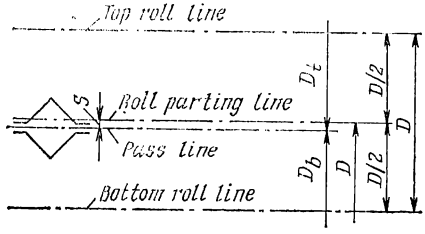


Fig. 5.4. Position of pass line in case of top pressure

To prevent unpredictable bending of the workpiece upon leaving the rolls (as the result of a nonuniform temperature distribution across the workpiece section) conditions are specially created under which the workpiece bends in a definite direction (upwards or downwards), this making it possible to ensure the normal straight delivery of the workpiece by using a top or bottom delivery guide. Rolls of different diameters are used for this purpose. Upon leaving the rolls, the workpiece will bend towards the roll of the smaller diameter. When the diameter of the top roll is greater than that of the bottom one, the so-called top pressure acts (see Fig. 5.4), and vice versa, when the greater is the diameter of the bottom roll, the rolling is called to be done with bottom pressure.

Rolling with top pressure requires the use of a bottom delivery guide, while a top guide is necessary when rolling is done with bottom pressure. If there is neither top nor bottom pressure or if these are negligible, both the top and bottom delivery guides are required. Rolling with bottom pressure is used in large cogging mills (blooming and slabbing mills). In these mills the billets of large cross sections cannot collar the rolls and the bottom pressure is applied merely to prevent the striking of the ends of blooms or slabs against table rollers. In case of rolling with bottom pres-

sure in blooming mills the difference between the diameters of the bottom and top rolls $m = D_b - D_t$ is usually equal to 10-12 mm. Rolling in two-high section mills is usually done with top pressure and in this case the difference between the roll diameters $m = D_t - D_b$ is 2-6 mm. Neither top nor bottom pressure is applied when rolling in final and

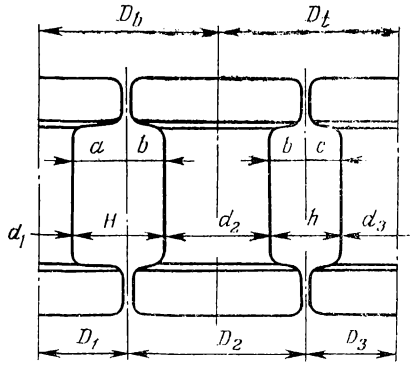


Fig. 5.5. Roll pass arrangement in three-high mill

leader passes because the difference between the effective roll diameters may give rise to additional internal stresses in the metal.

If there is neither top nor bottom pressure, the pass line coincides with the roll parting line, i.e. $S = 0$ (see Fig. 5.4). The neutral lines of passes must coincide with the pass line.

In case of rolling with top pressure the distance between the roll parting line and pass line (see Fig. 5.4) is

$$S = \frac{D}{2} - \frac{D_b}{2}$$

and

$$D_b = D - \frac{m}{2}$$

then

$$S = \frac{D}{2} - \frac{D - \frac{m}{2}}{2} = \frac{m}{4}$$

In simple symmetrical passes the neutral line of the pass coincides with its axis of symmetry. The neutral line of forming passes usually runs through the centre of gravity of the pass shape. The centre of gravity is determined analytically, graphically or practically (by suspending a pass template cut out of cardboard or sheet iron).

In three-high mills the roll passes (Fig. 5.5) are often made coupled, i.e. the groove cut into the middle roll is common for both the top and bottom passes. Such an arrangement ensures better use of the roll barrels, especially in cogging stands with rectangular (box) passes, but is disadvantageous in that it has an increased difference between the effective diameters of the rolls, thus giving high top or bottom pressure.

5.4. Initial Roll Diameters and Roll Redressing Coefficients

Imaginary diameters which make the rolls contact each other without any gap are referred to as the initial diameters. In redressing the diameters of new rolls $D_{max.}$ are reduced down to the minimum permissible value $D_{min.}$ A diameter corresponding to the distance between the axes of the pinion shafts is sometimes referred to as rated diameter D_0. But more often D_0 is determined from the formula:

$$D_0 = \frac{D_{max} + D_{min}}{2}$$

The roll redressing coefficient is expressed as

$$k = \frac{D_{max} - D_{min}}{D_0}$$

Hence,

$$D_{max} = \left(1 + \frac{k}{2}\right) D_0$$

$$D_{min} = \left(1 - \frac{k}{2}\right) D_0$$

The roll redressing coefficient amounts to 0.08-0.12 for section mills and to 0.05-0.08 for sheet and plate mills.

The minimum diameter of the rolls is determined by their strength.

The mean effective diameter of the rolls is such a diameter at which the peripheral speed of the rolls corresponds to the speed of the workpiece leaving the rolls:

$$v = \frac{\pi n}{60} D_{m.e}$$

where v = speed of the workpiece delivery (without accounting for forward slip)
$D_{m.e}$ = mean effective diameter
n = rpm of the rolls

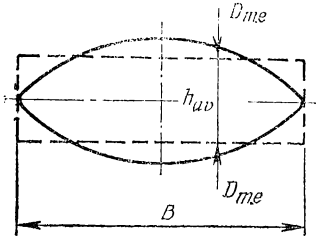


Fig. 5.6. Determining average pass height

For box passes the mean effective diameter corresponds to that at the pass bottom. The effective diameter of an intricate pass is difficult to determine, and in practical calculations the average referred height of the pass (Fig. 5.6) is usually used for this purpose:

$$h_{av} = \frac{F}{B}$$

where F = cross-sectional area of the pass
B = width of the pass

Then

$$D_{m.e} = D - h_{av} = D - \frac{F}{B}$$

5.5. Distribution of Deformation Among Passings

The roll pass design is aimed at the determination of optimum deformation in each passing and of the number of passings, as well as at the creation of rolling conditions necessary to obtain rolled sections which will meet the prescribed requirements. It must provide for high output of the rolling mill, reduction of labour consumption in rolling and improvement of the technical and economical characteristics of the rolled metal production (consumptions of metal, power, fuel, rolls, etc.).

For passes where the reduction of metal is different in different sections of the workpiece along its width average

height h_{av} of the pass is determined:

$$h_{av} = \frac{F}{B}$$

Average reduction in height Δh_{av} may be obtained as the difference between average heights or between the heights of rectangles which have their areas equal to that of the given workpiece section:

$$\Delta h_{av} = H_{av} - h_{av} = \frac{F_0}{B_0} - \frac{F_1}{B_1}$$

The most typical cases of nonuniform distribution of deformation along the width of the section being rolled are considered below. By dividing the given cross section of the

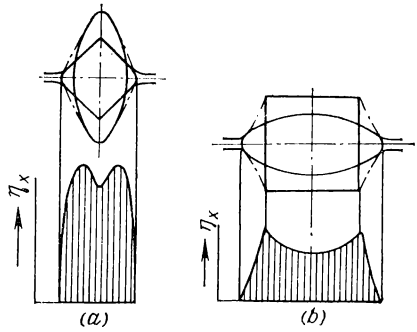


Fig. 5.7. Distribution of draughts along pass width in square pass (*a*); and in oval pass (*b*)

whole workpiece into a number of vertical sections, let us determine conditionally the draught of each of them. These draughts are called the natural draughts. Let us construct a diagram of natural draughts η_x along the width of the workpiece for the square-oval and oval-square methods of rolling (Fig. 5.7). The draughts are distributed nonuniformly. In both cases the draught for the whole section will correspond to the ratio of the initial to the final section $\frac{F_0}{F_1}$ and its value will be intermediate between the largest and smallest natural draughts in different vertical sections.

With the given combination of the shapes of the pass and the workpiece section, a tendency to greater spread or to

greater elongation is observed, depending upon the distribution of the natural draughts in the extreme and medium sections of the workpiece along its width. A square workpiece, rolled in an oval pass (Fig. 5.7*b*) tends to spread, but an oval workpiece rolled in a square pass (as well as a diamond workpiece rolled in a square pass) tends to lengthen (Fig. 5.7*a*).

The foregoing formulas are used to solve the problem of the distribution of deformation among passings, the number of the passings being preset, or to determine the number of passings proceeding from the total deformation required and the average deformation per passing.

Similarly, we may establish relationships between the total and average coefficients of reduction η_t and η_{av} on the one hand and the number of passings n on the other hand:

$$n = \frac{\log \eta_t}{\log \eta_{av}} = \frac{\log H_0 - \log H_n}{\log \eta_{av}}$$

If the average deformation per passing is known, the number of passings may be determined approximately. In modern rolling mills the number of passings, as a rule, is equal to that of the stands engaged in the given system of rolling, because only one passing is done in each stand.

The selection of the amounts of deformation (draughts) for individual passings is a very complex problem in the roll pass design. In the line-type rolling mills, whose output depends on the number of passings, one tries to use the maximum deformations (reductions, draughts) in each passing. When selecting the maximum (optimum) draughts per passing, the ductility of the rolled metal at the given temperature, the conditions of the metal biting by the rolls, the shape of the pass, the roll wear and strength, the power of the main drive motor, as well as the requirements for the quality of the metal and the dimensional accuracy of the rolled section, are taken into consideration.

The draught in rolling is not only the most specific characteristic of the degree of deformation, but also the most convenient quantity which determines the change of the cross-sectional areas of the rolled piece in passings. It is the most important parameter used in designing the roll passes and rolling schedules.

The distribution of deformation among passings depends on the increase of the resistance to deformation with the decrease of the metal temperature and the reduction of the workpiece thickness during rolling.

The change of the degree of deformation (draught) from passing to passing is set in accordance with a linear (Geuze) or parabolic (Kirchberg) law (Fig. 5.8). However, such regularities of the change of deformation, which do not account for actual conditions of rolling in each passing (especially in the first and the last ones), cannot serve as natural criteria in the roll pass design. One should take into consideration the capabilities of the main rolling mill equipment and the requirements for the dimensional accuracy and structure of the rolled section. In the majority of cases in the first passings, when the cross section of the workpiece is large and the temperature is high, a considerable degree of deformation can be provided for, its maximum being limited mainly by the conditions of the metal biting by the rolls. The old opinion that great draughts are impermissible as causing destruction of the metal has been refuted by practice.

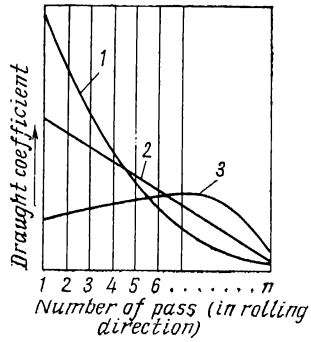


Fig. 5.8. Distribution of coefficients of draught among passings

1—according to Geuze; *2*—according to Kirchberg; *3*—generally adopted

The application of light reductions in the first passings when rolling cast ingots is not always justifiable because the plastic properties of cast metal depend on the techniques of its making and casting, the content of deleterious impurities, the character of heating (the use of prolonged homogenizing soaking), etc. After the first passings, when the conditions of the metal biting are no longer a limiting factor, the deformation can be increased, its maximum being now limited by the strength characteristics of the rolls and other units of the stand, as well as by the capacity of the main drive motor. In the last passings the reductions and draught

are decreased since the resistance to deformation increases as the metal temperature drops and the workpiece thickness decreases. To decrease the groove wear and ensure the required surface finish and dimensional accuracy of the final shape only light reductions are applied in the finishing passes.

In rolling mills with individual working stand drives the distribution of reductions among the stands is controlled by making direct measurements or by using ammeter readings.

When designing the passes for rolling alloyed steels, it is necessary to decrease the draught as compared to that adopted for common carbon steels. In this case a correction is made to take into account the relation of the ultimate strength of a given special-purpose steel or alloy at the rolling temperature to that of carbon steel. A. Protasov recommends the following simple formula for the calculation of the draught in rolling alloyed steels:

$$\mu_t' = 1 + (\mu_t - 1)\frac{\sigma_t}{\sigma_t'}$$

where μ_t = draught coefficient for common steel at the rolling temperature

σ_t and σ_t' = ultimate strength of carbon and alloyed steels at the rolling temperature

5.6. Finishing Pass Design

When designing the shape and calculating the dimensions of the finishing passes, corrections are made to take into account the thermal expansion of the workpiece, the irregularity of the temperature distribution across the rolled shape section and the pass wear. The coefficient of linear expansion of iron $\alpha = 0.000012$. The linear dimensions of a hot section a_h at the temperature corresponding to the end of rolling may be determined from the formula

$$a_h = a_c(1 + \alpha t)$$

where a_c = linear dimensions of the cold section.

In this formula the monomial $(1 + \alpha t)$ is equal to 1.010 at 800 °C, 1.012 at 1000 °C, 1.013 at 1100 °C and 1.0145 at 1200 °C.

Individual portions of the rolled section are unevenly cooled during rolling, and therefore, their dimensions change by different amounts. Thus, a leader diamond may be distorted along the diagonals during rolling in a finishing square pass (Fig. 5.9). Since the sharp corners of the diamond are cooled more rapidly than the rest of its section, the horizontal (larger) diagonal has a greater shrinkage during cooling and the dimensions of the square rolled from such a

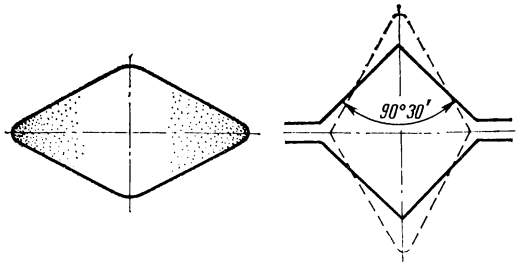


Fig. 5.9. Cooling of corners of diamond-shaped piece

diamond will be disturbed. To prevent this distortion the horizontal diagonal of the square pass is made somewhat longer by increasing the angle at the apex. Some alterations are also made in the configurations of the finishing round passes.

Individual parts of a roll pass do not wear evenly, this being caused mainly by the nonuniform deformation and cooling. This circumstance should be taken into consideration when designing the finishing passes.

The dimensions of the final shape must meet the requirements of the corresponding standards. The standards permit deviations both above and below nominal size (plus and minus tolerances). But to decrease the mass of a running metre of rolled products only minus tolerances are used. If the minus tolerance is presumed to be of the order of 1.5% of the shape size, which corresponds to the decrease of the nominal size by a factor of 1.015, then this is much the same as the temperature correction factor used when designing hot sections. Therefore in many cases to obtain the final section dimensions with minus tolerances the fi-

nishing pass dimensions are made to be equal to those of the cold section, i.e. no corrections are made to take the rolling temperature into account.

When designing the finishing passes, the mill springing and wear on the pass surface are also taken into consideration.

5.7. Specific Features of Roll Pass Design for Continuous Mills

In case of continuous rolling the workpiece is rolled simultaneously in all or several stands of the mill (Fig. 5.10). The basic condition of the process continuity is the equality

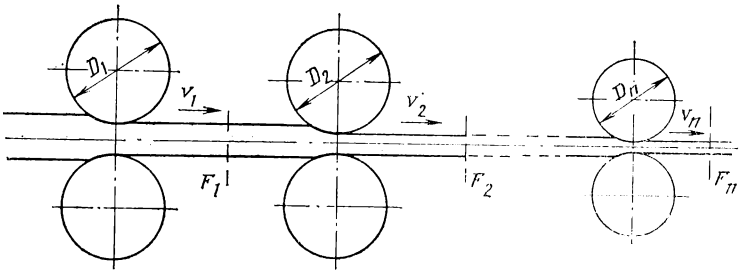


Fig. 5.10. Schematic representation of continuous rolling

of the second volumes of metal:

$$v_1 F_1 = v_2 F_2 = \ldots = v_{n-1} F_{n-1} = v_n F_n = \text{const}$$

or in a general form

$$vF = \text{const}$$

Expressing the speed in terms of the roll diameter and rpm and taking into account the forward slip:

$$\frac{\pi D n}{60} (1 + S) F = \text{const}$$

and after cancelling by constant $\pi/60$ we have

$$DnF (1 + S) = C$$

where C = roll pass design constant.

If the forward slip is disregarded, then

$$DnF = C$$

By calculating from this formula the value of the constant for some stand, the roll rpm for any other stand may be determined:

$$n = \frac{C}{DF}$$

or the cross-sectional area of the pass

$$F = \frac{C}{Dn}$$

where D = working (effective) diameter of the rolls.

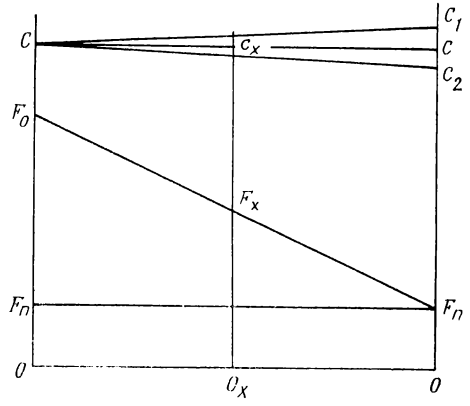


Fig. 5.11. Logarithmic pattern of continuous rolling

The equality of the second volumes of metal may be expressed in a logarithmic form:

$$\log D + \log n + \log F = \log C$$

or

$$\log v + \log F = \log C$$

To illustrate the continuous rolling process and to make the selection of its main characteristics more convenient the author has suggested a logarithmic pattern (Fig. 5.11), where

OF_0F_nO is the diagram of the workpiece cross sections, $OCCO$ — the diagram of the roll pass design constant, and F_0CCF_n — the diagram of the compensating speed component of the second volumes of metal.

Speed component F_xC_x providing for the equality of the second volumes of metal may be found at any point along the horizontal axis for any variable cross section of the workpiece O_xF_x. This component is expressed as $\log v = \log D + \log n$ or, in a general form, as $\log v_x$ (in this case the forward slip may be taken into account). When a slight tension is preset, the constant will take position CC_1. Constant CC_2 characterizes looping.

5.8. Effect of Tension on Section Pulling Down in Continuous Rolling

The roughing-stand groups of continuous mills, as well as their finishing stands, operate with tension. A tension of up to 1-2% prevents looping. When operating with a preset tension the changes of the metal temperature and the workpiece cross section, as well as deviations from the adopted rolling speed schedule, etc. are compensated. In continuous rolling (in contrast to endless rolling) the tension grows gradually as the head end of the workpiece moves from the first stand to the last one. When the workpiece is rolled in all the stands simultaneously, the tension is at its maximum and remains constant until the tail end of the workpiece leaves the first stand. Since the tension changes the character of the state of stress of the deformed metal, the conditions of lateral deformation are also changed. The spread tends to zero or becomes even negative as the tension increases.

In hot rolling lateral contraction (pulling down) of the workpiece section may occur not only in the zone of deformation, but also between the stands. The shorter the distance between the stands, the smaller the interstand contraction.

The differences between the cross-sectional dimensions at the ends and the middle of the rolled piece must be within the tolerances.

CHAPTER VI. ROLL PASS DESIGN FOR COGGING AND ROUGHING STANDS

6.1. Fundamentals of Rectangular (Box) Pass Design

The rectangular or so-called box passes are used in cogging and billet mills, as well as in the first cogging stands or roughing-stand groups of section mills. Several passings with different reductions can be performed in box passes of two-high cogging mills where the top roll is movable in the vertical direction. These passes differ from other ones in that they provide for uniform deformation and have shallow grooves in the rolls.

The box passes are disadvantageous in that they ensure the reduction of the workpiece in one plane only and make it impossible to obtain straight edges on the workpiece because of their great taper of sides.

A box pass has the following main dimensions: depth of groove, overall height, width, and overall width at the roll parting line. The latter is determined by the amount of the side taper and the depth of the groove. The side taper (unilateral) is tolerated within the limits of 10-50% of the groove depth. Large side tapers are used in roughing stands of light-section and wire-rod mills. A side taper within a range of 10-20% is generally advisable. When determining the pass width at the roll parting line, provision is made for an increased spread.

The pass opening or gap (t) varies over a wide range depending on the workpiece thickness and mill characteristics. For section mills it is

$$t = (0.1\text{-}0.3)\, h$$

In blooming and billet mills the pass bottom is sometimes made convex (up to 5 mm) which makes for the stability of the workpiece on the rollers of the mill table.

In the box-pass design, as well as in other cogging- or breakdown-pass design systems, the most difficult

problem is the determination of optimum intermediate sections.

It is common for the breakdown pass series to have square or round initial passes (odd-numbered in most cases) and intermediate sections in the form of rectangles, oblong hexagonal diamonds and ovals. Therefore, the main problem of the roll pass design, i.e. determining the geometrical shapes and dimensions of the given series of breakdown passes, may be presented as follows: first the dimensions of the initial and final squares are determined and the number of passings, specified (proceeding from the average draught throughout the mill and the arrangement of the mill stands) and then the sections of all other squares, as well as of other shapes intermediate between them (rectangles, diamonds, ovals, etc.), are calculated. The determining of the intermediate sections involves the calculation of spread. When calculating the spread in rolling between plain rolls, maximum deviations from the experimental data come to 15-20%. These deviations further increase under the conditions of nonuniform deformation. In their theoretical calculations of the optimum shape and dimensions of intermediate passes B. Bakhtinov and M. Shternov, M. Mutyev, N. Litovchenko, B. Diomidov, and V. Kurdyumova largely depend on experimental data.

As far as the general calculation scheme is concerned, B. Bakhtinov and M. Shternov suggest the following method for determining the dimensions of intermediate rectangles and distributing the draughts among rectangles and squares:

1. First the dimensions of intermediate square passes are determined.

The total draught is divided into paired draughts from a larger square to the next smaller square:

$$\mu_{sq} = \frac{F_0}{F_1} \frac{F_1}{F_2} = \mu_1 \mu_2 = \mu_{av}^{\prime 2}$$

where $\mu_{av}^{\prime}$ = preset average draught for the given pair of adjacent square passes.

The draughts for each pair of passes are corrected so as to ensure that their product corresponds to the total draught for the given series of passes.

The areas of the squares for the rectangle—square series (Fig. 6.1) are

$$F_0 = F_2\mu_{sq} \quad \text{and} \quad F_2 = \frac{F_0}{\mu_{sq}}$$

and draught

$$\mu_{sq} = \frac{A^2}{a^2} = (\mu'_{av})^2$$

$$A = a\mu'_{av}; \quad a = \frac{A}{\mu'_{av}}$$

$$\mu'_{av_1} \cdot \mu'_{av_2} \cdot \mu'_{av_3} \ldots \mu'_{av_n} = \frac{A_{in}}{a_{fin}}$$

According to this calculation scheme the draughts and sections of squares are determined beginning from the first

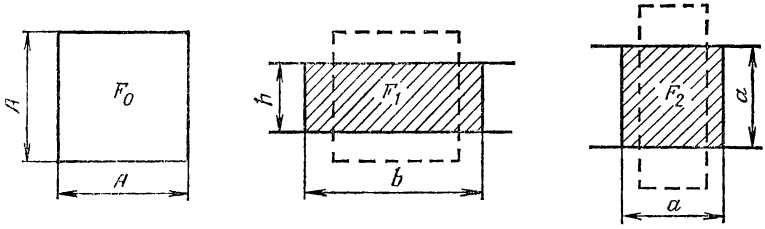


Fig. 6.1. Rectangle—square pass series

passing. The distribution of deformations among the passings is checked proceeding from the equality of rolling power conditions in each passing.

2. Then the dimensions of the intermediate rectangle (between the squares of the given pair) are determined.

For the sake of simplicity let us assume that the rectangles and squares have sharp right-angle corners and are rolled between plain rolls (see Fig. 6.1.). The area and dimensions of the rectangle are determined on the basis of the selected dimensions of the squares (A and a), provided the rectangle and square are completely filled with metal.

The draught for the rectangle

$$\mu_r = \frac{F_0}{F_1} = \frac{A^2}{bh}$$

that for the square

$$\mu_{sq} = \frac{F_1}{F_2} = \frac{bh}{a^2}$$

their ratio

$$\frac{\mu_r}{\mu_{sq}} = \frac{A^2a^2}{b^2h^2} = \frac{F_0F_2}{F_1}$$

whence

$$bh = \frac{Aa}{\sqrt{\frac{\mu_r}{\mu_{sq}}}}$$

It is adopted that

$$bh = \frac{Aa}{\sqrt{\frac{\mu_r}{\mu_{sq}}}} \approx \text{const}$$

The area of the intermediate rectangle is found from the ratio μ_r/μ_{sq} which is designated as $K_{\Delta b}$ (dependent on the spread):

$$K_{\Delta b} = \frac{\mu_r}{\mu_{sq}} = \frac{A^2a^2}{b^2h^2}$$

$$F_1 = \sqrt{\frac{F_0F_2}{K_{\Delta b}}}$$

Hence it follows that

$$Aa = bh\sqrt{\frac{\mu_r}{\mu_{sq}}}$$

and

$$Aa = bh\sqrt{K_{\Delta b}}$$

It may be easily shown that if $K_{\Delta b} = 1$, then

$$\frac{A^2a^2}{b^2h^2} = 1$$

$$Aa = bh$$

$$A = b - \Delta b_1$$

$$h = a - \Delta b_2$$

where Δb_1 and Δb_2 = spreads in rectangular and square passes.

Whence

$$(b - \Delta b_1)\, a = b\,(a - \Delta b_2)$$

$$a\Delta b_1 = b\Delta b_2$$

$$\frac{\Delta b_1}{b} = \frac{\Delta b_2}{a}$$

Consequently, when the draughts in the adjacent rectangular and square passes are equal, the relative spreads in these passes are also equal.

Usually $K_{\Delta b} > 1$. To make the calculations more simple B. Bakhtinov and M. Shternov suggested diagrams for determining $K_{\Delta b}$ under various rolling conditions (Fig. 6.2).

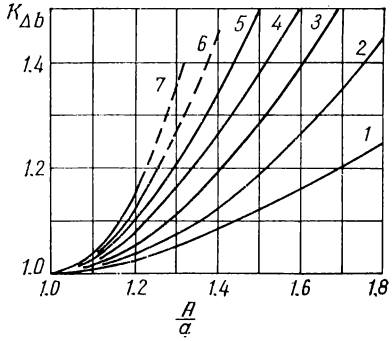


Fig. 6.2. Relationship between $K_{\Delta b}$ and average draught A/a in box pass series (unrestricted spread) for different roll diameters (D) and coefficients of friction (f):

Number of curve	A, mm	D, mm	f
1	120	400	0.50
2	60	300	0.40
3	60	350	0.50
4	30	300	0.35
5	20	250	0.32
6	15	250	0.30
7	10	250	0.28

When rectangular passes have sufficient space for spreading, the dimensions of intermediate passes are calculated on the basis of the condition that the spread is proportional to the reduction (taking into consideration the possibility for controlling the spread by changing the reductions):

$$\Delta b = k\Delta h$$

An approximate roll pass design method which uses coefficient k as a characteristic of spread whose value is equal in a number of box passes has found wide application. Assuming that the side of the larger square is

$$A = a + \Delta h\,(1 - k)$$

and that of the smaller square,

$$a = A - \Delta h\,(1 - k)$$

and substituting

$$k\Delta h = \Delta b,$$

we have

$$A = a + (\Delta h - \Delta b)$$

$$a = A - (\Delta h + \Delta b)$$

With high rolling temperatures reductions in cogging stands are chosen to be as great as it is possible to achieve with the maximum angles of bite, the given strength of rolls and the main drive capacity. It is recommended that the maximum angle of bite α be 22-26° for steel rolls and 18-22° for cast-iron rolls. In continuous billet mills with forced feed of the workpiece the maximum angles of bite may be as great as 26-32° for steel rolls and 22-26° for cast-iron rolls.

The maximum reductions are related to the value of the coefficient of friction, the latter being determined from Ekelund's empirical formula. In such cases the use of the following simplified formula is recommended

$$\Delta h_{\max} \approx \alpha^2_{\max} R \approx \frac{D}{2} f^2$$

For comparatively small angles of bite and assuming that the contact arc is equal to its chord

$$R\alpha \approx \sqrt{R\Delta h}$$

whence

$$\alpha \approx \sqrt{\frac{\Delta h}{R}}$$

This method may be used for preliminary calculations.

6.2. Diamond—Square Series

The diamond—square series of breakdown passes shown in Fig. 6.3 is widely used for producing finished square sections. This series ensures comparatively high coefficients of draught and provides for the stability of the workpiece during rolling. With the passes of this series the corners of squares are well filled with metal and hence, there is a possibility for obtaining good intermediate squares. Squares of different dimensions are sometimes produced in a single pass by changing the roll lift.

The passes of the diamond—square series are disadvantageous in that their grooves are cut deeper into the roll bodies

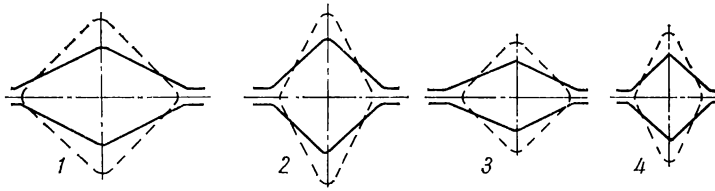


Fig. 6.3. Diamond—square series of passes

(as compared to flat or box passes) and wear more intensively due to the nonuniform deformation of the workpiece and uneven cooling of its corners.

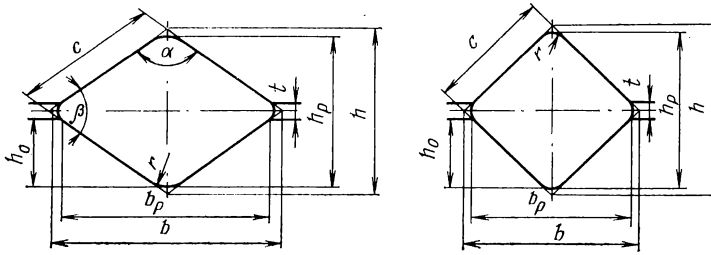


Fig. 6.4. Dimensions of diamond and square passes

The dimensions of diamond and square passes of this series (Fig. 6.4) are determined from the following geometrical relations.

The width of the diamond groove cut

$$b_p = b\left(1 - \frac{t}{h}\right)$$

The height of the diamond with rounded corners

$$h_p = h - 2r\left[\sqrt{1 + \left(\frac{h}{b}\right)^2} - 1\right]$$

The cross-sectional area of the diamond

$$F = \frac{1}{2}bh$$

Angle β is determined from the formula

$$\tan \frac{\beta}{2} = \frac{h}{b}$$

then

$$\alpha = 180 - \beta$$

The cross-sectional area of the square

$$F = \frac{1}{2} bh$$

The side of the square

$$a = \sqrt{F}$$

When $b = h$

$$a = \sqrt{\frac{1}{2} b^2} = 0.707b$$

The width of the square groove cut

$$b_p = b - t$$

The height of the square with rounded corners

$$h_p = h - 2r(\sqrt{2} - 1) = h - 0.83r$$

The groove depth

$$h_0 = \frac{h_p - t}{2}$$

The angle at the apex of finishing square passes is adopted to be 1.58 radians (90° 30′) and that of breakdown square passes, 1.62 radians (93°).

When designing the diamond—square and diamond—diamond series of breakdown passes the width of the next pass is usually taken to be much the same as the height of the preceding pass, without considering rounded corners, i.e. $b_n \approx h_{n-1}$. In such cases, the pass corners are rounded off and the pass is somewhat widened out to provide free space for metal spreading. Other design methods are also used.

In modern rolling mills, the diamond—diamond series is used rarely because of shallow draughts with deep groove cuts. The gothic pass series (diamond passes with convex sides) is even less popular due to shallow draughts and instability of the workpiece in the passes.

Diverse methods of calculating the intermediate passes for the diamond—square series are employed. In calculating the dimensions of sections the spread or the relations between the spread and reductions (or draughts) are usually determined. The nonuniformity of deformation and intricate shape of the zone of deformation has brought about the need for simplifications, which make it possible to take rough account of the amount and character of the spread in passes of different shapes.

A. Golovin has determined the spread by the method of correlated rectangles. The same method has been used by a number of other authors (B. Bakhtinov, M. Shternov) as well.

V. Smirnov has established the relation between the spread in oval and square passes and the ratio of sides (b/h) of the corresponding rectangular workpiece.

B. Bakhtinov and M. Shternov have suggested the method of replacing more intricately shaped passes by circumscribed rectangles. Then, the determination of the section dimensions and interrelations between the draughts in intermediate intricately shaped passes is made by the method described above for rectangular (box) passes. The area of the intermediate rectangle is found from the ratio μ_r/μ_{sq} as a function of the spread:

$$K_{\Delta b} = \frac{\mu_r}{\mu_{sq}} = \frac{A^2a^2}{b^2h^2}$$

$$F_1 = \sqrt{\frac{F_0F_2}{K_{\Delta b}}} = \frac{Aa}{\sqrt{\frac{\mu_r}{\mu_{sq}}}}$$

Approximate values of $K_{\Delta b}$ are determined from the diagram shown in Fig. 6.2. When calculating the distribution of draughts among the diamond and square passes by this method we obtain:

$$K_{\Delta b} = \frac{\mu_r}{\mu_{sq}} \approx \frac{\mu_d}{\mu_{sq}} \approx 1$$

i.e. the draughts are approximately equal. The size of the intermediate diamond section may be determined as the geometric mean of the areas of the large and small squares (Fig. 6.5):

$$F_1 = \sqrt{A^2a^2} = Aa$$

The height of the diamond pass is determined from a semi-empirical formula as proposed by B. Bakhtinov and M. Shternov:

$$h = (1.3\text{-}1.38)\ a - \frac{A-a}{2A}\left(\sqrt{R(A-a)} - \frac{A-a}{2f}\right)$$

When the height and area of the diamond pass are known, its width may be found from the formula:

$$b = \frac{2F}{h}$$

If the spread is preset or calculated, the diagonals of the diamond may be determined with due regard for the spread

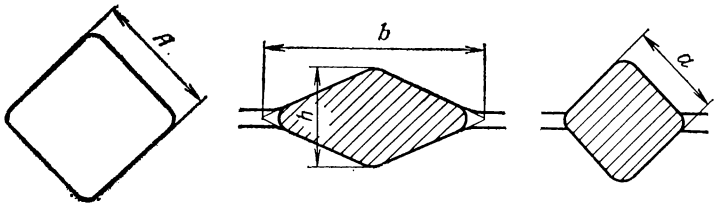


Fig. 6.5. Calculating dimensions of diamond passes

in the diamond and square passes:

$$b = 1.4A + \Delta b_d$$

$$h = 1.41a - \Delta b_{sq}$$

i.e. the width of the diamond pass is larger than the diagonal of the square workpiece entering it and the height of the diamond obtained is smaller than the diagonal of the next square pass by the value of the spread.

M. Mutyev has deduced the following rather simple formulas for calculating the spread in diagonal and square passes. The spread in a diamond pass may be expressed, using designations as proposed by M. Mutyev, as

$$\Delta b = \left(0.2 + 0.08\frac{l_x}{b_{av}}\right)\frac{b_1'}{h'}\Delta h$$

where $\Delta h = h_1' - h_d'$ is the absolute reduction in the diamond pass.

The spread in a square pass may be expressed as

$$\Delta b = \left(0.2 + 0.12 \frac{l_x}{b_{av}}\right) \frac{l_2'}{b_d'} \Delta h$$

where $l_x = \sqrt{0.5 \Delta h R_e}$ is the average length of the zone of deformation

$b_{av} = \frac{b_1 + b_2}{2}$ is the average width of the workpiece

b_d' = diamond width

h_1' = square height

b_2' = square width

Δh = absolute reduction in the diamond pass

$\Delta h = h_d' - h_2'$ is the absolute reduction in the square pass

6.3. Oval — Square Series

For a long time the oval—square series (Fig. 6.6) has been extensively used as a system of breakdown passes ensuring maximum draughts with a minimum number of passes. In the oval—square series of passes the edges of the workpiece undergo alterations. Since the square workpiece is fed flatwise into an oval pass (Fig. 6.7) its edges pass to the sides of the next smaller square. Such an alteration of the edges contributes to a more uniform working of the rolled section.

To analyze the operation of this series of passes the method of consecutive cross-sectional deformations in the zone of deformation (Fig. 6.8) is commonly used.

During deformation the sides of the square workpiece are maximally reduced, this being conducive to the formation of folds followed by the development of surface seams.

The horizontal components of the roll pressure arising on the roll-to-metal contact surface during rolling of a square workpiece in an oval pass restrain the lateral deformation of the metal. This explains decreasing of the spread and increasing of the elongation.

When rolling an oval workpiece in a square pass, the lateral deformations with low lateral stresses in an elongated, narrow zone of deformation develop normally, and therefore the draughts in such passes are smaller than in the oval ones,

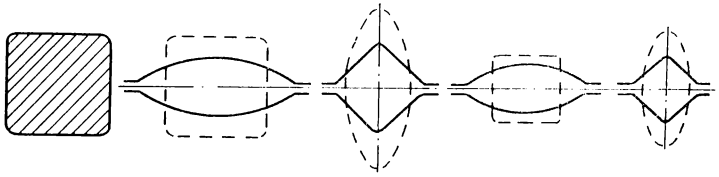


Fig. 6.6. Oval—square series of passes

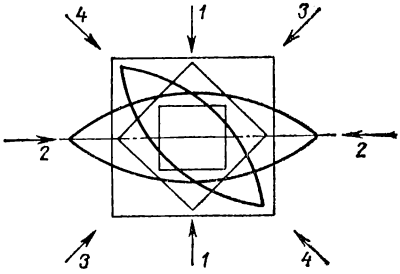


Fig. 6.7. Diagram of changes in direction of reductions for oval—square method (numbers show operation sequence)

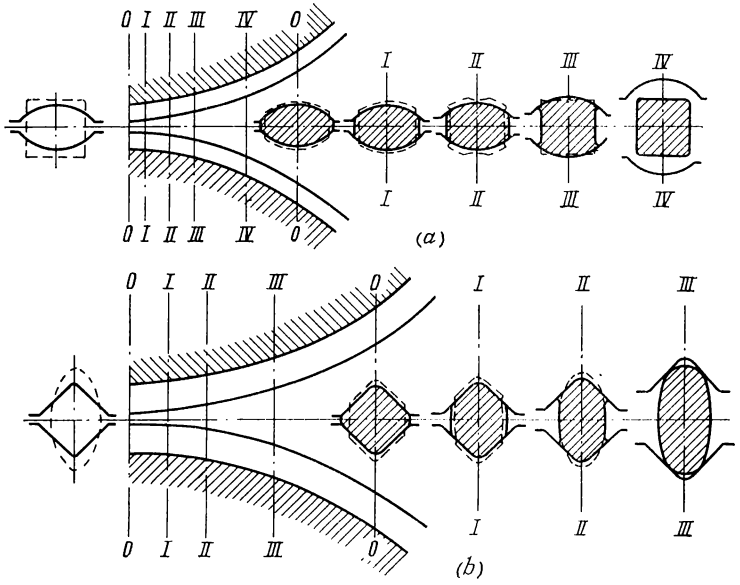


Fig. 6.8. Reduction sequence of square piece in oval pass (*a*) and oval piece in square pass (*b*)

The disadvantages of the oval—square series include an increased wear of the passes resulting from the nonuniform deformation across the workpiece.

The ratio of the width of the oval to its height b/h determines the distribution of draughts among the adjacent square and oval passes. P. Polukhin has established that with small ratio b/h the increment of the draught in the square pass ($\mu_{sq} - 1$) is 0.9 of that in the oval pass ($\mu_{ov} - 1$). With a higher b/h the increment of the draught in the square pass diminishes to 0.3 of that in the oval pass. It is thought that on the average $\frac{\mu_{sq}-1}{\mu_{ov}-1} = 0.7\text{-}0.75$.

The geometric shape of an oval pass and its construction may differ, depending upon the rolling conditions and its purpose. In the simplest and most common instances the oval pass is formed by two segments (Fig. 6.9). Height h, width b and radius R of the oval can be determined from the following geometric relations:

$$R^2 = OB^2 = OC^2 + CB^2$$

$$R^2 = \left(R - \frac{h}{2}\right)^2 + \left(\frac{b}{2}\right)^2$$

$$R = \frac{b^2 + h^2}{4h}$$

Whence

$$\frac{R}{h} = \frac{\left(\frac{b}{h}\right)^2 + 1}{4}$$

For different values of b/h the values of $\frac{F}{h^2}$ and $\frac{R}{h}$ may be found from the table below.

The filling of an oval pass with metal depends on a wide variety of factors.

M. Mutyev recommends to check the spread in an oval pass by using the Petrov—Zibel formula:

$$\Delta b = K \sqrt{\Delta h_{av} \cdot R_{m.e}} \cdot \frac{\Delta h_{av}}{C_1}$$

where C_1 = height of the square

$\Delta h_{av} = C_1 - h_{ov.\,av} = C_1 - \frac{F_{ov}}{b_{ov}}$ is the average reduction (in the oval pass)

$h_{ov.av}$ = average height of the oval

$R_{m.e} = \frac{D - h_{av}}{2}$ is the mean effective radius

K = coefficient accounting for other rolling factors

The values of K based on the data furnished by M. Vratsky and P. Polukhin are in the range from 0.28 to 0.42. Commonly K is taken at 0.35.

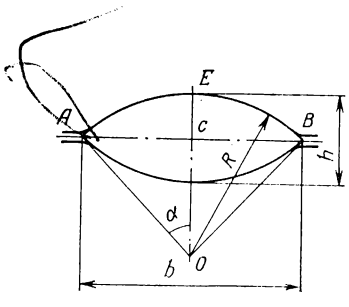


Fig. 6.9. Constructing of oval pass

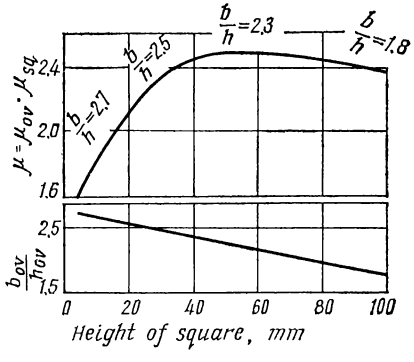


Fig. 6.10. Diagram for determining $\mu = \mu_{ov} \cdot \mu_{sq}$

In the roll pass design wide application has found the method of determining the coefficient of spread from diagrams and nomograms constructed on the basis of theoretical and experimental data.

Table 6.1

Values of $\frac{b}{h}$ and Corresponding Geometric Relations for Dimensions of Oval $\frac{F}{h^2}$ and $\frac{R}{h}$

$\frac{b}{h}$	$\frac{F}{h^2}$	$\frac{R}{h}$	$\frac{b}{h}$	$\frac{F}{h^2}$	$\frac{R}{h}$
1.30	0.962	0.672	2.60	1.784	1.94
1.50	1.084	0.812	3.00	2.044	2.50
1.80	1.271	1.06	3.30	2.240	2.97
2.00	1.398	1.25	3.50	2.371	3.31
2.30	1.590	1.572			

The paired coefficients of draught between adjacent square passes $\mu = \mu_{ov} \cdot \mu_{sq}$ are also found by consulting pertinent diagrams.

Such a diagram for determining μ and b_{ov}/h_{ov} is presented in Fig. 6.10.

Depending upon rolling conditions, modifications other than simple oval pass are used.

A flat-type oval pass is used in the oval—round series of breakdown passes (Fig. 6.11). The use of the flat oval pass

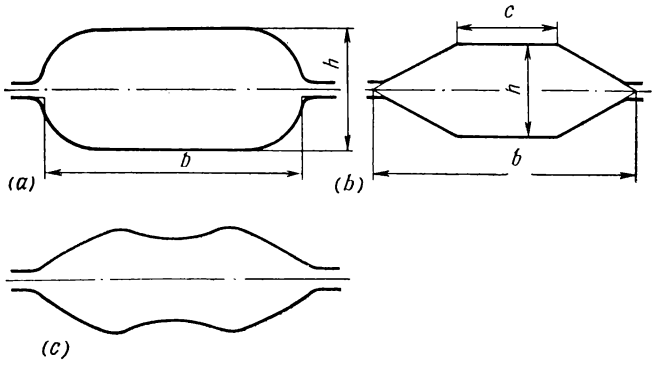


Fig. 6.11. Oval passes: (*a*) flat; (*b*) hexagonal; (*c*) concave

in the oval—square series provides for a more uniform distribution of draughts between the square and oval passes, however, in this case a tendency to overfilling of the pass with metal and to undercutting is observed. The area of the flat oval pass is determined from the following formula:

$$F = bh\left(1 - 0.215\frac{h}{b}\right)$$

A hexagonal flat-type oval pass is used to obtain comparatively large draughts equal to those in the square pass. The area of hexagonal oval pass is

$$F = \frac{bh}{2}\left(1 + \frac{c}{b}\right)$$

Concave oval passes are employed to prevent overfilling of the next square passes and to level out the draughts.

6.4. Oval — Round Series

This series of passes was previously used as a breakdown series in rolling metal with low plastic properties. At present it has found wide application in light-section mills, as well as in continuous and semi-continuous wire mills, and supplants the oval—square pass series. The advantages of the oval—round series are the following: lower stresses

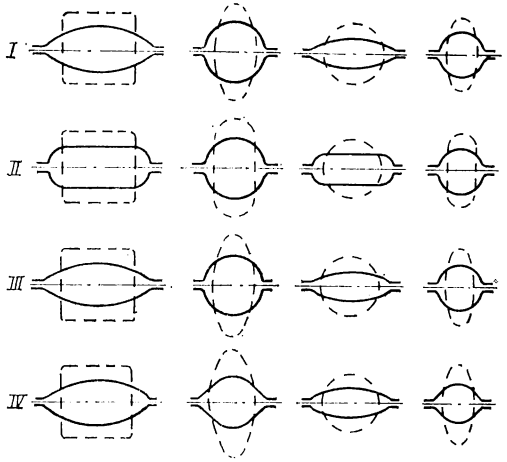


Fig. 6.12. Oval—round series of passes

explained by the smooth shape transitions in these passes, uniform distribution of temperatures across the section and better conditions for preventing laps and seams and for descaling. Various types of the oval—round series are illustrated in Fig. 6.12. It has been found that the use of this roll pass design method provides for an increased dimensional accuracy of round sections.

The disadvantage of this series is the reduced coefficients of draught (1.3-1.55) as compared to the oval—square series. Besides, the filled ovals are not stable during rolling in round passes. The workpiece turns even in tight passes.

The areas of intermediate, rough rounds with radius R are calculated with some underfilling:

$$F_r = 0.98 \cdot \frac{\pi}{4} R^2$$

To ensure the stability of an oval workpiece in a round pass it is made underfilled. Side height m of the underfilled oval is of a particular importance. Taking into account the

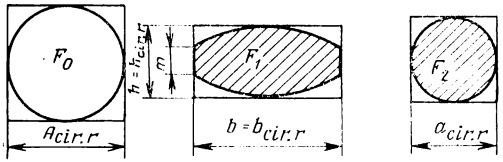


Fig. 6.13. Application of the method of circumscribed rectangles for oval—round series of passes

side height (Fig. 6.13), the area of the oval is

$$F_{ov} = \frac{1}{3} \cdot bh \left(2 + \frac{m}{h}\right)$$

The dimensions of the intermediate rounds and the values of the paired coefficients of draughts are selected proceeding from the total and average draughts, with the considerations discussed above for other series of breakdown passes being taken account of. The sections of the intermediate ovals are selected by the method of circumscribed rectangles.

The amount of truncation m may be taken at:

$$\frac{m}{d} = 0.3\text{-}0.4 \text{ for cast-iron rolls}$$

and

$$\frac{m}{d} = 0.4\text{-}0.5 \text{ for steel rolls}$$

where d = diameter of the round pass.

Experience has shown that a considerable underfilling of oval passes is desirable because in this case, apart from the better stability of the truncated oval workpiece in a round pass, the fluctuations of the spread, depending on various rolling factors (including the distribution of the tempera-

ture along the workpiece) do not have any appreciable effect upon the accuracy of the finished rolled sections. The use of oval workpieces with optimum truncation is especially important for leader passes.

6.5. Oval — Oval Series

The oval—oval series (Fig. 6.14) has not found wide application due to the deep groove cuts and increased wear of the passes. The oval—oval series was employed for rolling steels with a low plasticity.

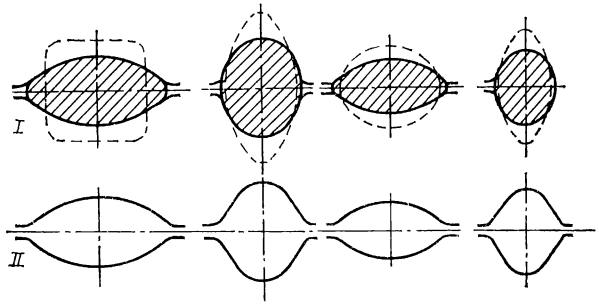


Fig. 6.14. Oval—oval series of passes

Nowadays the oval—oval series is used in continuous section mills equipped with edgers (Fig. 6.15).

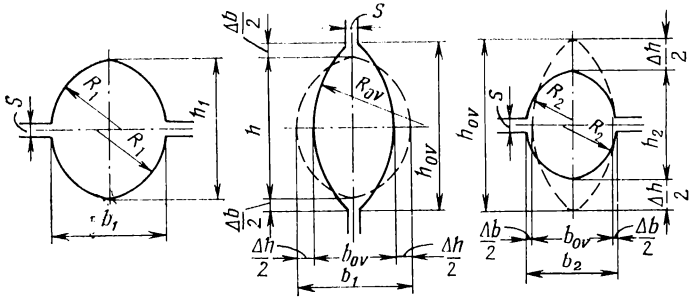


Fig. 6.15. Oval—edging oval series of passes

According to the data obtained by M. Mutyev, the ratio $\frac{h_{ed.ov}}{b_{ed.ov}} \approx 1.2$ for an edging oval pass and $\frac{h_{ov}}{b_{ov}} > 2.0\text{-}2.5$ for

an oval pass. The coefficients of draught are respectively:

$$\mu_{cd.\,ov} = 1.2\text{-}1.35 \text{ for edging oval passes}$$

and

$$\mu_{ov} = 1.2\text{-}1.4 \text{ for oval passes}$$

The spread in oval passes is determined from the well-known Petrov—Zibel formula:

$$\Delta b = k \sqrt{\Delta h R} \frac{\Delta h}{h}$$

The coefficients of draught for the oval—edging oval series of passes are determined from the ratio h/b with due regard for the coefficient of spread k taken from Tables 6.2 and 6.3.

Table 6.2

Coefficients of Draught for Rolling Edged Oval Workpieces in Oval Passes

$\frac{h_{ov}}{b_{ov}}$	Coefficients of draught μ_{ov} as function of coefficient of spread $k_{ov} = \frac{\Delta b}{\Delta h_{av}}$							
	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1
1.5	1.16	1.14	1.13	1.11	1.10	1.09	1.07	1.07
2.0	1.43	1.37	1.33	1.29	1.27	1.23	1.20	1.18
2.5	1.69	1.61	1.55	1.49	1.45	1.39	1.35	1.31
3.0	1.95	1.86	1.77	1.68	1.63	1.55	1.49	1.44

Table 6.3

Coefficients of Draught for Rolling Oval Workpieces in Edging Oval Passes

$\frac{h_{ov}}{b_{ov}}$	Coefficients of draught $\mu_{ed.ov}$ as function of coefficient of spread $k_{ed.ov}$						
	0.2	0.3	0.4	0.5	0.6	0.7	0.8
1.5	1.17	1.12	1.09	1.07	1.04	1.04	1.03
2.0	1.37	1.27	1.21	1.14	1.09	1.05	1.02
2.5	1.53	1.37	1.26	1.16	1.10	1.04	—
3.0	1.65	1.43	1.27	1.17	1.08	1.01	—

In new and modernized rolling mills a number of modifications and combined series of breakdown and final shaping passes are used, depending upon specific requirements for the quality of rolled products, operational conditions and economical reasons.

As it has been indicated above, the production of various rolled sections requires an optimum number of final shaping passings. Therefore, the total number of passings, upon which the rolling scheme for the given mill depends, is equal to the sum of the roughing passings (from the selected section of the billet to the last preparatory section of the workpiece) and the number of the shaping passings required for the given finished section, including the last two passings (leader and finishing).

6.6. General Roll Pass Design Schedule

The roll pass design sequence is the following. Taking into consideration the distinguishing features of the given rolling process, the design of the main and auxiliary equipment of the rolling mill and the requirements for the quality of the produced range of rolled sections with respect to the shape, dimensions and steel grade:

(1) an approximate scheme of rolling is selected;

(2) the dimensions of the billet and the total and average coefficients of draught are determined and the number of passings is specified;

(3) an approximate distribution of temperatures among passings is determined, as well as the coefficients of draught for the finishing, leader and shaping passes, i.e. starting from the final pass and then proceeding to the previous one and further to the preleader pass (the last of the breakdown passes);

(4) the series of breakdown passes is selected, as well as the draughts in the intermediate and paired passes and the distribution of the draughts among each pair (or a group) of passes; depending on the character of deformation, the spread is determined which is necessary for the analysis of the roll pass design and for verifying calculations (in practical calculations the shape and dimensions of the passes are determined from diagrams, nomograms or tables with due regard for the coefficient of spread $\Delta b/\Delta h$);

(5) the nominal and effective diameters of the rolls and the rolling speeds are specified;

(6) the centres of gravity of sections are found by using the roll pass templates, the roll parting line is located and the top or bottom pressure is determined; the pass line, to which the neutral lines of the passes are matched, is defined; the roll drawing, which shows the arrangement of grooves, is made and the optimum gap is selected, depending on the stand springing;

(7) the metal-to-roll contact area is determined, as well as the rolling pressure, the work done in rolling and the rolling torque and power, and the capacity and other characteristics of the drive (the strength of the main units of the working stands) are checked;

(8) the output of each rolled section is determined from the mill loading diagram, taking into account the rolling rhythm.

CHAPTER VII. ROLL PASS DESIGN FOR COGGING MILLS

7.1. Roll Pass Design for Blooming Mills

The roll pass design for a blooming mill includes:

(1) selection of rolling schemes, including the determination of the number of passings, the distribution of draughts among the passings and the number of ingot tilts;

(2) selection of the rational shape and geometric dimensions of passes and their arrangement along the roll bodies.

Since the three-high cogging mills are now in operation only at old iron and steel works, below are considered the essentials of the roll pass design for two-high reversing blooming mills which are most commonly employed.

Optimum reductions and rolling speeds are determined, depending upon the design of the main working stand units and upon the capabilities of the main drive. The selection of rational roll pass design system for blooming mills is complicated by the fact that the same rolls must provide for rolling ingots of different size, chemical composition

and properties and for obtaining square (blooms) and flat (slabs) billets of different cross-sectional dimensions. When designing the roll passes, it is difficult to take into account the whole variety of factors upon which the efficiency of the blooming mill operation depends. Therefore the initially designed roll passes are usually corrected on the basis of test and research data.

The selection of maximum reductions Δh_{max} which are possible to obtain in the given passing is determined by the strength of the rolls and other working stand components, by the capabilities of the drive and by the conditions of the metal biting by the rolls as well:

$$\Delta h_{max} = D\left(1 - \frac{1}{\sqrt{1+f^2}}\right)$$

where D = effective diameter of the rolls
f = coefficient of friction

Under favourable conditions the angle of bite may be as great as 0.56 radian (32°). Many authors recommend light reductions for the first passings, this being explained by the peculiarities of the cast structure of ingots (destruction of dendrites, etc.). However, undue caution in selecting the reductions often results in an increased nonuniformity of deformation and bad working of the ingot. The plastic properties of the ingot cannot be ignored, but there are no reasons for decreasing the reductions in the first passings when the ductility of the ingot is high. The reduction in each passing is selected taking into consideration the capacity of the main drive motor, the conditions of the metal biting by the rolls and the dynamic and speed conditions as well.

When rolling is done with increased speeds and great reductions and angles of bite, the rolls may begin slipping thus causing dangerous dynamic stresses in all the units of the drive. In practice this was the very cause of break-downs which occurred when the modern blooming mills were being put in operation.

As regards the reductions, two different rolling methods are known. Some iron and steel works apply moderate reductions (up to 80 mm) under the conditions of reliable metal biting by the rolls, but use forced rolling speed schedules,

thus seeking to save the productive time. Other works operate with heavy reductions and cut down the number of passings, but carry out the rolling with reduced speeds. The first method is considered to be more preferable.

The whole range of ingots, blooms and slabs of various steel grades is usually broken down into groups and since the rolling is done with the same rolls, the rolling and reduction schedules are drawn up individually for each group.

The dimensions of ingots are selected so that sound ingots can be produced under the particular plant conditions. The total reduction for the entire rolling cycle (Fig. 7.1), without taking spread into consideration, is $B - b) + (H - h)$. The total spread is

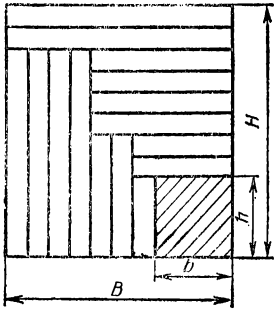


Fig. 7.1. Determining of total reduction in blooming mill

$$\Sigma\Delta b = 0.10\text{-}0.15\,(B - b) + (H - h)$$

If the spread is taken into account the total reduction is

$$\Sigma\Delta h = [(B - b) + (H - h)] + [0.10\text{-}0.15\,(B - b) + (H - h)] =$$
$$= 1.10\text{-}1.15\,[(B - b) + (H - h)]$$

Average reduction Δh_{av} is taken at 65-80 mm for steels with low resistance to deformation and at 60-70 mm for those with high resistance to deformation.

The number of passings

$$n = \frac{\Sigma\Delta h}{\Delta h_{av}}$$

To provide for the working of the ingot across the whole section, P. Alexandrov recommends that the optimum reductions set for each passing should be directly proportional to the square of the workpiece height and inversely proportional to the roll diameter.

The permissible angles of bite, depending upon the rolling speed, may be determined from Table 7.1.

Table 7.1

Permissible Angles of Bite (according to the data obtained by Tyagunov)

Roll characteristic	Angles of bite (deg) at rolling speed, m/s								
	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.7
Plain barrel-shaped . . .	25.4	24.5	23.5	22.5	19.5	16.0	12.5	12.0	—
Grooved . . .	29.0	27.5	26.0	24.5	21.0	17.0	13.0	12.0	—
Fluted	33.0	32.0	31.0	30.0	28.0	26.0	24.0	21.0	20.0

The permissible reduction is determined as a function of the roll strength from the well-known formulas.

The permissible rolling pressure

$$P_{per} = 0.1 D_e^3 R_r \frac{L + l}{c\,[(L + l) - c]}$$

where c = distance from the centre of the roll neck to the vertical centre line of the pass

D_e = effective diameter of the pass

L = length of the roll body

l = length of the roll neck

$R_r = \frac{\sigma_r}{5}$ is the permissible roll stress

For forged steel

$$\sigma_r = 60\text{-}65 \text{ kgf/mm}^2$$

$$R_r = 12\text{-}13 \text{ kgf/mm}^2$$

For cast steel

$$\sigma_r = 50\text{-}60 \text{ kgf/mm}^2$$

$$R_r = 10\text{-}12 \text{ kgf/mm}^2$$

Equating the total rolling pressure to the permissible pressure

$$pb\sqrt{R\Delta h} = P_{per}$$

we obtain the maximum reduction as a function of the roll strength

$$\Delta h = \frac{1}{R}\left(\frac{P_{per}}{pb}\right)^2$$

The permissible reductions according to the overload torques may be checked by different methods. Usually, the calculated maximum rolling torque for the given passing is compared with the rated torque of the main drive motor:

$$M_r = \frac{0.716N}{n} \text{ tf·m}$$

where N = power rating of the motor, hp
n = rated rpm of the rolls

The rated number of revolutions per minute (50, 40, 60) denotes in each case (0-50-120; 0-40-100; 0-60-120) the working speed of an adjustable-speed reversible motor at the full rated voltage.

The number of passings in single-stand blooming mills must be odd. The spread, which is hardly appreciable in the first passings, increases in the course of rolling the ingot.

The following approximate values of the spread are recommended: $\Delta b = 0$ for the first passings, $\Delta b = 5$ mm for intermediate passings and $\Delta b = 10$ mm for the last passings.

The coefficient of spread is

$$k = \frac{\Delta b}{\Delta h} = 0.1\text{-}0.25$$

When drawing up the rolling scheme, of great importance are the choice of the number of the workpiece tilts in the course of rolling and their distribution among the passings. Increasing the number of tilts, as a rule, improves the surface finish and structure of the rolled metal. However, the greater the number of tilts, the lower the blooming mill output. The following scheme of the workpiece tilting is practised on a large scale: the first tilt is done after the first two passings, the second, after four passings, the third and subsequent, after either four or two passings, and the last, after the leader passing.

When selecting the number of tilts, it should be borne in mind that the tilter is usually located on the entry side of

the rolling mill and therefore the tilts are done after the even-numbered passings.

Rolling without tilting the workpiece is not advisable when the ratio B/H is up to 1.3 approximately in the first pass and 1.7 in the subsequent passes.

There are two methods of arranging the passes along the bodies of the rolls for single-stand blooming mills. The consecutive pass arrangement (Fig. 7.2) ensures repeating of

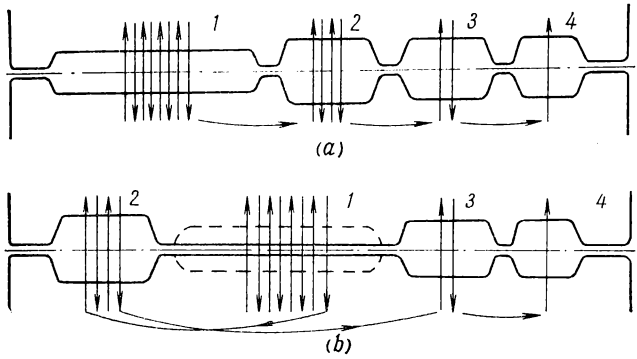


Fig. 7.2. Pass arrangement on rolls of blooming mill: (*a*) consecutive; (*b*) symmetrical

the workpiece only in a single direction. When the passes are arranged symmetrically, the first, wide pass having the maximum effective diameter is at the centre of the roll body. At works with a mixed structure of the rolled metal production (bar sections and sheets) the share of flat billets (slabs) for sheet mills in the blooming mill products is great. In such cases the second roll pass arrangement scheme is preferable.

The number of passes usually varies between 3 and 5. As great number of passings as possible is performed in the first pass which frequently is of the bullhead type. This facilitates the workpiece alignment when feeding it into the rolls, diminishes the roll wear and increases the output. The bullhead rolling is, however, limited by possible turning of ingots in the pass due to their instability, especially when tilting ingots with a height to width ratio of the order of 1.3.

The pass reductions are set by moving the top roll in the vertical direction. Therefore, the height or the depth of the groove cut of the box passes used in blooming mills is determined by the number of passings done in the given pass and the total reduction in this pass.

The depth of the groove cut for the first pass is adopted in the range from 0 to 200 mm. When rolling slabs in blooming mills, the depth of the groove cut for the first pass is

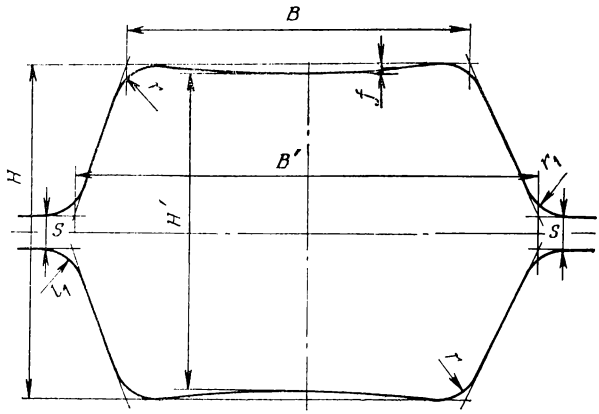


Fig. 7.3. Constructing of roll pass for blooming mill

selected according to the preset minimum thickness of the slabs. This for subsequent passes ranges from 130 to 250 mm. The box passes of blooming mills are made with great tapers of sides and corner rounding radii. When rolling blooms, the width of the bullhead corresponds to the maximum ingot width.

The roll pass design for blooming mills is illustrated in Fig. 7.3. The pass width at the bottom of the groove cut is selected so that it is slightly less (by about 10 mm) than the minimum width of the fed workpieces. The pass width at the roll parting line must be by 5-10 mm greater than the width of the fed workpieces. The taper of sides for these passes amounts to 5-30%. The pass convexity comes to approximately 5% of the pass height (it is absent in the first and last passes).

The corner rounding radii are

$$r = (0.12\text{-}0.20)\,B$$

and

$$r_1 = (0.08\text{-}0.12)\,B$$

The width of the collars amounts to not less than 50% of the groove cut depth, gap S = 10-15 mm. The bottom pressure is usually applied, the difference between the roll diameters being equal to 10-15 mm.

Below is given a roll pass design version for 1150-mm blooming mill rolling 300 × 300 mm blooms from ingots 7 tons in weight and 760 × 680 mm in cross section. The mill is powered by a 7000 hp motor running at 0-50-120 rpm. A medium carbon steel with a carbon content of 0.35-0.50% is adopted for the case being considered.

Basing upon the metal biting conditions in the first pass, we find the maximum reduction. Proceeding from the groove depth, which depends on the thickness of slabs (140-150 mm), the effective diameter in the first pass is

$$D_e = 0.9\,(1150 - 150) = 900 \text{ mm}$$

The coefficient of friction is determined from the Ekelund formula for steel rolls. When the rolling speed is equal to 50 rpm, rolling temperature $t = 1250$ °C and $K = 1$

$$f = K\,(1.05 - 0.0005t) = 1.05 - 0.0005 \cdot 1250 \approx 0.45$$

The maximum reduction

$$\Delta h_{\max} = D\left(1 - \frac{1}{\sqrt{1+f^2}}\right) =$$

$$= 900 \cdot \left(1 - \frac{1}{\sqrt{1+0.45^2}}\right) \approx 75 \text{ mm}$$

The maximum reduction is checked against the rated power of the motor:

$$\Delta h_{\max} = \frac{N_{def}}{p_m b_{av} v_0}$$

Mean specific roll pressure p_m is determined from the Ekelund formula:

$$p_m = K_f\,(\sigma + \eta U)$$

Assuming that the average value of $H_0 + h_1$ in the first pass is equal to 1100 mm:

$$K_f = 1 + \frac{1.6 \cdot 0.45 \sqrt{450 \cdot 75} - 1.2 \cdot 75}{1100} \approx 1.04$$

$$\sigma = (14 - 1.01t)(1.4 + \text{C\%} + \text{Mn\%}) =$$

$$= (14 - 12)(1.4 + 0.4 + 0.6) = 4.8 \text{ kgf/mm}^2$$

When $C = 1$

$$\eta = 0.1(14 - 0.01t)C = 0.1(14 - 12) \approx 0.2$$

The deformation rate

$$U = \frac{2v\sqrt{\frac{H_0 - h_1}{R}}}{H_0 + h_1} = \frac{2 \cdot 2\sqrt{\frac{0.075}{0.45}}}{1.1} = 1.45 \text{ s}^{-1},$$

when $n = 40$ rpm

$$v = \frac{\pi D n}{60} = \frac{3.14 \cdot 0.9 \cdot 40}{60} \approx 2 \text{ m/s}$$

Then

$$p_m = 1.04(4.8 + 0.2 \cdot 1.45) \approx 5.2 \text{ kgf/mm}^2$$

we assume $p_m = 6$ kgf/mm².

At rated motor power N of 7000 hp and a short-term overload factor of 2

$$N_{\max} = 14{,}000 \text{ hp} = 14{,}000 \cdot 75 =$$

$$= 1.05 \cdot 10^6 \text{ kgf} \cdot \text{m/s} \approx 10.3 \text{ MJ/s}$$

Assuming that the total power loss in the drive equals about 20%, the power consumed in deformation is

$$N_{def} = 0.8 \cdot 1.05 \cdot 10^6 = 0.84 \cdot 10^6 \text{ kgf} \cdot \text{m/s} \approx 8.25 \text{ MJ/s}$$

$$\Delta h = \frac{0.84 \cdot 10^6}{6.0 \cdot 10^6 \cdot 0.76 \cdot 2.0} = 0.092 \text{ m} = 92 \text{ mm} \approx 90 \text{ mm}$$

We assume that the reductions in the first passings are equal to 70 mm.

To determine the number of passings we find

$$\Delta h_{av} = \frac{90 + 70}{2} = 80 \text{ mm}$$

The number of passings

$$n = \frac{1.15\,[(B - b) + (H - h)]}{\Delta h_{av}} =$$

$$= \frac{1.15\,[(760 - 300) + (680 - 300)]}{80} = 12$$

We take an odd number of passings of 13. Then the average reduction is

$$\Delta h_{av} = \frac{966}{13} = 74 \text{ mm}$$

Then the reduction schedule is drawn up. The following workpiece tilting sequence is chosen: first, when reducing a tapered (on both sides) ingot, the tilting is done after every two passings, then, twice after four passings, and prior to the last, thirteenth passing one more tilting is done to obtain the required dimensions of the rolled square. The corrected average reduction $\Delta h_{av} = 74$ mm.

Reduction Δh in the first two passings is taken at 65 mm, then it is increased up to 70-80 mm and thereupon, decreased down to 70 mm in the last passing. Usually, several rolling scheme versions have to be drawn up, these being correlated with the adopted pass dimensions and tilting schedule.

Table 7.2 gives an actual version of the reduction schedule which corresponds to the selected rolling conditions.

Ratio b/h amounts to 1.18 when the first tiltings are done (the permissible ratio is equal to 1.3), and to 1.65 for subsequent tiltings (the permissible ratio is equal to 1.7). The spread in the first passings is taken at its minimum and then corrected by slightly changing the reductions after tilting, if it does not correspond to the selected one.

In some cases the total spread is checked to determine the width of the passes:

$$B_p - b_p \geqslant \Sigma \Delta b$$

The value of Δb in passings may be checked by using the Petrov—Zibel formula, coefficient k being taken at 0.1 for the first passings and at 0.3 for subsequent ones. The dimensions of the passes for the given roll pass design version are selected in accordance with Table 7.2. These for the first passes are determined by the width and thickness of

Table 7.2

Distribution of Reductions Among Passings

Passing number	Section dimensions, mm	Δh, mm	Δb, mm	Pass number
Ingot	760 × 680	—	—	
1	695 × 680	65	0	
2	630 × 680 *T*	65	0	
3	610 × 635	70	5	
4	540 × 640 *T*	70	5	1
5	560 × 545	80	5	
6	480 × 550	80	5	
7	410 × 555	70	5	
8	340 × 565 *T*	70	10	
9	490 × 345	75	5	
10	420 × 355	70	10	2
11	350 × 365	70	10	
12	290 × 370 *T*	60	10	
13	300 × 300	70	10	3

Note. *T* — tilting.

the rolled slabs. Taking into consideration the design of the feed rolls, one tries to select approximately the same depths for subsequent passes.

7.2. Roll Pass Design for Continuous Billet Mills

Continuous mills of two types are used in combination with blooming mills for rolling billets. Continuous billet mills with group-driven stands (the peripheral speed of rolls in each stand is unadjustable) continue operating. There are only horizontal roll stands in these mills and the rolled piece is tilted between the stands. Recently, continuous mills with horizontal roll stands and vertical edgers have been designed. In these mills the speed of each set of rolls is adjustable (individual drive) and the rolled piece is not tilted.

In rolling mills with constant roll speeds in each stand, the selection of the workpiece sections and effective diameters of the rolls to retain the roll pass design constant ($C = FDn$) is a very difficult problem. The roll designer

has to analyze and compare different design versions in order to find an optimum solution. This selection is complicated by the fact that it is necessary to roll pieces of different sections in different series of passes cut into the rolls of a single set, when quick change-overs from one section to another are required. An actual roll pass design for 720-mm continuous billet mill is shown in Fig. 7.4 and its reduction schedule is given in Table 7.3.

To raise the output of blooming mills one tries to increase the sections of blooms for billet mills, provided all the

Table 7.3

Reduction Schedule for 720-mm Continuous Billet Mill

Number of stand	Roll speed, r.p.m.	Roll diameter, mm	Roll gap, mm	Piece dimensions, mm		Reduction, mm	Spread, mm	Roll pass design constant C
				thickness	width			
			120 × 120					
I	14.5	587	60	183	265	67	15	412
II	19.4	554	40	196	196	69	12	412
III	23.15	580	15	140	214	56	12	405
IV	31.6	554	15	152	152	62	12	412
V	35.8	640	10	158	228	56	14	412
VI	45.1	625	10	171	171	57	13	412
			150 × 150					
I	14.5	560	85	235	310	60	15	594
II	19.4	505	40	247	247	63	12	595
III	23.15	547	15	178	262	67	15	594
IV	31.6	508	15	192	192	70	14	590
V	35.8	570	10	140	207	52	14	592
VI	45.1	562	10	153	153	54	13	592
			210 × 85					
I	14.5	578	60	190	265	60	15	428
II	19.4	550	40	200	200	65	10	427
III	23.15	583	15	142	220	58	20	423
IV	31.6	611	15	89	240	53	20	414
V	35.8	634	10	76	245	13	5	422
VI	45.1	535	10	210	85	35	9	429

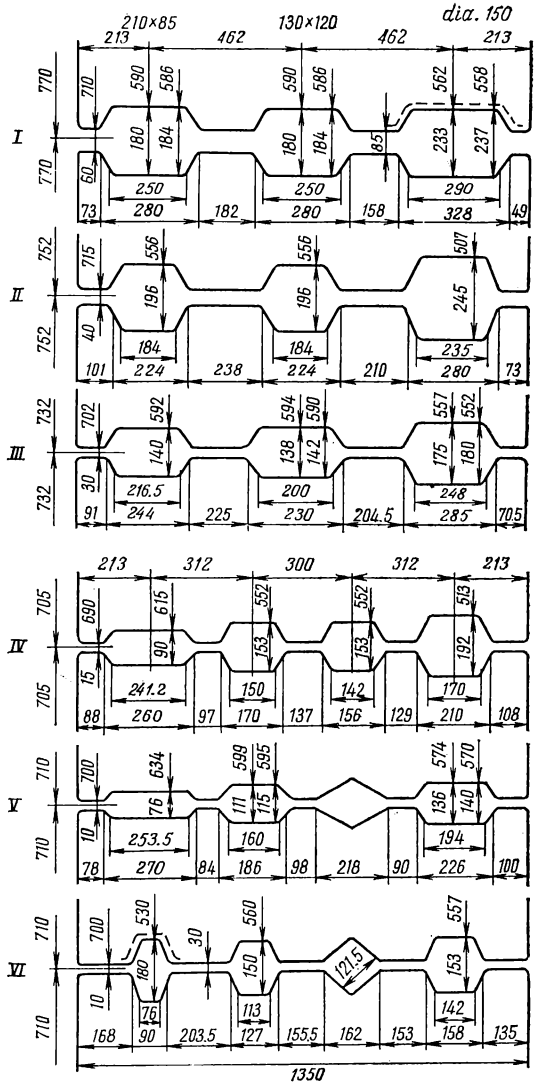


Fig. 7.4. Roll pass design for 720 mm continuous billet mill

stands of the mill (sometimes equipped with additional stands) are fully charged.

The roll pass design for continuous billet mills with adjustable roll speed is substantially simpler.

The rolling scheme for 850/700/500 continuous billet mill is given in Table 7.4.

Table 7.4

Scheme for Rolling Billets 80 × 80 and 120 × 120 mm in Cross Section from Blooms 370 × 370 mm in Cross Section, 8.6 m in Length and 9.3 t in Weight

Number and type of stand	Number of passing	Passes	Cross section of piece, mm	Reduction, mm	Draught	Length of piece, m	Rolling speed, m/s	Productive time, s
I*H*	0		370 × 370	—	—	8.6	—	—
	1		305 × 376	65	1.21	10.5	0.30	35
II*H*	2		210 × 388	95	1.40	15	0.43	35
III*H*	3		315 × 219	73	1.19	18	0.51	35
IV*H*	4	Box	228 × 234	87	1.27	23	0.645	35
V*V*	5		152 × 242	82	1.425	32	0.92	35
VI*H*	6		162 × 164	80	1.39	45	1.28	35
VII*V*	7		102 × 184	62	1.42	63	1.82	35
VIII*H*	8		120 × 120	64	1.32	84	2.40	35
IX*V*	9	Diamond	123 × 158	—	1.19	100	2.75	36
X*H*	10	Square	100 × 100	—	1.18	118	3.24	36
XI*V*	11	Diamond	97·5 × 133	—	1.29	152	4.18	36
XII*H*	12	Square	80 × 80	—	1.24	189	5.20	36

Note. *H* — stand with horizontal rolls.
V — stand with vertical rolls.

CHAPTER VIII. ROLL PASS DESIGN FOR ROLLING SIMPLE SECTIONS (SQUARES, ROUNDS AND FLATS)

8.1. Roll Pass Design for Rolling Squares

Dimensions of square steel sections range from 5 to 150 mm. The old method of rolling squares with small draughts in several or even one square pass to obtain sharp corners is

now out of use. All present-day square pass designs provide for rolling a rough square section, which has been prepared to the required dimensions, in one or two leader diamond passes.

The finishing square pass is designed with vertical diagonal $h = 1.41a$ and horizontal diagonal $b = 1.42a$.

The temperature expansion is taken into account by using a correction factor:

$$a_{hot} = a_{cold}\,(1.011\text{-}1.015)$$

To obtain accurate squares with sharp corners it is necessary to correctly select the geometrical shape and dimensions of the leader diamond passes.

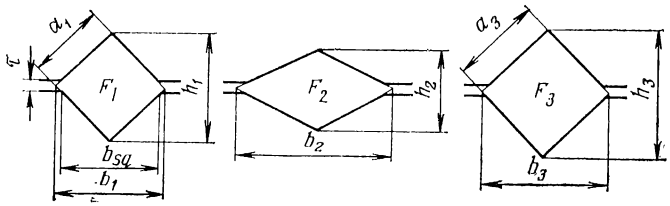


Fig. 8.1. Designation of dimensions of finishing and leader passes for rolling square steel section

The procedure of calculating the dimensions of the intermediate diamonds and squares does not differ from the calculation methods described above for the diamond—square series. Several calculation methods are known. According to B. Bakhtinov (Fig. 8.1)

$$F_2 = \frac{a_1 a_3}{\sqrt{K_{\Delta b}}}$$

where $K\Delta_b$ is determined from the diagram (see Fig. 6.2).

When the draughts are small or equal, it may be assumed that

$$F_2 = a_1 a_3$$

The height of the diamond is determined from the semi-empirical formula deduced by B. Bakhtinov and M. Shter-

nov:

$$h_2 = (1.3\text{-}1.37)\, a_1 - \frac{a_3 - a_1}{2a_3}\left(\sqrt{R\,(a_3 - a_1)} - \frac{a_3 - a_1}{2f}\right)$$

where expression $\frac{a_3 - a_1}{2f}$ may be neglected since the product $\frac{a_3 - a_1}{2a_3} \cdot \frac{a_3 - a_1}{2f}$ is insignificant as compared to h.

The width of the diamond

$$b_2 = \frac{2F_2}{h_2}$$

The cross-sectional area of the square

$$F_3 = F_1 \mu_{sq}$$

The side of the square

$$a_3 = \sqrt{F_3} = a_1 \mu_{av}$$

Small coefficients of draught are taken to decrease the pass wear. Advisable coefficients of draught are given in

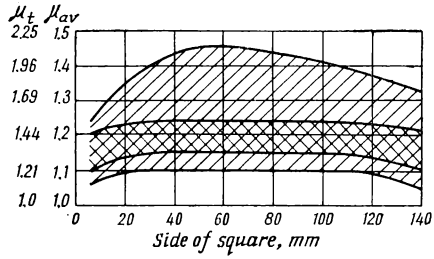
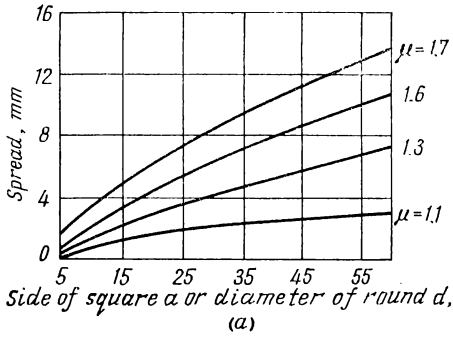


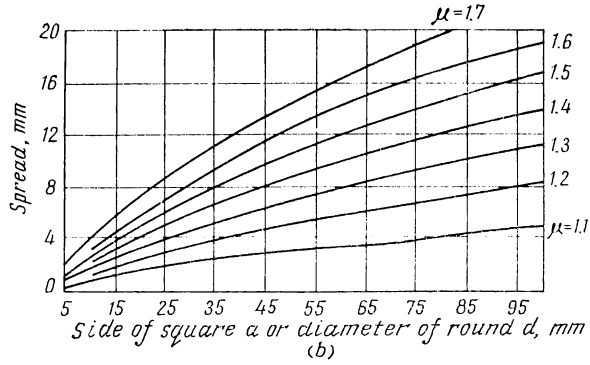
Fig. 8.2. Coefficients of draught for finishing and leader passes when rolling square steel section

Fig. 8.2, the most commonly used ones being indicated by double hatching. Figure 8.3 presents the diagrams obtained by A. Checkmarev while determining the spread for squares and rounds of different size at various coefficients of draught. The width and height of the leader diamond pass (Fig. 8.4) are

$$b = 1.41a_1 + \Delta b_{diamond}$$

$$h_1 = 1.41a_3 - \Delta b_{square}$$





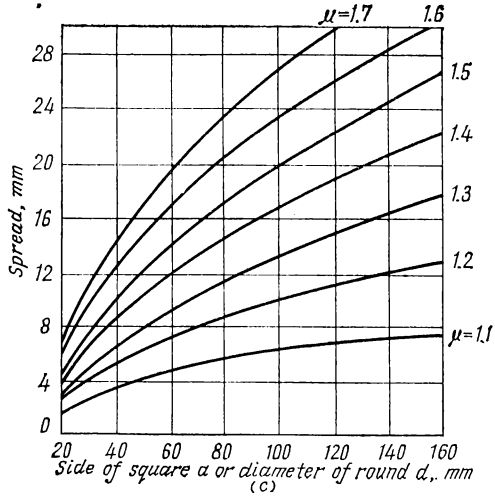


Fig. 8.3. Relationship between spread and dimensions of finishing square or round passes (with different coefficients of draught)
(*a*) 250 mm dia. rolls; (*b*) 350 mm dia. rolls; (*c*) 500 mm dia. rolls

To obtain sharp corners on the finishing square section the top and bottom corners in the leader diamond pass are made of a specific shape, these being sharpened to 90° (Fig. 8.4). In this case the diamond pass has the following dimensions:

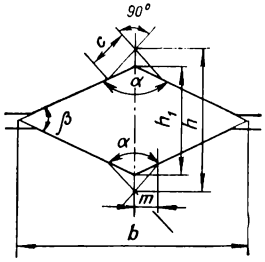


Fig. 8.4. Leader diamond pass

$$c = (0.25\text{-}0.35)\,a$$

$$m = 0.7c$$

$$h = \frac{b}{\tan\frac{\alpha}{2}} + \left(0.7c - \frac{m}{\tan\frac{\alpha}{2}}\right)$$

The leader square pass is designed in the same way as the roughing square passes of the diamond—square series. In some cases the angle at the apex of the top and bottom corners is taken at 1.59 radians (91°), the diagonals being $h = 1.40a$ and $b = 1.43a$ respectively. As a rule, these corners are not rounded off.

Below is given an example of calculating the dimensions of the finishing and leader passes for rolling squares 30 × × 30 mm in cross section in a 300-mm mill.

The finishing square pass

The dimensions of the hot section are determined

$$a_{hot} = a_{cold}\,(1.011\text{-}0.015) = 30 \cdot 1.012 = 30.4 \text{ mm}$$

Taking into account the minus tolerance

$$a_1 = 30.2 \text{ mm}$$
$$b_1 = 1.42a_1 = 42.8 \text{ mm}$$
$$h_1 = 1.41a_1 = 42.5 \text{ mm}$$

With roll gap $t = 2$ mm, the width of the groove cut is

$$b_p = b_1 - t = 42.8 - 2 = 40.8 \text{ mm}$$

The leader square pass

The average draught is taken from the diagram shown in Fig. 8.2:

$$\mu_{av} = 1.18$$

then

$$a_3 = a_1 \cdot 1.18 = 35.6 \text{ mm}$$

The leader diamond pass

The diamond area

$$F_2 = a_1 \cdot a_3 = 34.4 \cdot 35.6 = 1083 \text{ mm}^2$$

From the simplified Bakhtinov formula

$$h_2 = (1.3\text{-}1.37)\, a_1 - \frac{a_3 - a_1}{2a_3} \sqrt{R\,(a_3 - a_1)} =$$

$$= 1.3 \cdot 30.2 - \frac{35.6 - 30.2}{2 \cdot 35.6} \cdot \sqrt{150 \cdot (35.6 - 30.2)} = 37 \text{ mm}$$

and

$$b_2 = \frac{2F_2}{h_2} = \frac{2 \cdot 1083}{37} = 59 \text{ mm}$$

The roll passes for rolling square sections with the use of two pairs of finishing passes, i.e. two diamond and two square passes, are designed in a similar manner. Additional squared corners in the roughing diamond passes are not used in this case.

The use of four passes provides for the production of more accurate square sections, especially when the draughts are small.

8.2. Roll Pass Design for Rolling Rounds

There are several different roll pass design schemes for rolling round sections (Fig. 8.5).

The first and second versions are distinguished by the method of producing the leader square section (the square diagonal is exactly fixed and there is a possibility for controlling the height).

The second version is universal as it makes it possible to obtain a number of proximate dimensions of round steel.

In the third version the leader oval is replaced by a decagon. It is used for rolling large rounds.

The fourth version is similar to the second one, the only difference being in the shape of the edging pass which does

not restrict the metal spreading. This version is employed for rolling rounds in a wide range of sizes, but when the metal ductility is low, its use is inexpedient.

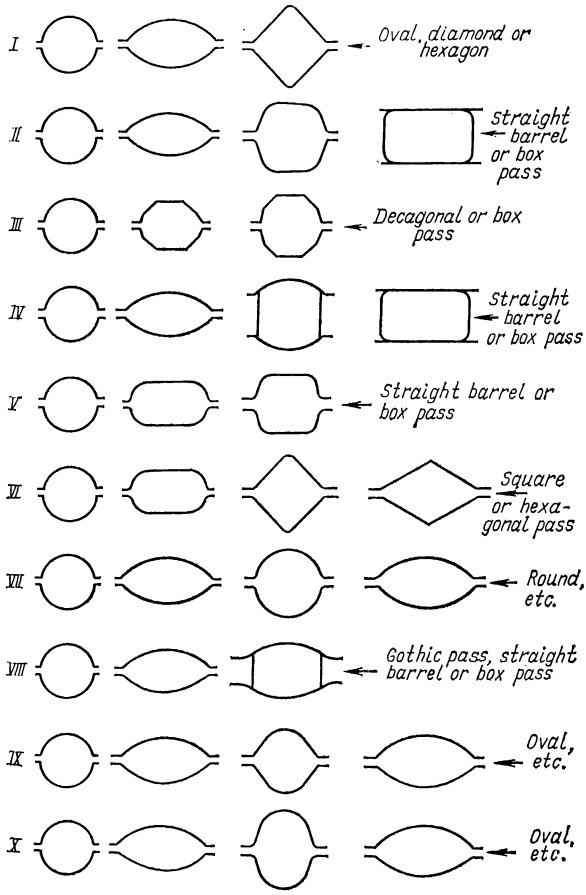


Fig. 8.5. Methods of rolling round steel sections

The fifth and sixth versions envisage the application of flat oval passes which offer considerable draughts and ensure better vertical stability of the rolled section in the leading-in passings.

The other four versions are based on the use of the oval — round series.

The first three versions have found the widest application.

When designing a finishing round pass, a correction is made to take into account the temperature expansion of the rolled section. Then, depending on the required tolerances, a part of the margin tolerance value at the roll designer's discretion is subtracted from the obtained value of diameter

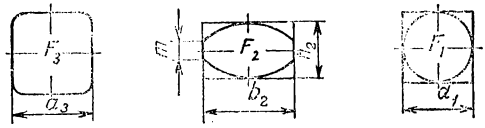


Fig. 8.6. Determining of dimensions of leader and finishing round passes

$d_{hot} = (1.012 - 1.015)\, d_{cold}$ in order to ensure the production of the rolled sections with the minus dimensional tolerance.

The nonuniformity of the temperature distribution across the workpiece section should be taken into consideration and some free space should be provided for the metal spreading. Therefore, the horizontal diameter of the round pass is often made slightly greater than the vertical one. This can be achieved by decreasing the roll gap, i.e. by adjusting the top roll lift. Sometimes, when turning a round pass, a small taper is made tangentially to the pass circumference at the pass parting.

Many methods of designing the leader oval and square passes are known. Empirical relationships between the principal dimensions of the leader passes (Fig. 8.6) are widely used. The size of side a_3 of the leader square is usually equated to the diameter of the finishing round or is made slightly greater:

$$a_3 = (1.0\text{-}1.1)\, d$$

Then, the total draught from the square to the round, when $a_3 = d$, is

$$\mu = \frac{d^2}{\frac{\pi}{4} d^2} = 1.275$$

when $a_3 = 1.1d$

$$\mu = \frac{(1.1d)^2}{\frac{\pi}{4} d^2} = 1.54$$

The average draught is respectively

$$\mu_{av} = 1.13\text{-}1.24$$

The draughts may be determined from the diagrams plotted on the basis of practical data, e.g. from the diagram obtained by A. Checkmarev (Fig. 8.7). The dimensions of the

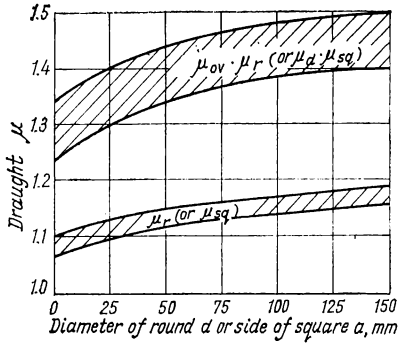


Fig. 8.7. Relationship between coefficient of draught and the size of side of square or diameter of round

leader square passes are determined by the methods which have been considered above for the intermediate square passes.

The design of an intermediate oval pass by the method proposed by B. Bakhtinov and M. Shternov, which is based on the regularities of the ratio of the draughts of circumscribed rectangles, leads to a number of practical relationships. Therefore we will utilize the final empirical formulas and diagrams for determining the leader oval dimensions. Oval truncation m/h_2 is determined from the diagram shown in Fig. 8.8. The ratio of the draughts in the oval and round passes is

$$\frac{\mu_0}{\mu_1} = (1.27\text{-}0.64)\frac{m}{h}$$

This ratio may also be determined from the diagram shown in Fig. 8.9.

The cross-sectional area of the leader oval pass is

$$F_2 = \frac{0.88 a_3 d_1}{\sqrt{\frac{\mu_2}{\mu_1}}}$$

where

$$\mu_2 = \frac{F_3}{F_2} \quad \text{and} \quad \mu_1 = \frac{F_2}{F_1}$$

The ratio of the height of the oval to the diameter of the round for round sections of various dimensions is found from

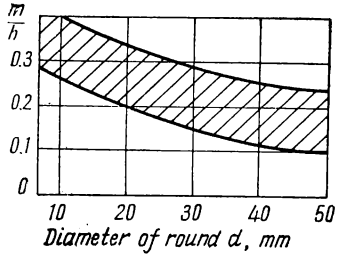


Fig. 8.8. Determining of the degree of oval truncation

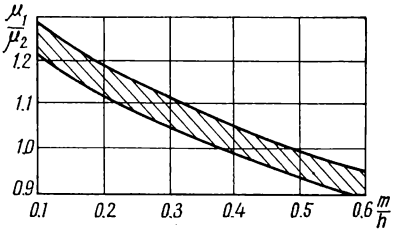


Fig. 8.9. Relationship between the ratio of draughts in oval and round passes and the shape of the oval

the diagram (Fig. 8.10) plotted on the basis of practical data:

$$\frac{h_2}{d_1} = c$$

The height of the oval

$$h_2 = c d_1$$

Taking into account the oval truncation, the width of the oval is

$$b_2 = \frac{3F_2}{h_2 \left(2 + \frac{m}{h_2}\right)}$$

The oval radius

$$R = \frac{b_2^2 + (h_2 - m)^2}{4(h_2 - m)}$$

Taking into account the roll gap, the width of the groove cut in the oval pass is

$$b_p = 2\sqrt{(h-t)R - \left(\frac{h-t}{2}\right)^2}$$

The presented data on the design of the leader square and oval passes refer to the first version of the rolling scheme for

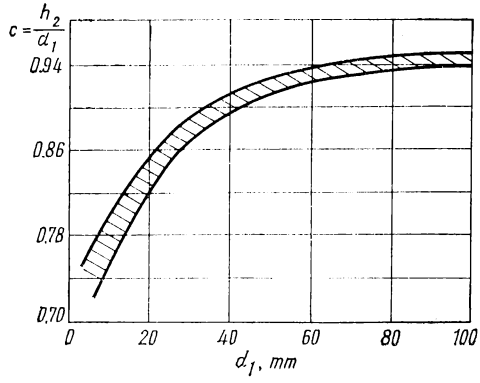


Fig. 8.10. Dimensions of leader oval passes h_2/d_1 for rounds of various diameters

rounds. With this version it is possible, within definite limits, to obtain rounds of different diameters by increasing the roll gap in the leader oval pass (in this case the size of the leader rectangular edging pass must also be changed).

With the second version a range of rounds of proximate dimensions can be produced rather simply (Fig. 8.11) by using the so-called edging square pass (adjustable as to the roll gap and height). The universality of this rolling scheme is achieved due to the margin of width in the oval leader pass and the possibility for changing the roll gap in the rectangular edging pass designed with a great taper of sides (up to 40-50%). The dimensions of the rectangular edging pass with the minimum (normal) roll gap should correspond to the minimum round section size in the given range. Rounds of larger diameters are produced by enlarging the roll gap in the rectangular pass, thus increasing its cross section. The

calculation procedure is similar to that for the first roll pass design version, *viz.* the leader square — oval — round series.

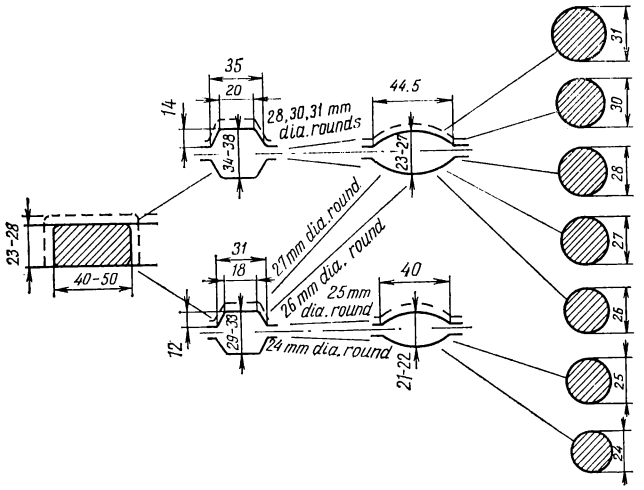


Fig. 8.11. Roll pass design for rolling round steel sections according to second version

Polygonal passes are used in place of the leader oval and square ones for rolling rounds of over 120 mm in diameter. A rectangular piece is shaped into a round via two-three intermediate polygonal sections, special attention being paid to the proper design of the pass and to its filling with metal. The design of the leader polygonal pass by the method proposed by B. Bakhtinov and M. Shternov (Fig. 8.12) may be recommended. The adopted height h of the pass is smaller than the diameter of the round by the amount of spread which ranges from 3 to 10 mm, depending upon the round size. Side a is adopted to be approximately equal to $0.5b$. The size of

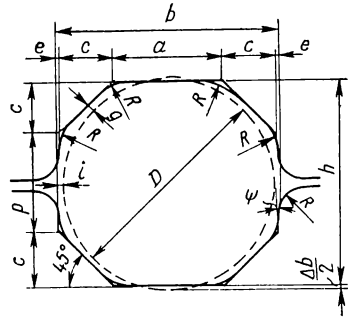


Fig. 8.12. Shape of leader pass for rolling rounds

this side should be selected so as to ensure the workpiece stability in passings. Sides adjacent to a are usually arranged at an angle (45°), those at the pass parting are made with small taper e. Width b of the pass is taken proceeding from the reduction in height in a round pass of diameter D:

$$\frac{b}{D} = 1.08\text{-}1.20$$

Other numerous roll pass design versions for round sections are comparatively simple and may be successfully used, should the need arise, provided the above described design principles are mastered well.

8.3. Some Features of Roll Pass Design for Die Rolling Reinforcement Bars

At present plain round steel for concrete reinforcement is seldom used, because it is being replaced by die-rolled section steel. Die-rolled section steel with the most popular configuration of projections and ribs is shown in Fig. 8.13.

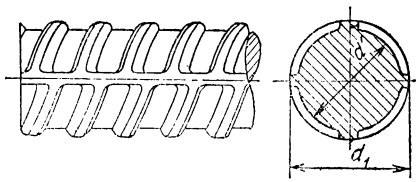


Fig. 8.13. Reinforcement steel bar

Modern constructional engineering widely employs low-alloy steel in place of previously used mild carbon steel. Therefore the hardness of the rolls has had to be increased and difficulties have arisen when milling the grooves. Anodic-mechanical and other methods of working high-hardness rolls have found wide application.

The roll passes for die rolling are designed in accordance with any scheme of rolling rounds. Intermittent flutes are only made in the finishing passes. Oval passes are designed with large truncations: $\frac{m}{h} = 0.4\text{-}0.5$. The leader oval passes

are often made flat. Draughts in the finishing passes are increased to ensure reliable filling of the longitudinal and helical projections with metal. So, these for die-rolled sections of various numbers amount to 1.4-1.6 in the finishing passes and to 1.2-1.35 in the leader oval passes.

Single-radius oval passes are used for rolling small sections. In flat oval passes the height is decreased by 1-4 mm for various section numbers. The ratio of the height of the oval pass to its width amounts to 0.50-0.55. The ratio of the width of the oval pass to the section number is equal to 1.6-1.7. Depending on the diameter of the round, the dimensions of the leader square passes are

$$a_3 = (1.2\text{-}1.25)\, d_1$$

8.4. Roll Pass Design for Rolling Flats

A considerably wide range of rolled products falls into the category of flat steel. This range includes: flats for constructional purposes with widely varied width and thickness, busbar sections 40-70 mm in width and 5-15 mm in thickness, hoop and band iron of more than 1.2 mm in thickness, plain and specially-shaped springs, strips (skelp feed for welded tubes) and other types of flat steel, such as plowshare and automotive rim sections, fire grate bars and so on.

Flat steel is rolled in open and tongue-and-groove passes. Rolling in open passes is widely used.

Step rolls in combination with edging passes intended for working the side edges of flats are applied for producing flats of various dimensions in line-type mills.

Some schemes for rolling flats in the line-type and modern continuous mills equipped with vertical rolls are shown in Fig. 8.14.

Strip, skelp and spring rolling mills are widely used. Edgings are most commonly accomplished in vertical roll stands (edgers). Flat billets are used as the mill feed, especially for wide flats.

Increasing the width of square or rectangular billets by making use of slitting (swabbing) passes is seldom used, because this decreases the product quality. In edging passes coefficient of reduction $\eta = \frac{B}{b}$ ranges between 1.05 and 1.20.

The last edging passes are the sizing ones which serve to obtain the exact preset width of the rolled strip, especially in case of thin strips.

The edging passes are designed in various ways. More often than not, they are cut to little depth into the rolls, this being dependent upon the design of the stand and the roll

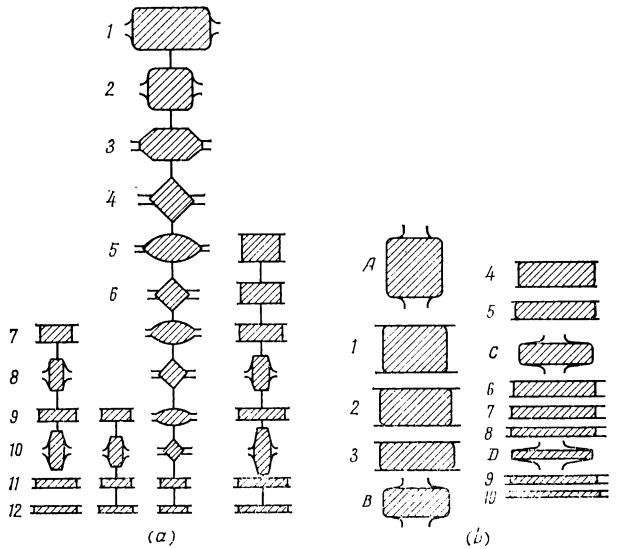


Fig. 8.14. Stages in rolling strip: (*a*) in line-type mill and (*b*) in continuous mill; *A*, *B*, *C*, *D* — vertical roll stands; *1-12* — horizontal roll stands

gap adjustment limits. The taper of sides in the edging passes amounts to 5-10%. The bottom of the pass for the first edging passings is sometimes made slightly convex. Plain rolls are used in strip mills to perform slabbing action (horizontal reduction). The uniformity of reductions across the workpiece is of exceptional importance for rolling strips. Otherwise the rolled strip will have a side camber or single-sided ridge buckles.

Maximum reductions are permitted in the first passings. The roll passes for flats are designed with due regard for

the power parameters of the rolling mill and the strength of the rolling stand units.

According to A. Vinogradov, the side of the leader square pass (when rolling narrow and thick strips) is determined from the following formula:

$$a = \frac{1.015b + kh}{1 + k}$$

where h and b = thickness and width of the finished strip

k = 0.2-0.4 is the coefficient of spread according to Geuze

The overall coefficient of reduction is

$$\eta_{ov} = \frac{a}{h}$$

The coefficients of reduction for each passing are determined from special tables and then corrected. When Δh is known, the spread in each passing is found from the Geuze formula.

B. Bakhtinov and M. Shternov also use the coefficient of reduction:

$$\eta = \frac{H}{h}$$

η is determined for each passing proceeding from the conditions of the metal biting by the rolls and the distribution of deformation among the passings. Then the total edge (lateral) reduction is determined as follows

$$\Sigma\Delta h = \Sigma\Delta b + (B_0 - b)$$

where $\Sigma\Delta b$ = total spread

B_0 and b = initial and final width of the strip

The spread of the strip is calculated either from the Bakhtinov formula, which gives satisfactory results when rolling plain strips, or from the Petrov — Zibel formula. The spread in edging passes is not taken into consideration.

CHAPTER IX. ROLL PASS DESIGN FOR ROLLING STRUCTURAL SECTIONS

The majority of rolled structural sections fall into two types, *viz.* angle sections (angles and zee-bars) and flanged sections (I-beams, channels, tees, piling bars and all kinds of rails).

Besides these two main types, there is a number of intricately shaped (most commonly nonsymmetrical) sections which are used in industry and construction.

9.1. Roll Pass Design for Rolling Angles

Various roll pass design methods for rolling angles are used depending on the distinguishing features of the employed rolling mill. Some of the most commonly employed methods are illustrated in Fig. 9.1.

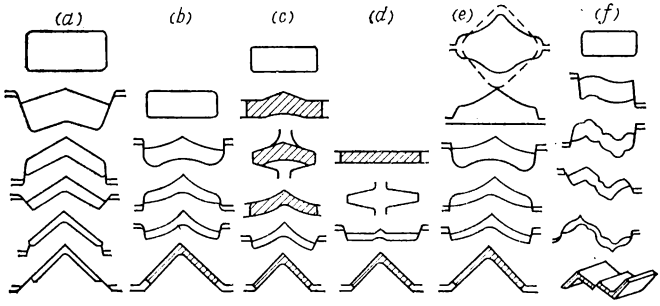


Fig. 9.1. Roll pass design methods for rolling angles

(*a*) straight flange design for tongue-and-groove passes; (*b*) butterfly design for tongue-and-groove passes; (*c*) for open passes; (*d*) with bending in finishing pass; (*e*) diagonal design; (*f*) with cutting of zee-bar in finishing pass

During the last few years a tendency is observed toward the reduction of the number of shaping passes to make the applied breakdown (roughing) passes universal. The straight flange method is seldom used due to a deep groove cut, a

great difference between the effective diameters and increased roll wear. The butterfly method of rolling angles in tongue-and-groove and open passes has found wide application.

The method of rolling angles in open passes is characterized by the ease of the rolling mill adjustment and provides for great durability of the passes. Rolling light angles with bending in the finishing pass was first accomplished in a semi-continuous mill (with looping in the finishing stands) installed at the Magnitogorsky iron and steel works. In this case was used an ordinary edging pass cut into horizontal rolls installed before the back repeater, as distinct from the vertical edging pass shown in Fig. 9.1*d*.

Leader passes of the tongue-and-groove type are employed in a number of rolling schemes to obtain exact flange dimensions in the open finishing pass.

The method of the diagonal rolling of a starting square piece in specially-shaped passes is used for making angles of the greatest possible dimensions.

The following characteristic features are common to all the above mentioned roll pass design methods for rolling angles:

(1) designing a conditional finishing pass in a flared-out form by the same method as the leader passes are designed;

(2) using the same roll pass design and calculation procedures for equal and unequal angles;

(3) determining the overall and particular coefficients of reduction by a method similar to that adopted in the roll pass design for rolling flats;

(4) determining the roll pass dimensions with respect to the centre line of the flanges;

(5) correcting the values of the estimated unrestricted spread to take into account noncontact deformation (when bending the section);

(6) changing the apex angle of butterfly passes in the direction of rolling.

The shaping passes are designed beginning from the last one, i.e. in a direction opposite to that of rolling, in accordance with the following scheme:

1. The dimensions of the finished section are determined with due regard for thermal expansion and minimum tolerances.

2. The shape and dimensions of the so-called conditional finishing pass are determined to design all the shaping passes by using the general geometrical principles. The conditional flaring out of the straight flanges of the finished section and determination of the design dimensions are carried out by using the centre line of the flanges as the basis (Fig. 9.2).

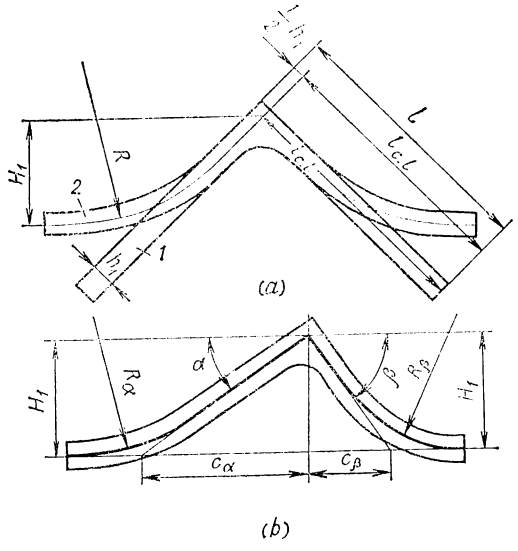


Fig. 9.2. Conditional finishing passes: (*a*) for equal angles; (*b*) for unequal angles
1—actual finishing pass; *2*—conditional finishing pass

The following relations are used for the determination of the geometrical parameters of conditional passes for equal and unequal angles.

Equal angles. The bisector of the angle pass is vertical (Fig. 9.2), therefore

$$\alpha = \beta = 45°$$

$$H_1 = (0.4\text{-}0.45)\, l$$

$$R \approx (0.8\text{-}0.95)\, l$$

The lengths of the flange centre lines to the point of their intersection are

$$l_{cent} = l - \frac{h_1}{2}$$

Unequal angles. If the passes for rolling unequal angles are cut into the rolls so that their bisectors are vertical, i.e. if $\alpha = \beta = 45°$, then great axial forces will arise during rolling and this will require a reliable lateral support for the rolls. Therefore most commonly used is such a finishing pass arrangement with which there is some angle between the bisector of the main section angle and the vertical.

According to Fig. 9.2, the main geometrical dimensions of butterfly passes for rolling unequal angles are the following:

$$H_1 = (0.4\text{-}0.45)\,\frac{l_l + l_s}{2}$$

where l_l and l_s = lengths of the centre lines of the large and small flanges

$$R_\alpha = (1.0\text{-}1.1)\,l_l = \text{const}$$

$$R_\beta = (0.7\text{-}08)\,l_l = \text{const}$$

$$\tan\alpha = \frac{l_s}{l_l}$$

$$\tan\beta = \frac{l_l}{l_s}$$

$$C_\alpha = \frac{H_1}{\tan\alpha} = \text{const}$$

$$C_\beta = \frac{H_1}{\tan\beta} = \text{const}$$

The design dimensions of the conditional finishing pass denoted by symbol 'const' are taken as a basis for designing all the actual butterfly passes.

3. The distribution of coefficients of reduction among the shaping angle passes is determined (in much the same way as for the schemes of rolling flats).

The general formula for determining the height of the leader (second) pass is the following

$$H_2 = H_1 - \frac{H_1}{h_1\eta_{oo} - h_1}\,\Delta h_1$$

where H_1 = height of the conditional finishing pass
h = flange thickness of the finished section
η_{ov} = overall coefficient of deformation (reduction)
$\eta_{ov}h_1 = h_{bil}$ is the height of the billet

From the Bakhtinov empirical formula the height of the billet is:

for rolling in tongue-and-groove passes

$$h_{bil} = h_1 + 2H_1$$

for rolling in open passes with unrestricted spread

$$h_{bil} = h + H_1$$

The height of any shaping pass of the nth serial number is

$$H_n = H_{n-1} - \frac{H_1}{h_{bil} - h_1}\Delta h_{n-1}$$

$$\frac{H_{n-1} - H_n}{H_1} = \frac{\Delta h_{n-1}}{h_{bil} - h_1} = \frac{\Delta h}{\Sigma \Delta h}$$

The tangent of the flange flaring angle is

$$\tan \alpha = \frac{H}{C}$$

The flange thickness in the nth pass is

$$h_n = \eta_{n-1}h_{n-1}$$

Coefficients of flange reduction η for each passing are determined from a diagram plotted on the basis of practical data (Fig. 9.3). It should be borne in mind that when rolling large angles by the butterfly method, 6-8 shaping passes are used and the apex angle of the section changes from 145-150° to 90°, i.e. by 5-10° per pass, depending on the number of passes.

4. The length of the pass flanges is determined, depending on spread. The length of the centre line in the nth pass is determined as the difference between the length of the centre line in the n-1 pass and the spread in this pass:

$$l_{cent(n)} = l_{cent(n-1)} - \frac{\Delta b_{n-1}}{2}$$

where Δb_{n-1} = total spread of the rolled piece.

The spread in an unequal angle pass is distributed proportionally to the width of the flanges.

For the larger flange

$$\Delta b_l = \Delta b_n \frac{l_l}{l_l + l_s}$$

where Δb_n = total spread.

For the smaller flange

$$\Delta b_s = \Delta b_n \frac{l_s}{l_l + l_s}$$

However, the spread calculated from the given formulas does not correspond to the actual spread in rolling angles.

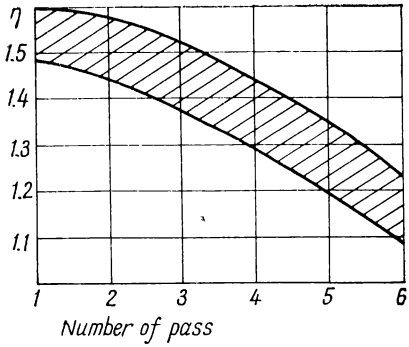


Fig. 9.3. Coefficients of flange reduction in shaping passes when rolling angles

In this case the spread is dependent upon two opposing factors. On the one hand, the so-called noncontact deformation, which takes place when the workpiece is bent in the first (relative to the rolling direction) shaping pass and the flanges are straightened in the finishing pass, gives rise to some increase in the spread. On the other hand, the spread in the tongue-and-groove passes used is restricted.

The increased spread (as compared to the calculated one) is usually taken into account only for passes where the workpiece is largely bent or straightened. It is always the case in angle passes used for rolling the initial rectangular piece and also in finishing passes. In such cases the spread calcu-

lated from one of the well-known formulas is increased by a factor of 1.1-1.5, depending upon the workpiece bending or flange straightening angle, in order to obtain the spread for the centre line of the flanges.

When the spread is restricted, its calculated value is decreased in conformity with practical data:

$$\Delta b_r = (0.5\text{-}0.65)\,\Delta b_{cent}$$

It is adopted for a finishing pass that

$$\Delta b_r = \Delta b_{cent}$$

Table 9.1

Data on Roll Pass Design for Angles 50 × 50 × (5-6) mm in Size

Estimated roll pass design data	Number of pass (in direction opposite to that of rolling)				
	1 (finishing)	2	3	4	5
Coefficient of reduction η	1.13	1.32	1.46	1.50	1.50
Flange thickness, h . .	5.5	6.2	8.2	12.0	18.0
Reduction Δh	0.7	2.0	3.8	6.0	10.0
Pass height with respect to centre line H_n . .	—	21.0	19.5	16.0	11.0
Tangent of flange flaring angle	1.00	0.95	0.88	0.72	0.50
Flange flaring angle . .	45°00′	43°40′	41°33′	36°01′	26°40′
Spread (for strip) . . .	0.6	2.3	4.0	5.0	—
Overall spread taking into account noncontact deformation Δb_{ov}	1.2	4.4	7.2	8.0	—
Restricted spread Δb_r . .	1.2	2.6	4.2	4.7	—
Length of flange centre line l_{cent}	47.9	47.3	46.0	43.9	41.5
Corner radius at butt of flanges r	6.0	7.5	12.0	22.0	40.0
Radius for top flange surfaces	—	36.9	35.9	34.0	31.0
Radius for bottom flange surfaces	—	43.1	44.1	46.0	49.0
Corner radii at flange ends	3.0	4.0	2.0	6.0	6.0
Ordinate of centre of gravity	—	7.2	6.6	5.2	3.2

Table 9.2

Data on Roll Pass Design for Angles 125 × 80 × (7-8) mm in Size

Estimated roll pass design data	Number of pass (in direction opposite to that of rolling)					
	1 (finishing)	2	3	4	5	6
Coefficient of reduction η . . .	1.15	1.30	1.45	1.55	1.60	1.60
Flange thickness h	7.5	8.6	11.0	16.0	25.0	40.0
Reduction Δh . .	1.1	2.4	5.0	9.0	15.0	25.0
Pass height with respect to centre line H_n	45.0	44.0	42.0	38.0	31.0	20.0
Tangent of flaring angle for large flange	0.64	0.62	0.59	0.54	0.44	0.28
Flaring angle for large flange α	37°30′	32°00′	30°50′	28°20′	23°50′	15°50′
Tangent of flaring angle for small flange	1.56	1.53	1.46	1.32	1.08	0.69
Flaring angle for small flange β	57°20′	56°50′	55°35′	52°35′	47°00′	34°50′
Spread (for strip)	1.1	2.7	5.2	7.5	9.2	12.8
Overall spread taking into account noncontact deformation Δb_{ov} . .	2.2	5.2	9.4	12.7	14.7	16.6
Restricted spread Δb_r	2.2	3.0	5.4	7.3	8.5	9.6
Spread of large flange Δb_l . . .	1.3	1.0	3.3	4.5	5.2	5.8
Spread of small flange Δb_s . . .	0.9	1.2	2.1	2.8	3.3	3.8
Length of centre line of large flange $l_{c.l.f}$ . .	122.9	121.6	119.8	116.5	112.0	106.8
Length of centre line of small flange $l_{c.s.f}$ . .	77.3	76.4	75.2	73.1	70.3	67.0
Corner radius at butt of flanges	11.0	14.0	20.0	34.0	62.0	115.0
Radius for top surface of large flange	—	125.7	124.5	122.0	117.5	110.0
Radius for bottom surface of large flange	—	134.3	135.5	138.0	142.5	150.0
Radius for top surface of small flange	—	55.7	54.5	52.0	47.5	40.0

For equal angles the length of the flange centre line in the nth pass is

$$l_{cent(n)} = l_{cent(n-1)} + \frac{\Delta b_r}{2}$$

When determining the lengths of the flange centre lines of unequal angles, Δb_r is distributed among the flanges in proportion to their lengths.

The data on the roll pass design for rolling equal and unequal angles are given in Tables 9.1 and 9.2.

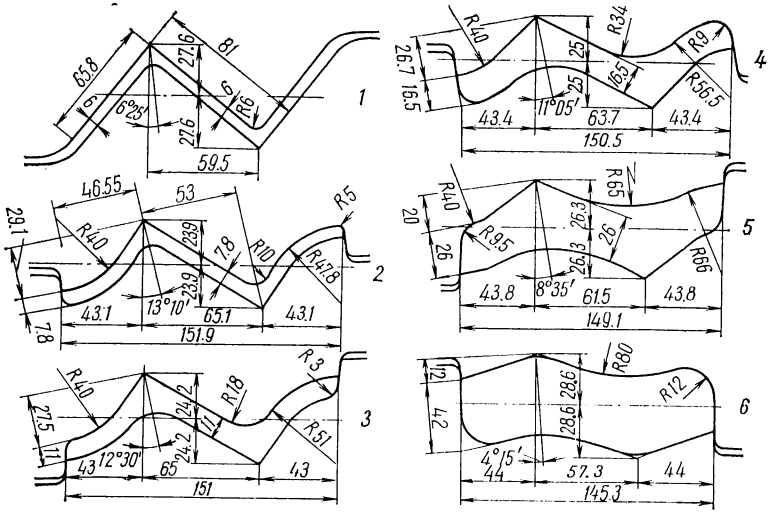


Fig. 9.4. Roll pass design for rolling zee-bars

The centre of gravity of the rolled section and, correspondingly, the neutral line of the passes is aligned with the roll parting line. The ordinates of the neutral line of angle passes are determined analytically by the method of dividing the pass sections into individual geometric figures whose areas and static moments are determined comparatively simply. Somewhat cumbersome formulas for calculating the ordinates of the neutral line of structural sections by this method are given in special roll pass design handbooks.

To provide for uniform working of all elements of the rolled section the top and bottom positions of the pass parting are alternated. The leader pass is usually made with its live hole cut into the top roll.

The roll pass design for rolling Z-bar sections is based on the same main principles as those of the roll pass design for rolling angles (Fig. 9.4). The Z-bar section can be regarded as two angles joined together by means of a common leg. The passes for Z-bar sections are designed for the butterfly method of rolling, use being made of the same constant parameters, coefficients of reduction and spread in passes as for angle passes. To make the design more simple, the bisectors of the two angles are located vertically and then turned a little around the centre of gravity of the section.

As a rule, Z-bar sections are rolled in tongue-and-groove passes. The exception is the finishing pass which is made open and with tapers to fix the rolls in the axial direction.

9.2. Roll Pass Design for Rolling I-Beams

Beams are the most typical flanged sections, if judged by the methods of rolling and roll pass designing. Procedures of designing the shaping passes for all flanged sections are studied using the beam as a model. Typical beam rolling schemes are illustrated in Fig. 9.5. Scheme *I* is used for beams of the lower range of sizes (Nos. 10-30); scheme *II*, for beams Nos. 30-60 (the shaping of the beam in live slitting passes begins in cogging mills); scheme *III* uses edging rolls which makes it possible to roll beams with tall parallel flanges (wide flange beams); scheme *IV* is based on the diagonal roll pass design.

Scheme *IV* provides for greater reductions and makes it possible to obtain parallel flanges. Due to the reduced friction in the pass the roll service life is increased. The surface finish of the final section is also improved due to a more uniform working of all elements of the section. The presence of substantial thrust loads is a disadvantageous feature of the diagonal pass design (scheme *IV*). This makes it necessary to have a special shape of collars (tapers) at the pass parting.

Schemes *I* and *II* are widely used, but they are distinguished by specific conditions of deformation of the flanged

section elements (dead and live flanges, web). These conditions are characterized by the fact that different points of the pass have different speeds. These speeds also differ from that of the rolled piece.

The web of the beam section is deformed as a plain piece under the action of the uniform normal roll pressure. The

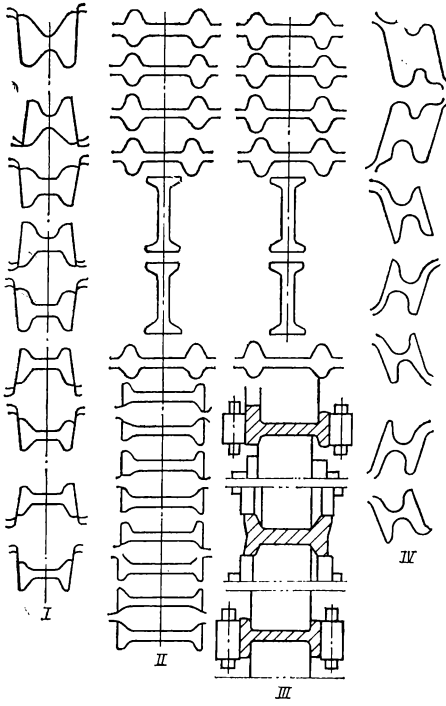


Fig. 9.5. Stages in rolling beams

speed of the web corresponds to that of the rolled piece. The live flanges are worked by the pass surfaces which are formed by two rolls and have different speeds.

The dead flanges are worked in the groove cut into the body of one roll only, the conditions of working being such that the speed of the workpiece delivery is higher than the

speed of the pass surface. In this case the flanges are pulled down when edge reduction is effected.

It should be noted that noncontact deformation also occurs in the last shaping passings.

As a rule, the nonuniform deformation in flange passes causes uneven wear on the pass surfaces, especially in the live flange holes, so that the surfaces of the latter wear out more rapidly.

Many investigators expressed different points of view on the relations between the degree of deformation of live or dead flanges and that of the web in the pass.

Thus, according to Metz, the coefficients of deformation are the same for the web and live flanges and are smaller for the dead flanges. According to Kirchberg, all elements of the section must be deformed to the same degree in the last passes while increased draughts of the web are recommended in roughing passes. According to Geuze, it is essential to provide a slightly increased (by 10%) draught of the web, the draughts of the live and dead flanges being equal. B. Bakhtinov and A. Chekmarev advise to have the same degree of deformation in all elements of the pass, but I. Kuchko is in favour of different ones. I. Kuchko believes that equal draughts are only feasible in the last passes where deformations are minimal. According to P. Polukhin, the distribution of deformations among the elements of the flange passes must correspond to the size of the rolled section since, depending on the section number, the ratio of the areas of the web and the flanges changes over a wide range.

Such a variety of opinions on one of the principal problems of the roll pass design for flanged sections points to the fact that the conditions of deformation in flange passes are very complex and are insufficiently studied. This also confirms the impossibility of using one and the same method for determining the deformation of different elements of the rolled section. Therefore, when designing roll passes for flanged sections, use is made of practical data and recommendations established on the basis of results obtained with well-performing roll pass series.

When working out the roll pass design problems and selecting schemes for rolling beams, the following should be taken into consideration.

1. The design of individual flange pass elements and the distribution of deformations among them should be based on the characteristic features of rolling the web and the live and dead flanges of the beam. As it has been mentioned above, the increase of the height of the live flanges and pulling down of the dead ones are caused by the difference between the surface speeds in the pass elements, this being dependent upon the difference in the effective diameters.

The pulling down and the height increment of the flanges are determined by using the practical data obtained by P. Polukhin, A. Chekmarev, I. Kuchko and N. Litovchenko.

In practice, the height increment of the live flanges is taken at 0.5-2.0 mm in roughing passes and at 0.5-1.5 mm in the leader and finishing passes. The pulling down of the dead flanges is taken at 4-8 mm in roughing passes and at 4-10 mm in leader passes. The pulling down increases with increasing the dimensions of the flanged sections. For the live flanges the coefficients of deformation of the edge portions of the section ($\eta_{l.a}$) are usually adopted to be greater than those of the base portions ($\eta_{l.b}$):

$$\eta_{l.a} > \eta_{l.b}$$

For finishing passes

$$\eta_{l.a} = (1.005\text{-}1.01)\,\eta_{l.b}$$

for other passes

$$\eta_{l.a} = (1.04\text{-}1.1)\,\eta_{l.b}$$

For the dead flanges the coefficients of deformation of the base portions ($\eta_{d.b}$) are greater than those of the edge portions ($\eta_{d.a}$):

$$\eta_{d.b} = (1.02\text{-}1.05)\,\eta_{d.a}$$

The coefficient of deformation of the web η_w in the last passes is slightly smaller than that of the base portions of the live flanges ($\eta_{l.b}$), but in roughing passes $\eta_w \geqslant \eta_{l.b}$. The coefficients of deformation of the flange edge and base for the nth pass are determined by the degree of the edge reduction of the flanges. For the live flanges the coefficients of deformation of edge a and base b are

$$\eta_{l.a(n)} = \frac{a_{d(n-1)}}{a_{l(n)}}\ ; \quad \eta_{l.b(n)} = \frac{b_{d(n-1)}}{b_{l(n)}}$$

For the dead flanges:

$$\eta_{d.a(n)} = \frac{a_{l(n-1)}}{a_{d(n)}}; \quad \eta_{d.b(n)} = \frac{b_{l(n-1)}}{b_{d(n)}}$$

For the web:

$$\eta_{w(n)} = \frac{d_{n-1}}{d_n}$$

These formulas for determining deformations (edge reductions) in different elements of flange passes are used by N. Litovchenko, B. Diomidov and V. Kurdyumova in their generalized practical calculations. The height increment and pulling down of the flanges (Δh_l and Δh_d) calculated according to the method proposed by B. Bakhtinov do not always correspond to those adopted in actual roll pass designs because they take no account of the nonuniformity of deformation, wear of passes, edge reductions (which are different for the edges and the base of the flanges) and some other parameters of rolling.

2. The deformations of the web and the dead and live flanges should be made as even as possible (especially in the last shaping passes) to eliminate the movement of the rolled metal from one part of the pass into another. But the live flange holes of the pass wear faster than the dead flange ones and therefore, the deformation (edge reductions) of the dead flanges is deliberately increased as compared to that of the live ones.

Also, in the first shaping passings (slitting and roughing), when the rolled metal is sufficiently ductile, the use is made of premeditated nonuniform deformations.

The optimum number of shaping passings and the roll diameter, depending on the size (number) of the rolled section, have been found practically. Thus, according to the data provided by A. Chekmarev:

Section number	10-12	12-18	18-24	24-60
Number of shaping passings . .	6-8	7-9	8-11	10-15
Roll diameter, mm	350-500	400-650	500-950	600-950

The number of passings is set, depending on the distinctive features of the rolling mill used. The first two or three passings are often intended to produce a rectangular billet

of the required dimensions which correspond to the shape and dimensions of the first slitting pass. Beams of large sizes (above Nos. 30-40) are rolled from a slitted billet delivered by a blooming mill.

Coefficients of deformation (draught, reduction) decrease in the course of rolling. The deformation is limited by the conditions of the metal biting in the first shaping passes (particularly in slitting ones) and by the pass surface wear in the last passes (particularly in finishing ones). The coefficients of draught (or reduction) practically amount to 1.5-1.7 in the first roughing flange passes and to 1.07-1.15 in the finishing passes. Average coefficient of draught μ_{av} amounts to 1.15 for heavy beams (No. 60) and to 1.30 for light ones (No. 10-12). For heavy beams the coefficient of reduction of the web is lower than the coefficient of draught of the entire section, while for beams of smaller numbers it is higher than the coefficient of draught of the entire section. This depends upon the degree of the web development in roughing passes, which in its turn influences the degree of the flange pulling down.

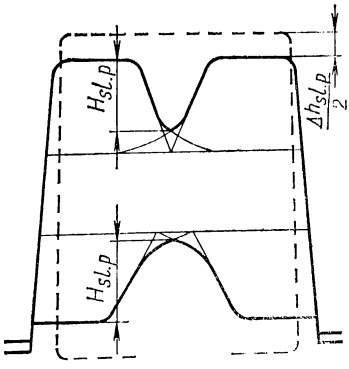


Fig. 9.6. Slitting pass ($H_{sl.p}$—height of slitting pass)

3. The cross section of the billet is determined empirically, depending on the dimensions and shape of the slitting pass (Fig. 9.6) obtained when designing the roll passes (this being in a direction opposite to that of rolling).

The shape of the slitting pass is defined by the angles of the slitting swells and their height. The use of a more acute swell is advantageous from the standpoint of obtaining a thin web in one passing, but this deteriorates the conditions of the metal biting and leads to dangerous slippages. Therefore the swells have to be rounded off, which makes it necessary to increase the height of the pass to take account of the flange pulling down in subsequent passes.

The spread in the slitting pass increases with increasing the angle of the swell. The spread in the slitting pass is

$$\Delta b_{s.p} = 5\text{-}10 \text{ mm}$$

consequently, the width of the billet

$$B_0 = B_{s.p} - (5\text{-}10) \text{ mm}$$

There are the following empirical formulas for determining the initial billet dimensions:

according to A. Chekmarev

$$H_0 = \frac{H_{s.p}^2 \cdot B_0}{F_{s.p}}$$

according to M. Mutyev

$$H_0 = \left(0.65 + 0.35 \frac{B_0 H_{s.p}}{F_{s.p}}\right) H_{s.p}$$

where H_0 = height of the billet
B_0 = width of the billet
$H_{s.p}$ = height of the slitting pass
$F_{s.p}$ = cross-sectional area of the slitting pass

4. Spread in flange (beam and channel) passes is a factor of definite importance which should be taken into consideration in order to design an efficient roll pass series and obtain the required surface finish of the rolled section. The spread in the finishing pass is adopted to be equal to 1-3 mm and then it is increased from 0.5-1 mm in subsequent passes (proceeding in a direction opposite to that of rolling) to 8-15 mm in the slitting pass. The following empirical formula may be used for calculating the spread in the nth pass (proceeding in a direction opposite to that of rolling):

$$\Delta b_n = 0.01 B_f + \left[\frac{n-1}{2} - (n-1)\right]$$

where B_f = width of the finishing pass.

The spread decreases with increasing the carbon content and degree of alloying of steel and increases with increasing the size (number) of beams and channels being rolled.

As is shown in Fig. 9.7, the normal position of the workpiece with respect to the width of the next pass it enters

will be ensured when

$$B' < B''$$

where B' = width of the workpiece across the ends of the flared-out flanges

B'' = width of the live hole of the pass across the ends of the live flanges

For the workpiece to enter the next pass normally without shearing the flanges the following condition must be satisfied:

$$A > C$$

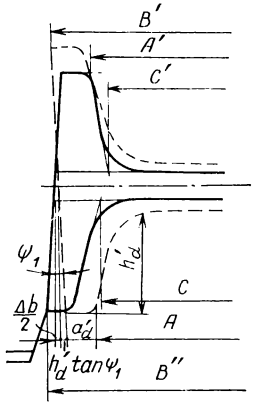


Fig. 9.7. Determining of conditions for section entering the pass

Designating the elements of a beam pass in accordance with Fig. 9.7, the following condition may be formulated:

$$b_l \geqslant a'_d + \frac{\Delta b}{2} + h'_d \tan \psi_1$$

or

$$\Delta b \leqslant 2 (b_l - a'_d - h'_d \tan \psi_1)$$

where ψ_1 = slope of the external surface of the flanges entering the live hole of the pass.

The corner radii in the groove cut of the dead flanges in all passes are selected so that

$$\frac{r}{a} = C = \text{const}$$

where a = thickness of the flange end.

There are no roundings on the live flanges because their height is increased as compared to the calculated value to prevent the formation of fins.

The tapers (slopes) of the external edges of passes for dead and live flanges are made differently. The external edges of the dead flanges are inclined inwards with a slight slope (2-4%). The taper of all live flanges is usually increased up to 4-6%. Sometimes, the live flanges are made flared-out

with large tapers (up to 10%) which are decreased down to 4-6% towards the leader pass.

A. Chekmarev suggests that the dimensions of intermediate flange passes be determined by using the overall and particular coefficients of draught in the passes and in their flange holes from the slitting pass to the finishing one:

$$\mu_{ov} = \frac{F_s}{F_f}$$

$$\mu_{ov} = \mu_1\mu_2\mu_3 \ldots\ldots \mu_{n-1}\mu_n$$

For flanges

$$\mu_{ov.\,fl} = \frac{F_{l.fl.sl}}{F_{f.fl}}$$

where $F_{l.fl.sl}$ = cross-sectional area of the live flange holes of the slitting pass

$F_{f.fl}$ = cross-sectional area of the flange holes of the finishing pass

Taking into account correction factor

$$m = \sqrt[n-1]{\frac{\mu_{ov.fl}}{\mu_{ov}}}$$

the draughts of the flanges are determined:

$$\mu_{fl(n)} = m\mu_n \quad \text{and} \quad \mu_{fl(n-1)} = m\mu_{n-1}$$

According to A. Chekmarev the estimated increment of the flange height is distributed in proportion to the flange draughts ($\mu_{fl\,(n)} - 1$), i.e. the height increment of a live flange amounts to 0.5 when the coefficient of draught for the flange $\mu_{fl\,(n)}$ is equal to 1.5.

The method proposed by B. Bakhtinov for determining the dimensions of the pass elements according to the overall and particular coefficients of draught is somewhat more complex. In accordance with this method the overall coefficient of deformation for flanges k_{fl} is adopted to be equal to the product of the coefficients of reduction in height η_{fl} and thickness λ_{fl}:

$$k_{fl} = \eta_{fl}\lambda_{fl}$$

For a live pass

$$k_l = \eta_l\lambda_l$$

For a dead pass

$$k_d = \eta_d \lambda_d$$

From the condition of the equality of deformations of all elements of the rolled section (according to B. Bakhtinov) it follows that $k_l = k_d = k_w$.

Figure 9.8 illustrates the roll pass design for rolling No. 16 beam in a 500-mm mill.

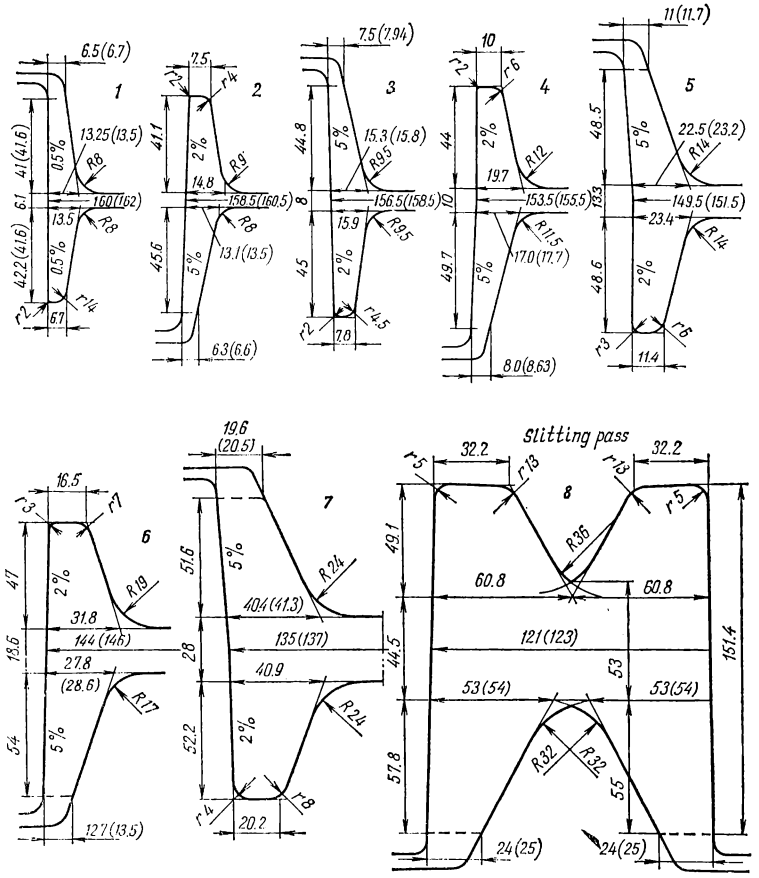


Fig. 9.8. Design and dimensions of passes for rolling beam No. 16

The rolling of thin flanged sections is characterized by opposing factors such as great deformations used and rapid drop of the rolled piece temperature. Experience has shown the undoubted advantage of continuous mills when rolling thin flanged sections, because only an insignificant drop of the workpiece temperature occurs in these mills.

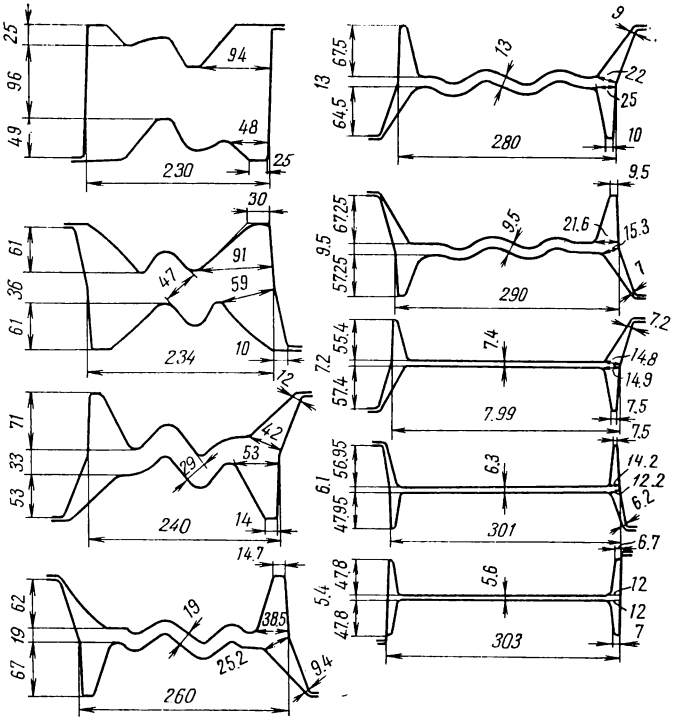


Fig. 9.9. Roll pass design for rolling thin-web beam No. 30

To fulfil the specific conditions of rolling thin flanged sections the total draught is decreased by forming a sinuous web (Fig. 9.9), which makes it possible to reduce the size of the billet, by using flared-out live flanges, etc.

In order to reduce the effect of the great nonuniformity of deformation the pass parting is made so that the live flanges are located diagonally.

9.3. Roll Pass Design for Rolling Channels

Channel sections are widely used in building structures and machinery.

The roll pass design for rolling channels is based on the same general principles as those of the roll pass design for rolling beams, but is distinguished by a great nonuniformity of deformation of the rolled section elements. As distinct from the roll pass design for rolling beams, there is no alternation of the live and dead flange holes in the roll pass

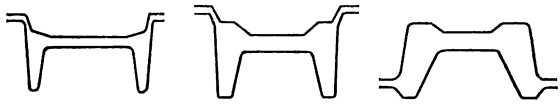


Fig. 9.10. Types of control passes

series used for rolling channels and several live flange holes follow one after another. Besides, use is also made of intermediate passes which differ in shape and in the character of deformation. The roll pass design for rolling channels uses the counter-flange method. The counter flanges serve to feed metal to the real flanges and make for maintaining the required temperature and hence, the ductility of the rolled metal at the corners of the section. They are reduced in the course of rolling, being completely rolled down in the leader and finishing passes. To provide for better filling of the rolled section corners with metal the counter flanges are deformed to a higher degree than the real ones.

The channel pass parting is usually made at the ends of the real flanges. In this case it is difficult to obtain the required height of the flanges and therefore the so-called control passes with dead or semienclosed holes for the real flanges are employed (Fig. 9.10). As a rule, the control passes are used for the leader and intermediate shaping passings.

The following three methods of rolling channels (Fig. 9.11) are most widely used:

1. Beam method of rolling channels.
2. Method of rolling channels in straight flange passes with increased tapers.
3. Butterfly method of rolling channels.

With the beam method the same roughing passes are used for rolling beams and channels which makes it possible to reduce the number of roll changes and the roll inventory. However, this method is characterized by a number of

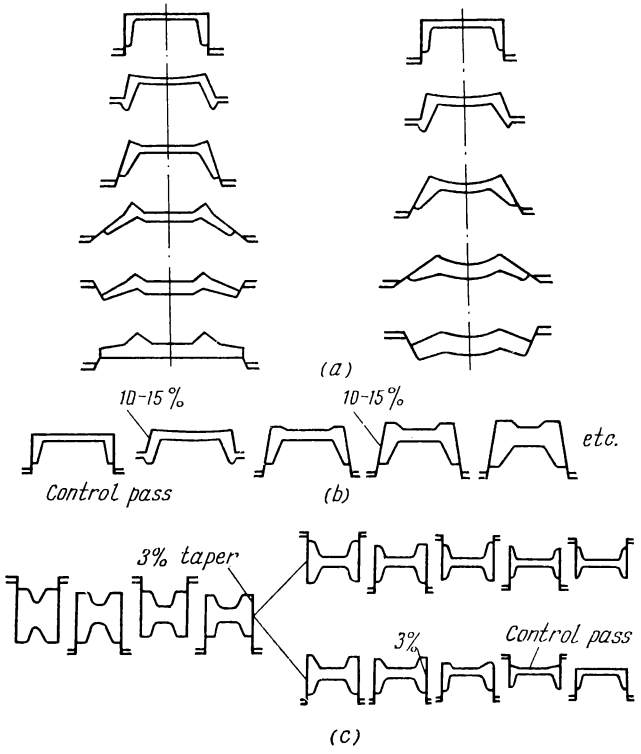


Fig. 9.11. Roll pass designs for rolling channels
(*a*) butterfly design; (*b*) straight-flange design with increased tapers; (*c*) beam design

disadvantages. The small tapers of sides decrease the number of roll redressing operations and thus increase the roll consumption. The nonuniformity of deformation results in an additional power consumption. Also, this method of rolling provides lower degrees of draught.

With other two methods of rolling the effect of these unfavourable factors is less pronounced and the number of shaping passes can be reduced.

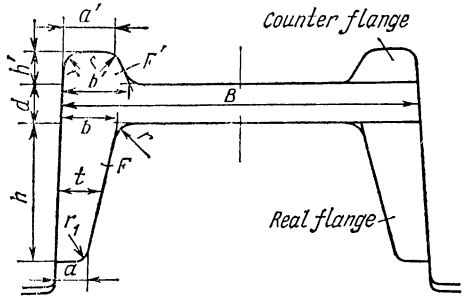


Fig. 9.12. Design of intermediate channel pass

When designing the channel passes the dimensions of the counter flanges have to be calculated (Fig. 9.12). According to B. Bakhtinov, the height of the counter flanges is

$$h' = (0.03\text{-}0.05)\, h\, (n-1)$$

where h = height of the real flange
n = serial number of the pass (counting in a drection opposite to that of rolling)

The thickness of the counter flange at the base is

$$b' = (1\text{-}1.3)\, b$$

where b = thickness of the real flange at the base.

The thickness of the counter flange at the tip is

$$a' = (0.6\text{-}0.65)\, b'$$

Small counter flanges are also used in the finishing passes (especially for heavy channels) in order to prevent the pulling down of the channel corners after cooling and improve the conditions of straightening the channels in the cold state. These flanges are made in the form of a rectangular triangle with a small leg 0.15-0.3 mm long and a large leg slightly longer than the base of the counter flange in the leader pass.

The general design scheme for channel passes is similar to that for beam passes.

1. Hot dimensions of the finished section are determined with due regard for minus tolerances. Counter flanges in the form of small rectangular triangles should be provided for heavy channels.

2. The number of flange passes and their designs are selected, depending on the type of the rolling mill used.

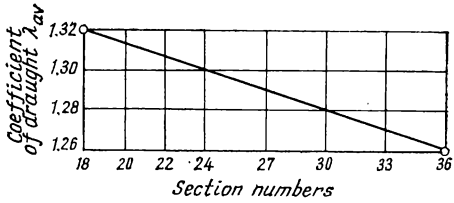


Fig. 9.13. Relationship between average coefficient of draught and section dimensions

3. The overall and particular coefficients of deformation are determined, as well as the dimensions of the passes, and appropriate corrections are made.

4. The dimensions of the slitting passes and of the initial billet are found.

5. The centres of gravity of the sections of all passes are determined and the roll drawings are made.

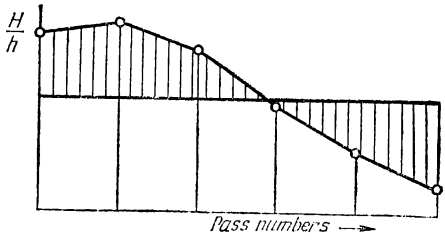


Fig. 9.14. Change of average coefficient of draught in passes

The recommendations given by I. Kuchko may be used for determining the average draughts, depending on the channel number and the distribution of the coefficients of draught among passes (Figs. 9.13 and 9.14). The practical data provided by P. Polukhin (Table 9.3) are used for determining the spread in channel passes.

Table 9.3

Change of Spread in Passes when Rolling Channels

Channel number	Spread, mm	
	in finishing pass	in other passes
5-8	0.5	$0.5+(n-1)$
10-18	1.0	$1+(n-1)$
20-27	2.0	$2+(n-1)$
30-40	3.0	$3+(n-1)$

Note. n — serial number of pass (counting in a direction opposite to that of rolling).

The pulling down of the dead flanges amounts to 4-7 mm. The height increment of the live flanges is equal to 0.5-1.5 mm. The calculations of the real flange dimensions are similar to those for the beam flanges.

9.4. Roll Pass Design for Rolling Rails

Rails of different shapes and dimensions are used in railroad transport (standard and narrow gauge tracks) and in industry (Fig. 9.15).

In different types of rails the metal is distributed between the elements of the section (head, web and base) dissimilarly. Stringent requirements for the quality, structure, strength and dimensional accuracy of rails dictate that a meticulous approach be taken in designing the roll passes for rails.

The difficulties which are met with in rolling rails are explained by the asymmetry of the rail section with respect to the vertical axis of the roll passes, this causing the development of considerable axial forces which act on the rolls and may affect the thickness of the base and the head of the rail in live flange holes.

To prevent the lateral displacement of the rolls special cones with tapers of 10-20% are provided at the parting of the roll collars. Lubrication is applied to decrease the cone

wear. The asymmetrical position of the rail section in passes and the presence of high flanges (which correspond to the wide base of the rail) have necessitated the use of the hat-type passes. According to the data provided by P. Polukhin and V. Gorobinchenko the shape of the hat passes determines the rolled metal structure, especially in the most critical element of the section, that is in the head.

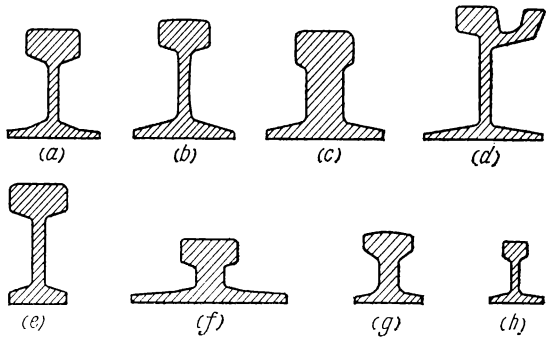


Fig. 9.15. Types of rails
(*a*) and (*b*) standard railroad rails; (*c*) point rail; (*d*) street-car rail; (*e*) guard rail; (*f*) and (*g*) crane rails; (*h*) narrow-gauge rail

Edging box passes are successfully used as preparatory ones (prior to the hat passes), which provides for better working of the rolled piece on the side of the future head.

Figure 9.16 illustrates eight schemes of rolling rails. The first four schemes are most commonly used for narrow-gauge track rails and light rails. The last two schemes suit best the present-day requirements for the quality of heavy rails. A diagonal arrangement of rail passes (diagonal pass design) is sometimes used in these schemes. This provides for better working of the metal, ensures increased coefficients of deformation and permits a greater number of roll redressing operations to be done. The inclination of the pass axes is 10-25% for roughing passes and 10-12% for finishing ones.

The following is the procedure of designing the rail passes.

1. The dimensions of the finishing pass are calculated as it is generally accepted in the flange pass design. The parting

of the finishing pass is made in the middle of the head (Fig. 9.17) to obtain a convex surface on the head. To avoid the jamming of the base in the pass an angle of not less than 1° is provided between the rolling and neutral lines. The taper of the tangents to the rounded corners of the head is taken within the limits of 5-25%. As mentioned above, the

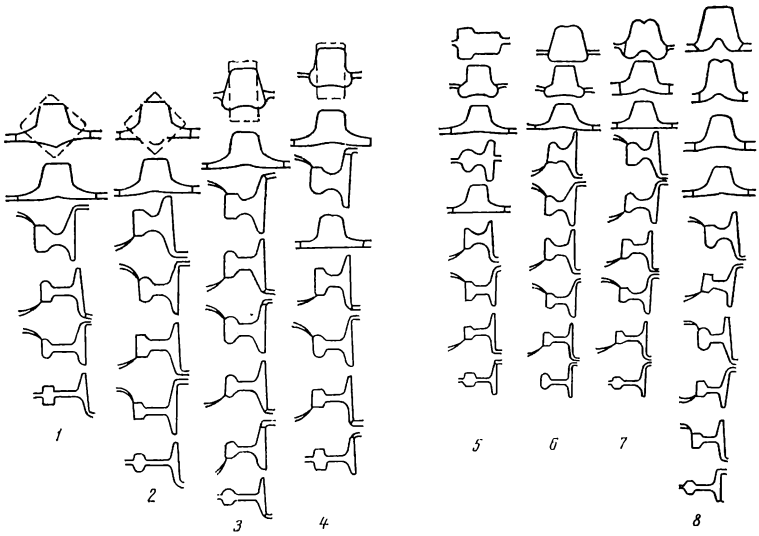


Fig. 9.16. Schemes of rolling rails

inclination of the axis, in case of the diagonal pass design, may reach 10-12% for the finishing pass and for roughing passes it may be even greater.

2. The leader, roughing and slitting rail passes are designed on the basis of general principles of the beam pass design. The dimensions of the section elements are calculated in accordance with the nature of deformation and the optimum number of passes (9-11) for the given rolling mill is determined. The overall draught in rail passes amounts to 2.7-3.2 (for 5-6 passes), the average coefficient of draught being equal to 1.19-1.21.

The draught amounts to 1.30-1.45 in the slitting pass and to 1.06-1.10 in the finishing pass. The degree of deformation

in intermediate passes constantly increases from the finishing pass to the slitting one.

P. Polukhin recommends that the coefficients of draught be equal to 1.05-1.15 for the web in the finishing pass, 1.5-1.6 for the web in the slitting pass, 1.3-1.45 for the flanges forming the base and 1.4-1.5 for the head.

The pulling down of the dead flanges of the base in the finishing pass amounts to 5-7 mm and reaches 10 mm in other passes. The height increment of the live flanges of the

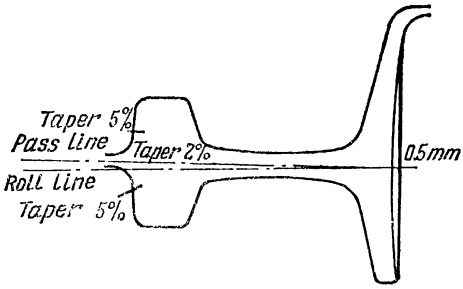


Fig. 9.17. Finishing rail pass

base amounts to 0.5-1.0 mm. Besides the edge reduction, the head is reduced in height by 1-3 mm in the finishing pass, by 3-5 mm in the dead holes of other passes and by 1-2 mm in the live holes. The spread in the finishing pass is taken at 2-3 mm and that in an intermediate pass of the nth serial number (counting in a direction opposite to that of rolling) is determined from the following empirical formula:

$$\Delta b = 0.01 \cdot B_1 + (n-1)$$

When analysing actual roll pass designs great nonuniformity of deformation, particularly in the first shaping passes, becomes evident. Consequently, the metal flows from one part of the rolled section to another.

When designing the slitting rail pass it should be borne in mind that the height of the rail head is greater than the thickness of the rail base and that the head becomes reduced to a greater degree than the base. Therefore, the position of the vertical axis of the pass should be shifted within the limits of 1.8-2.5 mm, depending upon the ratio between the

height of the head and the thickness of the base. The slitting swells are made more obtuse than in case of rolling beams. The height of the slitting swell (thickness of the web) is checked proceeding from the conditions of reliable metal biting by the rolls.

3. The hat passes must ensure the required width of the rail base. When there is a great nonuniform deformation the slitting of the lower part of the workpiece is done in the hat pass. This makes for the breaking of the initial columnar structure of the rolled metal. When the dimensions of the slitting pass are known, the height of the hat pass may be determined from the following formula:

$$H = B_{s.p} - \Delta b$$

where Δb = 8-10 mm is the spread in the slitting pass.

The taper of sides of the hat pass amounts to 8-10%. When several (2-4) hat passes are employed in the given roll pass series, provision in made for unrestricted spread of the base.

The cross-sectional area of the billet is determined by the average coefficient of draught with the given number of passings. The relation between the sides of a rectangular billet is determined, depending on the number and shape of the hat passes.

There are several empirical formulas for determining the ratio of the height of the billet to its width:

For three hat passes

$$\frac{H_b}{B_b} = 1.3\text{-}1.7$$

For four hat passes

$$\frac{H_b}{B_b} = 1.7\text{-}2$$

The dimensions of the billet are also determined by the dimensions of the finished rail:

$$H_b = (1.5\text{-}1.8)\, H$$

$$B_b = (2.0\text{-}2.5)\, B_h$$

where H = height of the rail

B_h = width of the rail head

The section of the billet is checked against the dimensions of the obtained slitting pass with due regard for the selected system of the hat passes.

PART THREE

TYPES OF METAL ROLLING OPERATIONS

CHAPTER X. ROLLING OF BLOOMS, SLABS AND BILLETS

10.1. General

Bloom, slab and billet or semi-finished steel production processes are intermediate ones between steelmaking and finish rolling.

Some part of the products of cogging mills is turned out as finished goods for use in the machine-building industry as billets for forgings and stampings.

The structure of cogging facilities, including the cogging and billet rolling mills, has been permanently changing as a consequence of the growth of the output of iron and steel works, process improvements and equipment modernization. The distinctive trends of this development are as follows.

The change-over to larger ingots and to making finished products from rolled billets rather than from small ingots is in progress. Following the ingot weight increase more powerful cogging mills have been constructed. These are specialized to suit production features, *viz.* blooming mills produce rectangular billets (blooms) for section mills; slabbing mills produce flat billets (slabs) for plate mills; if the production of rolled metal is mixed, i.e. bar and flat products are to be made, the blooming mill will then provide blooms and slabs for both of these production processes; blooming mills are also used to produce shaped, sectional billets for further rolling into large beams.

The raised output of blooming mills, brought about by diminishing the number of passings and thus increasing the cross section of the bloom, has been attained by using the billet mills as an intermediate rolling group between the cogging and finishing rolling mills. This production process

layout allows the bar mills to be supplied with billets of optimum cross sections.

The billet mill stands are generally arranged in series and such mills are named continuous billet mills. A definite tendency is to duplicate the main cogging facility, this bringing the entire complex of cogging and billet mill operations nearer to a continuous process.

Lately, as it has been shown above, the continuous steel-casting plants are combined with the billet mills.

Hence, it is possible to develop a continuous production process from the casting of the metal to the production of finished rolled steel.

At up-to-date iron and steel works the semi-finished steel production is developed in two directions, *viz.* rolling of ingots and continuous casting of steel, the former being at present predominant.

Rolling of heavy rectangular continuously cast billets in cogging mills with an intermediate heating in soaking pits or in end-charged end-discharged reheating furnaces is now considered as a promising process.

The most popular pattern of ingot cogging and billet production processes comprises heating the ingots in soaking pits, rolling in primary mills, cropping top and butt ends with shears to remove the top part of killed ingots which contains shrinkage porosity, rolling in billet mills and, finally, cutting the billets to length and cooling.

The cogging operations also include scarfing of hot blooms and slabs in continuous scarfing lines as well as preparing of billets for rolling, billet surface conditioning, stockpiling, and marking.

The general arrangement of cogging-mill shop equipment with a two-high reversing blooming mill is shown in Fig. 10.1.

The soaking pit bay is arranged longitudinally with respect to the axis of rolling. The soaking pit building is an extension of the blooming mill building which consists of a central, main bay, where the blooming mill is installed, and two adjacent bays, one for electrical equipment and the other for scrap handling. All electrical equipment of the mills and the shop itself is concentrated in the first bay and consists of a substation with the main and auxiliary panels, motor-generator sets and main drive motors. Crops, scale and some-

times cinder from the heating facilities are passed to the scrap handling bay.

The principal and auxiliary rolling equipment, including the main mill train with tables, manipulators and bloom tilters as well as approach, front, back and trailer mill tables, shears, crop handling device and a hot scarfing machine working in the production line, is installed in the main central bay. Usually continuous billet mills are also installed in this bay.

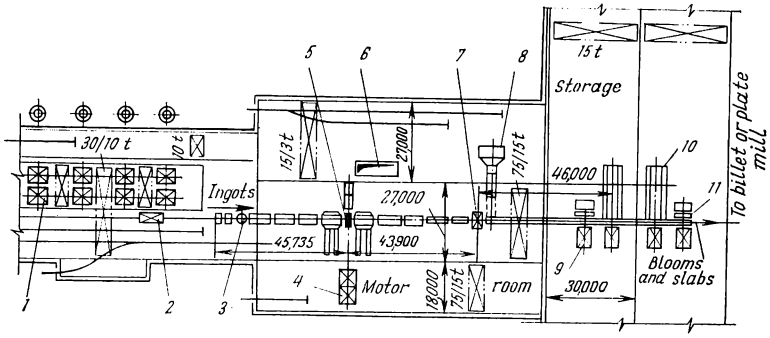


Fig. 10.1. Arrangement of 1150 mm blooming mill

1—soaking pits; *2*—ingot car; *3*—ingot turntable; *4*—main electric motors; *5*—working stand; *6*—scale pit; *7*—shears; *8*—scrap conveyor; *9*—pushers; *10*—bloom cooling bed; *11*—lifting table (slab piler)

The bays of the billet (blooms, slabs) yard are generally arranged perpendicular to the axis of the main bay. These bays are intended to handle rolled metal, this including cooling operations, preparatory operations for rolling in other mills (chipping, scarfing, mechanical surface conditioning with special machines, grinding-wheel dressing, etc.) and stockpiling in accordance with the types of steel produced and heat identification to feed the billets correctly to the receiving tables of rolling mills.

The operation management in the billet handling bays is characterized by the fact that it requires much labour, is complex and calls for an accurate, well planned system with permanent effective control. All bays of the shop are equipped with the necessary types of cranes; special designs are used in soaking pit and billet handling bays.

10.2. Ingot Heating and Soaking Pit Bay Management

Heating of ingots, placed in a horizontal position, in reheating furnaces is carried out at old plants (small ingots) or at plants producing high-grade steels in order to preheat cold alloy metal ingots prior to placing them into the hot holes of soaking pits.

At modern iron and steel works the soaking pit division is designed to provide an uninterrupted high-quality heating of ingots and to create reliable conditions for the organization of an accurate metal flow between the steelmaking and rolling plants. This is ensured by a rationalized equipment layout, more elaborate material handling equipment and a number of organizational measures which utilize present-day means of communication and control. Because of the increased production rate, much thought has been given to the problem of conveying quickly the heated ingots to the receiving section of the rolling mills. New types of high-speed ingot cars and circular conveyance schemes of unique design have been developed. To shorten the transportation distance of heated ingots a perpendicular (T-shaped) pit arrangement is used or the receiving table of the mill is introduced deeply into the soaking pit building.

The design of soaking pits has been changed to suit developments in the cogging processes.

In the thirties a change-over was made from the primitive single-ingot soaking pits to the regenerative soaking pits capable of accommodating 6-8 ingots (Fig. 10.2). At first blast-furnace gas was used to heat these soaking pits. Later, with forced heating, it became necessary to use fuel enriched with coke gas.

From the bottom of soaking pit holes dry cinder made up with coke-breeze was previously removed through cinder holes into boxes placed on rail-cars that moved through a cinder tunnel. This arduous work of removing the cinder and coke breeze make-up has been eliminated by using the new method of liquid cinder draining through cinder holes made in the end walls of the pits. These holes are heated with coke gas by means of nozzle-type burners.

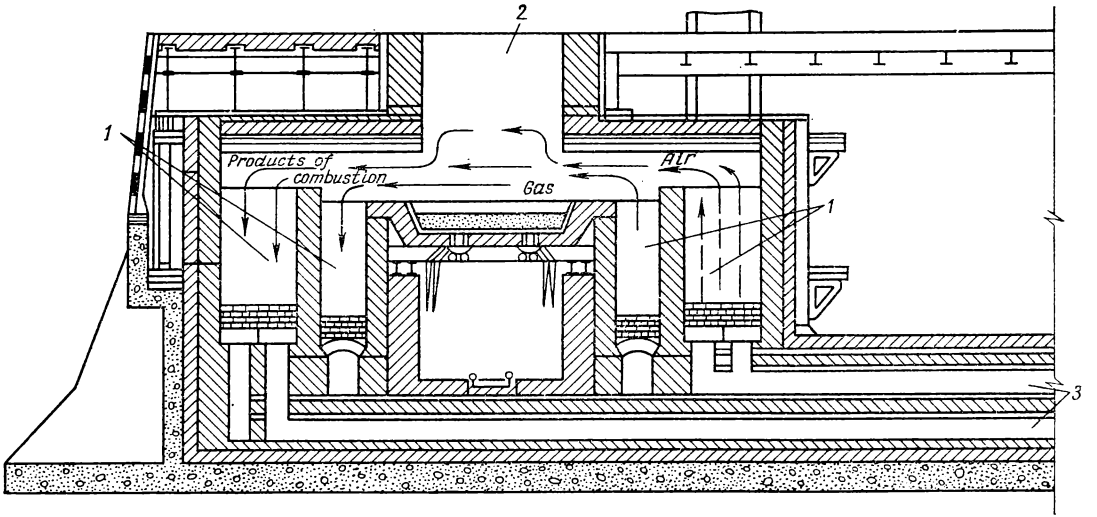


Fig. 10.2. Multiple-ingot regenerative soaking pit
1—regenerators; *2*—working chamber; *3*—flues

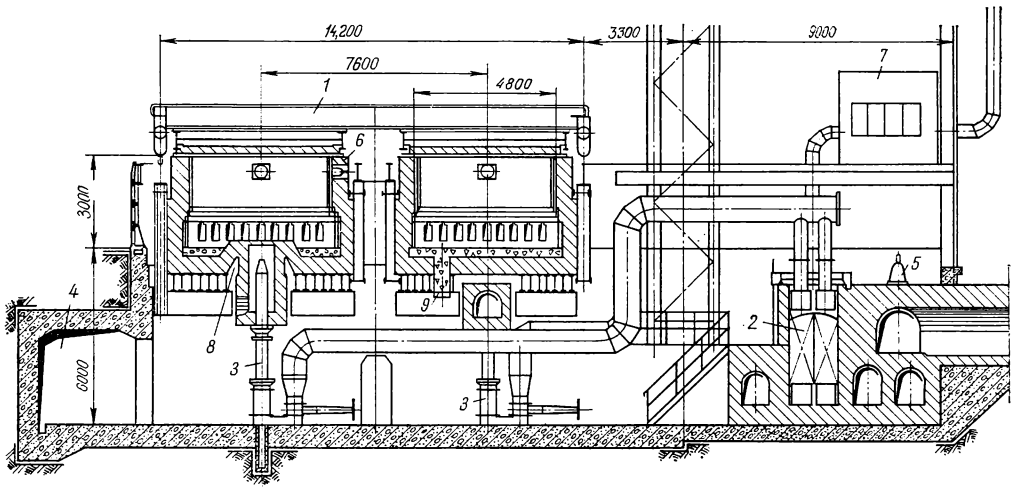


Fig. 10.3. Bottom centre-fired recuperative soaking pits

1—cover crane; *2*—metal gas recuperator; *3*—burner; *4*—cinder tunnel; *5*—fume valve; *6*—impulse thermocouple site; *7*—control room; *8*—air coolers to protect recuperators and burner hole against cinder; *9*—cinder hole

Liquid cinder draining is now mostly used in regenerative soaking pits. Previously the draining of cinder was carried out periodically, the use being made of fusible siliceous additives, including ferrosilicon. Now, continuous cinder drainage is applied. This method permits the soaking pit capacity factor to be greatly increased. With an average time of three hours for heating one ingot the number of pit hole cycles is equal to seven per day. The liquid cinder draining has increased the soaking pit productivity roughly by a factor of 1.4 as compared to the work on dry bottom.

With the growth of the output of primary mills, with an increase in the ingot weight and with more stringent requirements as to the quality of heating, small soaking pits heated with a low heating value fuel have been replaced by the multiple-ingot recuperative soaking pits which utilize fuel of a higher calorific value.

The principal drawback of regenerative pits is an uneven heating of ingots that results in surface fusion of those which are nearer to the flame. The unevenness of heating and fusion of ingot surfaces are aggravated if the pits are operated under forced conditions.

The use of multiple-ingot recuperative pits (Fig. 10.3) which have a burner in a central position at the bottom has created conditions for more uniform heating of a large number of ingots (up to 20 pieces, depending upon weight). The flame of the burner impinges onto the pit cover and then spreads over the pit chamber walls from top to bottom. The waste combustion products pass through side outlet ports and enter the recuperators where they heat the air delivered to the burners. The calorific value of a mixture of blast-furnace and coke-oven gases is 5800-8000 kJ/m^3 (1400-2000 kcal/m^3).

The calorific value of fuel may be decreased in up-to-date pits owing to the practice of using metallic recuperators for gas heating which are installed behind the ceramic recuperators for air heating.

Side heating is used in new recuperative soaking pit designs (Fig. 10.4).

Burners are arranged on one or both sides, at the top part of the pit. Products of combustion are removed, from one or both sides, at the bottom. As an automatic heating cycle is

used, the predetermined gas/air ratio and pressure are maintained, and the gas supply is cut off automatically or turned on, when the pit covers are removed or replaced.

Apart from the improved quality of heating, one may add to the positive features of the multiplace recuperative pits, the operational and constructive advantages, the absence of reversals, the ease of maintenance and repair, the reliability of auxiliary machinery of simple design, etc.

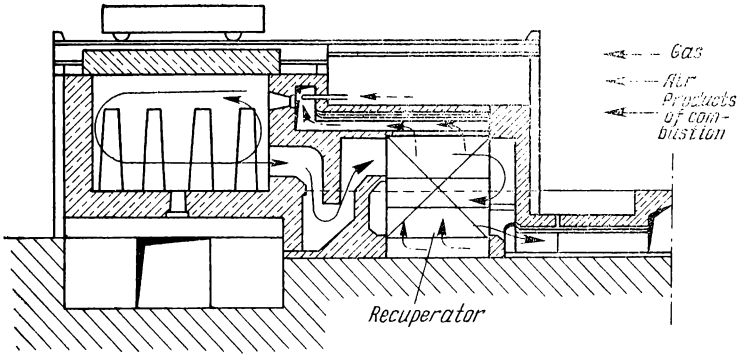


Fig. 10.4. One-way fired recuperative soaking pit

The duration of heating and its technical and economical characteristics depend upon the organization of hot charging, i.e. upon the mean temperature of ingots being charged, and also upon the number of cold ingots. Cold ingot charging depends upon the conditions existing at a given works, *viz.* the relation between the steelmaking and rolling-mill plant outputs and their operation schedules (which may include long delays in the operation of blooming and slabbing mills), as well as upon the management level of the production processes.

The mean temperature of hot ingot charge reaches 850°C, provided that a well-regulated schedule of transportation and handling of ingot mould trains, as well as that of pit charging, are carried out.

The time from the finish of pouring rimmed ingots to their delivery into the soaking pit building is 50-55 min, that for killed ingots is about 2.5 hours. Charging pits with rimmed

ingots out of one ingot mould train (25-30 ingots) requires 10-15 min, that for killed ingots, 15-20 min, provided that two cranes operate in unison.

Rimmed steel poured into big-end-down moulds is delivered without the mould, while big-end-up ingots of killed and semi-killed steel are delivered in moulds. Depending upon their specific heating conditions, all steels may be divided into groups. The different heating conditions which are used to heat the carbon, low-alloy, high-carbon, high-alloy and individual special steels take into account the temperature of the ingot surface at the time of charging. Cold ingots of steel containing more than 0.5% C are placed into previously cooled pit holes. The heating cycles at each works are determined by taking into account the construction of the soaking pits, their heat capacities, the weights of the ingots, the principal heating parameters (the temperature of pit chambers at the period of charging, the time for temperature increase, soaking periods, the total heating time) and the programme of gas and air consumption together with the conditions of their delivery, all this being set in special instructions.

A well established service of information, recording and control is required to carry out these instructions because the flow of large quantities of metal is continuously changing. Recording boards, written documentation, manual operation and visual control are nowadays replaced by automation with remote control and computer handling of continuously incoming information.

A number of measures are undertaken to alter the dimensions of the work chambers of regenerative and recuperative soaking pits, to improve and automate them, to use additional burners besides the central one, and to utilize the heat of waste gases in economizers and boilers.

Electric soaking pits equipped with carborundum heating elements filled with petroleum coke and preheated electrically are used to heat special-purpose steels. Electric pits are well controlled, occupy little space and provide non-oxidizing heating. When using elements filled with petroleum coke the power consumption is 60-70 kW·h per ton of steel.

The throughput of soaking pits is determined mainly by the heating time which depends upon the average tempera-

ture of the ingot charge, upon the quantity of cold ingots and upon the time for auxiliary operations (cinder removal, pit charging, etc.).

The theoretical hourly capacity of one pit chamber P_0 is

$$P_0 = \frac{Qn}{t}$$

where Q = average weight of the ingot
n = average number of ingots in the chamber
t = average heating cycle time which is equal to the sum of the average heating time (for the given steel grade and average temperature of the ingot charge) and the time for auxiliary operations in preparing the furnace chamber for charging, including the removal of cinder

In calculating the total throughput of soaking pits account must be taken of the time for heating the cold ingot charge and of the pit capacity factor which takes into consideration non-coordination of the ingot delivery, departures from the set schedule (of about 0.8-0.85) and the time for running repairs and general overhauls.

The soaking pit bay is equipped mainly with mechanical handling facilities, *viz.* stripping cranes of special design with a rigid lifting arm (swinging and disengaging operations of tongs are considered in the design) and slewing trolley, equipment for lifting pit hole covers and moving them sideways and devices for ingot delivery.

The timely supply of the ingots to the mill receiving facility is a principal problem for all modern highly efficient blooming and slabbing mills that have many groups of soaking pits occupying a total space of 300 m or more in length.

Self-propelled or electric loco-driven heavy two-axle ingot cars are widely used. These cars are equipped with ingot cradles that tilt ingots from their vertical position into the horizontal position onto the rollers of the mill approach table. To raise the rolling mill efficiency the average speed of ingot conveying has been increased to 5-7 m/s by increasing the running speed and acceleration and deceleration rates of the ingot cars.

Ingot cars are now generally remote controlled. A rope drive of the ingot cars which makes it possible to reduce

greatly the dead weight of the cars has found application at a number of works (especially in the USA).

With ingot rolling times of approximately 40 seconds and the total soaking pit lengths (ingot car track lengths) of 270-300 metres it is technically impossible to provide the required rate of the ingot delivery, even if two-seated cars are used, since in this case the car speed would have to be as great as 16 m/s.

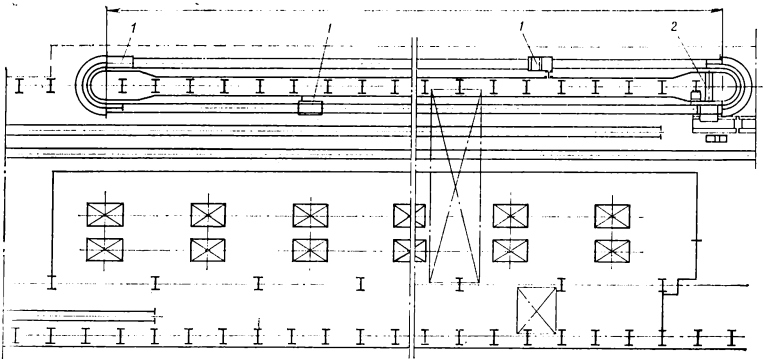


Fig. 10.5. Circular-track ingot delivery
1—ingot cars; *2*—ingot pusher

This problem has been barely solved by using two or more two-seated ingot cars and a mill approach facility which deeply penetrates into the soaking pit building.

A progressive and original method of delivering ingots by a circular track has found application with powerful blooming mills in the USSR (Fig. 10.5). Four ingot cars move in one direction along the track, one branch of which is outside the soaking pit building. As the car approaches the receiving table of the mill the ingot is transferred by a pusher from the car onto the table rollers. Electric power is delivered to the car via a telescopic trolley collectors.

10.3. Cogging and Billet Mills

One-stand two-high reversing blooming mills forming a part of the single-line rail-and-structural or heavy-section rolling mills may be considered as prototypes of modern

cogging mills. The output of the first blooming mills could not be high as they were driven by steam engines, were equipped with primitive hydraulically driven auxiliary equipment and rolled ingots of 2-3 ton weight. Later special mills to produce billets for further conversion became a part of the single-line rail-and-structural and heavy-section rolling mills.

The next stage of cogging mill development was characterized by the propagation of one-stand two-high reversing cogging mills. These electrically driven mills were provided with more elaborate equipment. Subordinate to the structure of the rolled metal production, the cogging-mill shops were equipped with continuous billet mills which worked in combination with blooming mills and rolled billets without reheating. Further growth of cogging-mill shop outputs was ensured by increasing the bloom cross sections (to 300 × 300 and 400 × 400 mm), installing additional stands in billet mills and duplicating the working stands of blooming mills. Thus, continuity was being introduced into the process of rolled billet production.

The development of plate and sheet production required the rolling of very many flat heavy billets (slabs). At first slabs were rolled in usual blooming mills but the rolling of wide slabs required greater lifts of the top roll for edging passings. Therefore the so-called blooming-slabbing mills were constructed. Specialized cogging mills were simultaneously developed to produce slabs of any width. These mills, called slabbing mills, have horizontal and vertical rolls. Ingots up to 30-40 t in weight can be rolled in present-day slabbing mills.

Blooming mills may be divided into the following two groups, depending on their output: large mills with rolls greater than 1000 mm in diameter and small mills with a roll diameter of 750-1000 mm. The latter have gained recognition as mills for special steel production or as cogging stands in heavy-section and rail-and-structural rolling mills.

According to the design of the working stands, small blooming mills are divided into two-high reversing or three-high mills. The latter are constructed with fixed or variable roll gaps.

As to the arrangement and number of their stands, bloomings are divided into one-stand, two-stand (tandem) and multi-stand mills. The latter are somewhat similar to semi-continuous or continuous rolling mills.

The peculiarities of the cogging process and the equipment of cogging mills are considered below. Heated ingots are delivered to the receiving facility of a primary mill which is usually a roll table equipped with ribbed rollers. In many cases the table is provided with a turnover device which turns the ingot with its top or bottom toward the mill stand and often serves also to weigh the ingot.

Ingots enter the mill by the front mill table. Provision is made for the installation of electrically- or hydraulically-driven stationary ingot tilting pots to return ingots or unfinished sections into soaking pits or deliver ingots by means of a crab crane without using the ingot car. The ingot reception bay was one of the first blooming mill section to be fully automated.

In modern rolling mills the receiving units and other pieces of equipment are provided with automatic devices that have reliable interlocking systems. These coordinate the operation of various units of the mill which may be controlled manually, semi-automatically or automatically.

With semi-automatic control the mill operator gives the control impulse signal that is required to start the coordinated working cycle of the whole complex of the mill units. When the mill control is fully automated the control signal is given by a computer.

The main unit of a cogging mill is the working stand with roll adjustment devices which ensure the reduction schedule set in accordance with the given roll pass design.

The working stand of a blooming or slabbing mill is serviced by auxiliary equipment to feed the workpiece into the rolls and to provide longitudinal, transverse and turn-over transfer operations (working tables, manipulators, tilters).

It should be kept in mind that the operation of a primary mill depends, to a great extent, on the pass arrangement on the roll body. Thus, the introduction of the symmetrical pass arrangement (with a plain barrel at the roll centre) has revealed a number of advantages as compared to the successive pass arrangement, *viz.* increased strength of the roll

and more uniform distribution of loads among the roll necks. Besides, the number of additional edge reduction operations has been reduced and conditions for the roll redressing improved.

Among the disadvantages of this pass arrangement is the increased time required for transferring and tilting the workpiece between the second and third passes which are on the opposite sides of the plain roll barrel. If the difference between the pass diameters is great, a more elaborate system of stepped rollers is required to ensure that the workpiece correctly enters the roll passes.

The surface finish of rolled killed steel is completely independent of the rolling schemes used. Good reheating and intensive deformation are factors that provide for the elimination of defects in the ingot macrostructure. As far as rimming steel is concerned, ingots which have blowholes near the surfaces (a thin skin zone) and especially those which have been partly fused in reheating should be rolled with limited reductions and an increased number of tilts.

The investigation of rolling process serves as a basis for checking the adopted pass arrangements and rolling schemes from the standpoint of energy and speed parameters, as well as of the optimum cycles of all the principal and auxiliary operations.

A multichannel oscillograph which checks simultaneously the transient performance characteristics of all the main and auxiliary motors makes it possible to obtain a detailed pattern of the rolling process.

The following methods are used to intensify the rolling process in blooming mills:

(1) Increasing the reductions by means of the roll adjusting device or screwdown at different periods of the rolling cycle, *viz.* after the moment of the metal biting by the rolls and while the workpiece passes through the mill. This method helps to combine operations (the working pass and operation of the screwdown), provided the screwdown has sufficient power and strength, and improves the conditions for the metal biting.

(2) Retarding the workpiece by means of the back side guards as it leaves the rolls at a higher speed.

(3) Transferring the workpiece with the side guards simultaneously with tilting.

(4) Advancing the beginning of the workpiece tilting by about 0.5 s with respect to the end of the working pass.

(5) Using the so-called tandem rolling method whereby two ingots are rolled in tandem, the principal and auxiliary operation cycles for both ingots being combined. This makes it possible to increase the mill output by 15-25%, provided the main drive motors are sufficiently powerful and the side guards, adequately long.

10.4. Design Features of the Main Cogging Mill Train

The main train of a primary rolling mill (Fig. 10.6) consists of one working stand and a mill drive comprising two electric motors, each being coupled to one of the working rolls by means of universal-joint spindles. If only one main drive motor is employed, a pinion stand with a main coupling is used. The rolls are changed by means of a special rig.

The blooming mill train is serviced by auxiliary equipment: working roll tables, manipulators and tilters.

The working stand serves to carry out the rolling proper and this unique set consists of a housing assembly, rolls with chocks, a roll-balance gear, screwdown, breast rolls and a roll-changing rig. As a rule, housings of the close-top type are used. These are heavy castings made in the form of quadrangular rigid frames. The height of the housing window is determined by that of the roll lift, and the width of the window, by the maximum diameter of the rolls (to permit the work rolls to be changed by moving them along their axes).

The housings are checked for the maximum rolling pressure proceeding from the strength and rigidity requirements and taking into consideration dynamic operating conditions. The safety factor for the housings is taken at 10, i.e. the maximum permissible stress for the housing material equals 500-600 kgf/cm^2. The housings are cast from steel with a carbon content of 0.30-0.40% and annealed after casting.

The roll force in the present-day blooming and slabbing mills may be as great as 2000-2500 tf. The rated roll force in

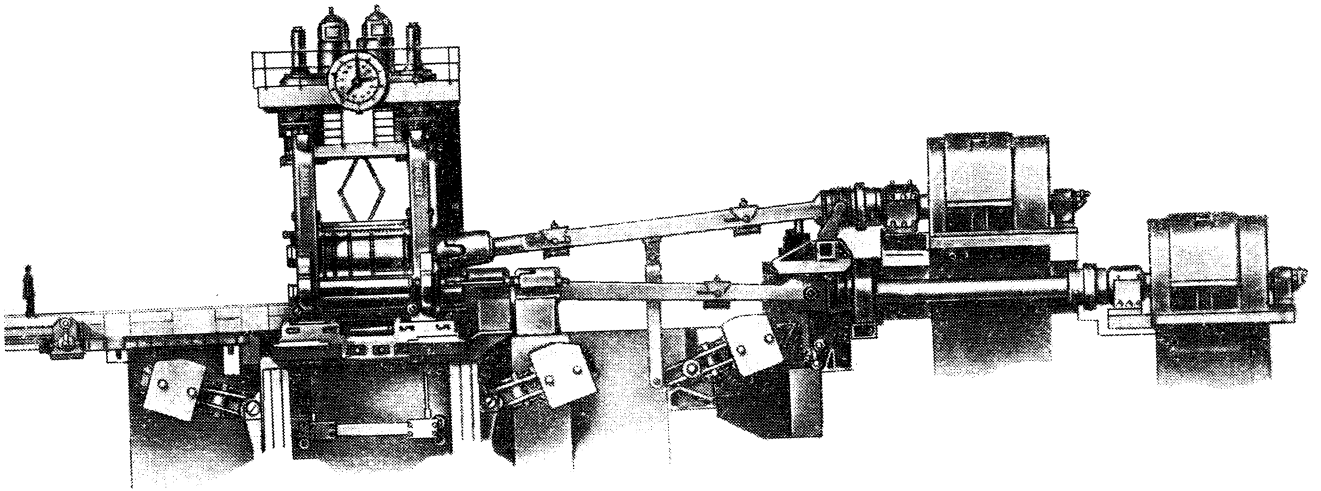


Fig. 10.6. Working train of modern 1300 mm blooming mill

the 1300-mm blooming mill is 2200 tf. Depending upon the pass arrangement, the force applied through the roll necks to one housing equals about 1540 tf which corresponds to a stress of 470 kgf/cm^2 for the top cross bar, 400 kgf/cm^2 for the bottom cross bar and 200 kgf/cm^2 for the posts. The dynamic loads applied to the housings are increased when the rolling is done with high accelerations.

The intensification of the rolling processes in modern blooming mills has made it necessary to increase the cross section and weight of the housings. Thus, the cross section of the posts has been increased from 3000-3800 to 4800-5000 cm^2 and the weight of one housing, from 60-80 to 100-105 t.

Textolite bearings with ample water cooling (0.75 l/min per 1 cm^2 of the textolite area) are usually used for the work rolls of cogging mills. The latest cogging mill designs employ all-in-one-piece textolite bearings; the roll necks are protected against corrosion by using a grease lubricant which is fed to the necks through holes in the chocks and wedges. When employing the fabric bearings, lower coefficients of friction are achieved (0.004-0.006), provided the roll necks are highly polished.

The work rolls of blooming and slabbing mills are usually made of forged carbon or chrome-nickel steel (grade 60XH).

The bending stress for the roll body is

$$\sigma_b = \frac{M_b}{0.1D^3}$$

where M_b = bending moment
D = roll diameter

The bending stress for the roll neck is

$$\sigma_{b.n} = \frac{M_{b.n}}{0.1d^3}; \qquad M_{b.n} = \frac{\frac{l}{2}Q}{0.1d^3}$$

where Q = maximum bearing reaction (due to rolling pressure) applied to the neck
l = length of the neck
d = diameter of the neck

The torsional stress for the neck is

$$\sigma_{t.n} = \frac{M_t}{0.2d^3}$$

M_t = torque applied to the roll at the drive end.

The total stress is

$$\sigma_{\Sigma} = \sqrt{\sigma_{b.n}^2 + 3\sigma_{t.n}^2}$$

The dimensions of the roll neck are determined by the following empirical formula:

$$d = (0.55\text{-}0.65)\, D$$

The length of the neck is

$$l \approx d$$

The mill screws are driven by two vertical flange-type motors through spur gears. When moving the top roll, steady contact of the mill screw with the top roll chock is ensured by a counterweight balance, this being accomplished by means of counterweights and a system of hinged levers and cross members which transmit a steady force to the top chocks through four vertical rods inserted into special slots in the housings.

The mill screws are made of forged carbon steel with a tensile strength of about 70 kgf/mm^2. With a safety factor of 5 the maximum rated stress is equal to 1200-1400 kgf/cm^2.

The thread of the mill screw is of the trapezoidal buttress type.

The mill screw nuts are made of bronze. The nut is tightly fitted in a bore in the housing and is fixed in place with bolts. Thick lubricating oil is forced into the threads to decrease the coefficient of friction and wear. Water is also delivered to the threads for cooling.

The diameter of the nut is about 1.5 times greater than that of the mill screw, the height of the nut is approximately equal to or slightly greater than its diameter. The specific pressure is checked by load Q applied to the mill screw:

$$P = \frac{4Q}{\pi (d_0^2 - d_1^2)\, Z}$$

where d_0 and d_1 = minimum and maximum diameters of the thread, respectively

H = height of the nut

h = pitch of the screw thread

$Z = \frac{H}{h}$ is the number of threads in the nut

The permissible variation of the roll diameter in redressing is 12%. The design of the roll-balancing devices and screwdowns of cogging mills makes it possible to move the top roll vertically over a considerable distance under severe high-speed, dynamic operating conditions. Various designs of such devices are found in existing blooming mills. Nowadays the counterweight balance and screwdowns driven by flange-type motors via simple spur-gear transmissions are most commonly used.

The results obtained by Soviet researchers and designers N.Baimov and E.Palmov have shown that optimum operating conditions of screwdowns are achieved with a triangular speed diagram, definite gear ratio of the drive, minimum gyrating masses (GD^2) of the motor and transmission and maximum pitch of the mill screw thread.

Such conditions are ensured by increasing the so-called coefficient of dynamism:

$$K = \frac{375h}{60i\left(GD^2_{mot} + \frac{GD^2_{trans}}{i^2}\right)}$$

where h = pitch of the mill screw thread
i = gear ratio
GD^2_{mot} and GD^2_{trans} = moments of gyration of the motor and transmission, respectively

Taking into account that some increase of the reduction during rolling may be necessary, the torque which must be applied to the mill screw is

$$M_{scr} = Q\left[\frac{d_{piv}}{3} f_{piv} + \frac{d_p}{2} \tan(\alpha + \varphi)\right]$$

where Q = pressure applied to the screw
d_{piv} = diameter of the screw pivot
f_{piv} = coefficient of friction at the screw pivot
d_p = pitch diameter of the screw thread
α = helix angle of the thread
φ = angle of friction of the thread
$\tan \varphi = f_{piv}$ ($\varphi = 5°40'$)

The screw torque referred to the motor shaft, which is generally called the static torque (M_s), is

$$M_s = \frac{M_{scr}}{i\eta}$$

where i = gear ratio from the motor to the screw
η = efficiency of the transmission

The dynamic torque that takes into account the dynamics of the system operating with accelerations and decelerations is

$$M_d = \frac{\Sigma GD^2_{m.sh}}{375} \cdot \varepsilon_{mot}$$

where $\Sigma\ GD^2_{m.sh}$ = moment of gyration of all the rotating masses (including the motor rotor) referred to the motor shaft
ε_{mot} = angular acceleration of the motor (300-600 rpm/s)

High dynamic loads and short-term duty-cycle operation of the screwdown demand much of its transmission and drive. In modern blooming mills the forward speed of the mill screws comes up to 180-220 mm/s, which makes it possible to carry out the screwdown operating cycle during a time out of rolls of 1.2 s. The capacity of the screwdown driving motors has been increased to 500-550 hp.

The roll changing, which in modern blooming mills takes less than two hours, is done through the housing window (opposite to the drive end of the rolls) by means of special rigs. In the 1300-mm blooming mill the roll changing rig comprises a hydraulically driven truck which travels parallel to the axis of the working stand.

The working stands of cogging mills are equipped with the so-called breast rolls which are installed on both the front and rear sides of the stand immediately before and after the work rolls and serve to feed the workpiece into the rolls. In the old mill designs these were actuated by a friction drive and in modern mills they are driven by individual electric motors.

Figure 10.7 shows a general view of a slabbing mill. The mill is provided with a vertical roll stand installed in front of the horizontal roll stand which is similar to that of the blooming mill.

The housings of the vertical roll stand are attached to those of the horizontal one by means of lugs. The adjustment of the vertical rolls is accomplished with horizontal mill screws. The rolls are driven through a bevel-gear transmission.

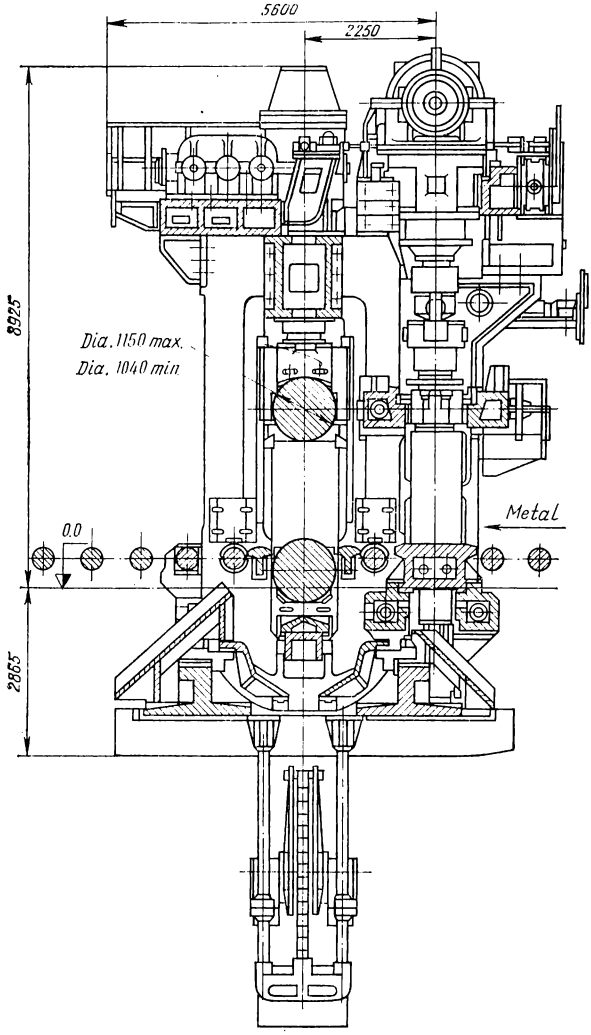


Fig. 10.7. Working stand of 1150 mm slabbing mill

Scale and sometimes cinder and metallic scrap may accumulate in large quantities under the working stands of the main line of blooming and slabbing mills. Formerly, special boxes were used for waste removal; hydraulic flushing equipment or special conveyors are now in use.

10.5. Main Drive of Cogging Mills

The rolling torque is transmitted to the work rolls via universal-joint spindles which are installed between the working and pinion stands or, in modern mills, directly between the rolls and the main drive motors. The length of the spindle depends upon the roll lift and the spindle inclination angle may be as great as 8-10°.

The motor speed may be increased above the rated value by changing the field current and decreased below the rated value by changing the motor voltage. To improve the dynamic characteristic of the motor one tries to have the moment of gyration of the motor armature GD^2 as small as possible.

The motor torque is expressed as

$$M_m = 0.716 \frac{N_m}{n_m}$$

In accordance with this expression, if the motor speed is increased above the rated value, the power delivered by the motor remains constant while its torque decreases. On the other hand, when the speed is decreased below the rated value, the torque remains constant and the power decreases.

Figure 10.8 shows speed and torque diagrams of a cogging stand, where n_1 is the speed at the moment of the metal biting by the rolls, n_2, that in the middle of the passing and n_3, at the end of the passing. The acceleration and deceleration of the motor take place in two periods, *viz.* when there is an ingot in the stand and when there is no ingot. When there is no ingot in the stand, the acceleration and deceleration periods form a part of the time out of rolls.

As previously explained, the motor torque is expressed as

$$M_m = M_s \pm M_d$$

where M_s = static torque of the motor
M_d = dynamic torque of the motor

In this expression the plus sign corresponds to acceleration and the minus one, to deceleration.

The dynamic torque during acceleration is

$$M_{d(a)} = \frac{GD^2}{375} a$$

and that during deceleration is

$$M_{d(b)} = \frac{GD^2}{375} b$$

The permissible overloads of the rolling mill motors are a factor of 2.5-3 greater than the rated motor power and torque.

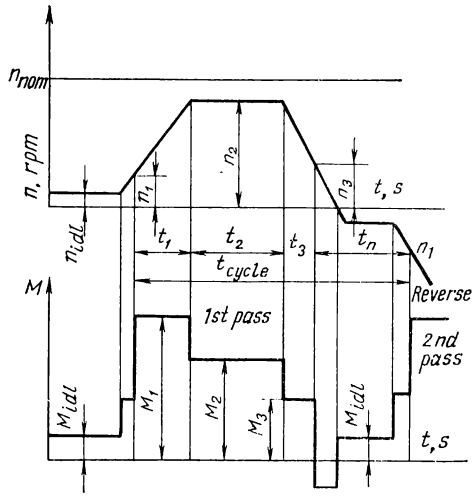


Fig. 10.8. Speed and torque diagrams for blooming mill

The maximum motor torque is

$$M_{\max} = k \cdot M_{rate}$$

where k = overload factor of the motor.

The rated torque and, consequently, the rated power of the motor is selected on the basis of the following condition:

$$M_{rate} \geqslant \frac{M_{\max}}{k}$$

The motors are checked for overheating.

Up to the end of the fifties the main drive motors of reversing mills were fed with direct current from flywheel convertors driven by induction motors with liquid-type slip regulators (Ilgner sets). Depending upon the load variations, the flywheel convertors can give back the stored kinetic energy and thus smooth out load peaks in the network. The slip (15-20% of the rated speed) of the induction motor with the liquid-type slip regulator results in power losses due to heating of the regulator.

As the time out of rolls has been reduced and the rolling process intensified, the flywheel convertors have become obsolete and now they are replaced by convertors without flywheels which are driven by synchronous motors. As is known, the use of powerful synchronous motors makes for an increase of the power factor ($\cos \varphi$) and provides for the supply voltage stabilization.

Recently, controllable mercury-arc (ionic) and thyristor (semiconductor) convertors have become widely used for feeding the main drive motors of reversing mills.

10.6. Main Roll Tables, Manipulators, Tilter

The main roll tables which serve to feed the rolled piece into the rolls and receive it from the rolls operate under severe conditions characterized by mechanical impacts, repetitive short-term duty cycles and dynamic transients (accelerations and decelerations).

On the whole, the requirements for the main roll tables, as well as for all the auxiliary equipment, are identical to those for the main drive and screwdown.

Main roll tables consisting of several sections which are group-driven through bevel gears are widely used. The design of the main roll tables is being improved by replacing the bevel gears by spur gears, by using individual drives and by improving the dynamic characteristics of transmissions and electric motors.

The main table of the 1300 mm blooming mill consists of nine all-forged rollers 500 mm in diameter which are supported by double-row spherical roller bearings (Fig. 10.9). The first two rollers are driven by individual 45 kW motors

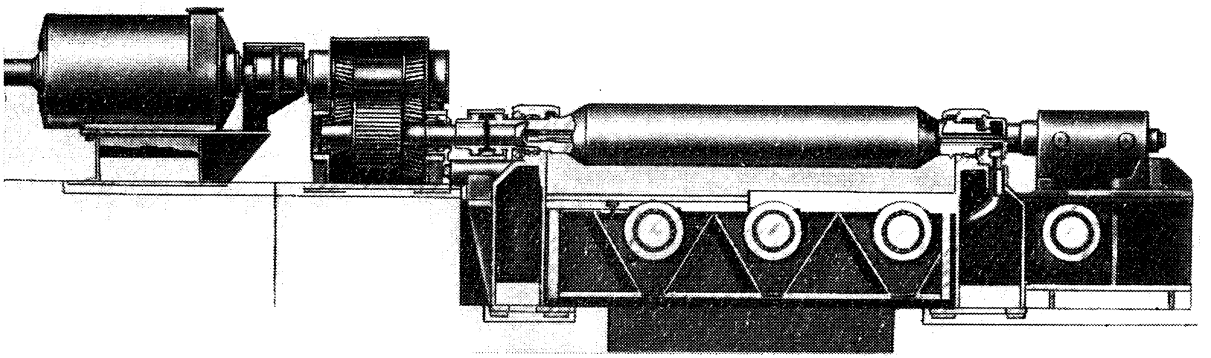


Fig. 10.9. Main table of 1300 mm blooming mill

and the others are group-driven by a 200 kW motor through a spur-gear speed reducer with seven output shafts. Gear clutches permit the rollers to be quickly disengaged from the drive by moving the clutch cages towards the drive. The rollers are internally cooled. In up-to-date blooming mills the speed of the main roll tables has been increased from about 2 to 4 m/s. With a coefficient of friction of a hot ingot against the rollers of 0.2-0.3 the workpiece acceleration comes to 2.0-2.5 m/s^2.

The manipulators shift the workpiece from pass to pass in a lateral direction and, besides, take part in tilting and straightening the workpiece. The force applied to a manipulator guard may be as great as 150 tf. The manipulator guards squeeze and retard the workpiece after it has left the rolls. In modern blooming mills they are also used to measure the width of the workpiece.

The manipulators comprise two pairs of guards made of cast steel. The right-hand front side guard usually carries a tilting device. The front and back side guards have a common drive. Most commonly the drive is a rack-and-pinion mechanism whose racks are hinged to the guard actuating rods. The right- and left-hand guards are driven by one or two motors; the two motors are sometimes paired off on a single shaft to reduce GD^2.

It is unadvisable to have too high a guard travel speed because with small guard displacements (about 500 mm) under short-term duty-cycle operating conditions the motors might have not enough time to pick up the rated speed and this might cause positional errors in the guard setting. In the new 1300 mm blooming mill the guard travel speed is 1.4 m/s. An electric manipulator drive devoid of speed reducer is successfully used (Fig. 10.10).

The tilter is an operating member of the manipulator. Three to five lifting dogs with a system of levers are usually mounted on a single shaft and installed on one of the front side guards.

Various tilter drive designs are known. At present two principal types of the drive are used, *viz.* a drive installed together with the tilter on the movable guard and a stationary drive. The latter type is used in new cogging mills. In this case the tilter is linked to a stationary motor by

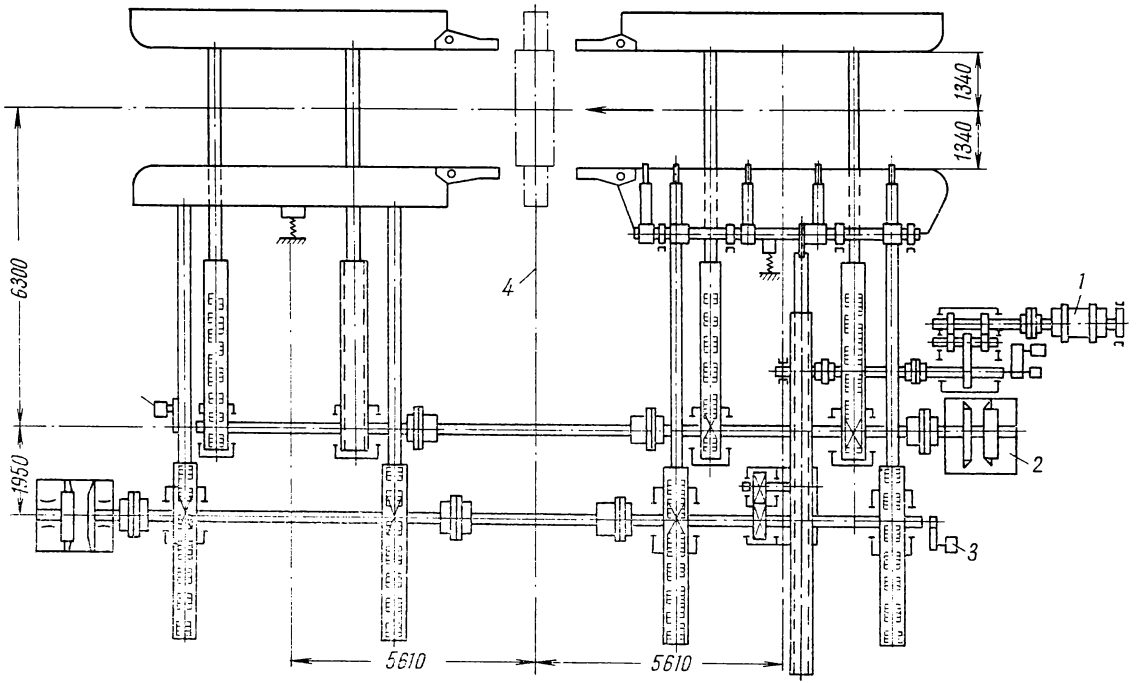


Fig. 10.10. Kinematic diagram of manipulator and tilter of blooming mill
1—tilter drive; *2*—manipulator drive; *3*—master controller; *4*—axis of working stand

means of a rack-and-pinion and a planetary reducing gear.

In new tilters the lifting dogs in their upper position are sunk in the slots of the guard which enables the tilting to be done during one half of the tilter operating cycle. Figure 10.11 shows the kinematic diagram of the new tilter. With this

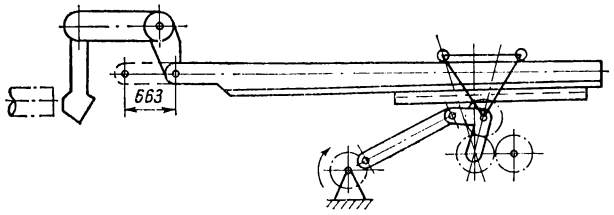


Fig. 10.11. Kinematic diagram of modern tilter

tilter the tilting time is reduced because after tilting the lifting dogs sink into the slots and do not interfere with the movement of the workpiece, and it is only after the workpiece passing that they lower into their initial position.

10.7. Production Process in Cogging Mill Shop

Below is described the sequence of operations in a cogging mill shop which are done after rolling metal in a blooming or slabbing mill.

If rolling of billets in continuous billet mills is not provided for, the rolled piece delivered by the blooming mill is cut to length with shears in accordance with the order specifications, the defective head and tail ends of the piece being cropped preliminarily. Individual billets are marked (stamped) and delivered by special cranes to racks for cooling or packing. Then the billets are delivered to the billet storage or subjected to slow cooling (specified for individual steel grades) in tunnel car-bottom hearth furnaces, pits or other facilities.

If there are continuous billet mills in the shop, the head and tail ends of the rolled piece delivered by the blooming mill are cropped with the first shears and then the piece is

rolled in one or two continuous billet mills (groups of stands) and cut to length with stationary shears (similar to the shears installed after the blooming mill), which are equipped with roll tables and gauge stops, or with flying shears which "fly" at the same speed as that of the piece delivery and cut the piece as it leaves the final stand of the continuous billet mill. The remaining operations are similar to those previously described for blooms and slabs.

The characteristic features of the equipment intended for cutting, surface conditioning and handling semi-finished rolled products are considered below.

The high output of modern cogging mills demands much of the shears designed for the hot cutting of blooms, slabs and billets. The capacity, output and reliability of the shears are determined by the maximum cutting force (the maximum cross-sectional area of the metal of the given chemical composition being cut at the given temperature) and also by the maximum possible number of cuts per minute.

The cutting force is determined from the following formula:

$$P = k_1 k_2 \sigma_B \frac{F}{1000}$$

where $k_1 \approx 1.3$ is a coefficient taking into account the cutting blade dulling

$k_2 \approx 0.6$ is the ratio of the shear strength of the metal to its tensile strength

σ_B is the ultimate tensile strength of the metal at the temperature at which it is cut

F is the maximum cross-sectional area of the piece being cut, mm^2

The cutting force of shears of the old design was rated at 900-1000 tf. As the cross-sectional area of blooms and, especially, of slabs has been increased the cutting force of shears has had to be increased to 2000-2500 tf.

The first section of the roll table after the shears is designed to tilt or shift in order to facilitate the transfer of crops to a conveyor installed under the shears. In the latter case the frame of this roll table section is shifted longitudinally by means of a system of levers. The scrap falls onto a horizontal conveyor installed under the floor and then is delivered to an inclined conveyor.

Both conveyors are simple chain conveyors which slide in guides. The scrap drops via a receiving chute into massive water cooling tanks or into special cars for delivery to steelmaking plants.

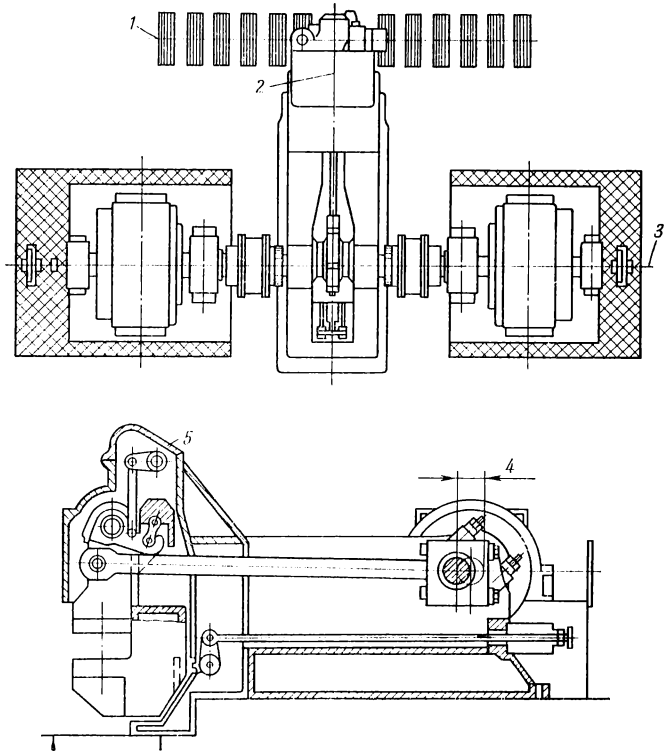


Fig. 10.12. Kinematic diagram of blooming mill shears devoid of speed reducer

1—roll table axis; *2*—shears axis; *3*—motors axis; *4*—eccentricity; *5*—shears

Lately, shears with drives devoid of speed reducer have come into use. These modernized shears (Fig. 10.12) at one metallurgical plant gave very reliable operation at 12-14 cuts per minute (instead of 4) and maintained their cutting force of 900 tf, this required the motor capacity to be increased from 200 to 515 kW.

The shears are equipped with a gauge stop and there are facilities for scrap disposal. On existing cogging mills the position of the piece to be cut to the desired length is set by means of the movable gauge stop in the shears, the head and tail ends of the piece being cut by eye. On up-to-date mills the length of the piece to be cut off is measured automatically and the piece positioning is carried out without using the gauge stops.

Not long ago the required markings were stamped by hand on the end of billets. Now special marking machines are used for this purpose. In the latest marking machine designs the marking stamps are changed automatically.

Hot scarfing is used extensively in the modern rolled metal production. Hot scarfers (see Fig. 3.6) are installed in an opening in the processing roll table before the shears. The machine consists of a vertical shaft and gas-cutting heads mounted on carriages whose horizontal and vertical position can be adjusted by means of pneumatic cylinders and balancing devices. The working part of the machine is designed as a follow-up system which operates as follows. When the cylinders are activated, the lower head moves upwards and the upper head simultaneously moves downwards. When the lower head reaches the surface of the workpiece, only the upper head continues lowering until it abuts on the top of the hot workpiece. The horizontal carriages independently move until they abut against the edges of the workpiece which then begins moving along the table.

The machine is moved in a direction perpendicular to that of rolling by means of a feeding screw and is equipped with solenoid valves, hydraulic scale breakers and head cooling devices.

Acetylene or coke-oven gas at a pressure of 20-25 at are used as fuel. The machine operates with technically pure oxygen and its gas system is filled with nitrogen after gas shut-off to prevent backfire. First the fuel gas is ignited and the metal, preheated and then scarfing begins, the hydraulic scale breaker being simultaneously put into operation.

The scarfing speed and thickness of the metal layer being removed are governed by the oxygen and fuel gas consumption and also by the rate at which the workpiece moves. The flame is directed opposite to the direction of the metal

movement at an angle of 39-45° (angle of attack) to the metal surface. The time of the metal preheating is 12 s, the scarfing speed is 30-45 m/min and the thickness of the removed metal layer is up to 2.5 mm.

Economic scarfing (from the standpoint of metal losses) is of extreme importance and because of the great waste of metal (due to the long perimeter of the cross section) the scarfing of slabs in four-sided scarfers is seldom done. Sometimes only the sides of the slabs are scarfed. The scarfing of cold slabs is carried out in scarfers installed separately.

The use of two hot scarfers in the blooming mill train provides for the continuity of the scarfing process.

In universal hot scarfers contact rollers maintain a constant gap of 3-5 mm between the gas-cutting heads and the metal. This considerably simplifies the machine adjustment and makes it possible to follow the metal surface, especially when scarfing warped pieces. With these machines the scarfing speed is 0.25-0.70 m/s, and the oxygen consumption comes to 2.5-5.0 cubic metres per ton of scarfed metal. Hand scarfing is five times more expensive than hot machine scarfing.

Although the equipment used in cogging mill shops for handling semi-finished products is rather simple, the design of its drives and lubrication systems must be capable of coping with high mechanical and thermal loads. This equipment includes racks with transfer beds or walking beams, stockpiling devices, self-propelled cars for transporting metal between the bays, and special cranes (crab cranes) for taking hot and cold metal from the racks and for stockpiling and delivering metal to the slow cooling section or to the receiving facilities of rolling mills. These cranes are equipped either with special grippers for handling hot metal or with powerful electromagnets for cold metal.

In modern cogging mill shops there is a large continuous flow of metal and therefore the mechanical handling equipment must have more than adequate reserves.

10.8. Billet Mills

The most popular cogging mill shop production pattern for an iron and steel works that produces a variety of bar sections integrates in one complex the cogging stands with

billet stands which roll without intermediate reheating the required range of square, flat and sometimes round (tubular) billets.

Three-high single-line or two-high tandem billet mills are still in use. At some works the billet rolling is done in section mills intended for rolling finished products. But continuous billet mills are most justifiable economically and therefore have found the widest application. Depending on the rolled product range, these mills may have one or two continuous groups of stands designed for rolling billets of large or small cross sections.

After rolling in reversing cogging stands and cropping the head and tail ends with the shears, the piece enters the first group of stands of the continuous billet mill which usually delivers billets with cross sections greater than 100 × 100 mm. These billets may be handled in a separate flow line, cut to length and delivered to the billet storage bays. To roll billets of smaller cross sections the piece delivered by the first group of stands is directed to the second group. When an ingot is rolled into a billet of a small cross section, the billet produced is very long and can only be cut when in motion with special flying shears. After cutting the billets are accumulated on a stop by means of a skew table and then delivered by roll tables to the corresponding cooling bed section.

The first continuous billet mills rolled billets from 45 × × 45 to 150 × 150 mm in cross section. These mills had 4-6 horizontal roll stands with a common drive in a group. The stands were driven by a common synchronous motor via spur- or bevel-gear speed reducers, i.e. the rolling speeds in the stands were preset in accordance with the adopted gear ratios of the speed reducers. Billets of intermediate cross sections were delivered from intermediate stands (the rolled pieces passed the following stands without reduction) or from different rows of passes.

The roll pass design constants for the continuous billet mills were selected individually for each pass in accordance with its cross section and effective roll diameter; sometimes small variations of the constant were permitted for individual stands. The roll pass design parameters for different rows of passes were selected so as to ensure that a change

from one billet cross section to another would not cause delays in the operation of the blooming mill.

Some factors which have affected the development of the roll pass designs for the continuous billet mills, as well as of the designs of the mills themselves, should be noted.

The continuous billet mills with helical twist guides whose speed cannot be controlled (Fig. 10.13) have the

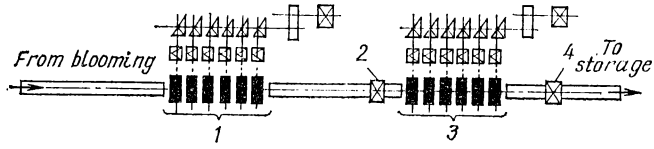


Fig. 10.13. Layout of continuous billet mill with group drive
1—first continuous group; *2*—pendulum shears; *3*—second continuous group; *4*—flying shears

following essential drawbacks: the roll diameters must be carefully selected so as to exactly ensure the required rolling speed in the given stand and the mills must be provided with an appropriate number of duplicate rolls for replacing the worn out or otherwise damaged rolls; the mills are difficult to adjust; their product range is limited and, lastly, the conditions of the workpiece tilting in the twist guides vary when changing the reductions.

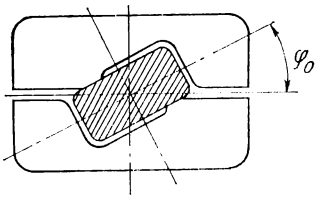


Fig. 10.14. Helical twist guides

Large forces acting on the contact surfaces of the twist guides cause not only additional power losses, but also rapid wear on the guide surfaces which in its turn leads to the deterioration of the surface finish of rolled products.

The twist guides (Fig. 10.14) consist of two lock joined halves which form a helical cavity whose cross-sectional dimensions correspond to those of the workpiece. The angle of the workpiece twist φ_0 at the exit of the guide is calculated, depending on the total angle of the workpiece twist between two adjacent stands φ, length of the guide l and distance between the rolls of the stands L, from the follow-

ing formula:

$$\varphi_0 = \varphi \frac{l}{L}$$

The rollermen of the Magnitogorsk iron and steel works have suggested a method of tilting the workpiece with special grooved rolls which form diagonal passes corresponding to the workpiece cross section and the required angle of its twist (Fig. 10.15).

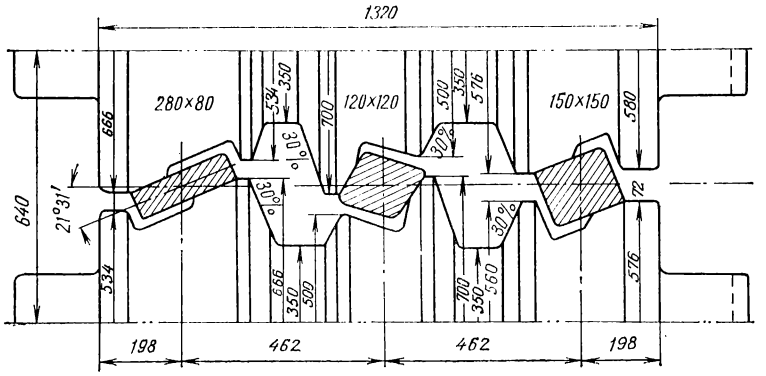


Fig. 10.15. Tilting rolls

The use of the tilting rolls in continuous billet mills and in continuous roughing stand groups of section mills has shown their considerable advantages over helical twist guides as to the cost, ease of adjustment and surface finish of rolled products.

The working stands of modern continuous billet mills are driven by individual adjustable-speed motors; vertical roll stands are used in place of tilting devices.

Figure 10.16 shows the equipment layout of the modern 900/700/500 type continuous billet mill which consists of 14 stands. In front of the first stand there is a turn-table device which serves for turning the workpiece with its tail end forward and after the second stand the workpiece is tilted with a tilter installed in front of the third stand. The gap between the second and third stands makes it possible to manage without helical twist guides in the first

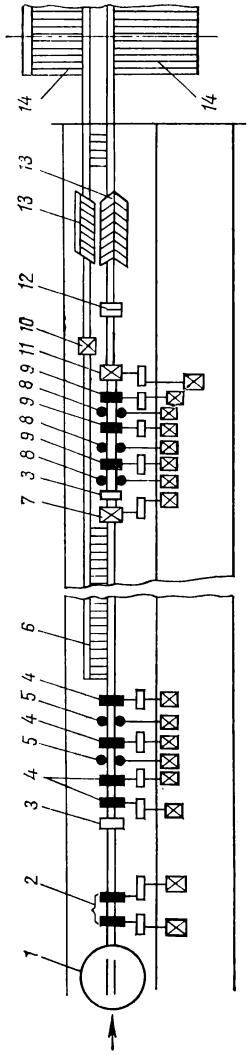


Fig. 10.16. Layout of 900/700/500 type continuous billet mill

1—turn-table; *2*—stands with horizontal rolls 900 mm in diameter; *3*—tilters; *4*—stands with horizontal rolls 700 mm in diameter; *5*—stands with vertical rolls 700 mm in diameter; *6*—transfer beds to by-pass table; *7*—pendulum shears; *8*—stands with vertical rolls 500 mm in diameter; *9*—stands with horizontal rolls 500 mm in diameter; *10*—shears with a cutting force of 1000 tf; *11*—flying shears; *12*—marking machine; *13*—gathering tables; *14*—cooling beds

four horizontal roll stands; further, horizontal and vertical roll stands are arranged alternately. The billets from the first group of stands may be transferred by transfer tables to a by-pass roll table equipped with stationary shears.

They are packed by a gathering skew table, transferred to a runout table and then delivered to the cooling beds of the first mill group.

In front of the second mill group there are installed pendulum shears for emergency cutting of the workpiece or for cropping of its head end and a tilting device. After the fourteenth stand the piece moving at a speed of up to 7 m/s is cut by a lever-planetary type flying shears into 5-12 m long billets which are then packed on gathering skew tables, marked and delivered to the cooling beds.

The rated annual production of the mill is 5,500,000 t.

10.9. Specific Features of Continuous Billet Mill Equipment

The two-high working stands with horizontal rolls only slightly differ from

the ordinary stands of modern rolling mills which do not require the work rolls to be considerably moved in the vertical direction. The billet mills are equipped with the vertical roll stands of various and fairly complicated designs. The top work roll drive through a speed reducer coupled to a motor installed in the motor room has found wide application.

The vertical rolls are provided with roller bearings, two horizontal mill screws and a spring-type roll balance device. The entire set of these units (the working stand itself) is built in external vertical housings which are rigidly mounted on the mill shoes. To change-over from one pass to another the inner working stand may be moved vertically by means of vertical mill screws. Roll changing on the vertical roll stand is carried out by changing completely the inner working stand unit.

In early continuous billet mills the stands of the second group delivering small-section billets were equipped with steam- or air-driven lever-type flying shears to cut the billets in motion. The latest prototypes of these shears were distinguished by their excellent reliability. With the increased delivery speeds (up to 5-7 m/s), greater cross sections of the pieces to be cut and the necessity for cutting the pieces efficiently and without scrap into small measured lengths, electrically-driven shears of the crank-eccentric and lever-planetary types have become widely used.

Turning the workpiece lengthwise through 180° between the shears and continuous billet mill has been found to be justifiable economically since this reduces the croppage and ensures a good quality of the end of the piece which is important to provide for the reliable metal biting by the rolls (especially for rimming steel). A turn table installed on an inverted U-type support is used for this purpose.

In front of the second group of stands of the continuous billet mill there are installed pendulum shears for cropping the head end of the piece when necessary or for emergency cutting. An entry twist guide box is sometimes installed in front of the second group of stands to feed square pieces diagonally.

CHAPTER XI. MANUFACTURE OF SECTION STEEL

Of the section steel manufacture is characteristic a very wide range of products and features specific to its technology and equipment.

The production of section steel may be divided into two distinct branches, *viz.*, that of the rail-and-structural steel and of the section steel and wire rod. Occasionally the production of formed shapes is also referred to the section steel manufacture.

11.1. Production of Rail-and-Structural Steel

To the domain of the rail-and-structural steel production belongs that of standard-gauge rails, large-sized beams, channels, angles and other sectional shapes, as well as of large-sized squares and rounds. The production of narrow-gauge rails, beams and channels of the size numbers up to 24-30 and angles, up to 130-150 is conditionally included in the branch of the heavy-section steel manufacture.

After rolling the rails are subjected to laborious finishing processes, i.e., straightening, milling of the end faces, drilling of holes and testing. It is very difficult and often inexpedient to bring the capabilities of the rail finishing facilities in line with the output of the rail-and-structural mills and therefore a certain quantity of flanged, shaped and bar sections, which do not require such a laborious finishing, is also rolled in these mills.

Principal Types of Rail-and-Structural Mills

Two methods of rail-and-structural mills operation are known, *viz.*, rolling of finished sections directly from ingots without reheating and rolling of finished sections from billets with reheating. The first method is in use with the old rail-and-structural mills, while the second one is employed with all of the new mills. Earlier the rail-and-structural

mill included a two-high reversing roughing stand (blooming) and a two-high reversing finishing mill train. The equipment layout of such a mill is shown in Fig. 11.1. The two-high reversing stands arranged in a single line have two drives and the finishing stand is individually driven. With the old mills the ingot reheating was commonly done in fired soaking pits, more often of the single-ingot ground type, and in batch-type or continuous reheating furnaces.

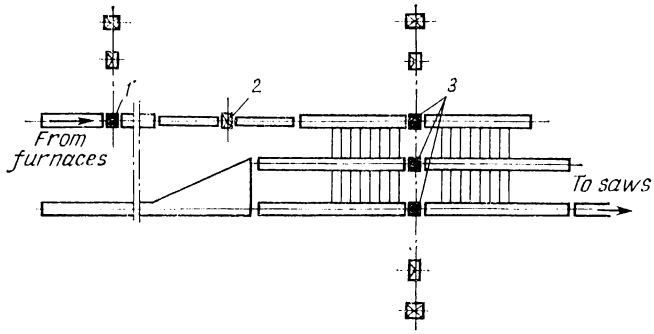


Fig. 11.1. Layout of 850 mm two-high reversing rail-and-structural mill

1—working stand of 1150 mm blooming mill; *2*—shears; *3*—working stands of the rail-and-structural mill

New rail-and-structural mills (Fig. 11.2) receive hot billets 250 × 250-300 × 300 mm in cross section from a separately installed blooming mill. The billets are placed into modern batch-type or continuous reheating furnaces and then delivered to a two-high reversing stand of the blooming type with rolls 900-950 mm in diameter.

The mill consists of one or two lines. The first two three-high stands and the finishing stand are either arranged in a single line or the finishing stand is installed in a separate line at a distance equal to the maximum length of the workpiece. The finishing two-high stand is equipped with an individual drive. The three-high stands (the roll diameter is 800-850 mm, the barrel length is about 1800 mm) incorporate lifting tables and are usually driven by adjustable-speed reversing d.c. motors which make it possible to adjust the rolling speed in a wide range.

The finishing stand is of the two-high type and is equipped with rolls of the same diameter as that of the rolls of the first two stands, but of a shorter barrel length (1100-1200 mm) since the stand is designed for one passing only. The stand is driven by an adjustable-speed d.c. motor. All the stands of the mill are of the open-top type. The roll changing is

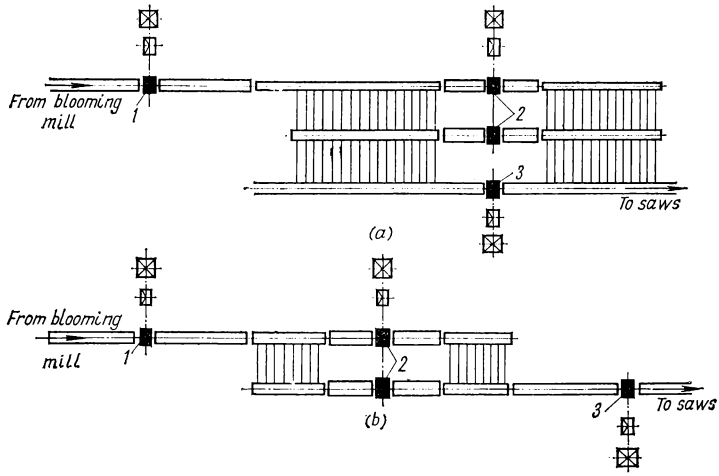


Fig. 11.2. Stand arrangements of new rail-and-structural mills in two lines (*a*); and in three lines (*b*)

1—900 mm two-high roughing stand; *2*—800 mm three-high roughing stands; *3*—800 mm two-high finishing stand (*a*) or 750 mm two-high finishing stand (*b*)

accomplished by replacing the assembled stands. Roller bearings are occasionally used in the finishing stands.

Universal specialized mills designed for rolling large-sized wide flange beams include a reversing stand (blooming) which reduces heavy ingots to I-shaped billets. Following the blooming the workpiece enters a universal mill consisting of three lines: roughing, intermediate and finishing (Fig. 11.3). Reheating of billets is commonly used between the two-high reversing blooming mill and the universal mill, which is employed only for rolling lighter beams.

A sharp increase in the production of wide flange beams in highly efficient universal combined rolling mills is

observed in many countries during the last decade. There are up to 40 such mills with an aggregate output of 20,000,000 tons per year.

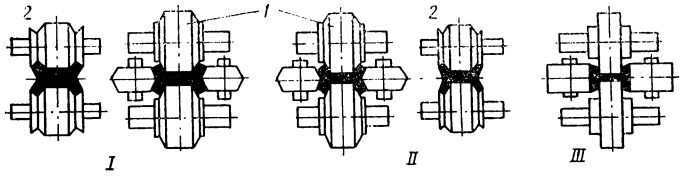


Fig. 11.3. Scheme of rolling in universal beam rolling mill
I—roughing line; *II*—intermediate line; *III*—finishing line; *1*—main stand; *2*—auxiliary stand

The universal combined mill (Fig. 11.4) is a conventional rail-and-structural or heavy-section mill to which one or two working stands for rolling wide flange beams are added.

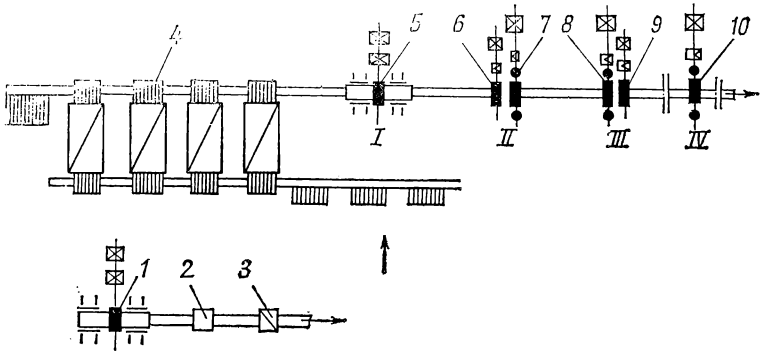


Fig. 11.4. Layout of universal beam mill
I—billet line; *II*—roughing group; *III*—intermediate group; *IV*—finishing line; *1*—1500 mm blooming mill; *2*—flame scarfing machine; *3*—blooming shears; *4*—walking-beam furnaces; *5*—billet mill; *6*—auxiliary two-high stand; *7*—universal stand; *8*—universal stand; *9*—auxiliary stand; *10*—universal finishing stand

The designs of the universal and auxiliary stands are usually similar, the only difference being in the diameter of the rolls. The diameters of the horizontal rolls in the universal stands are 1250-1500 mm, those of the vertical rolls are 950-1150 mm and those of the rolls in the two-high auxiliary stands, 1000-1200 mm. All the stands are provided

with hydraulic descaling equipment. The ends of the finished workpiece are cut with hot saws, then it is delivered to a cooling bed and after cooling is cut with cold saws. On modern heavy-section and universal rolling mills the saws are installed in soundproof rooms and are operated by remote control.

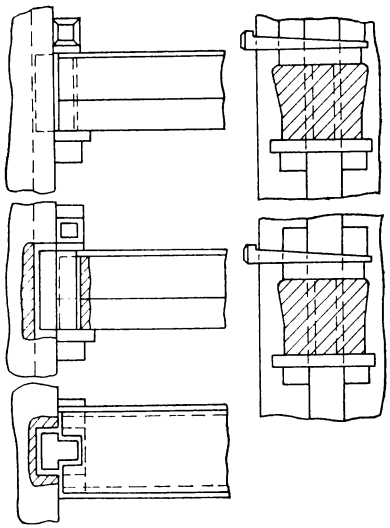


Fig. 11.5. Shape and mounting of guide bars

The output of some universal mills comes to 800,000-1,000,000 tons per year. The annual output of a universal mill for rolling wide flange beams, which is being projected in the USSR, will reach 1,500,000 t.

The process of rolling in rail-and-structural mills is determined first of all by the adopted scheme of deformation of the shaped (mainly flanged) sections. Blooms of a rectangular (sometimes shaped) cross section weighing up to 3-3.5 t are used as initial billets for modern rail-and-structural mills. The piece obtained after rolling an ingot in the blooming mill is cut into suitably sized billets so that 12.5 or 25 m long rails are rolled from them. Reheating of the billets in batch-type or continuous furnaces is generally used.

Successful rolling of shaped sections in the rail-and-structural mills depends not only upon the proper roll pass design, but also upon the design of roll fittings, their correct positioning and rational maintenance.

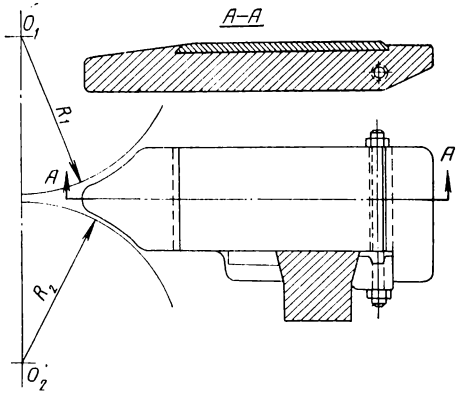


Fig. 11.6. Shape and mounting of guards

The roll fittings include guide bars, guards and guides. The guide bar (Fig. 11.5) has a dovetailed cross section with grooves in the end faces. It is fixed by means of a block inserted in a special slot in the mill housing and held in

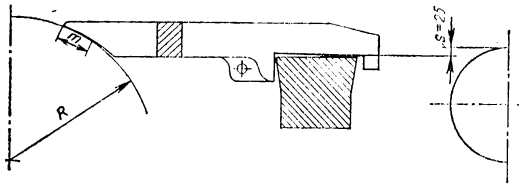


Fig. 11.7. Shape and mounting of guides

place by a gib. The entry and delivery guards (Fig. 11.6) are installed at the collars of the rolls. The working surface of the guards is thoroughly ground.

The operation of the mill to a great extent depends on the guides (Fig. 11.7). The correct selection of the guide shape and length, accurate grinding of the guide face which closely

fits the roll pass surface and proper mounting of the guide are very important operations, the success of which is determined by the organization of the roll fitting service in the shop and by the skill of the rollermen, guide fitters, foremen and head foreman of the mill. The working surface of the guide should be 5 mm below the horizontal tangent line to the surface of the roll pass, its nose being below this line by 10 mm. Grinding and positioning of the guides should provide for their close contact with the roll pass surface, but they should not touch the side walls of the pass.

The layout of the finishing part of a rail-and-structural mill shop is shown in Fig. 11.8. The rolled pieces are cut to the required length with reciprocating-type rotary saws which slide along a roll table with the aid of rack-and-pinion gears. Railway rails are cut into lengths of 12.5 or 25 m, an allowance being given for linear thermal contraction. Rack-type crop-end pushers are installed on the sliding frames of the front and back saws.

Rails should invariably bear a marking indicating the manufacturer's name, steel production method (for example, the first letter M stands for the open-hearth steel), date of production, type of the rail and kind of its heat treatment. The marking is done during rolling in the finishing roll pass which carries appropriate marking stamps on its surface. After cutting, an additional marking (the number of heat and the rail number along the height of the ingot) is stamped on the rail web in a marking machine. Then the hot rails are delivered to transfer cooling beds equipped with tilters. Since the rail flange cools faster than the head, the rail bends first in the direction towards the flange, but then, as the head cools it bends in the opposite direction. To produce straight rails and decrease stresses during cold straightening the hot rails are specially bent in the direction toward the flange on special roll machines.

From the cooling beds beams and other shapes (excluding rails) are delivered to roll machines and presses for cold straightening. After inspection they are transferred to stacking devices and then to the finished products store prior to shipping.

In accordance with high requirements for the quality of rails (macro- and microstructure, freedom from flakes,

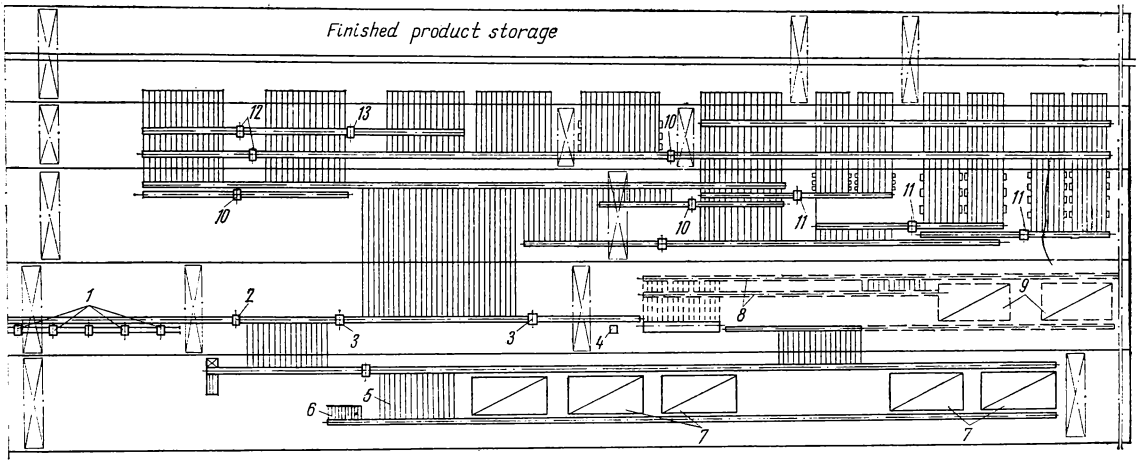


Fig. 11.8. Finishing section of rail-and-structural mill shop

1—saws; *2*—marking machine; *3*—bending machine; *4*—saw for taking samples; *5*—cooling beds; *6*—rack: *7*—isothermal holding furnaces; *8*—sorbitizing unit; *9*—normalizing furnaces; *10*—straightening machine; *11*—vertical gag press; *12*—horizontal gag press; *13*—shears

toughness, hardness and wear resistance) it has become necessary to use comparatively complex methods of heat treatment and cooling of rails. In addition, in a special rail finishing section cold rails are subjected to machining: milling of the ends and drilling of holes.

Rails are also subjected to more complicated tests as compared to other products (as a rule, a sample is taken for impact testing).

Outlined below are some features of the general-purpose equipment used for the entire range of products.

The efficiency of the reciprocating-type rotary saws depends upon the disc speed and also upon the shape, dimensions and steel grade of the rolled rails, which determine the disc feed rate when cutting. The peripheral speed of the disc is limited to 100-120 m/s and its feed rate is 130-270 mm/s. Discs are cooled with an ample supply of water. The saw discs should always be kept in good working order and changed in due time. Modern saws intended for cutting hot metal incorporate automatic devices for adjusting the disc feed rate in accordance with the load, this increasing the service life of the discs.

The rail-and-structural mills are equipped with transfer cooling beds which are designed for the cooling of heavy beams, channels and other sections. Rails, as a rule, are slowly cooled in isothermal soaking furnaces. The total cooling bed area is about 2000 m^2, the specific area being equal to 10-12 m^2 per ton of the hourly mill output.

All shaped sections after cooling, and rails after heat treatment and finishing, are subjected to cold straightening.

Straightening is a very important operation in the rail-and-structural steel production. It is carried out in roll straightening machines with horizontal rolls or in combined machines equipped with horizontal and vertical rolls. Finish straightening is accomplished in eccentric (gag) presses. The straightening rate is 0.8-1.6 m/s and the required number of straightening machines is calculated proceeding from this rate.

The milling of rail ends and drilling of holes are continuously carried out in the rail finishing bays. Several processing lines (Fig. 11.9) are usually installed, depending

upon the shop output. These lines include transfer tables for transferring rails from the roll table to the chain conveyor, tilters, as well as horizontal single-spindle milling machines with hydraulic vices having a capacity of 40-45 pieces per hour and three-spindle drilling machines of the same capacity.

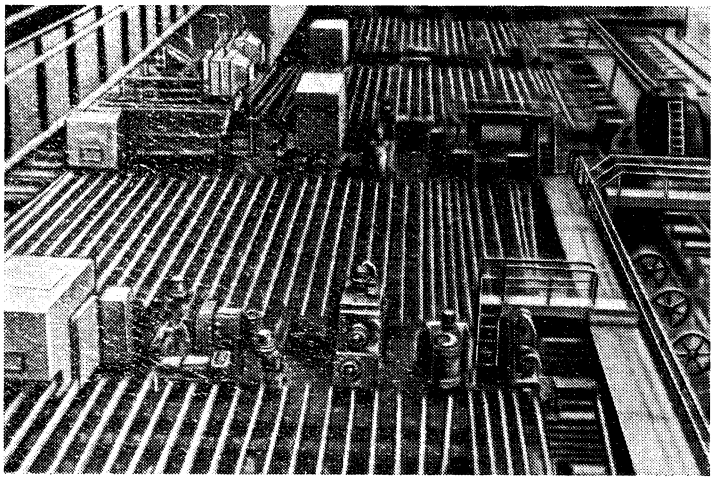


Fig. 11.9. Rail finishing line

Both ends of the rail are milled simultaneously; then the rail is automatically fed to the drilling machines. Sometimes equipment for high-frequency hardening of the rail ends is installed in this processing line in order to increase the wear resistance of the rail joints. For this purpose a high-frequency inductor 100-150 mm long is fitted on the rail end and its surface is heated for 40-45 s and then cooled for about 30 s with water or emulsion which is supplied periodically (to ensure self tempering). The hardened layer of not less than 4 mm in depth has a sorbitic structure with a good transition zone and its hardness is 300-400 Brinell.

Basic Trends in Improving the Technology of Production and of Heat Treatment of Rails

The intensified operation of the railway transport (greater loads and higher running speeds) has posed before the metallurgists a complex problem, *viz.*, to sharply increase the reliability and service life of rails and their fatigue strength in particular. The progress in the steelmaking technology, as well as in the processes of metal heating, deforming, cooling and finishing, is paralleled by the introduction of new heat treatment methods and of new grades of alloyed steel.

New methods of deoxidation and treatment of steel with synthetic slags and rare earth elements are used to eliminate non-metallic inclusions (oxides, aluminates and silicates). The optimum temperatures at the end of rolling rail steel are fixed as follows: 1100°C for the blooming mill, 1070-1080°C for the 900-mm rolling mill and 920-1000°C for the 800-mm rolling mill line. Increasing the temperature at the end of the finish rail rolling above 1050°C results in grain growth. The heat treatment of rails by using the rolling heat, which was employed previously, is not recommended since this may cause the formations of flakes.

The conditions of the metal deformation in the process of shaping the rail section are of great importance, therefore it is very essential to choose the optimum draugh schedules, as well as the design of roll passes forming the rail flange, etc.

The heat treatment of rails is one of the important technological factors which make for a considerable improvement of the rail performance. Two methods of heat treatment are used: surface hardening of the rail head after recrystallization furnace heating or after high-frequency surface heating and full oil hardening.

The method of hardening the rail head by using the recrystallization furnace heating is as follows. Rails are heated in the furnace to 820-840°C and then the rail heads are hardened in a special roller machine with a jet of water emulsion. The hardened rails are cooled intermittently so as to ensure self tempering. The rail travel speed in the hardening machine is 0.5-0.65 m/s.

The high-frequency hardening of the rail heads is carried out as follows. With the rail moving at a speed of 20.8 mm/s, its head is heated to 875-900°C in inductors with a high-frequency current of 2000 Hz, being thereafter cooled with water sprayed through special nozzles.

The full oil hardening of rails includes the following operations: heating before hardening, hardening in oil and tempering. Rails are heated in a gas-fired roller-hearth furnace to 910-950°C, then soaked for 30-60 min to equalize the temperature and thereafter cooled until the mean temperature of the rail shape elements drops to 840-860°C. Then the rails are fed to a hot straightening machine and after that, to a hardening machine where they are immersed in oil for 4-6 min. Mineral (spindle) oil with a moisture content of not more than 0.2% at a temperature of 60-80°C is used. Finally, the rails are tempered for 2 hours at 450-480°C in a roller-hearth furnace.

Warping of rails occurs with all methods of hardening. The amount and direction of the warping depend upon the temperature and duration of individual operations, the position of rails during hardening, etc. After the heat treatment the rails are straightened in both the horizontal and vertical planes.

The full hardening and straightening in the roll straightening machines result in some change in the rail length. Depending on the chemical composition, rails lengthen by 0.6-2.0 mm/m when hardening, shorten by 0.15-0.3 mm/m when tempering and also shorten by 0.3-0.6 mm/m when straightening. This change of the rail length resulting from phase transformations has to be taken into account when cutting rails to length.

The layout of the principal equipment of the heat treatment division of a rail-and-structural mill shop is shown in Fig. 11.10. After finishing and inspection the rails are delivered by crane *1* to stock rack *2*, where 8-11 pieces are packed together in such a manner that they lie on their sides. These packs are fed at 5-7 minute intervals into seven-zone roller-hearth furnace *3* whose rollers are made of heat-resistant steel and have water cooled journals.

After heating the rails are delivered piecemeal, turned so that the flange is underneath and fed into roll straighte-

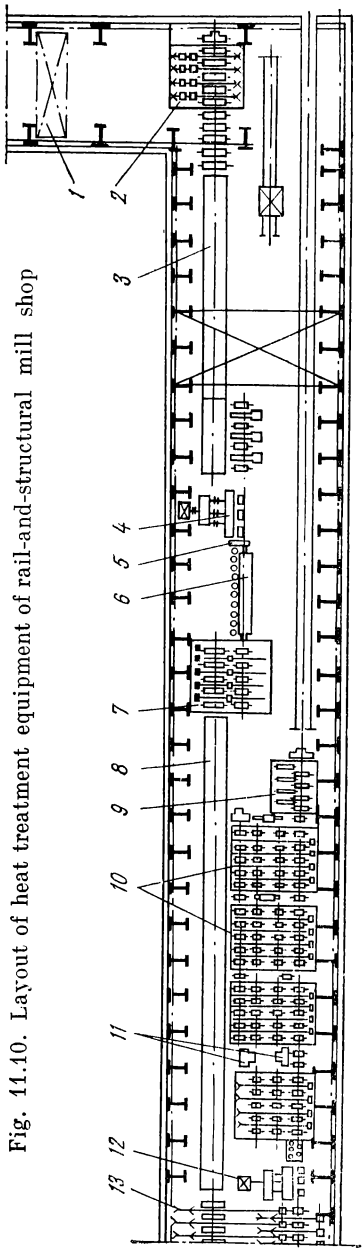


Fig. 11.10. Layout of heat treatment equipment of rail-and-structural mill shop

ning machine *4* and then, by rollers *5*, into hardening machine *6* which comprises a 26 m long cylinder equipped with suspension-type hardening carriages. When the cilynder turns through 30° the rail placed on the carriages on one side of the cylinder is immersed in oil while that located on the opposite side of the cylinder is extracted from it. Then the rails are delivered to closed bank *7* where they are packed into batches of 10-18 pieces which are then fed into tempering furnace *8*. The roller-hearth tempering furnace consists of ten sections.

Afterwards, the rails are cooled on cooling bed *13* and fed into combined roll straightening machine *12*, which consists of horizontal and vertical units, each having six rolls. If necessary the finish straightening of rails is carried out in two presses *11*. Finally the rails are delivered to inspection tables *10* for visual inspection, hardness checking, etc. The rails are then packed into batches of 25 pieces on packing table *9* and the packs are transported to a store or loaded directly in railway cars by lifting magnet cranes.

The most important regulations for accident prevention in the rail-and-structural steel production are given below.

Maintenance of roll fittings on two-high reversing stands is not allowed unless the mill is stopped. The rolling areas and roll tables must be equipped with guard fences up to 400 mm in height. Safe passages are to be made along the shop between the roll table lines. Passages and personnel gangways should also be made in the hot rack bay between the rack sections and at their edges. The slow cooling pits must be guarded with fences not less than 1.5 m in height.

The rails must be stacked and unloaded by lifting magnet cranes; manual operations with chains or other lifting appliances are not permitted. The work in the rail finishing bays should be completely mechanized, including operations such as feeding the rails into straightening machines and their tilting during straightening and inspection.

11.2. Production of Section Steel and Wire Rod

For many years there has existed a conventional classification of section mills into heavy-section, medium-section, light-section and wire-rod ones. This classification has now, to a certain extent, lost its validity because the development of section mills capable of rolling a very wide range of products has resulted in that modern medium-section mills, for example, produce besides the medium-section steel proper also heavy and some light sections.

As regards their capacity, all types of section mills may be divided into those of low, medium and high output. There are not only old, but also new low-output mill designs (non-continuous, line-type mills). Since modern section mills operate on the principle of "one passing only in each stand", the length of the roll body is reduced and this makes it possible also to reduce the rated roll diameter and yet maintain the same roll strength.

The diameters of section mill rolls for the entire range of section steel products (from light wire rod to heavy sections) usually range from 250 to 750 mm.

The range of products rolled in a general-purpose section mill includes 5-150 mm dia rounds, 8-150 mm^2 squares, 12-200 mm wide flats 4-60 mm in thickness, equal and unequal angles with a flange length of 20-150 mm and other sectional shapes of corresponding dimensions.

Section steel of all sizes is supplied in the form of straight bars and wire rod, in coils. Skelp steel is mostly supplied in coils. Rounds 9-22 mm in diameter and squares up to 14 mm^2 in cross section may be optionally supplied in coils.

Initial Billets and Their Heating Before Rolling

In most mills section steel products are rolled from billets of various cross sections which are produced in blooming or billet mills; ingots are used as initial billets for section mills only at very old iron and steel works.

The optimum cross section of billets is chosen by taking account of a number of considerations governed by general demands on the economically expedient heating and rolling operations, provided the high output and quality of products are maintained. In most cases it is profitable to have the minimum possible number of passes in a section mill, especially when this determines the number of working stands of the mill. Each definite group of section steel products has the optimum billet cross section of its own.

As to the section mill output, not only the cross section of the billet, but also its length is a factor of great importance. The length of billets for section mills was previously governed by the width of the reheating furnace used.

Recently, with the introduction of endless rolling billets of the maximum possible length are used in order to lessen the number of welding operations before rolling.

The present day high-speed section mills ensure the required accuracy of the rolled section when rolling pieces of any length (including endless one) owing to the fact that the temperature of the tail end of billets heated in a continuous or barrel-type high-speed reheating furnace is maintained at a high level and stiff stands are used.

In case of usual piecemeal rolling cross section, length and weight of the billet are selected, depending upon the specific features of each rolling mill, including the size and output of the reheating furnace, the arrangement and number of stands, the rolling schedule, the power of the drive, the design of the auxiliary equipment of the rolling mill, the area of the cooling bed, etc.

Most commonly billets of the following dimensions are used. For light-section and wire-rod mills with rolls 250-300 mm in diameter: 50, 60, 70, 80, 90 and 100 mm^2 squares 4, 5, 9 and 12 m in length; for medium-section mills with rolls 350-400 mm in diameter: 100, 120 and 150 mm^2 squares 3-6 m in length; for heavy-section mills with rolls 500-600 mm in diameter: 150, 200 mm^2 (and greater) squares and 150 × 170 mm rectangles 4-5 m in length. At a number of works the cross section of billets for wire-rod mills is not less than 60 × 60 mm, billets of intermediate cross sections being also used. Specialized strip and skelp rolling mills are provided with flat billets of various dimensions.

Various furnaces are used for heating billets for section mills, their design being dependent upon the billet dimensions, the mill output, the range of steel grades, and the design of the head part of the mill. The furnaces must have appropriate thermal and technical characteristics and meet the present day requirements as to reliable remote control with complete mechanization and automation of the entire complex of operations.

The heating rate depends on the steel grade and the conditions of heat transfer. The heating rate at first should not be very high, but as the billets get warmed it may be increased. This method is used in high-rate heating, various methods being used to improve the transfer of heat.

The optimum heating conditions necessary to minimize iron losses and decarburization of the metal surface are determined by the heating schedule and furnace atmosphere (the content of oxidizing elements, such as O_2, CO_2, H_2O, SO_2 or reducing elements, such as CO, H_2, CH_4).

Decarburization increases as the heating temperature and time are increased and also depends upon the content of carbon, tungsten, aluminium and cobalt in the metal.

Continuous end-charged end-discharged reheating furnaces (Fig. 11.11) equipped with refractory air recuperators are used in most cases to heat the billets for medium- and heavy-section mills. Injection burners are employed in furnaces of recent design; air is preheated to 600°C in refractory air recuperators and gas, to 350°C in metallic-tube gas recuperators. The furnace operates on a mixture of blast-

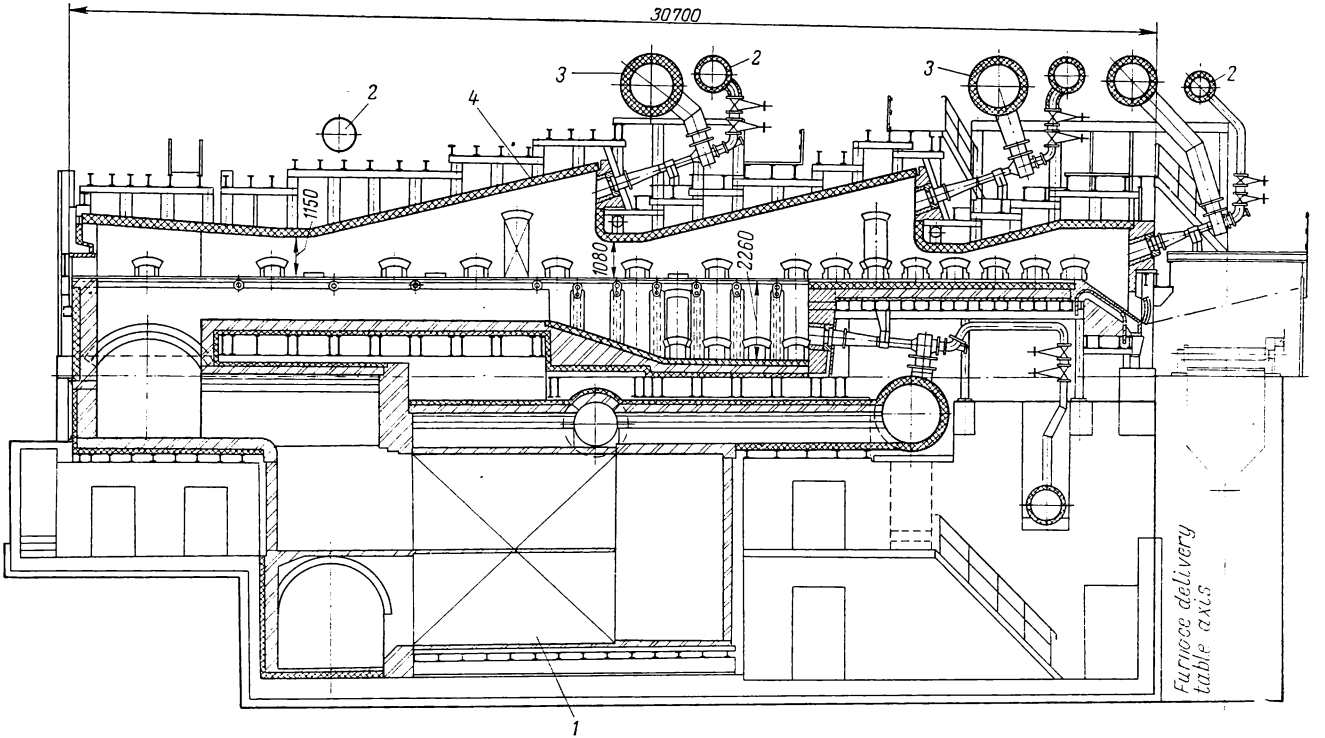


Fig. 11.11. Continuous end-charged end-discharged furnace

1—refractory air recuperator; *2*—gas pipeline; *3*—air pipeline; *4*—suspended roof

furnace and coke-oven gases with a comparatively low calorific value of 900-1500 kcal/m³.

Billets with a cross section of 100 × 100 mm or less for light-section and wire-rod mills have been heated in a single, wide (billet length is 9 m) side-charged side-discharged top-fired furnace with a monolithic hearth. In these furnaces refractory recuperators are used to preheat air to 300-350°C. Billets move along the inclined hearth and then, on bottom skids in the treating and soaking zones of the furnace. The movement of billets through the furnace is accomplished by a lever pusher.

The side charging of billets is carried out by means of a pusher carriage installed over the charging roll table or by furnace pull-in rolls installed beyond the charge door. The billets are discharged from the furnace by a pusher. Furnaces of this type are fired with a mixture of blast-furnace and coke-oven gases with a calorific value of up to 2500 kcal/m³. With hot charging their output is 70-80 t/h.

The output of these furnaces has been sufficiently raised by increasing the supply rate and calorific value of fuel and also by intensifying the burning process and enlarging the overall dimensions of the furnace. The increased width (within the maximum possible limits) of the working zone has made it possible to increase the billet length up to 12 m; further increase of the billet length entails difficulties in handling the hot metal. There are also limitations to the length of the furnace as the billets may tilt or climb one upon another when being pushed along the inclined (~7°) hearth.

The effective length of recently designed furnaces is up to 18 m. The hearth is made concave (curved) along its longitudinal axis to provide for reliable pushing of billets 60 × 60 mm in cross section. In the cold part of the furnace where the billet tilting is more likely to occur the hearth inclination is increased up to 10° and subsequently decreased.

Furnaces with injection burners installed in two top and one bottom zones have proved to be a very successful design. These furnaces have an output of up to 80 t/h and can be fired with blast-furnace gas almost without any additions. These furnaces are equipped with refractory recuperators for heating air to 600°C. The effective length of the hearth

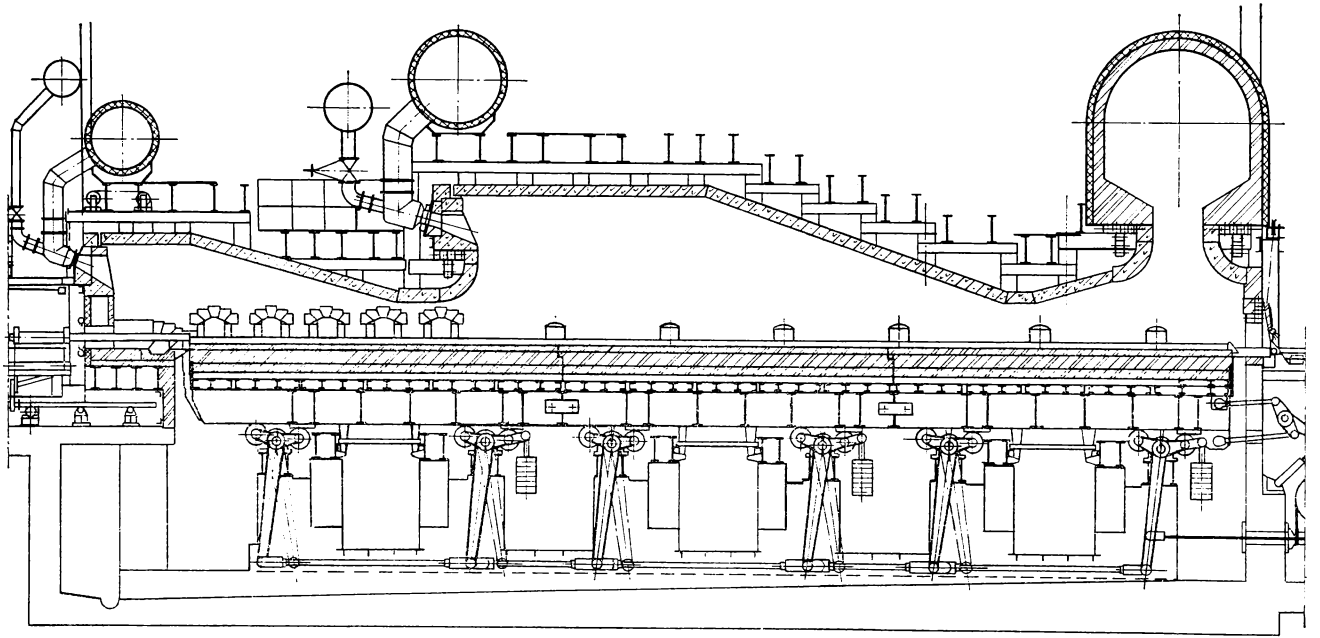


Fig. 11.12. Walking-beam furnace

is 16.5 m, the length of billets being 9 m. In later designs of these furnaces the billets length has been increased to 12 m and that of the concave hearth, to 18 m. Thermal capabilities are raised by applying injection burners and gas and air preheating. The air recuperator is of the refractory type and the gas one is made of metal tubes. These top-fired furnaces have an output of up to 140 t/h.

The bottom-fired reheating furnaces equipped with water-cooled skids have two principal disadvantages, *viz.*, substantial heat losses with cooling water and local cooling of the pieces being heated at the points of contact with cold skids, this being especially undesirable in case of high-carbon and alloy steels. From this standpoint the top-fired furnaces with monolithic hearth are preferable. Also, these furnaces provide for a lower specific fuel consumption.

A principally new billet heating method is used with new section mills, especially with light-section ones with endless rolling, i.e., when welding heated billets to make an endless piece. Here, the billets are first heated in a walking-beam furnace and then in a high-heating rate furnace installed before the first mill stand, which serves to compensate for the temperature drop in the course of welding and to equalize the temperature of the billet along its length. Such a heating pattern enables the distance between the furnace and the mill to be minimized and the arrangement of two instead of one furnace, secured, thus creating a surplus metal heating capacity for high-output mills. The use of two walking-beam furnaces secures a mill output of up to 200-220 t/h, with high level of mechanization and automation of heating facilities.

From the standpoint of heat transfer the walking-beam furnace has an advantage, for the intervals between pieces in these furnaces are about 200 mm which provides for their heating on three sides. In walking-beam furnaces the heating time is reduced and this makes for lesser iron losses and decarburization.

Figure 11.12 illustrates a double-zone end-charged side-discharged walking-beam furnace. Combustion products in the cold part of the furnace escape upwards, the metal recuperator is set up on the side since the lower part of the furnace is occupied by the walking-beam mechanisms.

Heavy-Section Mills

Heavy-section mills with rolls more than 600 mm in diameter have much in common with rail-and-structural mills.

The equipment layout of a 650 mm heavy-section mill is shown in Fig. 11.13. Billets 300 × 320 and 200 × 200 mm in cross section, up to 6 m long and weighing up to 4 t delivered from a blooming mill are heated in three continuous end-charged end-discharged furnaces. A 800 mm roughing stand of a small two-high reversing blooming type is driven by a 4550 kW d.c. motor at a speed of 0-70-120 rpm.

The second line consists of two three-high 650-680 mm stands with a 4550 kW, 0-90-180 rpm d.c. motor and a two-high stand with an individual 1980 kW, 0-125-250 rpm d.c. motor. The three-high stands are equipped with lifting tables, electrically driven rack manipulators and hydraulic tilters. The finishing stand has a box-type twist guide. Mill and transfer tables are installed in the working area of the second mill line. After rolling in the finishing stand the piece is cut by six reciprocating saws and then delivered to a cooling bed consisting of three sections equipped with transfers.

Provision is made for the installation of a universal finishing stand to roll wide-flange beams. The existing conventional rail-and-structural and heavy-section mills are often equipped additionally with universal stands or even with separate universal-mill lines to roll light-weight I-beams.

Tandem or staggered 500-550 mm mills are intermediate between heavy- and medium-section mills. Each of two-high stands of these mills is designed for one passing only and the distance between the stands exceeds the maximum length of the rolled piece.

In contrast to heavy-section mills supplied with billets coming from a continuous billet mill a new heavy-section 600 mm mill incorporates a continuous billet-rolling group of stands with rolls 730-850 mm in diameter (Fig. 11.14). Billets with a cross section of 280-300 mm and up to 6 m long are heated in three continuous reheating furnaces with an effective hearth length of about 32 m. The productive capacity of each furnace is 110 t/h. Squares 125-180 mm^2 in cross section or rectangles of the same cross section leave

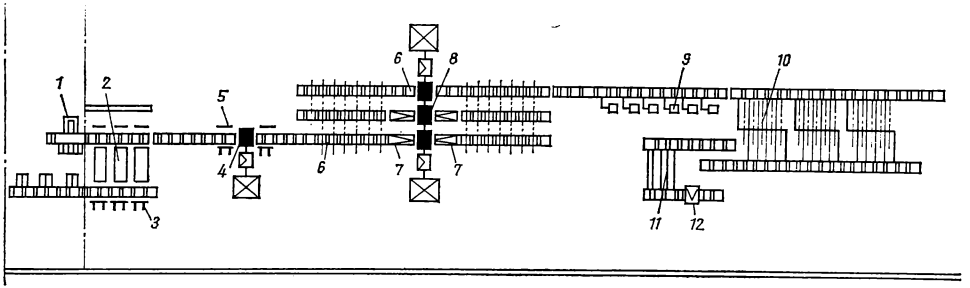


Fig. 11.13. Layout of heavy-section mill equipment

1—charging rack; *2*—continuous furnace; *3*—pusher; *4*—roughing stand; *5*—manipulator; *6*—roll table; *7*—lifting table; *8*—finishing line; *9*—sliding frame saws; *10*—cooling bed; *11*—racks; *12*—shears

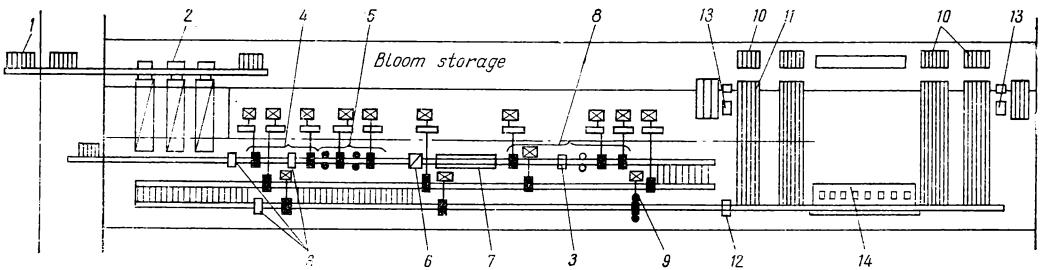


Fig. 11.14. 600 mm heavy-section mill

1—charging racks; *2*—furnaces; *3*—tilters; *4*—group of 850 mm stands; *5*—group of 730 mm stands; *6*—shears with a cutting force of 800 tf; *7*—reheating furnace; *8*—group of 580 mm stands; *9*—universal stand; *10*—bar piler; *11*—right- and left-hand cooling beds; *12*—stationary hot saw; *13*—right- and left-hand straightening machines; *14*—hot saws room

the first continuous group of stands made up of four horizontal 830 mm and two vertical 730 mm stands. Right after the first continuous group of stands is installed a long (107 m) roller-hearth continuous reheating furnace with a productive capacity of 350 t/h designed to raise the metal temperature by 65-80°C. The second continuous group of stands includes three horizontal and one vertical (730 mm) stands. In the wake of the two continuous groups of stands come stands arranged in tandem in two lines with a staggered transfer of the rolled piece from one line to the other (by means of transfers).

The range of products put out by the mill includes: 50-120 mm dia rounds; 50-100 mm^2 squares; beams of No. 10-20 sizes; mine rails and props; equal and unequal angles of No. 8-16 sizes and 14-50 × 160-200 mm strips.

Medium-Section Mills

Modern medium-section mills put out a fairly wide range of products: rounds with a diameter of 30-120 mm, strips with a thickness of 50-150 mm, angles with 40-120 mm long flanges, standard channels and beams of up to No. 20 size and thin-walled ones of up to No. 30 size.

The diameter of finishing rolls is 370-600 mm. Medium-section mills of the line type with an output of 150,000-250,000 tons per year are in the widest use.

The first mill line usually includes a three-high roughing stand and the finishing line has three to four two-high (or three-high) stands with a common drive. In some mills the last two-high stand is furnished with an individual drive. A vertical roll stand is occasionally installed before the finishing stand to roll strips and sectional shapes. A semi-continuous arrangement of stands is also in use. Continuous rolling is common in intermediate stands.

Wide application have found mills with a roughing two-high reversing stand and top roll adjustment, the remaining stands being arranged in tandem or in continuous groups, more often than not comprising 2-3 stands with horizontal and vertical rolls.

In a number of new medium-section mills are employed two-high non-reversing roughing stands, the other ones

being arranged along three axes of rolling, or all the stands run along a single line, the finishing group of stands for strip rolling running parallel to the main axis of rolling (Fig. 11.15).

The roughing group consists of eight stands with rolls 550-800 mm in diameter including two stands with vertical rolls. This group comprises two sections each counting four stands. The intermediate group has 450 mm dia rolls and consists of three stands (the central stand with vertical rolls). The finishing group for section rolling with rolls 450 mm in diameter also has a vertical roll stand.

The line for strip rolling includes nine stands: six four-high horizontal roll stands and three vertical roll stands. The roll diameters in the four-high stands are: 310 mm for work rolls and 610 mm for backup rolls. All stands of the mill are individually driven by 800-1200 kW motors.

In the forefront of each group of stands for section rolling there is a tilter; anterior to the strip rolling group are installed flying shears to crop the ends of the strip and cut it to length along with a hydraulic descaler.

The maximum section rolling speed is 8 m/s, the strip rolling speed amounting to 13 m/s. The range of products put out by the mill includes: 30-80 mm dia rounds; 28-75 mm^2 squares; angles with an unfolded flange length of 50-120 mm; beams and channels of No. 6-12 sizes and 4-4.5 × 100-500 mm strips. The initial billets are: 90, 120, 140 and 160 mm^2 squares and 70-100 × 120-500 mm rectangles; the billet length is up to 7 m. The mill output is 80-140 t/h of sectional shapes and 30-140 t/h of strips.

It should be noted that it is expedient to use not more than three stands in the continuous groups of medium-section mills, since this ensures stable rolling without tension; the rolling speed in the end stands is adjusted conformable to changes in rolling parameters. The introduction of continuous rolling in medium-section mills in combination with looping between the groups secures, in addition to rolling sectional shapes without tension, the minimum temperature drop and creates conditions necessary for rolling thin-walled light-weight section steel.

The layout of a semi-continuous medium-section 350 mm mill with an annual output of 900,000 t is shown in Fig. 11.16. The range of products put out by the mill is as follows:

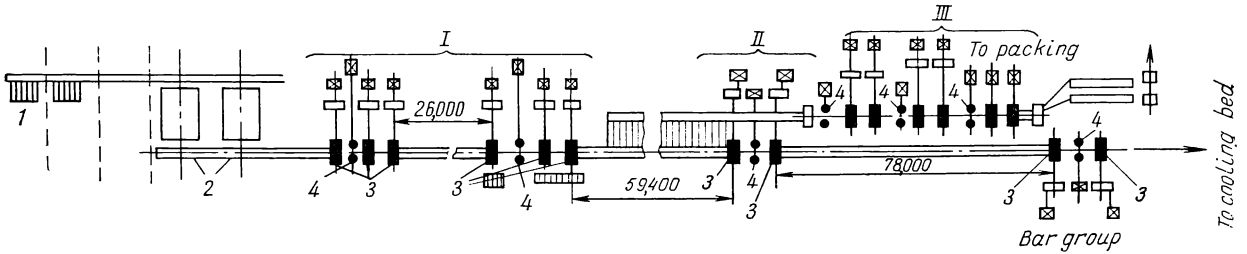


Fig. 11.15. 450 mm medium-section mill

I—roughing group; *II*—intermediate group; *III*—strip-rolling group; *1*—charging racks; *2*—furnaces; *3*—horizontal roll stands; *4*—vertical roll stands

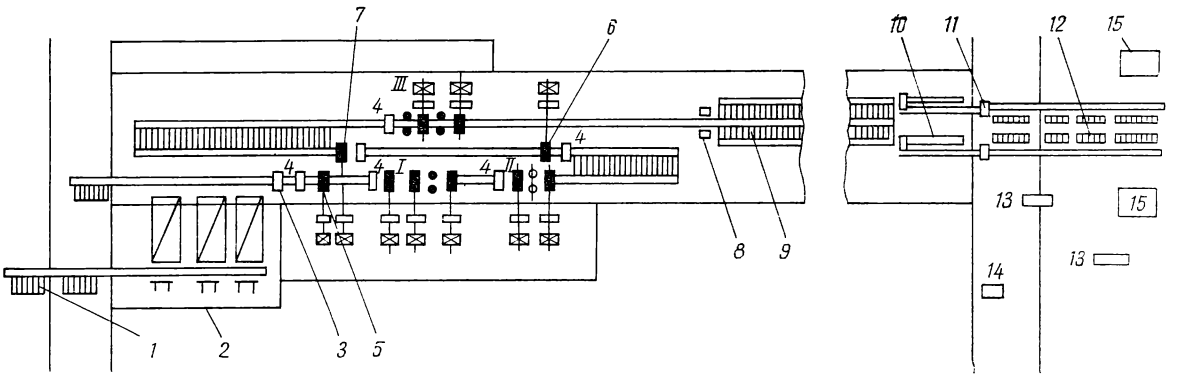


Fig. 11.16. Layout of 350 mm semi-continuous rolling mill

I, *II* and *III*—continuous groups: *1*—charging racks; *2*—reheating furnaces; *3*—dividing shears; *4*—tilters; *5*—530 mm horizontal roll stand; *6*—420 mm horizontal roll stand; *7*—420 mm horizontal roll stand; *8*—sampling saws; *9*—cooling beds; *10*—cold shears and transfer devices; *11*—straightening machines; *12*—cradles; *13*—separately installed cold shears; *14*—straightener with cross rolls; *15*—assorting units

20-75 mm dia rounds, 18-65 mm^2 squares, angles of No. 4-9 sizes, channels of No. 5-10 sizes, beams of No. 10 size and 40 × 40-120 mm strips. The initial billets are 100-170 mm^2 squares up to 6 m long. The mill includes 14 stands, 10 with horizontal and 4 with vertical rolls, arranged in three lines. It is equipped with three continuous furnaces, each of 40 t/h productive capacity. In the forefront of individual stands and groups of stands are installed tilters. The mill is fitted out with a two-side cooling bed, straightening machines and cold shears.

A trend is now observed towards increasing the cross section, length and weight of billets. The rolling speed usually does not exceed 9 m/s. With due consideration for the range of products put out by medium-section mills the exclusive use of continuous rolling has proved unjustifiable. Further progress in the design of medium-section mills involves the development of combined mills. Occasionally medium-section mills also produce light-section steel.

Light-Section Mills

Light-section mills are characterized by a great variety of designs and equipment layouts. The annual output of these mills is 50,000-800,000 t.

The following types of light-section mills may be distinguished according to the layout of their stands: line-type mills (Fig. 11.17), tandem mills (Fig. 11.18), including cross-country and staggered-type mills, semi-continuous (Fig. 11.19) and continuous (Fig. 11.20) ones.

Principal requirements which precondition the development of light-section steel production are as follows: higher dimensional accuracy and improved surface finish of section steel, possible production of light-section steel in straight bars and in coils of a large weight, with high production efficiency and complete automation. The specialization of light-section mills depends upon the scope of rolled steel production and the available range of section mill types. High-output light-section mills are usually of specialized nature. As a rule, these are continuous mills with a rolling speed of up to 20 m/s. In these mills light-section steel is rolled according to a double- or multistrand rolling pattern.

Along with the progress in the coiled light-section steel production the industry also turns out light-section steel in the form of straight bars. The application of long cooling beds (120 m and more) for straightening and cooling the bars cannot ensure full-length stretching of the bar rolled from a single billet. Cutting the billet into lengths is undesirable

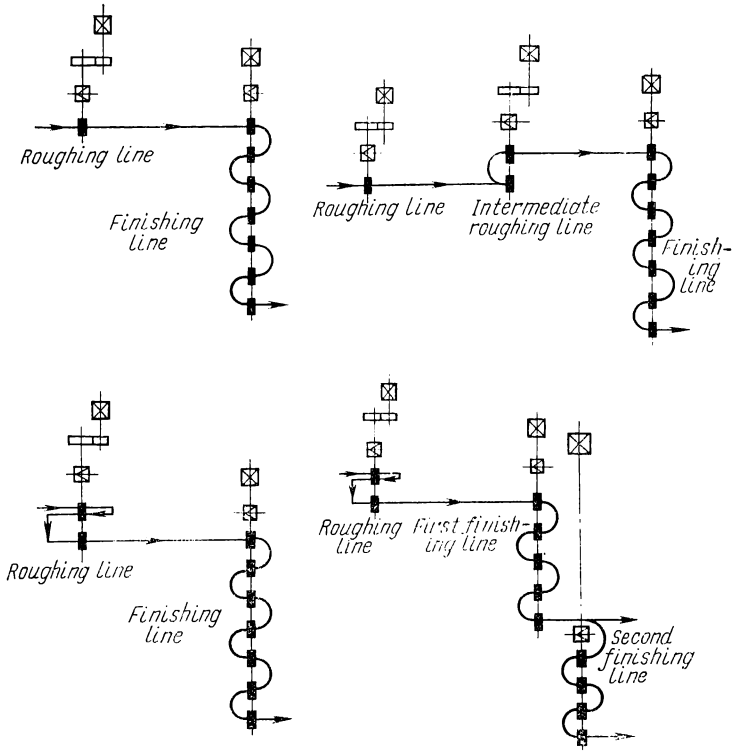


Fig. 11.17. Line-type light-section mills

from the standpoint of the efficiency and reliability of the rolling process, therefore the bar is cut to length on the run before reaching the cooling beds.

High speed of rolling in cross-country, line-type and semi-continuous mills is attained by the introduction of loop repeaters at both ends of the rolling mill line.

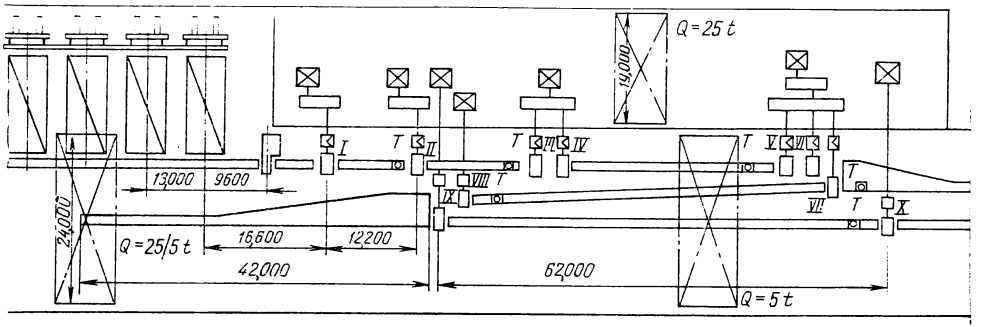


Fig. 11.18. Light-section tandem mill (figures designate serial numbers of stands, *T* — tilter)

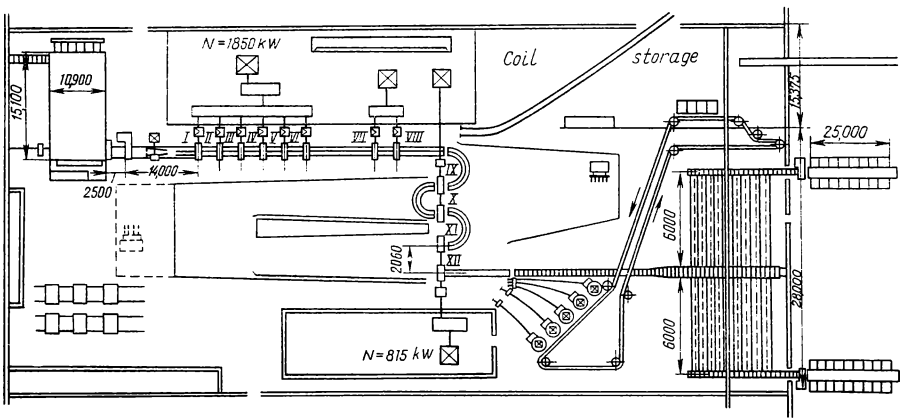


Fig. 11.19. Semi-continuous light-section mill

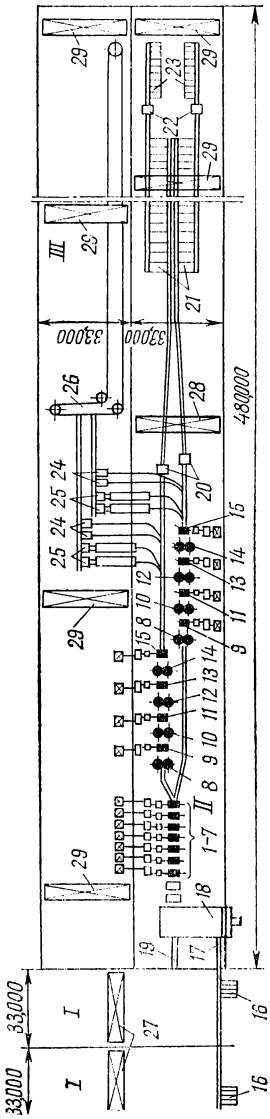


Fig. 11.20. Continuous two-strand light-section mill

I—billet storage; *II*—mill bay; *III*—scale removal and finished product storage bay; *1-7*—370 mm two-high roughing stands; *8-12*—320 mm prefinishing stands of the finishing group; *13-15*—270 mm two-high finishing stands; *16*—charging racks; *17*—pusher; *18*—reheating furnace; *19*—ejector; *20*—flying shears; *21*—cooling bed; *22*—shears; *23*—cradles; *24*—wire rod coilers; *25*—strip coilers; *26*—hook conveyor; *27*—15 t crane; *28*—20/5 t crane; *29*—10 t crane

The length of the loop between the stands of a line-type mill (Fig. 11.21) is

$$L = v_1 t_1 + \left(v_1 - \frac{v_2}{\mu}\right) t_2$$

where v_1, v_2 = rolling speed in the first and second stands respectively

t_1 = time to pass the bar from the first stand into the second one

t_2 = time of simultaneous rolling in both stands

μ = draught coefficient in the second stand

With the continuity of rolling, when

$$v_1 = \frac{v_2}{\mu}$$

the second term in the formula for the loop length will be

$$(v_1 - v_2)\, t_2 = 0$$

and the length of the loop

$$L = v_1 t_1$$

To increase the mill output and reduce the area required for the arrangement of special loop-storage pits the rolling should be

conducted under conditions closely approaching those of the continuity. This is possible only when the stands are provided with individual drives.

Stable rolling conditions along the entire length of the piece are achieved through uniform distribution of the temperature. To obtain the same cross sections of the head and tail ends of the piece the mill should be arranged in such a way, in which the tail end of the billet remains in the furnace whereas its head end goes to the cooling bed or to the coiler. This is also facilitated by the use of specially designed stands developing the minimum elastic deformation under the rolling pressure.

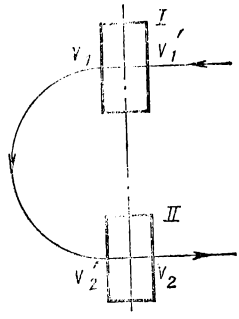


Fig. 11.21. Calculation of loop length

The quality of the rolled steel is influenced by the condition of the roll pass surfaces. For this reason a strict schedule for changing roll passes and rolls should be adhered to. A great effect on the quality of rolled products is produced by the rolling speed schedule and interstand tension. Continuous roughing groups with closely spaced stands and high longitudinal stiffness of the piece operate, as a rule, with a small predesigned tension, while the finishing groups of stands are adjusted for the minimal looping.

Below are given some characteristic features of the continuous rolling process.

The tension of the workpiece in rolling does not remain constant, but increases as its head end passes from the first to the last stand and correspondingly decreases while the tail end leaves the rolls. A piece rolled under tension will have a thinned central portion and overfilled cross section at the ends.

Great irregularity of the piece cross section in the first roughing group of stands makes it difficult to achieve steady operational conditions in the subsequent groups up to and including the finishing ones. It is difficult to obtain a uniform cross section of the workpiece in continuous mills equipped with group drives, while with new mills having indi-

vidual adjustable-speed drives one can achieve much greater uniformity in the cross section of the piece.

When rolling with tension the spread is decreased; it may become equal to zero or even negative (pulling-down). Under tension the reduction of the cross section in the vertical plane is insignificant and is caused by the change in the rolling pressure. In addition to pulling down in the zone of deformation this also occurs between the stands, where it has the nature of plastic tensile deformation.

Tension in rolling decreases the rolling pressure, the back tension being more effective in this respect.

The rolling moment changes depending upon the tension. The front tension reduces and the back one enhances the moment. With an excessively increased front tension the driving motor of the stand may start operating as a generator.

Continuous rolling places special demands upon the stability of operation, *viz.* good condition of the roll passes and fittings, proper dimension of billets, adequate rolling temperature and mill loading.

Of great importance is the stiffness of working stands which provides for but negligible changes in the cross-sectional dimensions of the rolled piece with variations of the rolling pressure due to temperature fluctuations and changes in the number of strands being rolled. A proper choice of the degree of stiffness of the speed-torque characteristic of the driving motors precludes a fall of the rolling speed after the rolls bite the metal. A less stiff characteristic facilitates self-regulation since with the load increasing under the effect of the back tension the speed of the motor declines.

Further progress in the design of continuous section mills proceeds along the line of introducing endless rolling, which eliminates transients of biting and ejecting the pieces.

Welding is done by fusing the metal and compressing the ends of the pieces under a high pressure, followed by the removal of dross at the site of the weld. Two welding methods are used: on the run of the billet and with stationary welding machines. In the latter case looping is resorted to so as to compensate for the loss of time in welding.

Figure 11.22 illustrates the equipment layout of a modern 250 mm light-section endless rolling mill. The range of pro-

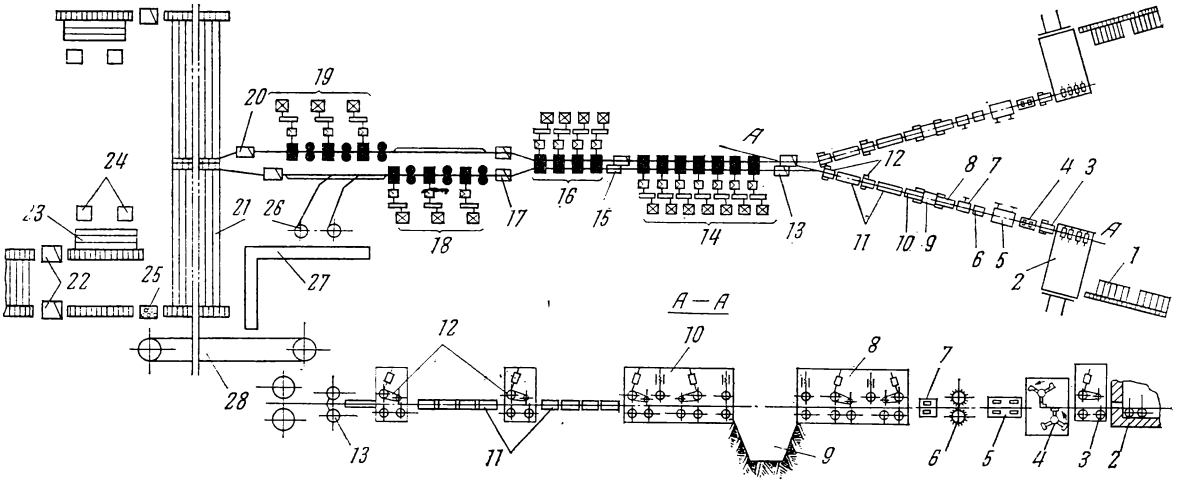


Fig. 11.22. Light-section endless rolling mill

1—charging racks; *2*—reheating furnace; *3*—accelerating roller device before welding machine; *4*—scale breaker; *5*—electric welding machine; *6*—descaming machine; *7*—correcting device; *8* and *10*—accelerating roller devices before and after looping pit; *9*—looping pit; *11*—individual furnace; *12*—accelerating roller devices; *13*—dividing shears; *14*—roughing group of stands; *15* and *17*—emergency shears; *16*—intermediate group of stands; *18* and *19*—finishing group of stands; *20*—double-drum flying shears; *21*—racking-type cooling bed; *22*—cold shears; *23*—cradles; *24*—pack-strapping machines; *25*—roller straightening machine; *26*—coilers; *27*—apron conveyor; *28*—hook conveyor

ducts put out by the mill is as follows: 8-30 mm dia rounds; 8-30 mm² squares; 4-20 × 12-70 mm strips and angles of No. 2-4 sizes. The initial billet is 80 × 80 × 2000 mm in cross section.

The mill is of the two-strand type. The rolling speed in the last stand reaches 5.4-20 m/s. The maximum output is 188 t/h. In the roughing and intermediate groups of stands

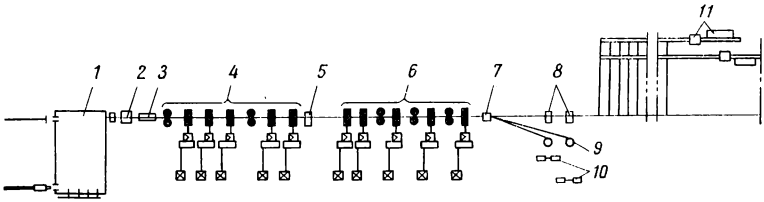


Fig. 11.23. Continuous single-strand light-section mill

1—reheating furnace; *2*—dividing shears; *3*—butt-welding machine; *4*—roughing group of stands; *5*—emergency shears; *6*—finishing group of stands; *7*—flying shears; *8*—band coilers; *9*—bar coilers; *10*—coil strapping units; *11*—cold shears

rolling is done with tension and in the finishing group, with looping. The working stands of the first two groups are equipped with horizontal rolls. The finishing stands are of the combined type, i.e. in each of these stands both vertical and horizontal rolls may be installed.

The speed in the finishing stands is controlled automatically (in accordance with the loop length). After leaving the finishing stands the piece is cut with flying shears. A 125 m long double cooling bed of the rack-type is used. Two Garret reels and a hook conveyor are employed to service the left-hand strand of the finishing stands. Coils are taken off the conveyor by two coil strippers.

Various versions of the above layout of a two-strand continuous light-section mill have gained popularity in the ordinary piecemeal rolling.

Figure 11.23 represents the equipment layout of a 300 mm single-strand rolling mill for making rounds 14-30 mm in diameter, die-rolled reinforcement steel of No. 14-25 sizes, strip and spring steel of 8-10 × 65-76 mm. The initial billet is a 104 mm² square 9-10 m long.

The mill consists of two continuous groups of stands: a roughing one with a scale-breaker, the group consisting of

six stands (stand No. 4 has vertical rolls), and a finishing one which includes eight stands, three of them with vertical rolls. The rolls are supported by roller bearings.

Right after the roughing group there are rotary shears intended to crop the head end of the piece and to cut it in case of troubles in the finishing group of stands. Prior to the delivery to the cooling bed the piece is cut to length corresponding to that of the cooling bed (125 m) with flying (rotary) shears. There is a provision for coiling strip and round bars. The rated annual output of the mill is 400,000 t.

The semi-continuous light-section 250 mm mill shown in Fig. 11.19 is characterized by a high output. The mill consists of three groups of stands: roughing, intermediate and finishing, the first two being of the continuous type and the last one, of the line type. Three strands are rolled simultaneously in the roughing group and up to four strands, in the finishing one.

The range of products turned out by the mill includes: 10-19 mm dia rounds; 12-16 mm^2 squares; 4-12 $\times$ 20-45 mm strips and angles with an unfolded flange length of 25, 28 and 32 mm. In addition to ordinary plain-carbon steel the mill also handles structural, free-cutting and alloy steel grades. The mill is equipped with five Garret reels and a hook conveyor for coils. A four-strand double-sided cooling bed 123 m long and 28 m wide is employed for handling straight bars.

Wire-Rod Mills

Specialized wire-rod mills produce wire (wire rod) 5-9 mm in diameter. Wire rod 6.0-6.5 mm in diameter is used for further cold drawing.

In recent years wire-rod mills with an annual output from 100-150 to 500-800 thousand tons have been built in various countries. However, in some of them it is considered expedient to employ low-output line-type mills for rolling wire rod of a high-accuracy class with a tolerance of ± 0.1 mm from special steels. But in most cases high-output continuous and semi-continuous mills are used for making wire rod of various steel grades (high-carbon). Only thirty years ago the delivery speed of continuous wire-rod mills was 20-22 m/s; in the forties first continuous wire-rod mills were

modernized with the rolling speed increased up to 30-35 m/s. Now the majority of new mills have a rated rolling speed of 45 m/s. It reaches 50 m/s in mills with multistand finishing blocks.

In accordance with the arrangement of their stands wire-rod mills may be classified as follows: staggered line-type mills, semi-continuous and continuous mills. The staggered line-type wire-rod mills are old mills in which the rolling speed is different in individual mill lines arranged in a staggered order. In the finishing line rolling is done with several (up to seven) strands by looping and repeating. However manual looping is still used quite often in handling oval pieces in the last passes. In this case the rolling speed in the last passes does not exceed 8-9 m/s. With these mills a comparatively high output (up to 120 thousand tons per year) is achieved by multi-strand rolling.

The semi-continuous wire-rod mills, which differ from the staggered ones in that they include continuous groups of roughing stands, are intermediate between the staggered line-type and continuous mills. Such a rolling stand layout helps cut down the labour consumption in attending the roughing passes, makes the mill more compact, improves the temperature conditions of rolling and enables the finishing stands to be charged more regularly. Single-strand finishing stands (Swedish-type mills) are also employed to produce high-carbon wire rod of a high-accuracy class (with tolerance of ± 0.1 mm).

The continuous wire-rod mills are characterized by a great variety of their stand layouts and also of their technological and design features.

A high output and constant trend towards increasing the rolling speed are common for these mills.

Wide use have found continuous wire-rod mills of the following types.

1. Mills with intermediate looping mill lines between the continuous roughing and finishing groups (so-called American method see Fig. 11.24).

The intermediate looping line is in a certain measure intended to level out the nonuniform cross section of the piece leaving the continuous roughing group. In such mills the accuracy of rolling is limited to $\pm(0.3\text{-}0.4)$ mm.

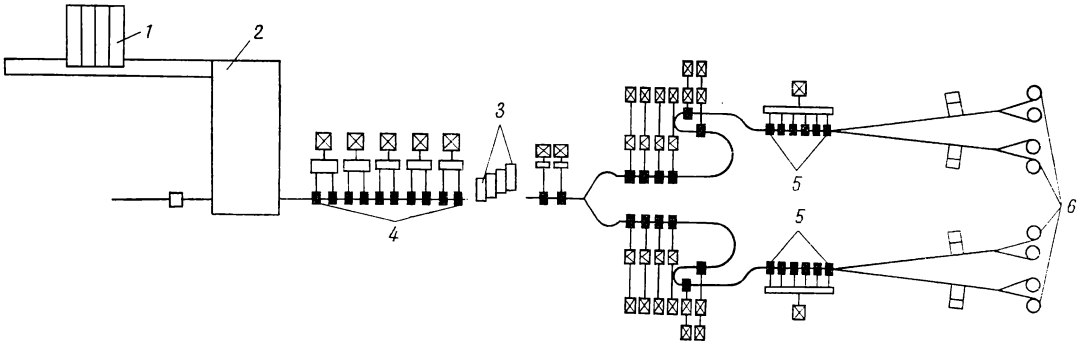


Fig. 11.24. Continuous wire-rod mill with intermediate looping line

1—charging racks; *2*—reheating furnace; *3*—flying shears; *4*—roughing group of stands; *5*—finishing group of stands; *6*—coilers

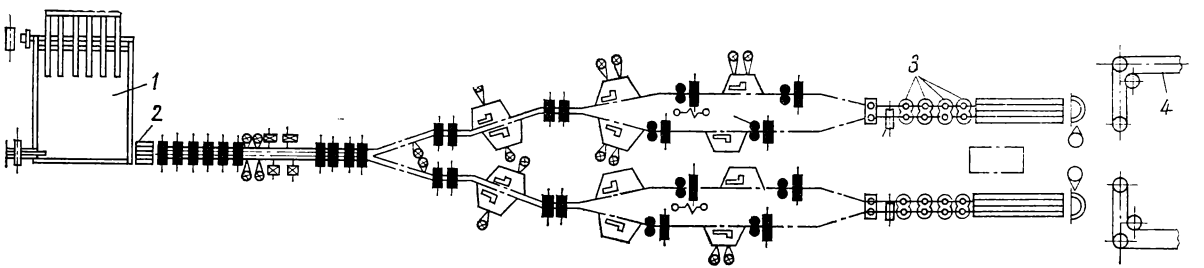


Fig. 11.25. 250 mm continuous wire-rod mill

1—furnace; *2*—distributor; *3*—coilers; *4*—hook conveyor

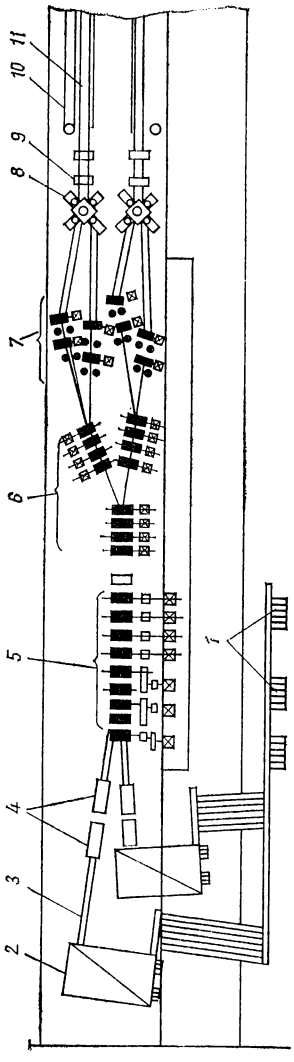


Fig. 11.26. Layout of modern wire-rod mill

1—charging racks; *2*—furnace; *3*—end-charged end-discharged roller-hearth furnace; *4*—welding machine and induction furnace; *5*—450/350 mm roughing group; *6*—360 mm intermediate group; *7*—250 mm finishing group; *8*—coilers; *9*—coil-strapping machines; *10*—hook conveyor; *11*—apron conveyor

2. Continuous mills with 2-4 single-strand finishing groups of stands equipped with vertical and horizontal rolls and individual drives (Fig. 11.25). In the roughing group rolling is done in 2-4 strands respectively. The roughing and branched intermediate (central) stands are arranged in continuous groups as follows. The first group includes 6-7 stands; the second one, 4 stands and the third and fourth groups, where the process flow is divided into two branches, are made up of 2 stands each.

In such mills rolling is done either with small loops between the groups and between the paired sets of stands with vertical and horizontal rolls, or with a low interstand tension. With the single-strand rolling in the finishing stands the wire rod obtained is accurate within 0.20-0.25 mm.

The output of modern wire-rod mills (of the European type, see Fig. 11.26) reaches 950,000 tons per year of wire rod 6-10 mm in diameter, the rolling speed being up to 45 m/s. The weight of a coil of wire rod is 800 kg. The stands are individually driven, only the roughing ones being driven in pairs.

3. In recent years an ever increasing popularity have gained multistand blocks which help cut down the equipment cost and reduce the overall dimensions of the rolling shop and yet retain all the advantages of the single-strand rolling at a speed of 30-50 m/s. The use of these blocks enables the wire rod to be rolled to a tolerance of ± 0.1 mm on the diameter with a maximum out-of-roundness of 0.1 mm.

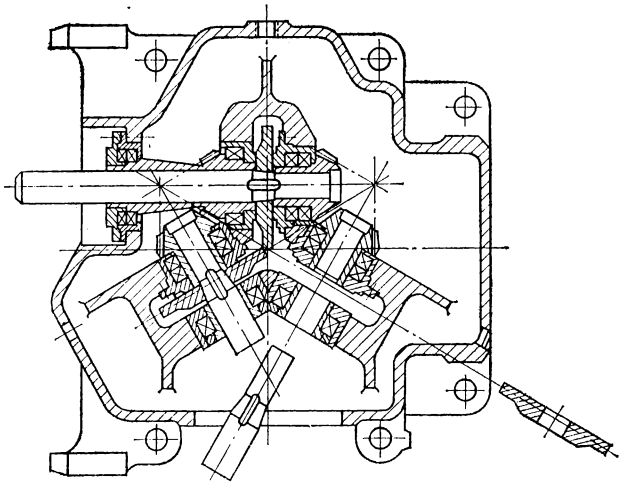


Fig. 11.27. Multistand block of modern wire-rod mills

The following two types of blocks are mainly used. The first type comprises up to 13 stands with three rolls whose axes intersect at an angle of 120°; the second type includes up to 16 two-roll stands (similar to those employed in tube-reducing mills) with the roll axes inclined at an angle of 45° to the vertical plane, the stands being arranged so that the axes of the rolls of adjacent stands are at right angles. The stands of the blocks are driven by speed reducers of special design, the drive being either common or sectionalized. The distance between the stands in the block is 300-400 mm (Fig. 11.27).

In the three-roll stand blocks an original system of triangular passes is employed, in which the position of the apices

of the triangles formed by the grooves of the three rolls is alternated so that the metal in two adjacent stands is worked on six sides. The apices of the triangles are underfilled in order to compensate for the spread in rolling. The rolling is done with tension. In the last three or four stands use is made of round passes formed by rolls up to 300 mm in diameter. Oval-oval and oval-round series are usually used in the two-roll stand blocks. With the multistand blocks the rolls are changed by replacing the entire block.

With high rolling speeds and rolled product output a major problem is the durability of rolls. The rolls for the multistand blocks are either made of white iron and special tungsten-base alloys or faced with hard metal.

The specific technology of continuous rolling in wire-rod mills does not practically differ from the above described technological principles accepted for rolling in continuous light-section mills. At high rolling speeds the process parameters must be stable, special importance being attached to the uniformity of the cross section of the rolled piece and to the stiffness of the mill structures. When rolling with tension efforts are made to reduce the distances between the stands, this creating the necessary conditions for shortening the lengths with increased cross section at the head and tail ends of the rolled piece.

Strip (Skelp) Mills

Specialized mills for rolling strip, band, hoop and skelp steel are, as a rule, of the continuous type. Rolling of thin strips in the line-type mills entails great technological difficulties (maintaining adequate temperature, rolling accuracy, etc.). As concerns their technological and design features continuous skelp mills occupy an intermediate position between section and sheet mills. There is a trend towards increasing the skelp width from 200 to 400-500 mm.

Experience gained in handling continuous skelp mills has shown their material advantages over section mills of the tandem type, mainly thanks to the possibility of keeping the rolled skelp at the required temperature. These mills, however, have their bottlenecks which can be eliminated and the efficiency of the mills thus raised by the following

measures: increasing the capacity of the heating facilities, using roller or film lubrication bearings for the horizontal rolls and replacing the last stands by four-high ones with more powerful drives.

Figure 11.28 shows a modern 15-stand mill for rolling skelp up to 400 mm in width. As compared to the first skelp mills the speed of rolling in this mill is slightly increased (from 10 to 12 m/s). The three finishing stands of the mill are of the four-high type. In contrast to earlier mills coiling of skelp with its cooling on conveyors is only provided for. There are no flying shears after the finishing stands and no cooling and packing of cut skelp is provided for because modern tube-rolling mills, which operate on the principle of furnace or electric welding of tubes, do not require cut skelp.

The technology of rolling in skelp mills has much in common with that of rolling in continuous section and sheet mills. The preset reductions in each stand are strictly controlled by means of instruments.

In the skelp rolling of paramount importance is a well regulated scheme of changing roll passes, working sections on smooth rolls and of rolls themselves. The rolls are changed by replacing either the entire stands or the sets of rolls together with their chocks. The working sections on smooth rolls are changed best of all by shifting the stands in the lateral direction (for rolling a narrower skelp when the roll body can accommodate two or more working sections). There is always a provision for shifting the vertical roll stands (edgers) in both the horizontal and vertical directions.

Wide skelp steel is rolled from flat billets. With a wide range of products turned out by a skelp mill it is very important to select the optimum cross section of the billet for each product size.

Some Design Features of Working Stands and Roll Fittings

The working stands of section mills usually have a permanent roll setting. The roll-gap adjustment should be accurate and smooth and there should be a provision for the individual adjustment of rolls by means of mill screws. Occasionally motor-driven screws are used, especially when the roll adjustment is effected automatically by the remote

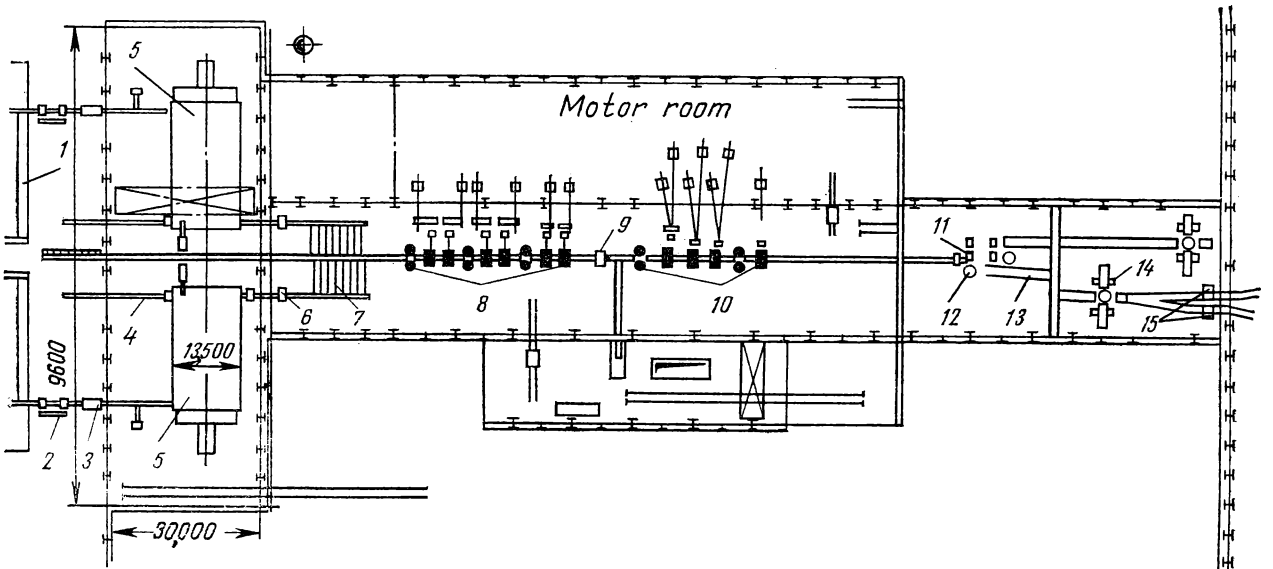


Fig. 11.28. Layout of continuous skelp mill

1—charging racks; *2*—tilters; *3*—pushers; *4*—ejectors; *5*—furnaces; *6*—shears; *7*—transfer beds; *8*—continuous roughing group of stands; *9*—emergency flying shears; *10*—continuous finishing group of stands; *11*—coilers; *12*—coil strippers *13*—coil conveyors; *14*—coil-strapping machines; *15*—coil packing machines

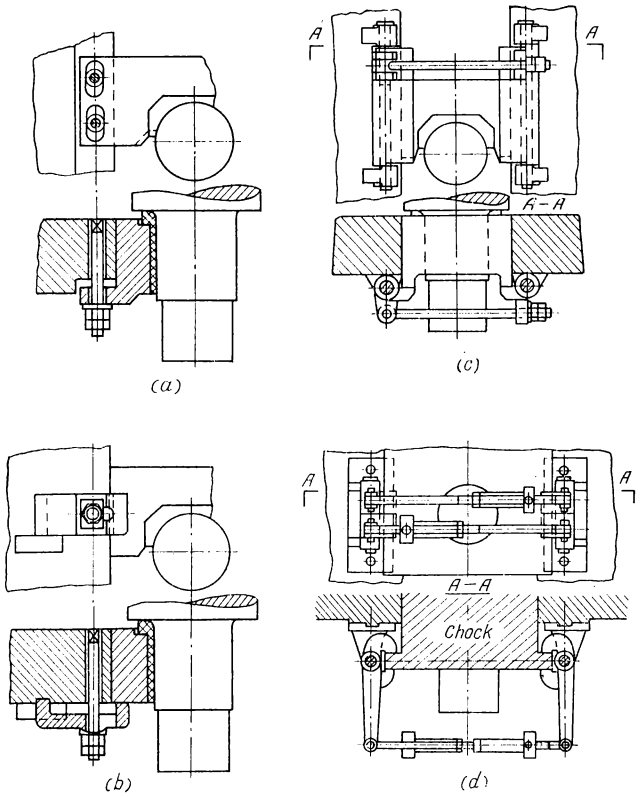


Fig. 11.29. Old (*a*, *b*) and new (*c*, *d*) means of lateral roll adjustment (*a*) bolts and chock lugs; (*b*) cleat; (*c*) single-lever system; (*d*) double-lever system

control system of the mill. The bottom roll is adjusted by means of a hand-operated screw (wedge) or shims. Of great importance are the lateral adjustment and fixing of rolls (Fig. 11.29). More often than not the top roll is spring balanced; in modern stands disc springs inserted into the bottom roll chocks are used for this purpose (Fig. 11.30). Hydraulic cylinders are sometimes employed in place of the springs.

In spite of their bulky structure and high cost the vertical roll stands have found wide application because with these stands there is no need for tilting the rolled piece.

The vertical roll stands are equipped with special mechanisms which ensure their vertical and horizontal movement (for changing the roll passes and altering the pass line in continuous mills). The rolls are changed by moving aside

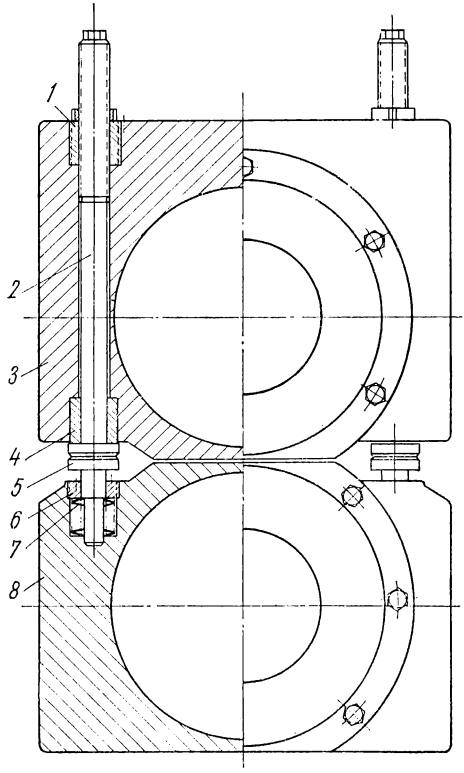


Fig. 11.30. Spring-type top roll-balancing mechanism of 250 mm continuous rolling mill

1—nut; *2*—adjusting screw; *3*—top roll chock; *4*—bushing; *5*—spring stop; *6*—adjusting nut; *7*—disc spring; *8*—bottom roll chock

the roll housing proper after the coupling spindles have been lifted. The design of the vertical roll stands in the finishing groups of some wire-rod and light-section mills is much simpler, since here one roll is idle and is only speeded up approximately to the rolling speed by a small flange-type

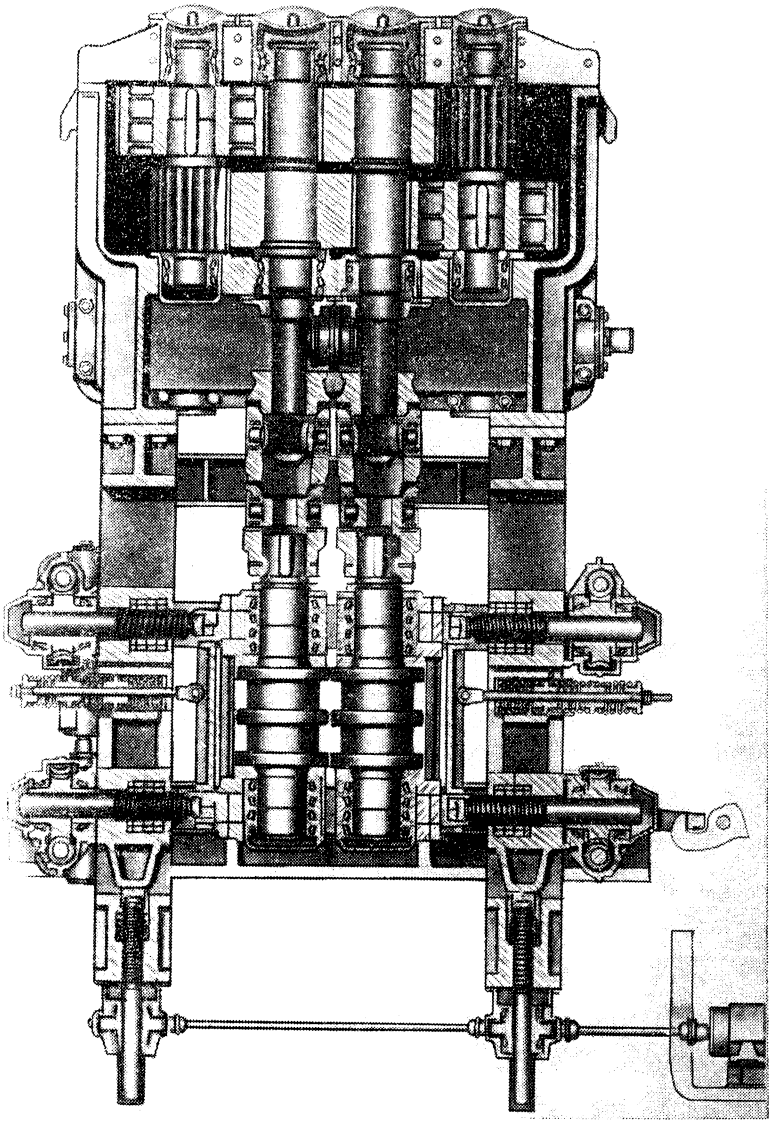


Fig. 11.31. Overdriven vertical roll stand

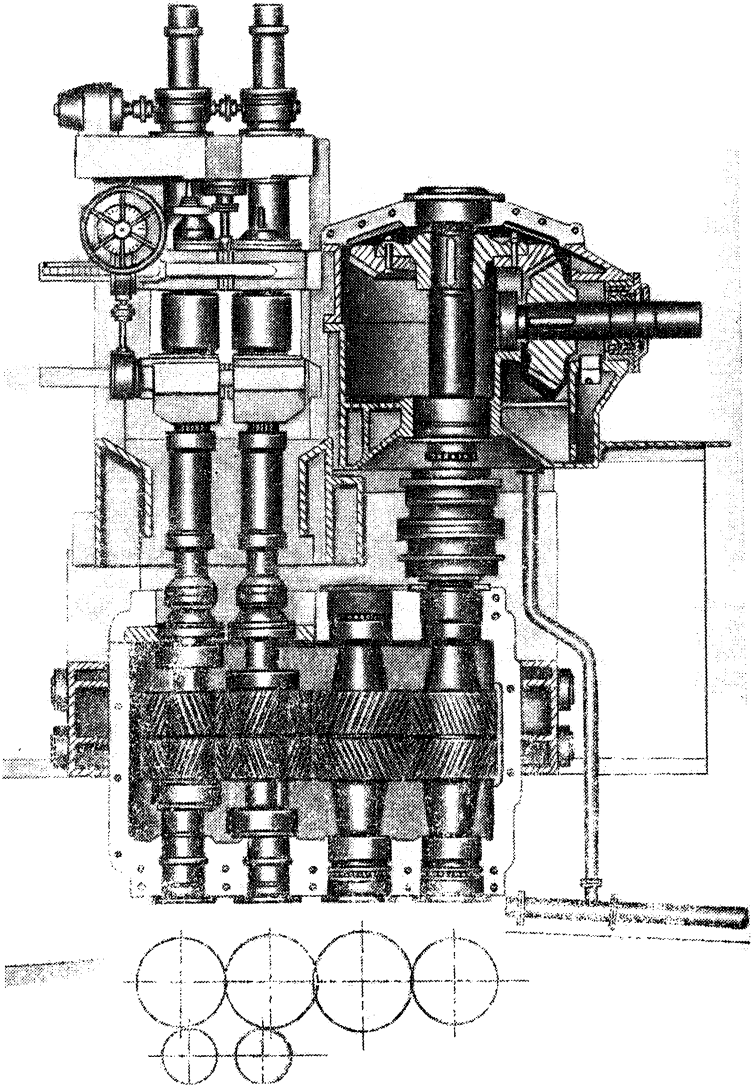


Fig. 11.32. Underdriven vertical roll stand

motor, the other roll being coupled directly to the vertical shaft flange-type main drive motor.

There are two main essentially different types of vertical roll stands, *viz.* overdriven (Fig. 11.31) and underdriven (Fig. 11.32). The underdrive of the work rolls is less convenient for it is difficult to maintain.

In some continuous section mills combination stands are used. These are designed as vertical roll stands, but can also accommodate a set of horizontal rolls, if necessary (Figure 11.33).

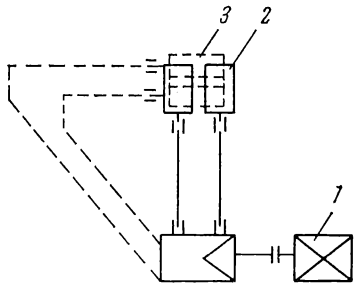


Fig. 11.33. Combination stand (schematic diagram)
1—electric motor; *2*—vertical rolls; *3*—horizontal rolls

The roll fittings of working stands serve to transfer the rolled piece from one stand to another and to reliably keep it in the required position on the entry and delivery sides of the stands. The working surfaces of the roll fittings should be wear resistant and should not damage the surface of the rolled metal. The roll fittings are mounted on the rest or guide bars and include guide boxes, guides, guards and, sometimes, twist guides and pipes. In modern mills the guide bars are made with special sockets which facilitate the adjustment and replacement of the roll fittings. Hinged guide bars are also used.

The entry roll fittings for feeding workpieces of a large cross section into box, oval or forming passes are made in the form of side guards which are attached to the guide bars. As a rule, inlet guides are used for oval (Fig. 11.34), diamond and square passes, their purpose being to keep the workpieces in the required position and prevent them from tilting in rolling. The inlet guides may be of the sliding type (whose cross section corresponds to the shape of the workpiece) and of the roller type.

In mills with a tandem and continuous arrangement of stands and also when employing repeaters the inlet guides are funneled on their entry side so as to catch and gradually center the head end of the workpiece. The width of the

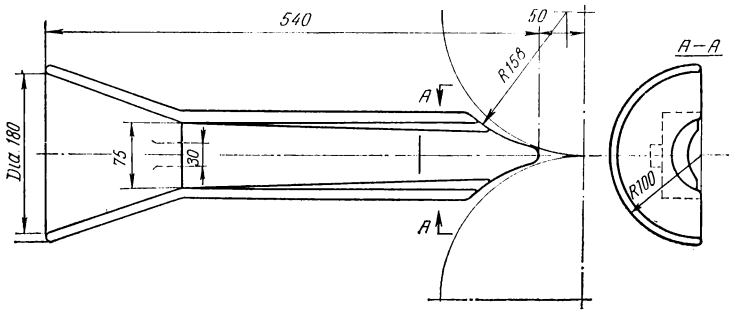


Fig. 11.34. Inlet guides for feeding oval bar into edging oval pass

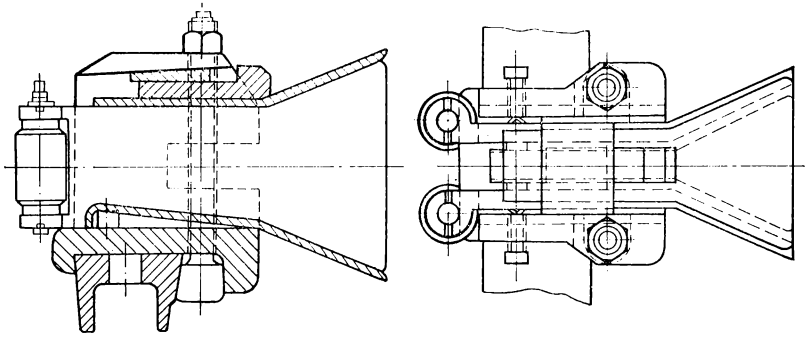


Fig. 11.35. Roller-type inlet guides

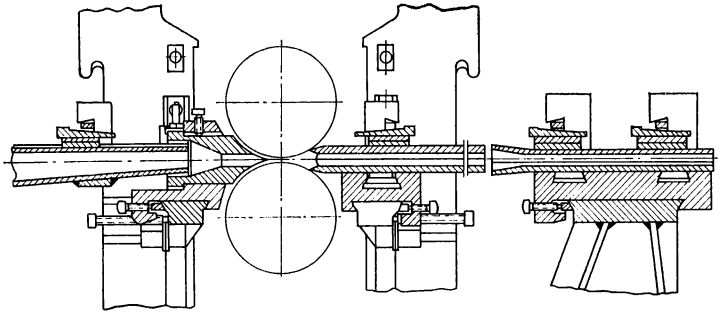


Fig. 11.36. Inlet and outlet roll fittings for finishing stand of light-section looping mill

bellmouth of the inlet guide is made 1.15-1.25 times greater than the height of the workpiece.

The inlet guides of the sliding type are made of chilled or alloy cast iron, steel and special alloys. The selection of materials for making the inlet guides is governed not only by their abrasion resistance, but also by the tendency of the rolled metal to stick to the working surfaces of the guide which is specific in rolling alloyed steel. Of wide use are the roller-type inlet guides which completely eliminate the metal sticking and secure reliable operation and good surface finish of the rolled metal (Fig. 11.35). The rollers for the inlet guides are manufactured of alloyed steel or of magnesium-inoculated cast iron. The rollers rotate in textolite or rolling bearings. The gap between the rollers is set by means of adjusting bolts. With high-speed rolling in continuous mills the metal biting by the rolls is accompanied by heavy impacts, this resulting in the formation of bulbs at the head end of the rolled piece, which should be taken account of in designing and setting up of the entry roll fittings.

The delivery roll fittings should prevent possible collaring of the rolls by the delivered piece and ensure its proper position in the course of rolling. The delivery roll fittings include guards, guides, pipes and twisters. Stripping guides of an adequately chosen shape are used in handling large-sized bars, while the delivery pipes are employed in dealing with light sections.

Figure 11.36 shows the arrangement of roll fittings on the finishing stand of a 250 mm light-section mill. In the figure on the right delivery side of the stand is seen the mounting of the delivery pipe with a wedge and cast iron blocks. The cross-sectional dimensions of the cross blocks are chosen so as to make them break, should the workpiece stick in the pipe, and thus prevent the damage of the guide bars and other more costly parts. The delivery twist guides and pipes are mounted in a similar fashion.

The angle of the workpiece twisting in a twist guide or pipe is calculated in a way similar to that described previously for continuous billet mills (see page 252), depending upon the distance between the stands and the length of the guide or pipe. For handling large-sized pieces in the roughing stands use is made of twist rollers. In dealing with smaller

pieces rolling-type fittings are employed in place of the sliding ones (Fig. 11.37).

At present, with the line or staggered arrangement of the mill stands looping of the workpiece is mostly realized by means of repeaters. For a long time repeaters have been

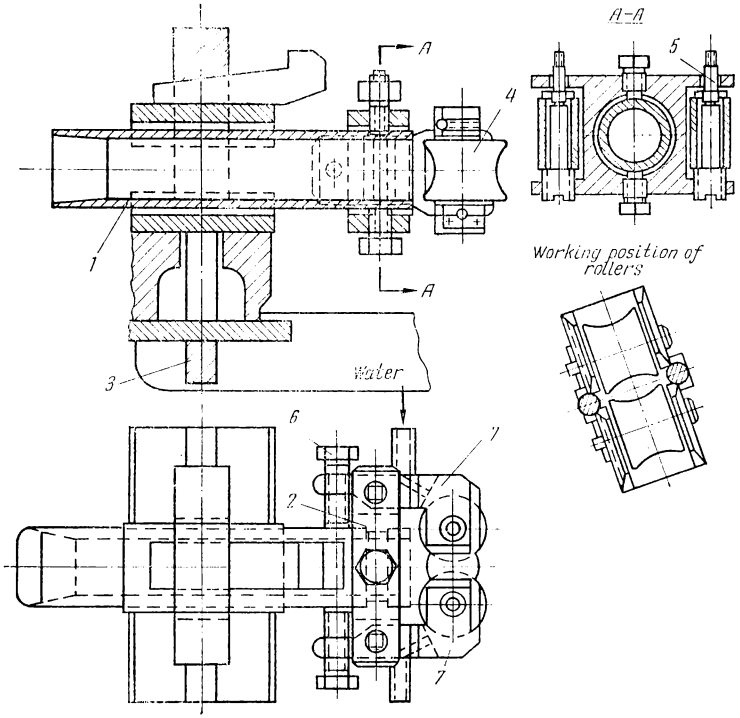


Fig. 11.37. Twist rollers for light sections

1—pipe; *2*—twisting head; *3*—special bracket; *4*—roller; *5* and *6*—special bolts; *7*—roller holders

applied in handling pieces of practically all sections, except for small ovals. Lately, thanks to the improvements in the design of the roll fittings, repeaters find an ever increasing use for pieces of all sections.

The repeater consists of a semicircular walled looping trough sloping in the direction of the metal movement and

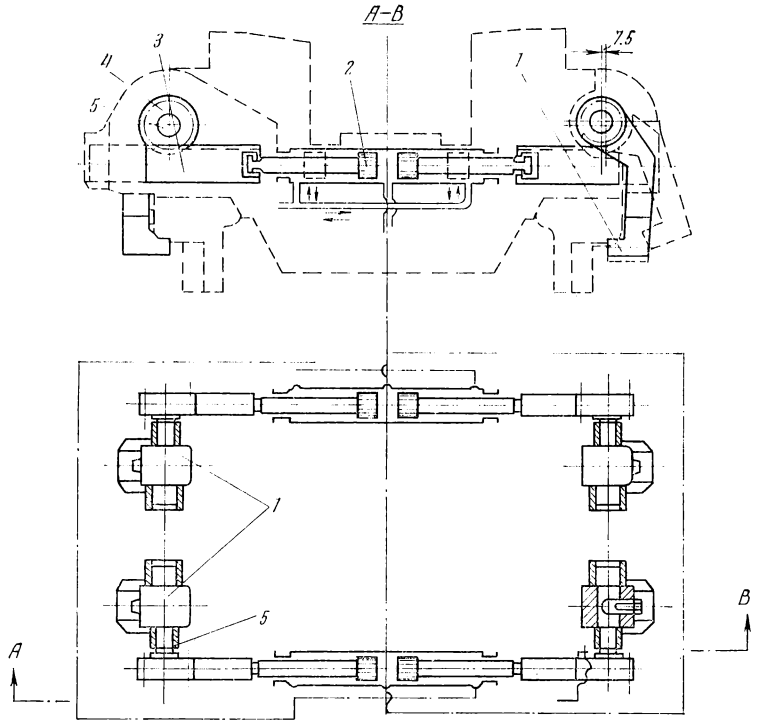


Fig. 11.38. Hydraulic device for joining the stand housings with mill shoes

1—feet; *2*—hydraulic cylinders; *3*—toothed racks; *4*—gears; *5*—shaft

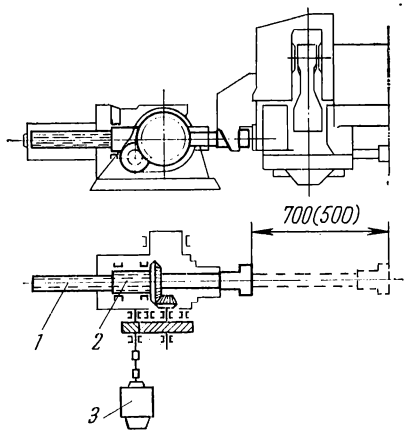


Fig. 11.39. Stand shifting mechanism

1—screw; *2*—nut; *3*—motor

of entry and delivery fittings. Prior to reaching the repeater the workpiece must often be tilted and, for this reason, twist guides or pipes with a strictly preset twisting angle are installed at the entry side of the repeaters.

Repeaters may be of the open- and close-top trough type.

Loopers of various designs are used between the stands of continuous looping section mills. In strip mills air-operated drop loopers are employed. In light-section and wire-rod mills electrically or air-operated kickers are applied to kick off the loop in the plane of the pocket. Use also find power-actuated loopers with tension rollers. Changes in the pressure on the roller caused by fluctuations of the workpiece tension between the stands may serve as a starting impulse for automatic loop control.

In continuous section mills a provision is usually made for lateral shiftings of the stands along the mill shoes, to adjust the pass line. This is accomplished by employing special devices for detachable fastening of the stand feet to the mill shoes, a sliding (splined-type) joining of spindles and a stand shifting mechanism. Figure 11.38 presents the diagram of a hydraulic device for joining the stand housings with the mill shoes and Figure 11.39 is illustrative of the stand shifting mechanism.

Roll-Neck Bearings of Section Mills

In view of high specific loads on the roll necks their bearings must meet particularly high demands as to the power losses and the accuracy of the rolled pieces. In roughing stands and in those requiring no high rolling accuracy use is still made of all-in-one-piece and assembled fabric bearings.

Fabric bearings are no longer used in the last stands of rail-and-structural and heavy-section mills and in all stands of light-section and wire-rod mills because of great elastic deformations of such bearings which result in disturbed rolling process. Modern rolling mills widely employ roller bearings (Fig. 11.40) along with film lubrication ones.

To keep the roller bearings in good working order great care should be exercised in their mounting and maintenance. Therefore a special space should be allotted in the rolling shop for their assembly and the servicing staff is to be pro-

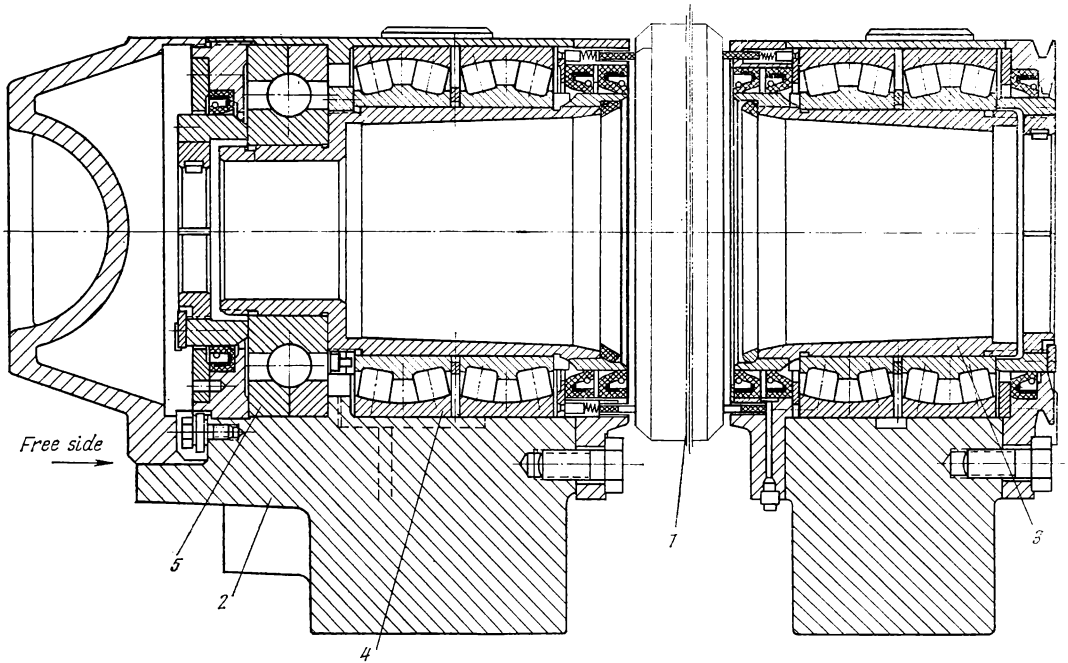


Fig. 11.40. Mounting of roller bearings

1—roll; *2*—chock; *3*—tapered sleeve; *4* and *5*—bearings

vided with the necessary equipment and instruments. Definite periods for the operation of each bearing should be fixed, on the expiration of which the bearing must be replaced.

The operation of film lubrication bearings is based on the principle of maintaining permanently an oil film between the roll neck and bearing sleeve, which creates conditions

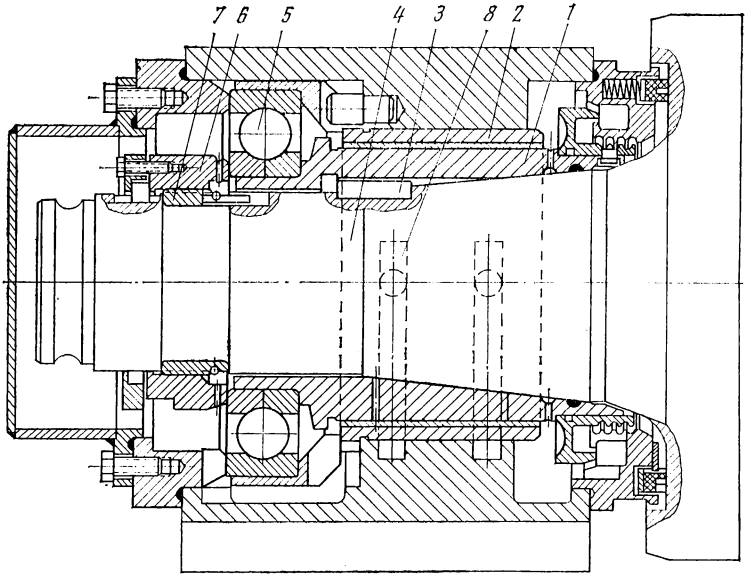


Fig. 11.41. Film lubrication bearing

1—tapered sleeve; *2*—outer sleeve; *3*—key; *4*—roll neck; *5*—bearing; *6* and *7*—rings; *8*—oil ducts

for liquid friction. The coefficient of friction in film lubrication bearings is as low as 0.003-0.005. These bearings have a high bearing capacity and provide for the necessary accuracy of rolling.

Such a bearing (Fig. 11.41) consists of two sleeves, one of them being tapered and the other, cylindrical. The cylindrical sleeve is lined with a thin layer of high-tin babbit grade Б-83. Oil supplied by a central system of lubrication is delivered under pressure to the rubbing surfaces of the bearing sleeves through oil ducts and thus a thin oil film

is permanently formed between the sleeves, this almost completely eliminating wear on the surfaces. A good sealing of the bearing assemblies is achieved by employing special oil seals made of oil-resistant rubber.

Stiff and Prestressed Working Stands

The dimensional accuracy of rolled steel depends in a large measure upon the total elastic deformation of all the elements of the working stand which take the rolling pressure.

The degree of yielding of any machine component depends on the modulus of elasticity of the material it is made of, its

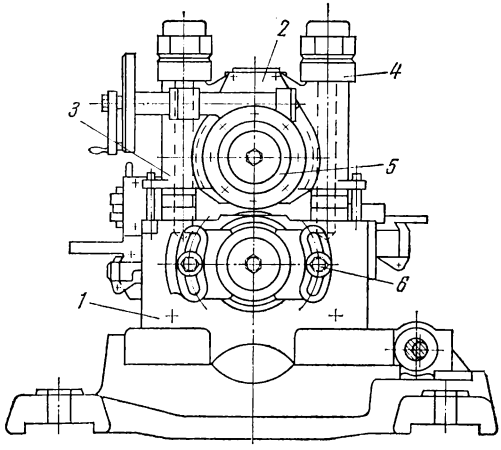


Fig. 11.42. Prestressed stand of Swedish design

cross-sectional area and the applied load. The total elastic deformation of the stand is determined by the number of mating components which take the rolling pressure and by their pliability (determined as the inverse of the stiffness factor). Of substantial importance are also the surface finish of the components, their hardness and the accuracy of their assembly.

The accuracy of rolling is improved by replacing cast-iron roll housings with steel ones, increasing the cross-sectional

area of the housing members, using close-top roll housings with the wedge-type roll adjusting device and by reducing the roll body length.

The method of prestressing the working stand components (roll chocks, housings, rolls) is employed to reduce elastic deformations of the stands. Figure 11.42 shows the prestressed working stand of a Swedish wire mill. The top and bottom roll chocks *2* and *1* are compressed with a force approximately corresponding to the rolling pressure by means of large-section screws *3* and nuts *4*. The roll gap is adjusted with eccentric bushings *5* and the lateral adjustment of the rolls is accomplished by means of coupling bolts *6*. The roll body length is only 350 mm. With prestressed stands high-carbon wire rod can be rolled with a tolerance of not greater than ± 0.1 mm.

Flying Shears of Section Mills

In section mills stationary hot and cold shears are employed as dividing ones for cutting the billets prior to rolling and also for cutting the finished rolled products to length. Their design does not differ from that of ordinary upcut eccentric- and crank-type shears.

The flying shears of section mills are installed before the first stand, to cut the billets in case of emergency (usually in continuous wire-rod mills), after the continuous roughing group of stands, to crop the head ends of the rolled pieces and also to cut the piece being rolled in the roughing group in case of emergency, and after the finishing stand, to cut the finished rolled products to length.

Simple emergency shattering shears have come into use in continuous wire-rod mills. These shears operate on the principle of free fall of a heavy blade onto the piece moving before a stand; the piece is broken as the result of its tensioning by the stand. Of a comparatively simple design are also air- or hydraulically-actuated shears installed on the entry side of the first roughing stand.

According to their kinematic principle flying shears may be divided into the following three types: rocker-type flying shears, rotary flying shears (drum-type shears with rotary blades) and flying shears with plane-parallel motio

of the blades (planetary-type, parallelogram-type, etc.). The rotary flying shears are the ones most frequently used.

The flying shears operate either continuously or on a start-and-stop duty cycle, when they are started to make each new cut and then stopped with the blades in their initial position strictly fixed by means of a selsyn. Cutting of the rolled piece on its run should invariably meet the following condition: the shear blades must "fly" at a speed equal to

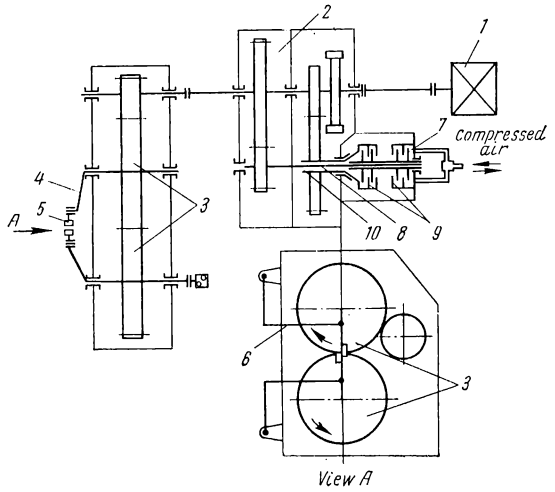


Fig. 11.43. Crank-type dividing shears

1—motor; *2*—reducing; gear; *3*—pinion stand; *4*—crankshafts; *5*—blades; *6*—rocking levers; *7* and *9*—friction disc clutches; *8*—shaft; *10*—hollow shaft

that of the moving piece. This condition is rather difficult to satisfy with rotary shears in continuous uniform motion, since the needed length of the cut piece is to be obtained simultaneously with the equalization of the speeds of the flying blades and the rolled piece.

More often than not two types of flying shears operating on a start-and-stop duty cycle, *viz.* drum- and crank-type, are used as dividing and emergency shears before the roughing group of stands and as crop and emergency ones after this group.

Figure 11.43 shows the diagram of crank-type shears for the roughing groups of continuous section mills. In the

shears of this design the driving motor and one of the speed reducers operate continuously. The shears are thrown in by means of air-operated friction disc clutches *7* and *9*. Shear blades *3* are mounted on cranks *4* which are joined with rocking levers *6* whose length is selected so as to ensure proper movement of the blades when cutting.

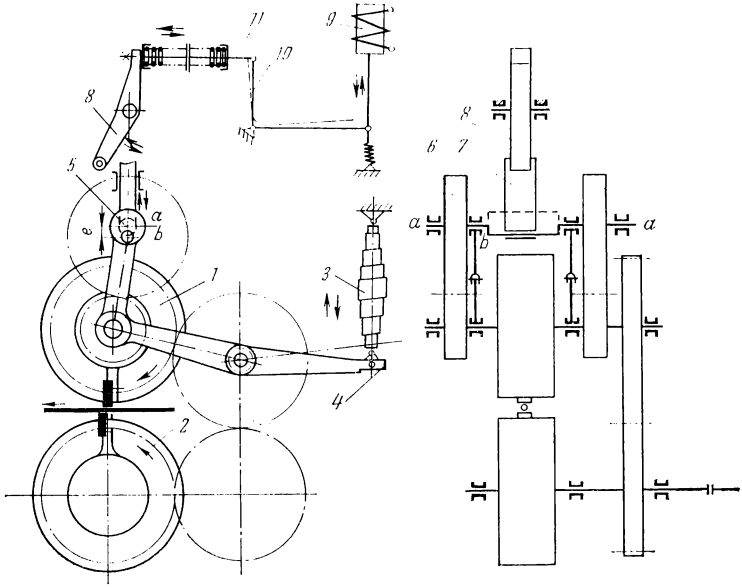


Fig. 11.44. Double-drum type flying shears of section mill

The finished rolled products moving at a speed of 15-20 m/s are cut to length with single- or double-drum type flying shears installed before the cooling bed, other types of shears, such as involute, rocker, planetary, etc., being also used for this purpose. The kinematic diagram of double-drum type flying shears is illustrated in Fig. 11.44. Upper drum *1* is lifted by spring *3* and lever *4* to ensure free passage of the rolled piece. When electromagnet *9* is energized lever *10* and connecting rod *11* shift detent *8*, thus arresting slide bar *7*; the axis of rotation of eccentric *5* descends

covering distance *e*, and top drum *1* together with its entire kinematic system is lowered over distance 2*e*, which is selected so as to ensure the engagement of the upper and lower shear blades for cutting. The rotation of drum *1* is transmitted to the shaft of eccentric *5* via gear train *6*. The axis of rotation of bottom drum *2* is stationary.

Single-drum shears comprise a drum rotating at a speed equal to that of the rolled piece. The drum carries two movable lever-operated blades held apart by a spring. The piece moves freely between the blades tangentially to the rotating drum and, when a starting impulse is given to the shears, the blades converges and cut the piece on the fly. Such shears are employed in multi-strand rolling.

Cooling Beds of Section Mills

Prior to the final cutting, straightening, packing and other finishing operations the hot rolled section steel, which is to be supplied in the form of straight bars, should be cooled so as to preserve its straightness; light-section steel should straighten when cooling. Some grades of high-carbon and alloy steel are slowly cooled on special mechanized cooling racks, after rolling.

Heavy-section and some medium-section mills use simple transfer-type cooling beds for cooling heavy-section steel. The area of these beds must be sufficient to cool all the heavy-section steel put out by the mill. The rolled metal which requires slow cooling is delivered to isothermal-cooling furnaces, pits or other special facilities.

The high rolling speeds in light-section mills have made it necessary to develop special highly mechanized cooling beds whose mass accounts for up to 50% of the total mass of the light-section mill equipment, the length of the beds reaching 120-150 m. The cooling beds may be single-sided, but more often than not double-sided beds are used. The cooling bed width is governed by the cooling area required with the given method of placing the metal and the given length of the bed. In consideration of the above circumstance and also with a view of creating two individual flow lines, one for straightening and the other for cold cutting of pieces, the double-sided cooling beds are preferable. In accordance with

the rolling method the cooling beds may be of the single- and multi-strand type.

Of prime importance in the mill layout is the adequate distance between the finishing stand with flying shears and the cooling bed, since this should provide for sufficient intervals between the rolled pieces and the braking length

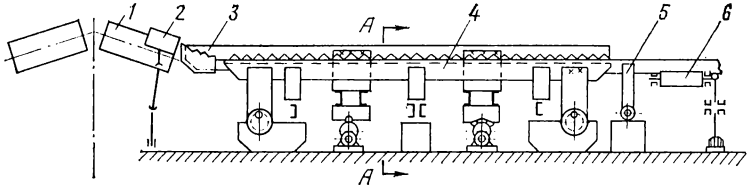


Fig. 11.45. Cooling bed of light-section mill

necessary for the piece to travel prior to reaching the kickoff devices of the cooling bed at higher speeds of the discharging roller table.

A typical cooling bed of a light-section mill (Fig. 11.45) consists of the following three main assemblies: run-in table *1* with kickoff *2*; dressing plates *3* and working section *4* comprising a system of movable and stationary racks, and run-out table *6* with transfer *5*. The run-in tables usually have shaped or slightly inclined rollers and are individually driven.

When slow cooling of the rolled pieces is required (for example, spring steel pieces) an additional packing device is used between the run-in table and the rack area, the pieces being packed standing on their edges.

Wide application find cooling beds with the transverse movement of the cooled metal effected by skew table rolls arranged at an angle of 60°. By regulating the speed of the rolls any desired distance between the cooled pieces can be set, thus making it possible to control the cooling time and utilize better the cooling bed area with the most diverse range of products. The final straightening and cold cutting of the pieces are accomplished with ordinary multi-strand straightening machines and shears.

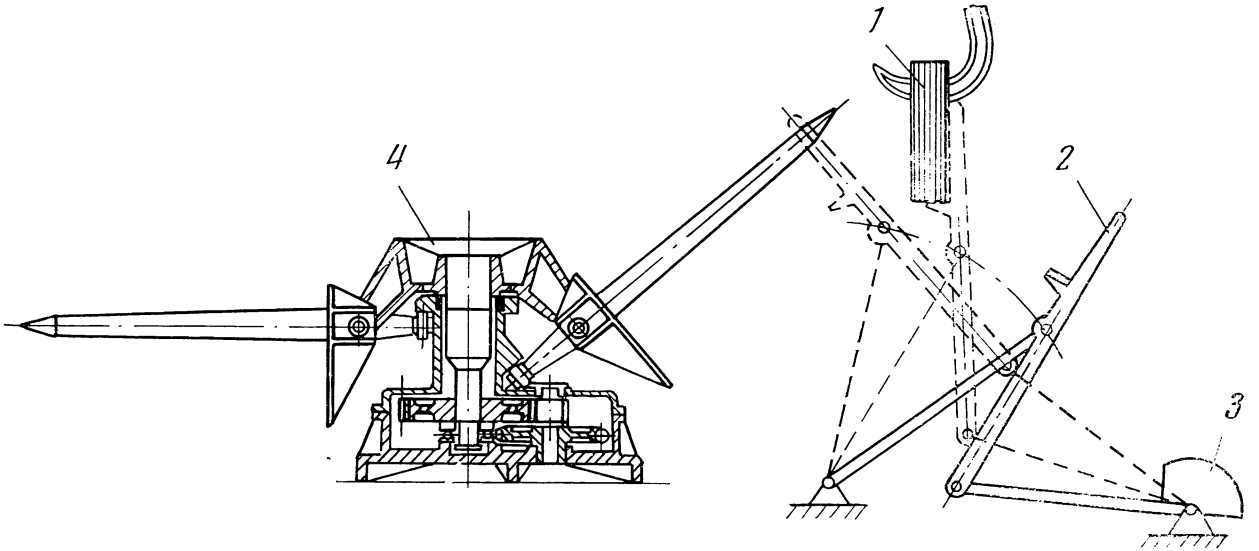


Fig. 11.46. Coil stripper

1—coil; *2*—tilting chair; *3*—counterweight; *4*—packing device

Coilers and Handling of Coiled Steel

Wire rod, light-section steel (up to 24-25 mm in diameter) and also all types of hoop, band and skelp steel are coiled. In light-section and strip mills a provision is made for handling metal either in the form of straight bars or in coils or else, both in bars and coils. The following two types of coilers are employed for wire rod and light-section steel: with

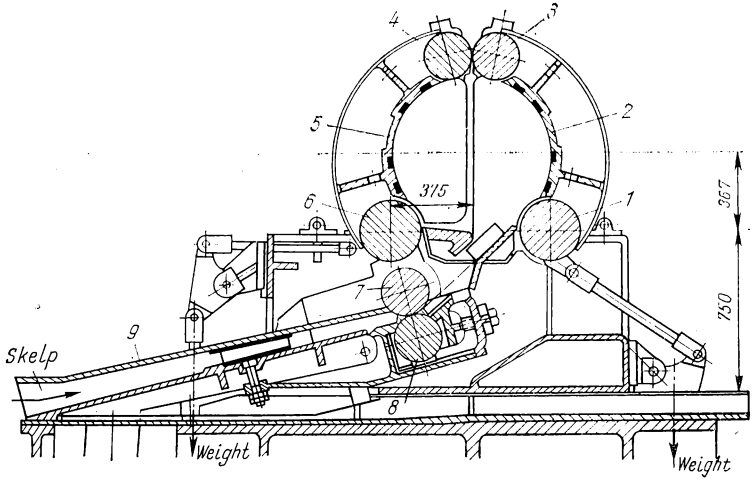


Fig. 11.47. Coiler of skelp mill
1 and *6*—driven rollers; *2* and *5*—arc segments; *3* and *4*—idle rollers; *7* and *8*—bending rollers; *9*—trough

axial and with tangential metal feeding, the first type having movable and the second, stationary drums.

When the coilers are installed at a comparatively great distance from the finishing stands special feeding devices are used to obtain the required velocity pressure of the rolled piece at the entry to the coiler.

To decrease the formation of scale and to harden the rolled metal the latter is cooled with water in special pipes or in coilers. Air cooling in coilers or in muffle conveyors is also used. Hot strapped coils usually go onto an apron conveyor wherefrom they are transferred onto a hook conveyor for

cooling. From the hook conveyor the coils are removed by means of a special coil stripper and delivered to packing machines (Fig. 11.46).

Lately the following method has been used for cooling high-grade wire rod to obtain a sorbite structure. On leaving the last stand the wire rod goes to a coiler with open flaps and then gets onto a moving apron conveyor where it is placed in loose turns which thereafter are made to pass through a special muffle with heated air, this securing uniform cooling of the wire rod.

The former practice was to use vertical-drum type coilers for handling strip steel. Since these had a disadvantage of operating not in pace with the mill they are now superseded by coilers with horizontal axis of coil rotation (Fig. 11.47), used with skelp mills. The strip is fed in-between the hinged coiling arms with rollers. As the diameter of the coil increases the arms open, but the pressure rollers keep the turns of the coil sufficiently tight.

Automation of Section Steel Production

With the introduction of high-speed continuous rolling, standard rolling technology and improved equipment, conditions favouring the automation of the section steel production have been created.

Apart from computerized integrated automatic process control, some control operations, such as weighing, temperature measurements, rolling pressure measurement, measurements of cross-sectional dimensions of the rolled products and checking the condition of their surfaces in the course of production, are sometimes automated individually.

In the USSR an integrated automatic section mill control system was first introduced in a light-section mill as far back as 1948. The main control element of the system was a rolling rate controller which set the optimum mill loading schedule; the billets were fed into the first stand at strictly controlled time intervals, thus providing for uniform, smooth loading of the entire complex of the mill equipment. The maximum possible rolling rate was determined by the capabilities of individual auxiliary mechanisms (tilters, skew roll tables, cooling bed kickers, etc.) some of which

had to be improved. Photorelays, flag switches, load indicators, limit switches and other interlocking devices were used as sensors in the control circuits of the system.

The present-day approach to the integrated automation of section mills by using improved methods and means of programming, along with control computers makes it possible to correct deviations from the standard rolling schedule on the basis of current information about the course of the process.

Experience gained in handling continuous sheet mills is now used as a basis for the development of automatic systems for controlling the operation of screw-downs and main drives in accordance with the workpiece tension between the stands. In recent years much has been done to automate the rolled metal cutting with flying shears.

Rolling Defects and Finishing of Section Steel

The profile defects of section steel caused by wrong roll pass design, improper rolling temperature, bad condition of roll passes, bad condition and wrong adjustment of roll fittings and wrong adjustment of rolls may be of the following types:

1. Underfilled section due to an insufficient cross-sectional area of the workpiece delivered from the leader stand or due to wrong adjustment of the finishing pass.
2. Unilateral underfilling of a round section due to an incorrect (asymmetrical) mounting of guides in the finishing stand.
3. Bilateral fins due to an excessive cross-sectional area of the workpiece delivered from the leader pass.
4. Unilateral fins due to an asymmetrical mounting of guides in the finishing stand.
5. Displacement of the upper part of profile relative to its lower part due to a lateral displacement of rolls.

Laps may develop as the result of worn roughing passes, wrong setting of roll fittings and bad condition of their surfaces.

Backfins are sometimes caused by tumbling or undercutting of the rolled section in the roughing and intermediate stands, by inadequate tilting of the workpiece, as well as

due to improperly mounted roll fittings and wrong roll adjustment.

To cut down the number of defects it is necessary to properly organize the roll and fitting service, strictly regulate the schedules for changing rolls, roll passes, fittings and bearings, to keep all the working stand components in sound condition, etc.

An indispensable condition for obtaining high-quality products is a continuous process control. This envisages not only taking, visual inspection and measurement of specimens, but also their torsional, bending and other tests. In a number of instances the specimens should be subjected to pickling and hot upsetting tests (for heavy-section steel).

In rolling high-grade steel prone to the formation of seams and other surface flaws piecemeal inspection and elimination of defects is organized in the finishing bays of section mills. It is not only the elimination of the defects, but also examination and filing for the purpose of detection and marking of such defects, that is an extremely labour-consuming operation. In some cases the bars undergo all-round surface conditioning with centerless grinding machines (rounds) or with surface-grinding machines (squares and strips).

To facilitate the surface defect finding outside the main process flow of the mill the bars are made to undergo pickling or shot-blasting; partial surface cleaning with grinding wheels (by grinding ring, serpentine or helical strips), and ultraviolet surface defectoscopy.

The surface defects of finished rolled products are eliminated with grinding machines, air hammers being also used for conditioning heavy-section steel.

Apart from surface conditioning, finishing operations also include straightening, cutting, packing, strapping and marking.

The finishing operations are, as a rule, done in the bays of the finished product store whose floor area should be sufficient to prepare the metal for shipment and should also provide for possible accumulation of products in case of an irregular supply of railway cars. The total floor area of the store, including that occupied by the equipment, transport facilities and communications, is determined at a rate of 50-100 tons of annual product output per square metre.

The storage, finishing and shipment bays of the store are equipped with special cranes having transverse beams. Wide application find cranes with a hoisting capacity of up to 15 t equipped with two trolleys or with slewing devices.

The technology of finishing operations and the needed equipment are selected and the required floor area determined so as to match the given range of products, account being taken of a possible increase of the products output.

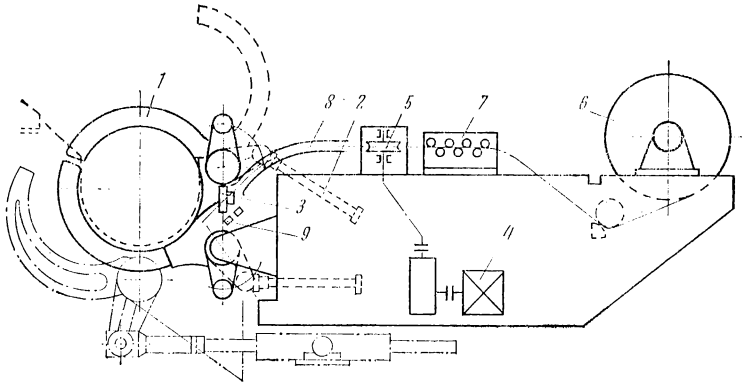


Fig. 11.48. Pack-strapping machine

Products rolled from ordinary and some high-grade steels undergo finishing (straightening, cutting, assorting and packing) in the main process flow of the mill. Medium- and large-sized pieces are straightened in nine-roll straightening machines equipped with vertical rolls at their entry and delivery sides. The speed of straightening heavy beams reaches 4 m/s, that for medium-sized sections is up to 6 m/s. Machines for straightening beams larger than No. 60 size and wide-flange beams with a flange height of up to 300 mm operating at a speed of up to 1.7 m/s are also available.

The assorting of products in the main process flow of the mill is done on inspection tables, prior to cutting to length, and also on discharging racks after cutting, the assorted products being loaded into corresponding cradles. Also finds application a processing pattern in which the rolled pieces are preliminarily cut into large lengths (25-30 m)

in the main process flow of the mill after they leave the cooling bed, and then the cut lengths are subjected to an additional straightening outside the main process flow, followed by cutting to length in accordance with order specifications. Wire rod is inspected while it is cooled on a hook conveyor.

Among finishing operations the most labour-consuming ones are packing and strapping. The diagram of a pack-strapping machine is shown in Fig. 11.48. Tongs *1* with hydraulic drive *2* compress the pack and then wire twisting head *3* is brought into its working position. The head has two 10 mm dia holes through which the tie wire passes prior to entering the lower jaw of the tongs and after leaving the upper jaw. The tie wire supplied from coil *6* mounted on a shaped payoff reel passes through roller straightener *7*, feeding unit *5* with drive *4*, guide *8*, the first hole in head *3*, tongs *1* and the second hole in the head and is held in grip *9*. When strapping, the upper jaw of the tongs is lifted and head *3* twists the ends of the tie wire. The wire is then cut with hydraulically-operated shears.

Technical and Economical Characteristics of Section Steel Manufacture

The rolling mill output depends on the rolling rhythm, range of products turned out by the mill, nominal working time, downtimes and rolling-mill use factor.

The annual output figures for modern rolling mills are as follows (millions of tons):

Blooming (ingots)	5.0-7.0
Slabbing (ingots)	6.0-10.0
Rail-and-structural mill . . .	2.0-2.5
Heavy-section mill	0.7-1.6
Medium-section mill	0.6-1.4
Light-section mill	0.5-0.8
Wire-rod mill	0.55-0.8

The elevated metal consumption in roll cogging (determined by the overall or particular metal consumption factors) is conditioned by high inevitable metal losses.

The iron loss in heating and rolling in blooming and slabbing mills is 2.5%. For rimming steel the metal losses due

to croppage and rejects come to 3-5% for blooms and 9-12.5% for slabs; those for killed steel amount to 13-16% for blooms and 15-17.5% for slabs. Correspondingly, the metal consumption factor for rimming steel is 1.08 for blooms and 1.18 for slabs, that for killed steel being 1.230 and 1.250 for blooms and slabs respectively. With continuous casting of steel the metal losses at the first rolling stage are considerably reduced.

For continuous billet mills the net yield amounts to 98.5%, the iron loss reaching 0.5% and additional metal losses due to croppage, approximately 1%.

The fuel consumption in heating ingots depends upon the average temperature of the metal charge, the number of cold ingots, the steel grade and the soaking pit design. With a cold metal charge amounting to 5% the consumption of reference fuel per ton of heated metal averages $270 \cdot 10^3$ kcal. The power consumption per ton of rolled metal averages 12 kW·h for blooming and slabbing mills and 15-20 kW·h for continuous billet mills.

The water consumption per ton of rolled metal comes to 0.5 m^3 for soaking pits, 2.5 m^3 for blooming mills and 3 m^3 for continuous billet mills.

The roll consumption per ton of rolled metal equals 0.12-0.15 kg for blooming mills and 0.25-0.30 kg for continuous billet mills (for both groups of stands).

The yield of the first-grade products is a very important characteristic of rolling mills. For the rail-and-structural mills it reaches 90-95%, therefore the ingot consumption per ton of the first-grade products comes to 1.39-1.47 t, the overall ingot consumption being equal to 1.325 t per ton of rolled products.

For reheating furnaces the fuel consumption per ton of rolled metal approximately equals $170 \cdot 10^3$ kcal.

In rolling rails the power consumption ranges from 25 to 50 kW·h per ton of rails, depending on the rail type, the water consumption equals 5 m^3 and the roll consumption, 2-2.5 kg per ton of rails.

For section and wire-rod mills the consumption figures per ton of rolled metal are as follows: billets — 1.05-1.03 t (greater figure is for larger sections); fuel — $500 \cdot 10^3$ kcal; power — 35-80 kW·h (greater figure is for wire-rod mills);

water — 8-15 m^3 (greater figure is for wire-rod mills); rolls — 0.35-1.2 kg (smaller figure is for light-section and wire-rod mills).

Accident Prevention Regulations in Section Rolling Shop

The design of roll fittings is of great importance in ensuring safe handling of rolling mills. The stands of hand looping mills should be well guarded by protective posts, combs and fenders. High-speed mills, as well as wire-rod ones are, as a rule, guarded by nettings which cover the stands on top and on both sides.

The mill tables should be fenced. All gangways at the site of mill stands must be guarded by nettings. The passage of men between the stands of continuous rolling mills is not to be allowed. The rolling fields of line-type mills should be completely fenced. Appropriate shock blocks should be used to prevent jumping out of the rolled metal loops and to protect personnel against injuries by the whipping end of the rolled piece. Intermediate troughs must be covered.

Scrap ballers (for light-section and wire-rod mills) should be installed at sites presenting no danger to personnel.

Internal water cooling of floors and forced humidity ventilation of workplaces should be arranged. Rolled product specimens are to be taken at safe sites only. Cutting, straightening and handling of rolled products should be remotely controlled.

CHAPTER XII. SHEET AND PLATE PRODUCTION

The percentage of sheet and plate products in the modern range of rolled steel is constantly increasing. This is conditioned not only by the expansion of production in some branches of industry, but also by an ever increasing introduction of sheet metal into the machine building industry (in place of labour-consuming castings and forgings), by the development of welded structures, by the expansion of pro-

duction of welded pipes, including those for main pipelines, etc.

The range of sheet and plate steel is greatly diversified and includes products, such as hot-rolled plate and sheet, cold-rolled sheet, sheet metal coated with tin, zinc and other metals and plastics, etc.

Rolled or cast (in continuous steel-casting machines) flat billets — slabs — serve as starting material for making plate and sheet steel. Rolling of plate steel directly from ingots (common to old plate mills), as well as rolling of sheet steel from strip, sheet bar, sheet billets and mill packs, has been sharply reduced.

A trend is now observed towards increasing the weight of rolled and cast slabs up to 25-30 t which makes for the production of built-up coils. The hot rolling of sheet steel is done in continuous and semi-continuous mills. At present there are more than 100 of such mills in the world, their aggregate annual output reaching 130 million tons. There is also a tendency to increase the weight of coils and reduce the thickness of hot-rolled sheet (down to 1.2 mm). High-output cold-rolling shops designed to produce heavy coils (up to 45 t), as well as multiroll stands for rolling extremely thin band and sheet steel have been built.

12.1. Plate Steel Production

In accordance with Standards plate steel is from 4 to 160 mm thick and from 600 to 3800 mm wide, the plate length ranging from 2 to 20 m.

Plate steel may be rolled from ingots and from slabs. Heavy plates for welded housings, frames and high-pressure boilers are rolled from ingots weighing 8-250 t.

The size of slabs for rolling plate is selected within a range of 150-500 mm in thickness, 600-2000 mm in width and 2-6 m in length so as to suit the size of the product to be rolled.

Ingots are heated in soaking pits or in car-bottom furnaces. The latter method is very laborious and inefficient and therefore it is now used only at old iron and steel works. At some works which have no cogging mills plate steel is rolled directly from ingots in combination mills.

In the manufacture of sheet and plate steel (especially in the latter case) of paramount importance is the correct consignment of heats, ingots or slabs in accordance with the chemical analysis data, qualitative characteristics of the slabs (taking into account the zonal structure of ingots from which the slabs are rolled) and their size, considered from the standpoint of meeting the order specifications on the finished, trimmed sheet or plate.

Empirical formulas are used at iron and steel works to correctly consign the heats in accordance with the chemical analysis data so as to obtain the required properties of plate steel of the given thickness, rolled at the set reduction schedule and temperature. The possible laminations and segregation in each zone along the height of the ingot are considered depending upon the order specifications.

To achieve the most economical cutting-up of the rolled piece into sheet or plate one must adequately select the mass and initial dimensions of the ingot and slab. Such procedures in the sheet and plate steel manufacture are within the purview of the so-called fabrication. There also exists a notion of the fabrication factor, i.e. the actual metal consumption in the manufacture of finished, trimmed sheet or plate, as distinct from the technological metal consumption factor which takes account of optimum iron loss and the end and side trimmings.

The net yield of sheet steel depends to a great extent on the shape of the end and side edges of the rolled piece to be cut into sheets, which in its turn is contingent upon the adjustment and the condition of rolls, as well as upon strict observance of the adopted rolling procedures. For the quality of sheet steel of great importance are the scale removal in the course of rolling, proper reduction schedule, adequate roll cambering and rolling temperature.

Methods of Obtaining the Required Width of Plate Steel

To obtain a rolled piece of a regular rectangular shape and of the required width with a stock provided for the edge trimming the following three methods are used, depending on the rolling mill design, the adopted rolling procedures

and the initial dimensions of the ingot or slab: when the width of the ingot or slab corresponds to that of the plate to be rolled the rolling is done lengthwise; when the width of the ingot or slab is less than that of the plate the required width of the rolled strip is obtained by two methods, namely, by broadside rolling and by rolling at an angle, both of them being used in the roughing passes.

The broadside rolling reduces the anisotropy (longitudinal fibering) of plate steel, but may cause laminations as the

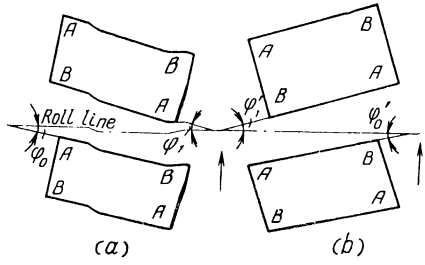


Fig. 12.1. Rolling at an angle
(*a*) workpiece feeding with corner *A* leading; (*b*) workpiece feeding with corner *B* leading

result of lateral deformation and changes in the shape of non-metallic inclusions and, therefore, special care should be exercised when using this method in rolling special-purpose plate steel.

When rolling at an angle, i.e. when the ingot or slab is fed into the rolls at an angle to the roll axis, both the longitudinal and lateral deformations occur simultaneously, their degree being dependent upon entry angle φ_0 (Fig. 12.1). The actual width of the rolled piece is determined by the formula proposed by I. Pavlov:

$$b_1 = \frac{b_0 \mu}{\sqrt{1 + \sin^2 \varphi_0 (\mu^2 - 1)}}$$

where b_0 = initial width of the slab
μ = coefficient of draught

After the first passing the rectangular workpiece fed into the rolls with one of its corners leading assumes the shape of a skewed parallelogram and then, after the next passing, a rectangle will be obtained again, if the workpiece is now

fed into the rolls with its other corner leading, provided the draught and the entry angle are the same.

Although rolling at an angle is advantageous in that it makes for levelling out of lateral deformations and ensures a gradual rise of the roll load, but with this method it is not always possible to exactly maintain the required entry angles and, consequently, obtain the needed width and rectangular shape of the plate.

Descaling

To obtain plate steel free from rolled-in scale it is essential that furnace and mill scale be removed from the workpiece surface prior to and in the course of rolling. The following descaling methods are employed:

1. Scale removal by scattering birch twigs, common salt or ground charcoal over the workpiece surface (old method).
2. Mechanical descaling with chains, fluted rolls, etc.
3. Compressed air or steam descaling with preliminary scale breaking in rolls.
4. Scale removal with a jet of water delivered at a pressure of 50-120 at (hydraulic jet descaling).

The lastly named procedure is the most effective current descaling method, but it can be used only when a scale breaker and high-pressure water supply (not below 100 at) are available. In the scale breaker the workpiece is rolled with a small reduction (5-10%), the hydraulic jet descaler being put into operation automatically for short periods of time. Water is delivered through nozzles with an orifice diameter of 1.2 mm, which are set at a small angle to the workpiece surface. More often than not the nozzles are mounted on the guide bars at a distance of 200-300 mm from the workpiece surface.

When rolling plate steel use is sometimes made of tilters, in order to ensure similar conditions for removing scale from the top and bottom surfaces.

Reduction Schedule

Among general factors limiting the degree of reduction are: the strength of the working stand members, the capacity of the mill drive, the roll bite conditions and the plasticity

of the rolled metal, the effects of these factors being dissimilar for the roughing and finishing passes.

In the last finishing passes temper rolling is frequently used to reduce lateral gauge variations and the sheet buckling. For some high-grade steels the degree of reduction in the finishing passes is determined by the conditions necessary for obtaining the required metal structure at the given rolling temperature. Generally speaking, in the finishing passes the roll bite conditions have no effect on the degree of reduction and this is limited only by the rise of the rolling pressure with a small thickness and lower temperatures of the rolled metal.

In the first roughing passes, where the plasticity of the rolled metal is high, use is made of the maximum reductions which are possible to achieve with the given strength of the rolls and the drive capacity, the roll bite conditions being also taken into consideration.

Proceeding from the roll strength the maximum possible reduction is

$$\Delta h = \frac{1}{R}\left(\frac{P}{p_m b}\right)^2$$

where R = roll radius
p_m = mean specific roll pressure
P = permissible rolling pressure
b = width of the rolled piece

The permissible rolling pressure is given by the following formula:

$$P = \frac{0.4D^3R_b}{a - \frac{b}{2}}$$

where D = roll diameter
a = distance between the roll neck centres
R_b = permissible bending stress

The temperature drop for each consecutive pass may be determined from the formula proposed by V. Tyagunov:

$$\Delta t = \frac{t-400}{16} \cdot \frac{z}{h}$$

where t = temperature of the rolled piece in the preceding pass

h = thickness of the rolled piece in the preceding pass

z = time in rolls plus time out of rolls for the given pass

In the first passes the same reductions are set for most grades of low- and high-carbon and alloy steel because at high temperatures they have approximately the same resistance to deformation.

The total reduction is distributed among the finishing and roughing passes so as to ensure uniform loading of the stands.

Roll Cambering

The shape of the sheet roll generatrix must compensate for the roll bending and also for its bellying as the result of uneven heating during hot rolling. The roll wear and the conditions necessary for securing the stability of the workpiece in rolling should also be taken into consideration when cambering sheet rolls.

When rolling in a four-high mill the rolling pressure is transmitted through the work rolls to the backup rolls. The character of the distribution of this pressure along the backup roll bodies depends on the width of the workpiece and the stiffness of the work rolls. If the rolling pressure is assumed to be transmitted uniformly, then the total camber of the backup roll will be

$$f = f_1 + f_2$$

where f_1 = roll camber due to bending moments

f_2 = roll camber due to shearing forces

In their turn

$$f_1 = \frac{P}{18.8ED_{b.r}^4}\left\{8a_1^3 - 4a_1L^2 + L^3 + 64c\left[\left(\frac{D_{b.r}}{d_{b.r}}\right)^4 - 1\right]\right\}$$

$$f_2 = \frac{P}{G\pi D_{b.r}^2}\left\{a_1 - \frac{L}{2} + 2c\left[\left(\frac{D_{b.r}}{d_{b.r}}\right)^2 - 1\right]\right\}$$

where P = total rolling pressure

E = modulus of elasticity of the roll material

G = modulus of transverse elasticity of the roll material

$D_{b.r}$ = diameter of the backup roll
$d_{b.r}$ = diameter of the backup roll neck
L = roll body length
a_1 = distance between the axes of the backup roll bearings
c = distance from the edge of the roll body to the bearing axis

The cambering of the work and backup rolls has a great effect on the distribution of the load between them. If the camber of the rolls is known, the gauge variation may be found.

The roll bellying due to uneven heating is determined proceeding from temperature difference Δt between the centre and the edges of the roll body. Diameter increment due to thermal expansion ΔD depends upon the coefficient of linear expansion α of the roll material and temperature difference Δt:

$$\Delta D = \alpha D \Delta t$$

where D = roll diameter at the temperature prevailing in the shop
Δt = 30-40°C for the work rolls
Δt = 5-10°C for the backup rolls

The roll camber relative to the width of the rolled plate is

$$\Delta D_{p.w} = \Delta D \frac{b}{L}$$

To compensate for the roll wear the roll bodies are made with a convex camber, this also securing better stability of the workpiece when rolling in plain rolls. In two-high roughing stands use is made of cylindrical rolls. In order to ensure the stability of the workpiece in four-high finishing stands the shape of the roll gap must be slightly convex (0.25-0.30 mm) when the rolling pressure is applied to the rolls. In this case the parabolic concavity of the original work roll contour comes to 0.40-0.51 mm. The backup rolls are cylindrical.

Most frequently the top and bottom rolls of the three-high Lauth stands are cylindrical, while the middle one is at first made with a concave camber (from 0.6-0.8 to 0.15-

0.05 mm), and then, with a heavier wear of the external rolls, it is redressed so as to have a slight convex camber.

To reduce the gauge variations new rolls are usually used first for rolling wider plates and then, as the roll wear proceeds, for rolling narrower plates.

Temperature Conditions of Rolling

The drop of the metal temperature in the course of rolling is determined either experimentally or by empirical methods, account being also taken of the effects of water cooling and hydraulic jet descaling.

In rolling some grades of steel the hydraulic jet descaling is used with a great care so as not to allow the metal temperature at the end of rolling to fall below 800-1050°C, since this causes work hardening of the metal. The optimum metal temperature at the end of rolling is 820-920°C. At higher temperatures the grain growth is observed, this deteriorating the plastic properties of the metal, especially its impact strength. A slight cooling of the workpiece prior to rolling in the last passes has sometimes to be resorted to. In this case it is very important to provide conditions for uniform cooling because uneven temperature distribution over the workpiece section leads to lateral gauge variations.

Plate Finishing

The plate finishing operations include hot levelling, cooling, inspection, marking out, trimming, taking samples, cutting to length, identification marking, surface conditioning and final inspection. If necessary, plates are subjected to heat treatment, pickling, shot-blast descaling and oiling. Prior to delivery the plates are assorted in accordance with order specifications and weighed.

Efforts are made to do all finishing operations in the main process flow of the mill, for this reduces labour consumed in finishing and the floor area required for the finishing equipment. After rolling the hot plates are levelled in roller levellers at a temperature of 680-700°C. Plates less than 8 mm in thickness are difficult to level. In this case to improve levelling the leveller is adjusted so as to carry out

the process under pressure, i.e. the gap between the rollers is decreased.

Plates up to 25 mm thick and those with a thickness of 25-40 mm are levelled in levellers equipped with 200-300 and 250-450 mm dia rollers respectively. In some cases levelling is done in stretching machines.

Prior to levelling the plates are cooled on tables equipped with disc-shaped or fluted rolls. When the metal temperatures at the end of rolling are high, the plates are cooled with water delivered through nozzles at a pressure of 10 at. After hot levelling the plates are cooled on cooling beds, conveyors and roll tables. Some grades of plate steel are slowly cooled to improve their plastic properties and prevent the formation of flakes, the plates being stacked at a temperature not below 400-500°C.

The inspection and cleaning of plates in the main process flow of the mill are done on inspection tables. More often than not the bottom surface of the plates is inspected and cleaned in the final inspection area so as not to impede the process flow.

Plates up to 20 mm thick are trimmed with rotary side trimming shears, those up to 50 mm thick, with guillotine shears, while plates with a thickness of more than 50 mm are trimmed with cutting torches outside the main flow line. Cutting of the plates to length is done with guillotine shears.

When the finishing operations cannot be organized in the main production flow, individual flow lines are set up which may include shears, levelling machines, etc.

Should it prove impossible to carry out on these lines all the finishing operations and assort the plates in accordance with the order specifications, special areas equipped with magnet cranes and additional finishing equipment (cleaning, cold levelling, cutting, heat treatment, pickling and oiling) are allotted.

Heat treatment, pickling (or shot-blast descaling) and oiling are used for some grades of special-purpose and alloyed steel. Heat treatment comprises all kinds of annealing (including homogenizing to level out chemical and structural inhomogeneity), hardening, tempering and normalizing. Plate steel is treated in car-bottom batch-type furnaces, cover-type annealing furnaces and roller-hearth continuous

furnaces. Hardening presses are used to prevent the plate warpage in hardening.

Pickling is done in acid or alkali baths and also by electrolytic methods, the latter being used for stainless and acid-resistant plate steel. Shot-blasting is the most efficient method of cleaning plate steel. Oiling of plates for the purpose of corrosion protection either is done with oiling machines equipped with flannel rollers or the protective oil film is applied by means of nozzles.

Plate Mills

In contrast to section mills the finishing rolls of plate mills have a greater body length, this being a factor determining the maximum width of the plate which it is possible to obtain.

The working stands of plate mills may be of the following types: two-high reversing and non-reversing, three-high (more often three-high Lauth stands with the idle middle roll of a smaller diameter), four-high reversing and non-reversing, and universal (stands with vertical rolls). Plate mills may also include individual vertical roll stands.

In accordance with the number and layout of their working stands plate mills are classed as single-stand (two-high reversing, four-high reversing and three-high Lauth mills), two-stand line-type and tandem, semi-continuous and continuous mills.

Single- and two-stand line-type mills are seldom used since their efficiency is low and they cannot produce high-quality products. Two-stand tandem plate mills are customarily referred to as specialized mills for rolling plate steel over 8-12 mm in thickness and over 1850 mm in width. Thinner and narrower plates are more economical to roll in semi-continuous and continuous mills. New specialized plate mills are equipped with rolls of 2000-5000 mm body length (more often than not, of 2800-4300 mm).

There exist various designs of two- and three-high roughing and finishing stands. At the currently operating mills three-high stands are being replaced by two-high ones. As a rule, new mills have four-high stands and, occasionally, a two-high roughing stand. Vertical roll stands have found

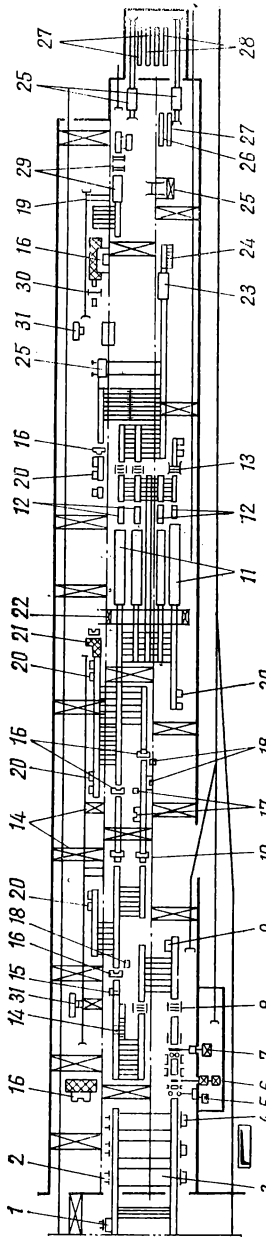


Fig. 12.2. 2800 mm plate mill

1—charging tables with pushers; *2*—furnace pushers; *3*—reheating furnaces; *4*—furnace shock-absorbers; *5*—vertical roll stand; *6*—two-high stand; *7*—four-high universal stand; *8*—leveller; *9*—hot plate piler; *10*—rotary disc shears; *11*—end-charged end-discharged roller-hearth furnaces; *12*—hardening presses; *13*—cold leveller; *14*—plate turnover; *15*—machine for marking out; *16*—guillotine cut-up shears; *17*—guillotine side shears; *18*—scrap shears; *19*—semi-gantry cranes; *20*—plate piler; *21*—guillotine shears with caster bed; *22*—magnet trolley; *23*—shot-blasting unit; *24*—plate turnover; *25*—transfer car; *26*—vat with melt; *27*—vat with water; *28*—vat with acid; *29*—scrubbing-and-drying unit; *30*—oiling machine; *31*—scales

wide application. These stands are installed before the horizontal roll stands, this arrangement ensuring the production of plates with good edges and increasing the net yield. The vertical roll stands are sometimes installed at a distance of up to 10 m after the four-high stands.

The layout of a 2800 mm plate mill with a two-high roughing stand and four-high universal finishing stand is shown in Fig. 12.2. The mill is equipped with a roller leveller which is installed in line with the main rolling equipment. The latter is arranged in three bays and to reduce the total mill length the plates are crosswise transferred from bay to bay. The transfer banks are also used for inspection. The shop is provided with modern finishing equipment including rotary side trimming shears, guillotine shears, four roller-hearth continuous furnaces installed in line with hardening presses, pick-

lers and oiling machine. Light-duty mechanical handling equipment is employed in the finishing bay to relieve the cranes operating in the main bays.

The output of modern plate mills (of the two-stand type) reaches 900-1200 thousand tons per year, but there are also specialized semi-continuous mills of a higher capacity intended to roll plates more than 3000 mm in width. They are equipped with a separately installed vertical roll stand, a broadside stand and a continuous finishing group of stands. The present-day specialized plate mills are fitted out with automatic programmed control systems for rolling, marking out and finishing of plates in accordance with order specifications.

In four-high stands the work rolls with a body length of up to 4000 mm usually have a diameter of 900-990 mm and their backup rolls are 1500-1600 mm in diameter; with the roll body length greater than 4000 mm the diameter of the work rolls is 960-990 mm and that of the backup rolls is 1700-1800 mm. In some mills the roll body length in the roughing stands is greater than that in the finishing stands this making it possible to obtain the required width of the plate by using longer slabs.

Plates up to 8-12 mm thick are often obtained in continuous and semi-continuous sheet mills in two ways: from the roughing stands (the rolled pieces being then delivered to a special plate finishing flow line) and from the finishing group of stands (mostly in coils). In the latter case suitable means are envisaged to convert the coils into plates according to order specifications.

12.2. Hot Rolling of Sheet Steel

Sheet steel up to 4 mm thick is put out in a wide range of products differing in their dimensions, application and grades.

According to the method of its production sheet steel is classed as hot- and cold-rolled. The thickness of hot-rolled steel is usually 1.5-2.0 mm and that of cold-rolled steel, 0.2-4 mm.

In modern sheet steel production the following two hot rolling methods are mainly employed: pack rolling (from mill packs) and coil rolling (from slabs).

The coil rolling of sheet steel is the most effective method, but the pack rolling is still used in the manufacture of tin plate, roofing iron, electrical grade sheet and special steel, especially when only light-weight billets can be used as the starting material.

The pack rolling of sheet steel is done in pass-over mills with unbalanced idle top rolls, three-high Lauth mills being also employed for this purpose. In these mills the stands are arranged in a single line. Batch- or conveyor-type furnaces for heating sheet bars, breakdowns and packs are installed close to the stands.

The pack rolling is done either with open packs (the Urals method) or with close packs (the European method). With the former method only one sheet is obtained from each billet (sheet bar), while with the latter, several sheets are rolled from one billet. The thickness of the finished sheet ranges from 0.33 to 1.5 mm, depending on the number of sheets in the pack (pairs, fours, eights). Rolling is done crosswise. The degree of reduction is controlled by following the length of the sheet obtained. The packs are rolled in pairs (while one pack is rolled, the other is dragged over the top roll). After folding or doubling, the doubles (pairs, fours, eights) are rolled in two packs as well. Folding is done by sheet doublers which are equipped with shears for trimming uneven ends of the folded sheets. In the final doubling the folds of the sheets in the pack are trimmed off and then the sheets are separated.

In some modern mills the pack rolling of high-alloy sheet steel is done in mechanized four-high reversing stands serviced by rotary-hearth furnaces.

The finishing operations are completed with close annealing and cutting to length and sometimes include dressing, planishing and levelling.

The coil rolling method is typical of semi-continuous and continuous mills, as well as of reversing mills with furnace coilers (Steckel mills) and planetary hot mills. The latter two types of mills have not found wide application, their use being limited to the rolling of special steel grades when the volume of production is small.

The reversing mills with furnace coilers usually have two stands, *viz.* a two- or four-high roughing stand and a four-

high finishing stand (Fig. 12.3). The coilers, which are located in furnaces at a temperature of 1000-1150°C, start to coil the rolled strip immediately after the first passing when the strip thickness reaches 20 mm. These mills are used for steels which require rolling in a narrow temperature range.

The planetary mills comprise two backup rolls rotating in opposite directions which carry cages with work rolls of

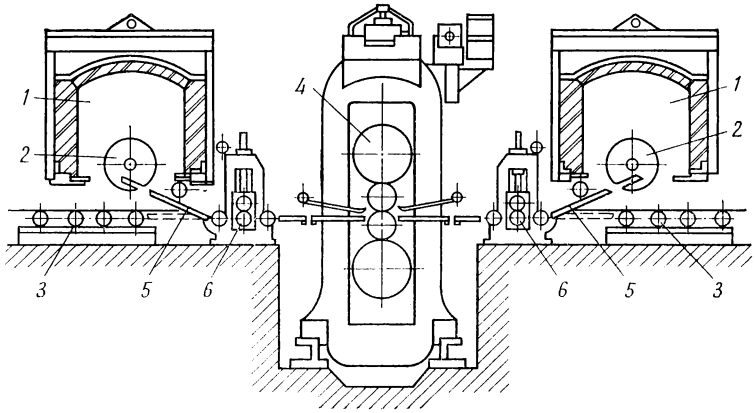


Fig. 12.3. Working stand of four-high mill with furnace coilers
1—hood-type annealing furnace; *2*—furnace coiler; *3*—roll table; *4*—four-high reversing stand; *5*—hinged guide; *6*—pinch rolls

a small diameter. The work rolls are rotated around their axes through friction by the backup rolls, the direction of their rotation being opposite to that of the backup rolls and to the rolling direction. The workpiece is fed into the mill by a special device (feed rolls, two-high stand). After the planetary mill there is a planishing stand. In such a mill a twentyfold elongation can be obtained in one passing. The metal deformation conditions specific to the planetary mill and its narrow rolling temperature range make it possible to use such mills for rolling high-alloy steel grades. Leading in the progress of hot rolling of sheet steel are continuous and semi-continuous wide-strip mills which are distinguished by their high efficiency and provide for the most effective further cold rolling.

With semi-continuous and continuous wide-strip mills the processing pattern is as follows: heating of slabs in continuous furnaces, rolling in the tandem roughing group of stands fitted out with a scale breaker and an edger, rolling in the continuous finishing group of stands and coiling of the finished strip.

The slabs are heated in double-fired multizone continuous furnaces equipped with water-cooled skid pipes or with heat-resistant skids. With the cold metal charge the productive capacity of the furnaces is 100-180 tons per hour.

Rolling in the Roughing Stands

The heated slabs are end-discharged from the furnace onto a roll table which delivers them to the scale breaker of the roughing group of stands. In the scale breaker comprising a horizontal roll stand the furnace scale is broken, the slab being reduced by 10-25%. Earlier it was thought that with great reductions scale might be rolled in, but the recent trend is to increase reductions in the horizontal roll scale breakers. The horizontal rolls are fluted to facilitate the scale breaking. Vertical roll stands are also used as scale breakers. Like in plate mills the scale is removed with water delivered at a high pressure, special attention being paid to the proper (automatic) control of the hydraulic jet descaler, the correct orientation of the descaler nozzles relative to the workpiece surface, and also to the adequate regulation of the water flow in accordance with the temperature of the rolled metal.

When the mill is fed with narrow slabs, the workpieces delivered from the scale breaker enter a four-high broadside stand, the pieces being turned through 90° by means of an air- or electrically-operated lift and turn device, installed between the fluted rolls of a roll table. In most cases the workpiece in the broadside stand is rolled in a single passing with a reduction of 40-50%, but with the broadside stands of the reversing type the required width of the strip is sometimes obtained in several passings. Powerful horizontal hydraulic presses (squeezers) were formerly used to eliminate the side camber of the workpieces delivered by the broadside stand by squeezing them on the sides. At

present these presses are almost out of use and the side camber is eliminated with vertical rolls. New mills are, as a rule, fed with wide slabs and therefore they include neither broadside stands, nor squeezers.

In semi-continuous mills the roughing and broadside rolling are accomplished in two- or four-high reversing stands, three-high Lauth stands being also used for the purpose.

The roughing group of continuous mills includes from four to six four-high stands. More often than not the roughing stands are of the universal type and their vertical rolls are usually installed at the entry side. Sometimes use is made of separately installed vertical roll stands. The stands of the roughing group operate with the maximum reductions which are determined mainly by the roll bite conditions. For ground cast-iron roughing rolls the maximum angle of bite equals 16-17°. To increase the achievable reduction the rolls are sometimes knurled or the cast-iron rolls are replaced by steel ones.

As the necks of the work rolls are their weakest points they are checked for torsion at the maximum rolling pressure:

$$P_{\max} \leqslant \frac{0.4 d_{n.w.r}^3 \sigma_t}{\sqrt{\frac{D_{w.r}}{2} \Delta h}}$$

where $d_{n.w.r.}$ = diameter of the work roll neck
$D_{w.r}$ = diameter of the work roll body
σ_t = permissible torsional stress of the work roll material
Δh = reduction

Since in the four-high stand the rolling pressure is transmitted to the backup rolls the adopted rolling parameters are sometimes checked proceeding from the bending strength of the backup rolls.

The percent reductions in horizontal roughing stands are usually as high as 40. When rolling without the broadside stand, the reductions in vertical roll stands come to 30 mm in the first stand and to 5-20 mm in the rest of them.

The strip delivered from the roughing group of stands is called strip plate. Efforts are made to keep to the minimum the number of strip plates of different sizes for the given

range of products turned out by the mill. In the majority of mills the rolling speed in the roughing stands is constant. The a.c. rolling motors are non-adjustable and at the given speed the rolling time in each stand is approximately the same.

To provide for stable operation of the finishing stands the temperature of the strip plate should be about 1160-1180°C with fluctuations of not more than 5-10°C. The temperature of the strip plate is measured by means of a photoelectric pyrometer on an intermediate roll table up to 150 m long installed between the roughing and finishing groups. If the strip plate temperature is above the predetermined level, the plate is held on the table. Occasionally, use is made of special devices for water- or air-cooling of the strip plate, which reduce the temperature and loosen the scale.

Rolling in Finishing Stands

The head end of the strip plate (which often has the shape of an extended tongue with a decreased temperature) is cropped with flying shears which are installed before the finishing group of stands. The cropped strip plate enters the finishing scale breaker (a horizontal roll stand) which operates with reductions of 2-10% (0.5-2 mm).

In the majority of mills the finishing group consists of six four-high stands. In new mills which produce thinner sheets the group includes seven stands, a provision being occasionally made for the installation of an eighth stand. The work roll diameters are increased from 600-650 mm to 700-800 mm to improve cooling of the rolls and prolong their service life. Backup rolls up to 1600 mm in diameter are used in mills with the roll body length of 1700-2000 mm. To increase the stiffness of the working stands the mill screw diameters are enlarged, and so are the cross-sectional areas of the housing members. To produce particularly thin strips the thickness of slabs is decreased to 120-150 mm and that of the strip plate delivered from the roughing group, to 16-18 mm.

Inasmuch as in the continuous finishing group the strip is deformed simultaneously in all the stands, of paramount importance is the in-process roll adjustment. To reduce the

longitudinal gauge variation (aggravated by some temperature gradient along the strip) efforts are made to decrease the strip tension between the stands, small loops controlled by loopers being sometimes used for the purpose.

The loopers comprising a horizontal tilting table tilt automatically through some angle while the head end of the strip passes over them, thus loop the strip and then return into their original position after the tail end of the strip has left the rolls. To reduce the strip tension the characteristic of the electric drive of the looper is selected so as to make the looper roller only support the loop without tensioning the strip. The automatic control of the rolling speeds and reductions in the continuous finishing group is sometimes effected by following the angle of the looping table tilt and the loop pressure. The metal reduction in each stand, as well as the overall reduction in the finishing group, is set so as to obtain the exact dimensions of the rolled strip with the minimum lateral gauge variation, to increase the durability of the working surface of the rolls in the last stands and to maintain the predetermined metal temperature at the end of rolling, necessary for securing the required metal structure (grain size). The percent reduction in the last stand of the six-stand finishing group of a ten-stand continuous rolling mill amounts to 10-15, that in the ninth stand is 20-25, in the eighth, seventh and sixth stands, 30-40, and in the fifth stand, up to 50.

The thickness of the strip leaving the last stand is controlled by means of non-contact micrometers operating on the principle of measuring the changes in the intensity of X-rays passing through the metal. The metal thickness is displayed on a calibrated scale installed at the last stand and is simultaneously registered by a strip-chart recorder.

In new mills the non-contact strip width control by the photoelectric method is also automated. The roll gap control is accomplished by means of a special device linked with the screwdown gear. The quantity of metal passing through the stands is indicated by output meters. The roll gap indicators are set so as to take into account elastic deformations of all the stand members.

The adjustment of rolls upon their change includes verification of their position in the horizontal and vertical planes.

To preclude the lateral gauge variation the finishing stands operate under the conditions of temper rolling, i.e. with small reductions. The roll camber is selected by taking account of the thermal expansion, wear and elastic deformation of the work and backup rolls. The work rolls of the finishing stands have a parabolic convex camber of 0.1-0.3 mm (sometimes work rolls of a smaller camber are used in the beginning of the backup roll service). The backup rolls are made cylindrical.

The work rolls are cambered on roll-turning or roll-grinding machines. At present, as a rule, automatic roll-grinding machines are used. In sheet steel production of paramount importance is an adequate organization of the roll service.

The rolling speed in the last stand of the finishing group amounts to 12-15 m/s and in new mills it reaches 15-20 m/s. The main drive of the finishing stands should be adjustable within a wide range of speeds. Some extra speed is also required to provide for a rapid disposal of large loops. It is usually recommended that the working speed should range from 1.08 v_{min} to 0.92 v_{max}.

The total rolling speed is limited by the maximum possible rate at which the coiler can be threaded with the head end of the strip. Therefore, to raise the rolling speed the existing practice is to speed up the mill from the threading to the top speed, once the coiler threading has been accomplished. This should be done gradually so as to maintain the setting of the continuous group, keep up appropriate roll cooling conditions, preclude growing longitudinal and lateral gauge variations, etc.

The rolling speed and the roll cooling conditions determine the metal temperature at the end of rolling, as well as that at the time of the strip coiling. With rimming low-carbon steel a good fine-grained microstructure with uniform equiaxial grains and fine-dispersed cementite, all this securing good formability, is obtained when the temperature at the end of rolling is 860-890°C, the percent reduction in the last stand being not less than 10 and the coiling temperature, below 680°C.

To obtain the required coiling temperature a water sprayer is installed on the outgoing table after the last stand. Water is sprayed through nozzles from above and from below.

Use is also made of controlled cooling, stepped cooling or of air cooling on special roll tables.

After cooling down to the predetermined temperature the strip is delivered to coilers, coiled and taken off the drum by the coiler kickoff; then the coiled strip goes onto a downtilter which places it onto a conveyor for transporting coils to a store or to the cold rolling shop. There also exist other methods of handling the rolled strip: either the strip is cut into sheets with flying shears installed after the last stand and then the sheets are delivered to sheet pilers or the whole strip is transferred onto side tables for cutting, levelling and packing. The rollers of the outgoing table are occasionally made concave in order to make the strip assume a trough-like shape after rolling (to render it more stiff).

Finishing Operations in Coil Rolling

Finishing of hot-rolled coils includes end and edge trimming, slitting into narrow strips or skelps and inspection of the slit strips which are to be shipped in coils. All these operations are carried out on special slitting and coiling lines consisting of a decoiler, straightening machine, crop and rotary disc shears and a coiling reel. If the coil-rolled steel is to be shipped in sheets, the strip decoiling, straightening, edge trimming and sheet shearing are done on sheet shearing lines (Fig. 12.4).

Before shearing on these lines the coils are placed onto a charging table which delivers them to a decoiler. The coil is opened up by a magnetic lifting device and then the strip is decoiled while passing through a straightening machine. After that the following operations are done on the line: descaling, thickness measuring, end cropping, temper rolling in a four-high stand to eliminate ridges and reduce the gauge variations, and edge trimming with rotary disc shears. The sheet shearing is done with flying shears. A provision is also made for sheet oiling and piling. A looping pit is used in the line to compensate for possible differences between the working speeds of individual units.

Heat treatment, temper rolling and pickling find an ever increasing use for improving the quality of sheet steel. Normalizing and soft annealing are employed to improve the

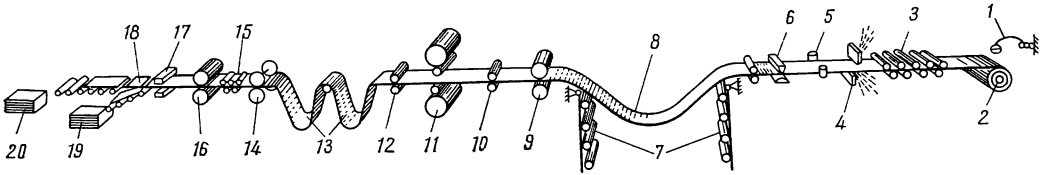


Fig. 12.4. Sheet shearing line

1—magnetic lifting device; *2*—decoiler; *3*—leveller; *4*—cleaning unit; *5*—vertical centring rolls; *6*—flying micrometer; *7*—hinge-type disappearing tables; *8*—looping pit; *9*—flying shears for end cropping; *10*—tension limiter; *11*—temper-rolling stand; *12*—pinch rolls; *13*—looping pits; *14*—rotary disc shears; *15*—leveller; *16*—flying shears; *17*—marking machine; *18*—reject gate; *19*—reject stacker; *20*—prime stacker

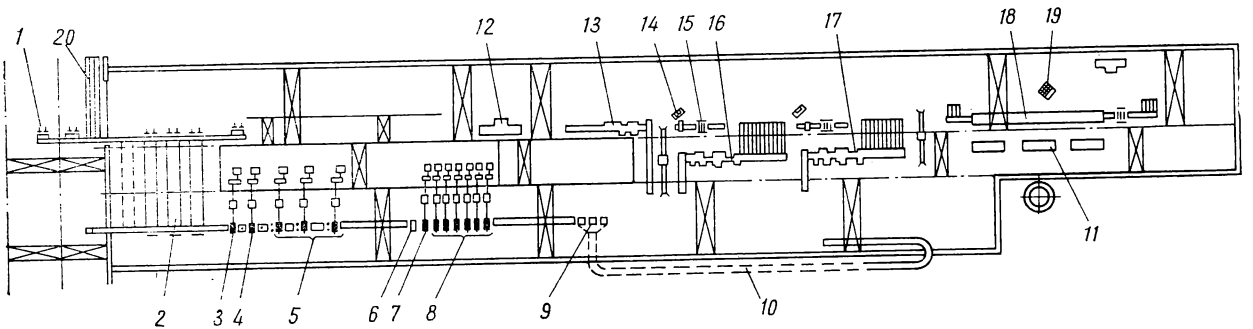


Fig. 12.5. 1700 mm continuous wide-strip mill

1—charging tables; *2*—reheating furnaces; *3*—roughing two-high scale breaker; *4*—universal four-high stand; *5*—four-high universal roughing stands; *6*—flying shears; *7*—finishing scale breaker; *8*—four-high finishing stands; *9*—coilers; *10*—conveyor for coil transportation; *11*—hood-type annealing furnaces; *12*—scales; *13*, *16* and *17*—cutting lines; *14*—separately installed shears; *15*—levelling and oiling unit; *18*—end-charged end-discharged normalizing furnace; *19*—separately installed shears with caster bed; *20*—inclined chain conveyor

plastic properties of the metal. The soft annealing with very slow cooling is done in roller-hearth furnaces, a roll table covered with refractory-lined hoods being used for cooling. Hardening by heating the metal up to 1150°C in continuous furnaces with its subsequent water-spray cooling or water quenching is mainly used for alloy and stainless steel grades. After hardening the sheets are levelled. Coiled steel is heat treated in hood-type annealing furnaces with an inert gas atmosphere.

Temper rolling is a process comprising a one-pass cold rolling with a light reduction. It is used to obtain the exact gauge of sheet, to eliminate ridges and to improve the formability of the metal through work hardening.

Pickling is done, mostly after heat treatment, on continuous pickling lines or in pickling vats, first in hot, strong solution of sulphuric acid, then in a weaker solution of the acid. The sheets to be pickled are loaded edgewise into the dipping baskets. After pickling the metal is rinsed in hot running water.

The layout of a 1700 mm sheet mill is shown in Fig. 12.5. The arrangement of the main equipment of this mill is typical of the majority of continuous and semi-continuous wide-strip mills. The charging facilities of continuous furnaces extend into the slab storage bays which are arranged perpendicular to or parallel with the mill bay. The structure and arrangement of individual pieces of equipment may vary, depending upon the range of products turned out by the mill, but one should always keep to the basic principle of modern technology, i.e. all the operations must be conducted within a single, continuous flow with individual units set up outside the main production flow to handle workpieces requiring additional treatment. Given the current trend towards increasing the weight of coils and slabs and reducing the thickness of finished sheets some correctives may be introduced into the layout under review. The layout of a new 1700 mm sheet mill is shown in Fig. 12.6.

Main Defects of Hot-Rolled Sheets

Defective sheets of wrong dimensions may be produced not only in the course of rolling as a result of failure to keep to the adopted rolling schedule, but also during their shearing,

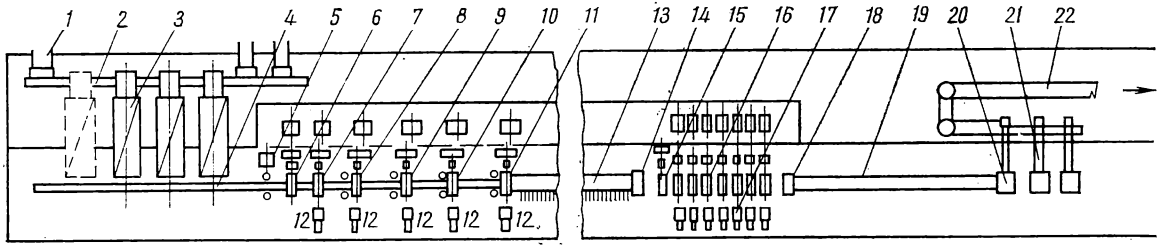


Fig. 12.6. Modern 1700 mm wide-strip mill

1—slab pushers; *2*—charging table with lifting devices and pushers; *3*—furnaces; *4*—furnace entry table; *5*—roughing scale breaker (with vertical rolls); *6*—two-high roughing scale breaker with horizontal rolls; *7*—four-high roughing stand $\left(\frac{D_{w.r.}}{D_{b.r.}} = \frac{1000\ mm}{1400\ mm}\right)$; *8-11*—four-high universal stands ($D_{v.r.}$ = 800 mm; $D_{w.r.}$ = 850 mm, $D_{b.r.}$ = 1400 mm); *12*—roll changing devices; *13*—intermediate table with strip plate kickoff; *14*—flying shears; *15*—two-high finishing scale breaker, D = 600 mm; *16*—continuous group of four-high finishing stands, $D_{b.r}$ = 700 mm, $D_{w.r}$ = 1400 mm; *17*—roll changing devices; *18*—strip gauge and width measuring device; *19*—delivery table with controlled sheet cooling; *20*—coilers; *21*—downenders and coil transfers; *22*—coil conveyors with tilting tables

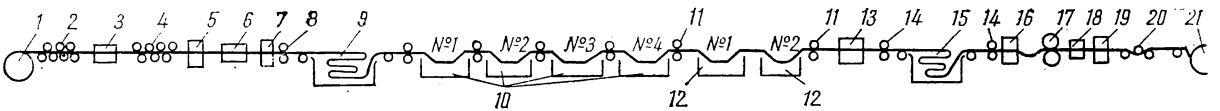


Fig. 12.7. Layout of continuous pickling line

which in sheet steel production requires adequate fabrication, correct marking out and good condition of shearing equipment.

As concerns the shape and appearance of sheets, the typical defects are buckles and camber which may arise from uneven cooling and lubrication (in pack rolling).

Folds on sheets result from differences either between the speed of the outgoing table and that of the sheet leaving the last stand or between the speeds of individual sections of the table. Mechanically damaged surfaces, edges and corners of sheets occur due to bad condition of mechanical handling equipment and also as a result of careless handling of sheets during piling and transportation.

A telescopic shape of coils may originate from poor condition of coilers, their incorrect threading and side strain of the strip. The surface defects include slivers, all kinds of cracks and tears, rolled-in scale, collar marks, flaws, dents caused by metal or scale particles welded to the roll surface or resulting from its bad finish, nicks, scratches, etc.

In the pack rolling defects arise from incorrect assembling of packs and their wrong feeding into the rolls and also from contamination of sheet surfaces with lubricants. A steady and well controlled rolling schedule along with proper organization of the production pattern and, above all, of the roll and roll fitting services are decisive factors eliminating various defects in sheet steel. Of paramount importance is also the routine maintenance of the equipment.

12.3. Cold Rolling of Sheet Steel

Cold rolling of sheet steel is one of the most progressive and rapidly developing branches of the metal rolling industry. Cold rolling makes it possible to produce sheets less than 1.5 mm in thickness with minimal dimensional tolerances, good surface finish, better mechanical properties (thanks to the characteristic features of cold-rolled metal structure) and also with special physical properties.

The process ensures good formability and other plastic and strength characteristics of sheets and makes it possible to produce sheets with predetermined properties required for their use in electrical engineering. The small thickness,

good surface finish and high drawability of cold-rolled sheet steel have offered great potentialities for the expanded manufacture of sheet with metal and plastic coatings.

The following two methods are used in cold rolling of sheet steel: pack rolling and coil rolling. The old pack rolling method is still used in the manufacture of small quantities of high-grade sheet.

In cold rolling of sheet steel the production process consists of the following main stages:

1. Preparing the strip plate for cold rolling including a preliminary heat treatment (in case of pack rolling and also when rolling stainless steel), and descaling.
2. Cold rolling and degreasing.
3. Heat treatment.
4. Finishing of cold-rolled sheets (temper rolling, levelling, shearing, assorting and packing).

Preparing the Strip Plate

The strip plate for cold rolling by both the pack and coil rolling methods must be thoroughly descaled. The low metal temperature at the end of rolling and rapid metal cooling which are characteristic of the method of hot rolling in pack form may cause work hardening of the sheets and also the development of internal stresses in them. In this case a preliminary heat treatment is required.

The coiled strip plate is slowly cooled while being transported by an underground conveyor and in the store. Therefore no preliminary heat treatment is necessary for coiled carbon and low-alloy steels. Soft annealing is used for stainless steels. The preliminary heat treatment of plain-carbon steels consists in normalizing (850 to 900°C) in continuous furnaces and process annealing (600 to 680°C) in hood- or tunnel-type furnaces. To prevent oxidation use is made of a shielding gas and hermetically sealed annealing boxes.

The metal is thoroughly descaled in order to avoid rolling of the scale in the strip and to prevent damaging of the rolls of cold rolling mills by the scale particles. The main descaling method is pickling in various acids and alkalies, sulphuric acid pickling being most widely used. Mechanical descaling is done by bending and unbending the strip in

rollers, by shot blasting and by temper rolling. When preparing the strip plate for cold rolling the mechanical descaling is usually used as a preliminary treatment to facilitate pickling.

Recently pickling in hydrochloric acid, as well as alkali pickling, especially for stainless and other chrome steels, has found an ever increasing application.

Sulphuric acid pickling involves the following chemical reactions:

$$FeO + H_2SO_4 = FeSO_4 + H_2O$$

$$Fe_2O_3 + 3H_2SO_4 = Fe(SO_4)_3 + 3H_2O$$

$$Fe(SO_4)_3 + H_2 = 2FeSO_4 + H_2SO_4$$

hydrochloric acid pickling:

$$FeO + 2HCl = FeCl_2 + H_2O$$

$$Fe_2O_3 + 6HCl = 2FeCl_3 + 3H_2O$$

$$2FeCl_3 + H_2 = 2FeCl_2 + 2HCl$$

Simultaneously occurs a partial dissolution of iron:

$$Fe + H_2SO_4 = FeSO_4 + H_2$$

$$Fe + 2HCl = FeCl_2 + H_2$$

To protect the metal against the attack of sulphuric acid use is made of special additives (inhibitors) which form protective films on the metal surface. Hydrogen released from the scale breaks away undissolved scale particles which fall down onto the pickling vat bottom in the form of sludge. Some diffusion of hydrogen into the metal during pickling results in the so-called pickling brittleness. The maximum pickling rate is obtained with a 25% sulphuric acid solution, however, in practice acid solutions of not higher than 22% concentration are used to lower the metal dissolution. To intensify pickling the acid solution is heated to about 90°C. In pack rolling the sheet packs are pickled in rotary batch-type picklers which usually consist of a vat with an acid solution and another one containing hot rinsing water. The sheets are loaded edgewise into dipping baskets suspended

from a turntable. The baskets are in turn lowered into the pickling and rinsing vats by means of a lift and turn device.

As a rule, in coil rolling use is made of continuous pickling (Fig. 12.7). A strip coil mounted on decoiler *1* is opened up by a magnetic lifting device and then the strip is fed into straightening machine *2* and scale breaker *3*, where, passing round rollers, it bends and the scale breaks up. Right after the scale breaker are installed straightening machine *4*, high-speed guillotine shears *5*, electric welding machine *6* with deseamer *7* and pinch rolls *8*. Stitchers are employed in old picklers and also in case of difficult-to-weld steels for joining the ends of successive coils. Stitching is done with a lap joint. Such a joint is not rolled and is cut out after pickling. To provide for the continuity of the process the pickler is equipped with looping pits *9* and *15*.

Pickling is performed in four pickling vats *10* and two rinsing vats *12* between which drying rollers *11* are installed to reduce acid losses with the moving strip. The pickling vats comprise welded metal tanks with covers. The tanks are rubberized and lined with acid-proof brick. The acid solution is heated with steam or by means of gas burners. Steam is fed either directly into the vats or into coil pipes set up therein. After rinsing the pickled strip goes into drying unit *13*. The concentration of a sulphuric acid solution in the first vat reaches 22% and that in the fourth vat, 11%. Rinsing is accomplished first in cold and then in hot (80-95°C) water.

The main pinch rolls *14* pull the strip through the pickler and provide for the process speed control. The second looping pit *15* is followed in a successive order by guillotine shears *16*, rotary disc shears *17* with scrap choppers *18*, oiling machine *19* and coiler *21* with pinch rolls *20*. The shears serve for cutting out the lap joints and defective strip sections and also for edge trimming.

In wide use is also the so-called cascade method of pickling. With this method a freshly prepared solution (15-25% concentration) is supplied continuously into the first vat wherefrom it is made to flow successively through all the vats to the last one either because of the difference in levels or by pumping, the acid concentration in the last vat being 3-6%. Such a procedure is conducive to a better utilization

of the pickler and helps maintain a constant concentration of the pickling solution and steady process conditions.

In modern picklers the strip speed varies within a range of 90-380 m/min. The output of high-speed units reaches 1200-1400 thousand tons per year. The total length of the pickler is 100-125 m. In pickling departments the recirculation of the pickle liquor is effected by extracting ferrous sulphate from the waste liquor by using vacuum-crystallizing and neutralizing apparatus.

In the last few years, thanks to the development of the method of complete regeneration of hydrochloric acid, hydrochloric acid pickling in horizontal and vertical (tower) picklers has found a wider application.

Picklers operating on hydrochloric acid are more efficient, their operation is less costly, and they provide for a higher quality of pickling. Hydrochloric acid reacts much more intensively with the scale-forming oxides than with the base metal, therefore the risk involved in overpickling is reduced. The disadvantages of the hydrochloric acid pickling are worse work conditions due to a high volatility of the acid vapours and the complexity of the pickling equipment because of the highly active pickle liquor.

The use of an aqueous solution of hydrochloric and nitric acids for pickling stainless steel has not justified itself because of the high cost and handling inconveniences (storage and transportation). More often use is made of sulphuric acid with additions of saltpetre and common salt, this resulting in the formation of a solution of sulphuric, nitric and hydrochloric acids.

Stainless steel pickling yields better results with preliminary alkali and subsequent acid treatment. The alkali pickling is done in a melt consisting of 75-80% NaOH and 20-25% $NaNO_3$ at a temperature of 400-520 °C. This melt causes destruction of the scale resulting from the change in its volume produced by the reaction of cromium oxides with the alkali and transformation of lower oxides into higher ones.

After rinsing the strip is dipped into a sulphuric acid solution with additions of saltpetre and common salt and then is rinsed again and brightened in a weak solution of nitric acid. In the manufacture of stainless steel use find combi-

nation heat treatment lines which include continuous pickling units, so-called quench-pickle units, operating at a speed of 3-18 m/min. After heating in a continuous furnace the strip is subjected to shot blasting and then pickled.

Cold Rolling and Degreasing

Cold rolling of sheet steel in pack form is usually done in two- and four-high reversing and non-reversing mills. Some modern mills designed for rolling high-grade metal employ the single-sheet rolling method.

Cold rolling of alloy steel grades is done mainly in reversing mills using the coil rolling method. Single-stand four-high reversing mills can operate on a thick strip plate (3-6 mm) and produce sheets 0.5 mm thick and sometimes even thinner ones.

Multiroll mills are used for rolling ultra-thin sheets and tin plate (less than 0.18 mm in thickness). With multistand mills thinner sheets (0.08 mm) are obtained by increasing the number of stands (up to 5-6) or by additional rolling in two- or three-stand continuous mills. Multiroll mills (twelve- and twenty-high) are also widely used in rolling low-formability alloy steels.

In pack rolling the mill packs are fed into the rolls by hand. With a non-reversing mill the reduction is not changed until a batch of packs is rolled. Thereafter the packs are delivered by means of a crane or conveyor back to the entry side of the mill for further rolling with a greater reduction.

The following is the way in which coil rolling is done in a reversing mill. From the decoiler the strip is fed into the rolls and the coiler is then threaded with the head end of the strip. This done the rolling with tension commences. Rolling in the reverse direction proceeds in the same manner, the tail end of the strip being now engaged. In reversing mills the rolling speed comes to 6-15 m/s, their output reaching 350 thousand tons per year.

In continuous mills the coil mounted on the decoiler is opened up and its tail is fed into the rolls by means of special magnetic lifting devices or stripper-type tail pullers, feeders and entry guide boxes, the mill being run at a low (threading) speed. Once the coiler has been threaded with

the head end of the strip, the mill is gradually speeded up until its normal working speed is reached. Continuous cold rolling mills with powerful main d.c. motors are adjustable within a wide range of speeds (2.5-3-fold changes of speed are possible). In four-stand mills the maximum rolling speed reaches 25 m/s and their output amounts to 1300 thousand tons per year.

Five-stand mills operate at a maximum speed of 37 m/s and have an output of 500-1000 thousand tons per year. Six-stand mills with a roll body length of up to 1420 mm which are designed to roll tin plate less than 0.16 mm in thickness from a strip plate about 2.0 mm thick operate at a speed of up to 35 m/s, their output reaching 700 thousand tons per year. Tin plate less than 0.076 mm in thickness is produced by secondary rolling in two- or three-stand mills. The rolling speed in these mills is up to 28 m/s and the output amounts to 300 thousand tons per year.

An ever increasing popularity gain multiroll mills which provide for great metal reductions in rolling and yet ensure high degree of surface finish and high dimensional accuracy of the rolled products. The annual output of these mills comes to 250 thousand tons when rolling 0.1 mm thick tin plate in a single passing and 50-90 thousand tons when rolling carbon steel in several passings.

As is known, when rolling with tension the roll pressure is reduced. Therefore in cold rolling it is very important to properly select the strip tension and to keep it constant. The strip tension is kept constant automatically by controlling the speed of the rolls and coilers, this being accomplished by an automatic control system which uses tensometric devices (tensiometer rolls) as sensors. The preset thickness of the rolled strip is also automatically maintained by means of the roll-gap control system incorporating nucleonic thickness gauges.

The total deformation of the metal, which it is possible to achieve in cold rolling, is limited by the growth of the metal resistance to deformation (proportional to its yield and ultimate strength) due to work hardening. At a reduction of 30% a sheet 1.5 mm thick made of soft steel has an ultimate strength of up to 60 kgf/mm^2, that of a similar sheet made of hard steel amounting to 90 kgf/mm^2.

Cold rolling is continued until the ultimate strength of the rolled metal reaches 90 kgf/mm^2 this being the case at a total reduction of up to 80% for soft steels, up to 30% for high-carbon steels and up to 15-20% for alloy steels. To make further cold rolling possible intermediate annealing is used.

Of great importance for continuous rolling is the coefficient of friction between the strip and the rolls which is determined by the quality of the lubricant used, the condition of the roll surfaces and the rolling speed.

For soft steels the reduction per pass usually does not exceed 35-38%, the total reduction amounting to 90%. For high-carbon steels the total deformation per 5-7 passes between intermediate annealing operations comes to 15-25%. The degree of pass reduction is determined not only by the plastic properties of the metal, but also by the strength and hardness of the rolls. The roll bending which may cause lateral gauge variations depends upon the rolling pressure. The work rolls are usually ground so as to make them somewhat convex, their convex camber being proportional to difference Δx between the roll deflections at the centre of the body and at the edges of the rolled strip. This difference is determined by the formula proposed by P. Polukhin:

$$\Delta x = 0.00002 \frac{Pb}{D^2} \left[1 + \frac{b}{8D^2} (12a - 7b) \right]$$

where P = rolling pressure, kgf
D = diameter of the rolls, mm
a = distance between the axes of the mill screws, mm
b = width of the strip, mm

Either both rolls are made with the same camber or only one roll is cambered so that its camber equals the sum total of both roll cambers (in this case the other roll is made cylindrical).

In cold rolling special attention is paid to the quality of the lubricant used which should reduce friction, ensure good cooling of the rolls and be easy to remove from the sheet prior to heat treatment. Various mineral oils (aviation, machinery) are used in cold rolling. Sometimes animal and vegetable oils also find application. For instance, palm oil is employed when rolling tin plate. Soluble oils in the

form of aqueous solutions reduce friction insignificantly, but are easily removed from the surface of the strip even by compressed air. Non-soluble oils in the form of hot aqueous emulsions are used when rolling extremely thin sheets, e.g. tin plate. These lubricants are somewhat difficult to remove after rolling and therefore in such cases use is made of chemical or electrolytic degreasing, the degreasers being either included in continuous annealing lines or installed separately. Bodied cotton seed oil is effectively employed in cold rolling of difficult-to-deform steels.

The hot roll profile is controlled by changing the temperature of lubricants and sometimes by means of special gas burners installed on the working stand. In a number of cases new cold rolling mills use hydraulic devices to bend the rolls in a direction opposite to that of their bending in the process of rolling.

Lately there has been a trend to increase the diameter of the rolls of continuous cold rolling mills. The diameter of the work rolls has been increased from 450 to 585 mm and that of the backup rolls, from 1200 to 1500 mm. As the work roll diameter is increased the removal of heat from the rolls improves, rolling speed increases, roll wear reduces and the surface finish of the rolled sheet improves. The greater diameter of the backup rolls makes the stand more stiff.

The work rolls are forged from chrome or chrome-molybdenum steels. The Shore hardness number of the hardened roll body surface is as great as 100.

The backup rolls are made of chrome steel grade 9X, the Shore hardness number of the roll body amounting to 85. Sleeved backup rolls are also used. To prevent the sheet sticking during annealing the rolls of the last stands in multistand mills are specially roughed somewhat by grinding with abrasive paper or by shot blasting.

Heat Treatment

Preliminary or intermediate soft annealing is used when making stainless steel sheet by the pack rolling method, as well as when rolling high-carbon and alloy steels, to soften the work-hardened metal. All grades of cold-rolled sheet steel are subjected without fail to final heat treatment,

which, in addition to softening of the metal, helps obtain a uniform fine-grained structure and improve the metal drawability (for deep-drawing steels).

The old-established method of final heat treatment consists in normalizing followed by low-temperature annealing, but it is cumbersome, inefficient and costly. Practical experience and research have confirmed that in cold rolling low-temperature annealing (600-700 °C) suffices when the total deformation of the metal is over 30% and high-temperature annealing (800-850 °C), when it is up to 25%.

Depending upon the type of products the heat treatment methods used are as follows. Annealing in a protective atmosphere at 600-720 °C followed by slow cooling is employed to obtain the required drawability of sheet steel and tin plate. Tin plate not intended for further drawing undergoes annealing in a protective atmosphere at 680-700 °C followed by a more rapid cooling. Sheet steel intended for further enameling is subjected to annealing in a protective atmosphere at 680-720 °C followed by slow cooling.

In accordance with the adopted heat treatment practice two types of furnaces are used in cold rolling shops, *viz.* hood-type and continuous furnaces. Batch-type box-annealing furnaces, formerly in wide use, are no longer constructed. Although continuous heat treatment processes are being developed hood-type annealing furnaces continue to be the main units for annealing carbon steel. They are more economical, provide for good quality of products and suit well for dealing with small batches of sheet products of different types.

Hood-type furnaces are used for annealing not only coils, but also mill packs. In the latter case use is made of radiation heating whereby fuel burns not under the furnace cover, but inside U-shaped heat-resistant radiant tubes placed all over the walls of the cover. The products of combustion are thus isolated from the hood and from the space between the cover and the hood.

The hood is filled with a protective gas. Radiant tube-fired hood-type annealing furnaces are of low productive capacity and therefore they are superseded by indirect-fired furnaces in which fuel burns between the hood and the cover.

A multistack hood-type annealing furnace is shown in Fig. 12.8. Coils are stacked in three-four pieces on the coil base. The coils in the stack are separated by ribbed spacers

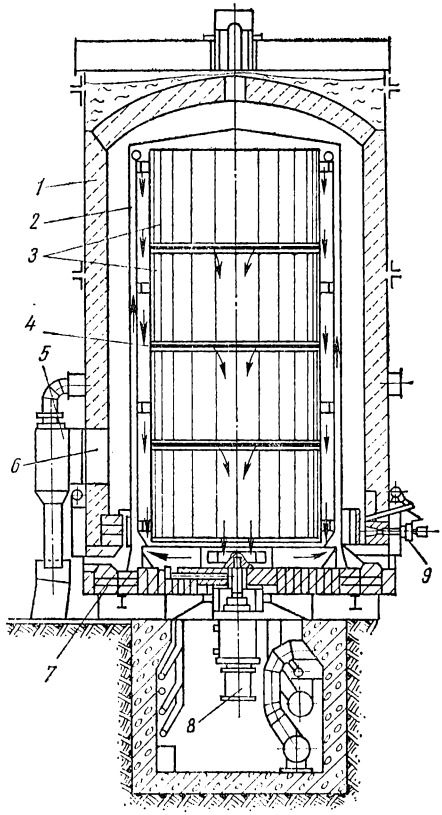


Fig. 12.8. Multistack hood-type annealing furnace
1—external cover; *2*—hood; *3*—coils; *4*—ribbed spacers; *5*—ejector; *6*—waste gas doors; *7*—coil base; *8*—fans; *9*—burners

and each stack is covered with a hood made of sheet steel and provided with a sand seal. The hood is filled with a protective gas which is made to circulate by a fan installed in the base to improve the transfer of heat. The furnace cover is lined with light-weight refractory brick.

In wide use are single-stack hood-type furnaces which operate on the same principle, but use individual covers for each hooded coil stack. This makes for better utilization of the covers, as well as of heat accumulated in their lining. Special covers equipped with fans to speed up the metal cooling are also used. The drawbacks of the hood-type furnaces are slow heating and cooling (at a temperature of 300-400 °C the cooling rate may be increased without deteriorating the mechanical properties of sheets), bulky structure and much labour required to operate them.

In furnaces of the latest designs particular attention is concentrated on accelerating the metal cooling and cutting down the attending labour. Annealing of expanded coils also finds application (prior to annealing the coils are rewound so as to form gaps between their turns and atfer annealing the loose coils are rewound into tight ones).

As concerns their productive capacity and controllability continuous annealing furnaces present advantages over hood-type ones. Continuous annealing lines usually include units for chemical or electrolytic cleaning (degreasing) of the strip. Chemical cleaning is done by washing the strip with hot soda solution and then rinsing it in hot water. With electrolytic cleaning the strip is passed through a vat containing a heated electrolyte (an aqueous solution of NaOH, Na_2CO_3 and Na_3PO_4), a voltage being applied between the vat electrodes (anode) and the strip (cathode). After electrolytic cleaning the strip is rinsed and dried.

The layout of vertical-type installation for continuous cleaning and annealing is shown in Fig. 12.9. Similar installations of the horizontal type are less common. With continuous annealing the speed of the strip passage is 5-6 m/s, higher speeds (up to 10 m/s) being also possible. The continuous method of annealing is widely used in the manufacture of tin plate.

The composition of the protective atmosphere is of great significance for the quality of cold-rolled metal. Modern installations are apt to keep the content of oxygen in protective gases within the range of 0.002-0.005%. Various protective atmospheres and vacuum of 10^{-2}-10^{-6} mm Hg are used in heat treatment, depending upon the specific requirements for the metal being treated (no surface oxidation and

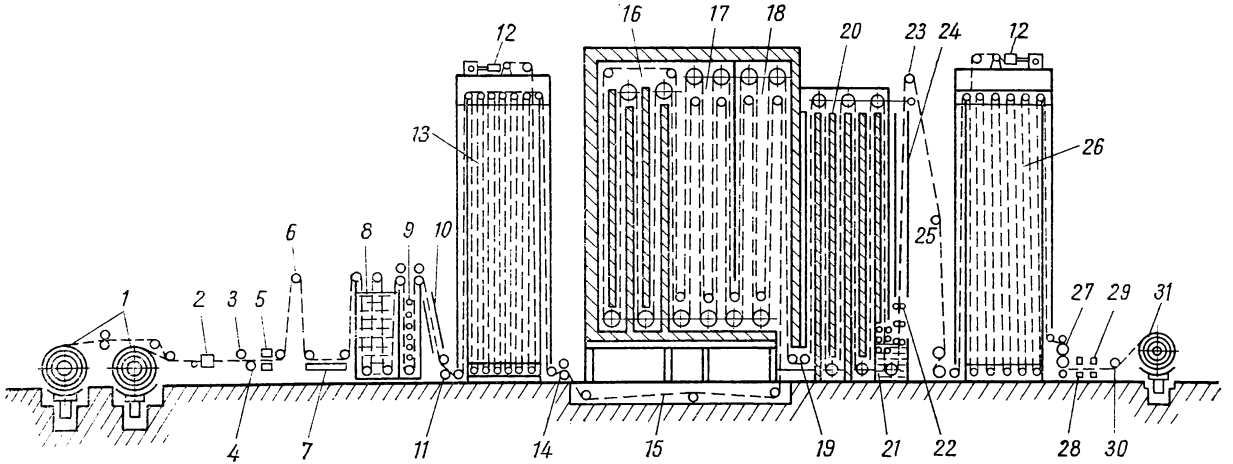


Fig. 12.9. Layout of installation for continuous strip cleaning and annealing

1—decoilers; *2*—welding machine; *3*—holddown rolls; *4*—guide rolls; *5*—photoelectric control device; *6*—looper; *7*—table for cutting out strip end welds; *8*—cleaning vats; *9*—rinsing vat; *10*—hot air drying; *11*—pulling unit; *12*—drive and strip tension regulator; *13*—entry looping tower; *14*—main pinch rolls; *15*—trench for strip; *16*—strip heating section; *17* and *18*—slow cooling and heat exchange sections; *19*—roller gate; *20*—cooling channels; *21*—hardening vat; *22*—wringer rolls; *23*—roll for strip bending; *24*—hot air drying; *25*—bending rolls; *26*—discharge tower; *27*—pinch rolls; *28*—sampling devices; *29*—photoelectric counter; *30*—guide roll; *31*—coiler

decarburization or, conversely, the need for obtaining a decarburized layer). A number of reducing and neutral gases—hydrogen gas, dissociated ammonia gas, exothermic gas and nitrogen gas—are used as a protective atmosphere.

Hydrogen is a strong reductant and is used to decarburize electrical sheet steels. Its thermal conductivity is high and therefore it improves the heat transfer conditions. In spite of its disadvantages, such as high explosiveness and cost hydrogen is widely used. It is produced in electrolyzers with alkali electrolytes and then cleaned from oxygen, water vapours and traces of compressor oils by using absorbents and catalysts. To provide for accident-free operation of electrolyzers, as well as of hydrogen apparatus used in heat-treating units, accident prevention regulations must be strictly adhered to.

Dissociated ammonia is also an active reductant. After dissociation and drying liquid ammonia contains 25% N_2 and 75% H_2.

Dissociated ammonia is characterized by its high heat conductivity and explosiveness. It is produced as follows. Liquid ammonia is fed from gas cylinders into an evaporator where its volume increases 860 times. The gaseous ammonia then flows into a dissociator wherein it dissociates at a temperature of 700-950 °C, heat being absorbed. As a result a gas of the above composition is produced. Steel chips and iron oxides are used as catalysts. The explosiveness of dissociated ammonia gas is reduced by burning it in the air, this reducing the hydrogen content down to 1-2%. The protective gas thus produced is then dried and cleaned.

Exothermic gas is produced by burning gases (blast-furnace gas, coke-oven gas, propane, butane, etc.), as well as kerosene. Burning may be complete or incomplete. In the latter case an exothermic gas with reducing properties is obtained. The heat conductivity of a rich exothermic gas is about three times less than that of dissociated ammonia gas, but its cost is comparatively low.

Nitrogen is usually obtained as a by-product at oxygen plants and therefore at modern works it is always available in sufficient quantities. However, nitrogen has a number of disadvantages. At high temperatures it combines with other elements, such as molybdenum, chrome, titanium and

niobium, thus forming nitrides. In addition, industrial nitrogen always contains some deleterious impurities, such as water vapours and oxygen, which cause oxidation of the metal surface. Therefore nitrogen is only used as a subsidiary protective gas.

Finishing of Cold-Rolled Sheet Steel

Finishing of cold-rolled sheet steel includes temper rolling, levelling, shearing, assorting, inspection and packing. The deep-drawing steel is invariably subject to temper rolling (0.5-3.0% reduction). In temper rolling the strength of the metal increases by 10-15%, its plasticity being sufficiently good. The sheet, at the same time, is finished to gauge and any required metal finish—bright, glossy, mirror, dull or rough—can be obtained.

Temper rolling is done in a single passing in one or two stands, the rolls being neither lubricated nor cooled. In some of modern two-stand temper mills intended mainly for tin plate the rolling speed is up to 30 m/s. The output of the two-stand mills amounts to 450 thousand tons per year, but there are also similar mills of higher capacity.

Particular attention is given to the accuracy and methods of machining the work rolls of temper mills, the rolls being polished to a high class of surface finish. In some cases the rolls are shot blasted with pellets of 0.5-1 mm size.

A temper-rolling line consists of a decoiler, feeding device (often provision is made for controlling the strip entry angle), dancer and tensiometer rolls to secure constant strip tension, one or two temper mills (four- or six-high) and coilers. Right after the decoiler there are installed end trimming shears. After temper rolling the strip passes through an oiling machine. The strip levelling and shearing are accomplished on a continuous processing line.

The finished products are shipped either in stacked sheets or in coils. In the first case sheet-shearing equipment is employed, while in the second case use is sometimes made of slitting equipment.

In cold rolling of sheet steel the quality control is effected in two stages. In the first of these the basic technological parameters (the rolling temperature in particular) are

checked with special instruments in the course of production, the dimensions being controlled by non-contact methods. At this stage overall assessment of the quality of a batch of sheets or of the whole of a coil is made and their destination is decided upon.

In the second stage the quality control is carried out on the shear lines by using the methods of automatic non-contact control of the sheet dimensions and punctures. Defective shape and appearance of the sheets are also detected visually and to do this the shear lines are often operated at lower speeds (1-2 m/s).

The main defects of cold-rolled sheet are size and gauge variations which arise as a result of changes in the strip tension occurring when changing over from the threading to the working speed. The gauge variations may be intrinsic, i.e. akin to the highly non-uniform thickness of the hot-rolled strip plate used, or they may be due to improper lubrication and cooling of the rolls, inadequate strip tensioning and bad condition of the roll surfaces. Variations in the length and width of the sheets result from wrong setting of flying and rotary disc shears.

The side strain of the strip is caused by irregular dimensions and shape of the hot-rolled strip plate used, as well as by wrong setting of the mill or rotary disc shears.

Ridge buckles and camber of sheet arise from bad surface condition and wear of the rolls and also from wrong setting of the mill.

Cuts (local squeezes involving discontinuity of a herringbone pattern) come from irregular reduction schedule, wrong roll profile, incorrect mill setting and also from the use of defective strip plate.

Among specific surface defects of cold-rolled sheet steel are stretcher strains (because of the failure to keep to the adopted reduction schedule in temper rolling) and temper colours (due to oxidation of the metal surface in annealing resulting from inadequate protective atmosphere).

In cold rolling of sheet steel particular attention is focussed on the transportation and storage of the metal so as to avoid mechanical damages (crimping of corners and edges, scratching). Special measures are taken to prevent corrosion of the metal during transportation.

Thin sheet metal and tie band are used for strapping. In wide use are minor mechanization facilities (in welding, punching, tightening and cutting of binding and packing materials).

The layout of a cold-rolling shop equipped with a 1700 mm four-stand continuous mill is shown in Fig. 12.10.

Figure 12.11 is illustrative of the equipment layout of a cold-rolling shop fitted out with a 1200 mm five-stand mill. This shop produces electrolytic tin plate, transformer steel and galvanized sheet.

12.4. Production of Sheet Steel with Protective Coatings

Corrosion-protection coatings of sheet steel (metal, enamel, varnish and plastic coatings) have found wide application. The manufacture of tin plate, galvanized and aluminized sheet is making a rapid progress. Rapidly developing is also the production of bimetals and clad metals. Different metals are strongly bonded together by casting, forging, welding, explosion and hot rolling. Metal spraying finds application as well.

Tin Plate

Tin plate (tinned steel sheet) is a black plate coated with a thin layer of tin. Cold-rolled tin plate is made of soft carbon steel of the following composition: 0.06-0.1%C, 0.3-0.5% Mn, 0.01% Si, S and P of not more than 0.05%.

The width of the coiled tin plate usually equals 710 mm, the size of the tinned steel sheets being 510 × 710 mm. The thickness of the steel base of tin plate is 0.075-0.7 mm. Tin plate 0.075-0.18 mm in thickness is called ultra-thin. The ultra-thin tin plate is cold rolled for the second time without annealing to improve its strength.

At the surface of the base steel sheet the tin coating forms an intermediate layer of $FeSn_2$. The total thickness of the coating is up to 0.025 mm.

Tin plating may be done either by the hot or electrolytic tinning method. The first operation in tin plating is electrolytic pickling in a 0.5-1.5% HCl solution, which is

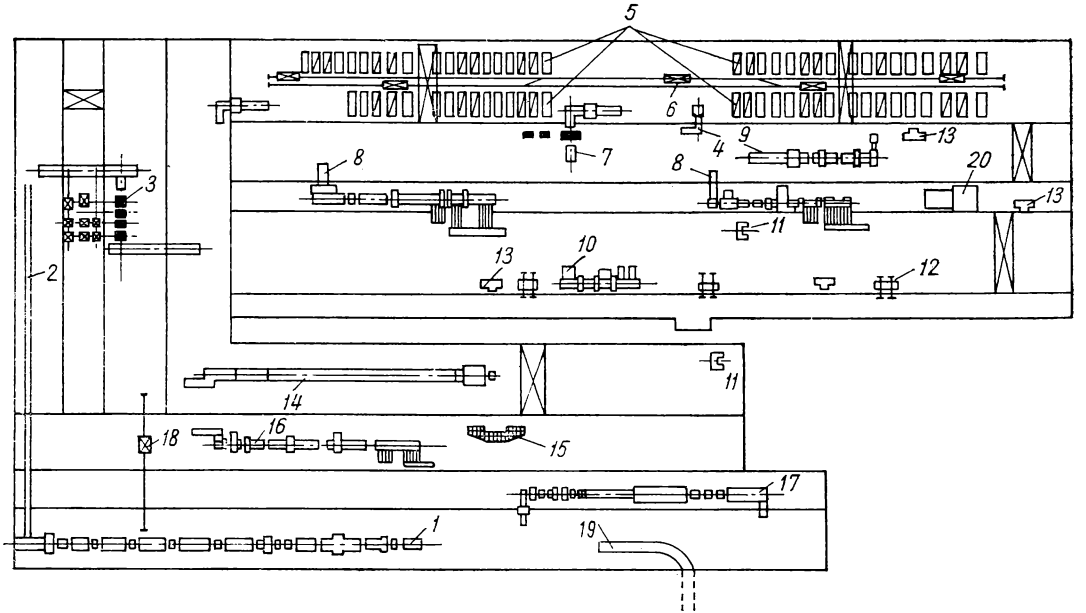


Fig. 12.10. Layout of cold-rolling shop equipped with 1700 mm four-stand mill

1—continuous pickling line; *2*—conveyor for pickled coils; *3*—1700 mm four-stand mill; *4*—conveyor with turnover device; *5*—hood-type annealing furnaces; *6*—transfer car; *7*—temper-rolling mill; *8*—continuous sheet-shearing line; *9*—continuous slitting line; *10*—sheet-metal grader; *11*—guillotine shears; *12*—transfer trolley car; *13*—scales; *14*—continuous galvanizing line; *15*—packing machine; *16*—combined cutting line; *17*—furnace for hot-rolled coil normalizing; *18*—transfer trolley car; *19*—conveyor for hot-rolled coils; *20*—packing press

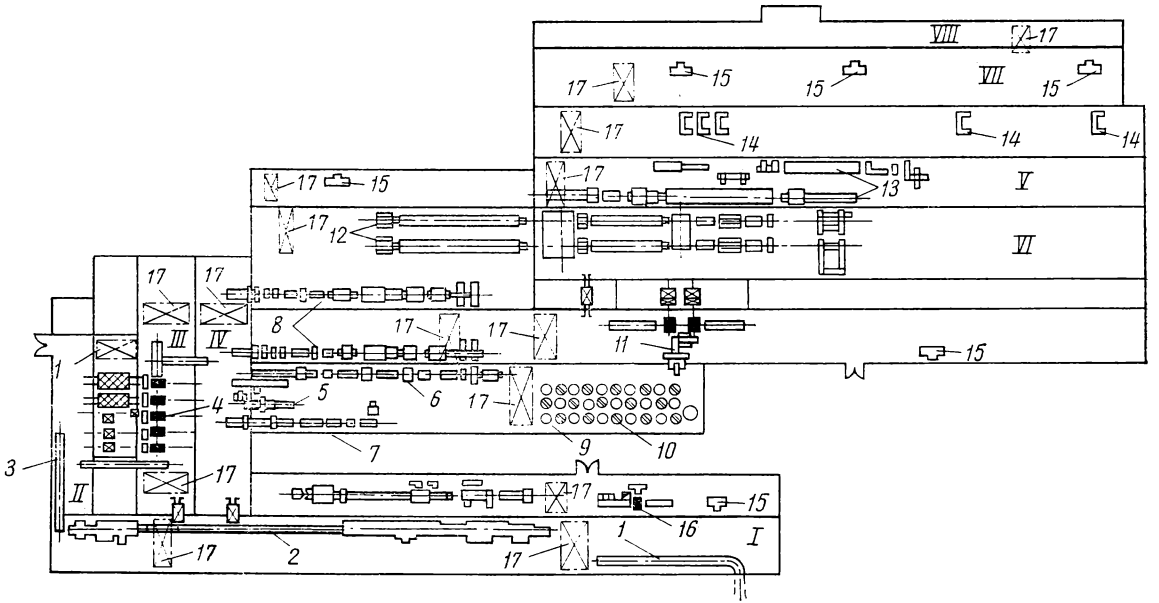


Fig. 12.11. Layout of cold-rolling shop equipped with 1200 mm five-stand mill

I—hot-rolled coil storage, *II*—pickled-coil storage; *III*—mill bay; *IV*—cold-rolled coil storage; *V*—hot tinning units site; *VI*—slitting and sheet shearing units site; *VII*—tin plate storage; *VIII*—shipping bay; *1*—hot-rolled coil conveyor; *2*—continuous pickling line; *3*—coil conveyor; *4*—1200 mm continuous five-stand cold-rolling mill; *5*—conveyor with downender; *6*—electrolytic cleaning unit; *7*—unit for applying protective coating before annealing; *8*—continuous annealing unit; *9*—hood-type annealing furnaces; *10*—electric furnaces for high-temperature annealing of transformer steel; *11*—two-high temper-rolling mill; *12*—varnishing unit; *13*—electrolytic tinning unit; *14*—guillotine shears; *15*—continuous galvanizing line; *16*—mill for corrugated sheet rolling; *17*—crane

done by using a direct current of 100-300 A at a voltage of 6-12 V. After pickling and rinsing the sheet enters a tin machine (Fig. 12.12). From the water tank the sheet is pulled out by magnetic rolls and then passed through the flux

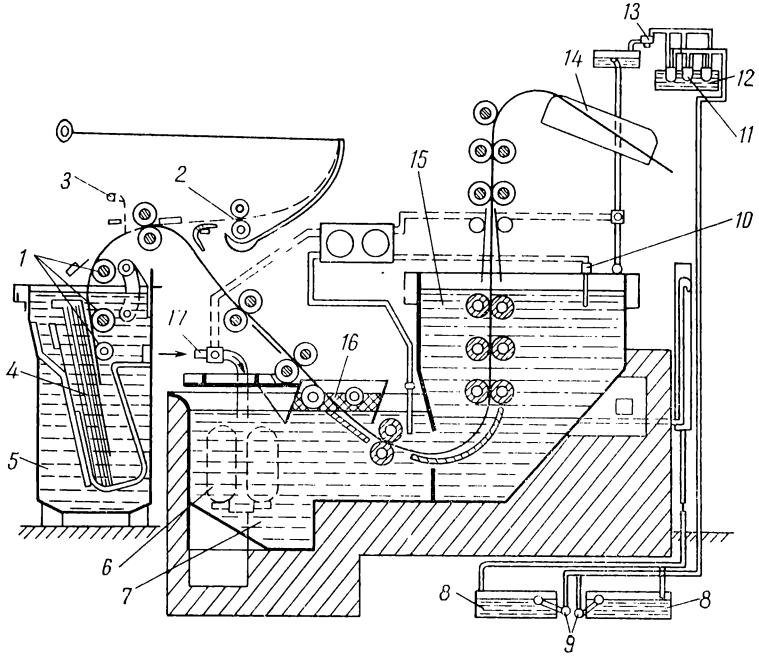


Fig. 12.12. Hot tinning unit

1—magnet rolls; *2*—doubled sheet removal; *3*—water; *4*—sheets; *5*—water; *6*—heater; *7*—molten tin; *8*—tanks for palm oil purification; *9*—pumps; *10*—thermocouple; *11*—steam heaters; *12*—palm oil heating; *13*—pump; *14*—sheets of tin plate; *15*—palm oil; *16*—flux; *17*—gas

box containing $ZnCl_2$ salt at 300 °C. In the flux box the sheet is dried and cleaned, mainly from oxides:

$$ZnCl_2 + 2H_2O = Zn(OH)_2 + 2HCl$$

$$2HCl + FeO = FeCl_2 + H_2O$$

Then the sheet is carried by rolls through the tinning vat with molten tin at 300-320 °C. By this time the temperature

of the sheet reaches 110-120 °C and tin at its surface first solidifies and then melts forming an intermediate layer of $FeSn_2$ alloy. The speed of the sheet passage through the tinning vat is of a particular importance. It should be sufficient to ensure that the first brittle layer of the tin-iron alloy is thin, but not too high so as to let the sheet acquire the optimum temperature, since at lower temperatures the thickness of the coating increases and becomes irregular. Before leaving the vat the sheet takes a vertical position and some of the molten tin flows down, provided the temperature is high enough.

From the tinning vat the sheet goes through a layer of palm or cotton seed oil at 230-240 °C. The oiling vat is installed above the tinning vat. The former is provided with rolls which squeeze out the excess tin.

Tinned sheets enter a washing machine equipped with felt rolls which are wetted with a weak solution of sodium salts to remove the oil. After rinsing and drying tin plate is cleaned and polished in a branner between whose rolls dry bran is fed to absorb residuals of oil and water. Continuous (coil) method of hot tinning is also used.

Electrolytic tinning is the most economical method. The consumption of tin with it is about twice as low as in hot tinning (7-8 as against 18-20 kg per ton of tinned metal) and the output of tin plate is higher.

The essence of the electrolytic tinning is as follows. After through cleaning and degreasing the strip to be tinned is immersed into an electrolytic tinning vat containing a solution of tin salts, the strip being used as the cathode and plates of pure tin, as the anode. To reduce the porosity of the tin coating the strip after plating is heated until the tin layer fuses and then passivated to produce a protective oxide film.

In accordance with the composition of the tinning bath the following three methods of electrolytic tinning are distinguished: alkaline, sulphoacid and haloid. The alkaline tinning bath is a solution of $NaSnO_3 \cdot 3H_2O$ salt heated to 80-90 °C. The sulphoacid bath consists of a $SnSO_4$ solution and paraphenolsulphonic acids. To obtain easy-to-fuse tin particles the bath also contains an addition of the so-called "arescap" which is a high organic surface active sub-

stance. The haloid bath consists of a solution of tin chlorides and chloro-fluorides.

The efficiency of electrolytic plating and the quality of the coating obtained depend upon the current density. It is just for this reason that the alkaline method is superseded by the sulphoacid one, which secures current densities of 0.08-0.3 A/m², tinning rate of over 5 m/s and an annual output of the order of 200,000 t, whereas with the alkaline method the respective figures are 0.05 A/m², 2 m/s and

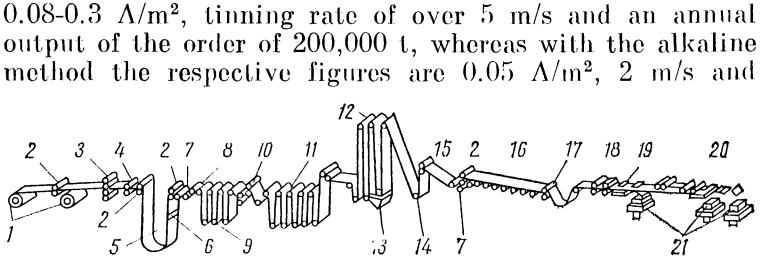


Fig. 12.13. Layout of machine for electrolytic tinning by sulphoacid method

1—decoilers; *2*—feed rolls; *3*—crop shears; *4*—welding machine; *5*—looping pit; *6*—feeding magnet; *7*—pinhole defectoscope; *8*—thickness gauge; *9*—alkali cleaning and pickling vats; *10*—brushes; *11*—tinning vats; *12*—unit for fusing tin coating and for chemical treatment of strip; *13*—water-spray washing; *14*—steam dryer; *15*—oiling machine; *16*—conveyor; *17*—flying shears; *18*—levelling machine; *19*—photoelectric sheet counter; *20*—sorting conveyor; *21*—prime and reject stackers

30,000-40,000 t. The haloid method of electrolytic tinning provides for an annual output of up to 290,000 t, current density of about 0.4 A/m² and a tinning rate of up to 10-12 m/s. However this method is disadvantageous in that harmful chlorine and fluorine fumes evolved during the process require that the equipment be hermetically sealed and arranged in a horizontal position only.

The sulphoacid method of electrolytic tinning is being used most widely.

Figure 12.13 is illustrative of the most common equipment layout of the machine for electrolytic tinning by the sulphoacid method. The units for cleaning, tinning, fusion, electrochemical treatment (passivation), washing, drying and oiling are usually arranged vertically.

The strip to be tinned is subjected to alkali cleaning and then to electrolytic pickling with a 5% sulphuric acid solution. Then it undergoes electrolytic tinning in vats, and thereafter the tin coating is fused by the contact method and

then chemically treated (passivated). The coating is fused by the Joule heating which is effected by passing an a.c. current through the coated strip moving between contact rollers. Immediately thereafter the strip enters a vat with warm water. As a result the coating cools retaining its lustre. Passivation is an electrochemical treatment in a sodium bichromate solution at a current density of 0.03-0.05 A/m^2. After passivation the strip is washed with hot water and then passed through a steam drier and oiling machine where a thin film of oil (cotton seed oil) is applied onto the strip surface in an electrostatic field.

Finished tin plate is sheared into sheets which are then inspected for the presence of pinholes and assorted in accordance with their gauge by using nucleonic defectoscopes and thickness gauges.

The overall dimensions of one of the operating electrolytic tinning lines are as follows: the total length is 120 m, the length of the strip being treated is 500 m, the height of the fusion unit tower is 7 m above the shop floor level and the depth of the looping pits is 27 m. Depending on the required surface finish the thickness of the coating ranges from 0.4 to 1.5 microns, the tin consumption being equal to 6-22 g/m^2.

It should be noted that the corrosion resistance of the electrolytic tin plate is less than that of the hot-dipped tin plate, especially when it is used for canning fish, meat or vegetables. Various organic varnish coatings (most often of phenolic resin varnishes) are used as an additional protection means for tin plate. Varnishing of tin plate is frequently done directly at can-making plants.

Galvanized Steel

The production of galvanized steel is rising rapidly. Galvanized steel is used for manufacturing utensils, in building and sanitary engineering and, recently, has found application in automobile construction. It is produced in sheets 500-1800 mm wide and 0.2-4.5 mm thick. Galvanizing of coiled steel is carried out on special continuous lines by the hot-dipping or, much more seldom, by the electrolytic method.

The equipment layout of a continuous galvanizing line provided with a horizontal annealing furnace is shown in Fig. 12.14. The entry and exit sections of this line are similar to those of the electrolytic tinning line (see Fig. 12.13). Two looping pits provide for the continuous movement of the strip through the middle section of the line.

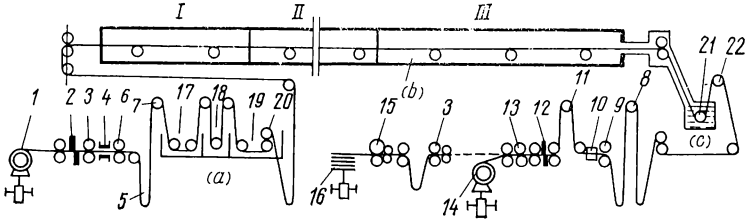


Fig. 12.14. Layout of continuous hot galvanizing line

The entry section consists of two decoilers *1* operating in turn, double-cut shears *2*, pinch rolls *3*, welding machine *4* for lap welding the ends of strips, small looping pit *5* placed before rotary disc shears *6* for edge trimming and first looper *7*.

The exit section includes second looper *8*, bridle rolls *9*, table *10* for the strip defectoscopy instruments, oiling device *11*, pinch rolls *12* followed by shears for cutting out the welds, levelling machine *13*, coilers *14* for coiled finished products, drum-type flying shears with levelling machine *15*, conveyors with sheet stacking devices *16* and a rack for rejected sheets.

The middle section of the line comprises degreasing section *a*, heat-treating section *b* and galvanizing section *c*. To economize on the floor area required for the shop the horizontal annealing furnace is installed on top of the line.

The degreasing section consists of vats *17* and *18* for electrochemical treatment in an alkaline solution, scrubbing machine *19*, washing and drying chamber *20*, bridle rolls and a small looping pit.

The strip annealing and final preparation of the strip surface for galvanizing are done in the annealing furnace. In zone *I* the strip is heated to 450-475°C and oil residues burn out. In zone *II* the strip is annealed at 730-850°C or

normalized at 950°C in a protective atmosphere. Zone *II* is radiant tube-fired. In zone *III* the strip is cooled to 500°C.

From the furnace the strip heated to 500°C goes on through an inclined duct filled with a protective gas into vat *21* containing molten zinc (without contacting the air). The temperature of zinc in the vat is 440-460°C (maintained by the hot strip entering the vat). The galvanized strip passes through squeezer rolls and air-cooling device *22*.

The speed of the strip passage through the furnace and the vat with molten zinc is 0.5-1.5 m/s. The annual output of the line is 100,000 t. The annual output of similar lines equipped with the tower-type continuous annealing furnaces amounts to 200,000 t.

Cold galvanizing (electrolytic zinc plating) is similar to electrolytic tinning. Coating of steel with other metals, such as aluminium, chrome, lead is carried out on continuous lines similar to the hot galvanizing or electrolytic tinning lines. Lately work has been in progress aimed at improving other methods of metal coating, such as vacuum and electrolytic plating and deposition of metals from molten salt baths.

Non-Metallic Coatings of Steel

Non-metallic coatings of sheet steel include plastics, paints and varnishes. Plastic film coatings allow for deep drawing of the coated metal and possess high electrical insulating properties and corrosion resistance. Painted and varnished strip steel is widely used in construction industry and also for the manufacture of household appliances. Varnishing and painting are fairly often carried out by the metal consumers themselves, but coating of steel with plastics, i.e. the manufacture of metal-plastic composite materials, is, as a rule, done at iron and steel works.

From among a great number of polymers and plastics PVC plastics in the form of films, pastes and powders, with additions of various plasticizers which improve their physical and chemical properties, are of most common use in coating steel. PVC resins yield a broad gamut of coating colours. Operations preparatory to coating the strip with a

plastic film include skin rolling in rough rolls, special pellet treatment, degreasing, pickling and passivation or phosphatizing. Then an adhesive layer is applied which is thoroughly dried and activated. Thereafter the strip and film enter a coating device provided with two rolls. The gap between the rolls should provide for a small reduction. The preliminarily coated strip is heated for the second time to fuse the coating and apply an embossed pattern when passing the strip through the rolls for the last time.

A superior method of coating steel with plastics consists in the application of powdered polymer materials. After the surface of the strip has been prepared the latter goes into a coater where a powdered polymer is applied onto it in an electric field of high intensity. Then the applied coating is fused (by induction heating) and treated in rolls to make it more dense.

12.5. Production of Electrical Sheet Steel

Rising capacities of power stations and systems, improved electric drives, and broad use of radio-electronics and automation facilities, all this has imposed higher demands upon the quality and special magnetic properties of electrical steels intended for making cores of transformers, electric machines and other magnetic systems.

Electrical grade sheet has long since been produced from low-carbon and silicon steels, but only the development of cold rolling processes and of modern means of heat treatment has made it possible to obtain the required magnetic properties.

In accordance with the methods of rolling and heat treatment, crystal structure and magnetic properties electrical steels may be divided into hot-rolled non-textured steels, cold-rolled low-textured steels and cold-rolled grain-oriented steels with cube-on-edge and cubic textures. The magnetic properties of sheet steel depend, within certain limits, upon the silicon content and the sheet thickness. The silicon content in low-alloy steels amounts to 0.8-1.8%, that in medium-alloy steels, 1.8-3.8%, and in high-alloy steels, 3.8-5.0%. The thickness of electrical grade sheet largely ranges from 0.05 to 1.0 mm. Special brands of electrical

steel may have a greater thickness (up to 5 mm) or a smaller thickness (down to 0.02 mm). The quality of electrical steels depends in large measure on the heat treatment, as well as on the presence and composition of electrical surface insulation (organic, inorganic, with ceramic additions).

In spite of a number of drawbacks of hot-rolled sheets (poor magnetic properties, heterogeneity, bad surface) the hot rolling of electrical steels continues to play a definite role, for this method makes it possible to obtain isotropic steel, provided the silicon content is sufficiently high (up to 4.8%).

Total unit losses and magnetic induction are the principal magnetic characteristics of electrical steels. The total unit power losses in magnetization due to hysteresis and eddy currents are measured in watts per kilogram (W/kg) and designated by the letter P with two subscripts. For example, $P_{10/50}$ stands for unit losses in a standard sample with a magnetic induction of 10,000 G (1T) at a frequency of 50 Hz. Of considerable significance are also other magnetic properties such as magnetic permeability, magnetostrictive properties, magnetic anisotropy, magnetodynamic properties in magnetization by an alternating current, etc.

Hot-rolled non-textured steels have the minimum unit losses $P_{10/50} = 0.80\text{-}1.0$ W/kg with a silicon content of 4.0-4.5% and a sheet thickness of 0.35 mm. A rise of the silicon content and a decrease of the sheet thickness make for the reduction of unit losses. Raising the annealing temperature and increasing the grain size contribute to the reduction of hysteresis losses.

A great proportion of hot-rolled non-textured electrical steels is used in the magnetic systems of power transformers, motors and generators operating in high-intensity fields [of more than 10,000 G (1T) at a frequency of 50-60 Hz].

Low-texture electrical steel may be hot- or cold-rolled (with critical deformation and annealing). The steel thus obtained has a weak texture wherein the grains are oriented along the axis of rolling, this resulting in a low-pronounced anisotropy (not more than 30%).

A sheet 0.35 mm thick made of low-texture dynamo or transformer steel with a silicon content of 3.0% has total unit losses $P_{10/50} = 0.85\text{-}1.05$ W/kg.

For the first time transformer steel with extremely high magnetic properties along the direction of rolling was obtained in the beginning of the thirties. This steel which contained up to 3.4% Si was obtained by Goss after two-fold cold rolling. The texture of this steel is shown in Fig. 12.15. With this texture easy magnetization axis [100] coincides with the direction of rolling while difficult magnetization

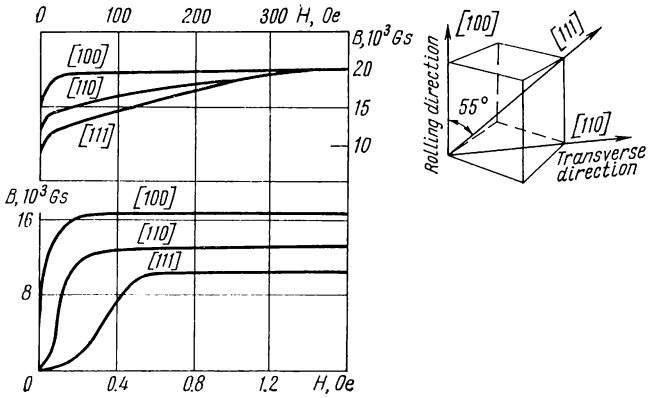


Fig. 12.15. Magnetization curves of single crystals of iron — 3.8% Si alloy

axis [111] and medium magnetization axis [110] lie at 55° and 90° to the direction of rolling, respectively. Plane (110) matches with the rolling plane. Such a texture bears the name of cube-on-edge or Goss texture.

To obtain a texture of this type one has to resort to a special method of smelting steel of a definite chemical composition (2.8-3.5% Si) and, mainly, to certain cold rolling and heat treatment procedures. Unit losses $P_{10/50}$ of a sheet 0.30-0.35 mm thick made of such a steel amount to 0.45-0.50 W/kg. The steel is highly anisotropic: the difference in the unit losses along and across the direction of rolling reaches 400%, and that in the magnetic induction, up to 700%.

Steels with cubic texture have the following grain orientation: plane (100) coincides with the rolling plane and edge [100] of the cube runs along the rolling direction. Medium

magnetization axis [110] lies at an angle of 45° to the rolling direction, and the difficult magnetization axis is altogether beyond the sheet magnetization plane.

Transformer steel with cubic texture is characterized by high magnetic properties along and across the rolling direction. The cubic texture is obtained by various methods, e.g. by changing the reduction ratio in two-fold (sometimes multiple) cold rolling or by creating particular conditions for the second recrystallization during high-temperature annealing within the range of 1100-1300°C which is usually done in the atmosphere of extra-pure electrolytic hydrogen acting as a potent reducing agent.

The best grades of electrical steel are produced by using the well-developed technology of obtaining the cube-on-edge structure. With a large volume of production the general processing pattern is as follows. The starting material comprises rolled or cast slabs of extra-pure electrical silicon steel which are heated to 1200-1250°C with a prolonged soaking. Then they are rolled by the coil rolling method in continuous or semi-continuous strip mills or in a reversing mill with furnace coilers. The optimum thickness of the hot-rolled strip plate thus obtained is 2.0-2.5 mm, the finishing temperature being equal to 840-860°C.

The most typical and currently adopted pattern for cold rolling of transformer steel includes the following operations:

1. Strip plate descaling on continuous pickling lines, sometimes involving the use of additions of strong acids (e.g. hydrofluoric acid) or of shot blasting units.
2. First operation of cold rolling in a continuous or reversing mill for the strip thickness of 0.62-0.85 mm.
3. Intermediate annealing at 925-980°C for 3-5 min on continuous annealing lines for the purpose of decarburization and recrystallization, a damp hydrogen containing gas being fed into the soaking zone.
4. Second operation of cold rolling for the strip thickness of 0.30-0.35 mm.
5. Second decarburize annealing in continuous or hood-type annealing furnaces.
6. Strip coating with non-organic substances to prevent sticking.

7. High-temperature hydrogen annealing for 25-30 h at 1150-1180°C for the purpose of second recrystallization. This operation is done in hood-type annealing furnaces, occasionally in recuperative ones. Vacuum annealing furnaces are also used.

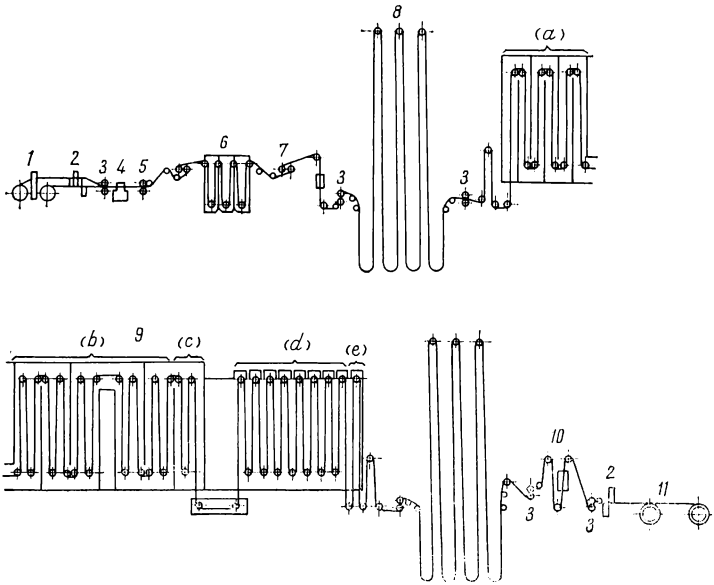


Fig. 12.16. Layout of line for heat treatment of transformer steel (*a*) heating chamber; (*b*) soaking chamber; (*c*) controlled cooling chamber; (*d*) accelerated cooling chamber; (*e*) forced air cooling chamber; *1*—decoiler; *2*—guillotine shears; *3*—pinch rolls; *4*—welding machine; *5*—spraying and brushing machines; *6*—electrolytic degreasing vat; *7*—dryer; *8*—looping pits; *9*—tower furnace; *10*—unit for applying protective coating; *11*—coiler

8. Annealing at 760-820°C in continuous furnaces to eliminate the strip camber resulting from the high-temperature coil annealing.

9. Strip coating with electrical insulating materials. This operation is done in the entry section of continuous final annealing lines so that drying of the coating takes place simultaneously with heat treatment.

At the present time continuous furnaces in combination with the in-line type equipment for cleaning, electrolytic

degreasing, drying and coating are successfully used for making transformer steel. Annealing with decarburization in the atmosphere of hydrogen, followed by controlled cooling, is the key operation. The reliability and safety are secured by using electric heaters with a capacity of 10,000 and more kW. Lately continuous annealing furnaces have been built in the form of vertical (tower-type) units, this helping economize on the floor area required for the equipment. The output of these furnaces amounts to 55-60 t/h.

Figure 12.16 shows the equipment layout of a continuous line for the heat treatment of a strip 600-1020 mm wide and 0.1-1.0 mm thick. The line is fitted out with a tower-type furnace which includes heating, soaking, controlled cooling, accelerated cooling and forced air cooling chambers. The furnace has a hermetically sealed sheet steel shell with stiffening ribs. The first two chambers are lined with lightweight refractory brick. Slow cooling from 950 to 700°C is effected by using air-cooled tubes. Accelerated cooling from 700 to 150°C is done by means of water jackets and from 150° to 40-50°C, by air blasting. The strip speed is adjustable within the range of 30-300 m/min.

Electrical insulation coatings with sufficient electrical resistivity, dielectric and mechanical strength and small thickness are available as oxide films, organic varnishes and paints, ceramic and phosphate compounds. They may be applied by dipping, rolling, air spraying or through electrophoresis.

12.6. Stainless Sheet Production

Stainless steel sheets find extensive application in chemical machine building and other branches of industry. Therefore its production becomes particularly important.

In accordance with the metal structure obtained after air cooling subsequent to high-temperature heating high-alloy steels are divided into the following classes: (a) martensitic (the principal structural constituent is martensite); (b) martensite-ferritic (in addition to martensite there is 5-10% of ferrite); (c) ferritic (the principal structural constituent is ferrite); (d) austenite-martensitic; (e) austenite-ferritic (in

addition to austenite there is more than 10% of ferrite); (f) austenitic.

Stainless sheet is produced mainly by cold rolling the steels of the ferritic, austenite-ferritic and austenitic classes.

The main grades of stainless steel contain not more than 0.08-0.14% C, not more than 0.6-0.8% Si, 0.6-2.0% Mn, 11.0-22.4% Cr, 4.8-11.0% Ni (in some grades), about 0.7-0.8% Ti, not more than 0.020-0.025% S and not more than 0.030-0.035% P.

Steels containing 18% Cr and less than 0.08% C are purely ferritic, while with a higher carbon content an austenite-ferritic structure is formed.

Heating of single-phase ferritic steels to above 600°C results in a sharp grain growth and the coarse-grained structure thus obtained cannot be improved by subsequent heat treatment. Therefore adequate mechanical characteristics can only be obtained by using the optimum combination of temperature and deformation. Ferritic chrome steels are characterized by rough surface defects in ingots and slabs, as well as by higher brittleness which may result in cracks. Research has shown that the development of brittleness depends upon the cooling rate (after rolling the slabs should be cooled in air).

Austenitic steels containing about 18% Cr and up to 9% Ni are more common. Apart from chrome and nickel these steels contain other alloying additions. To eliminate the susceptibility of austenitic steels to intercrystalline corrosion use is made of titanium and niobium which bind carbon and thus prevent the formation of chrome carbides.

Depending upon the chemical composition and heating temperature austenitic steels may occasionally contain from 5 to 35% of α-phase (ferrite). The plasticity of such steels depends upon the content of the excess phase. Ferrite and austenite have different properties and this may cause the development of internal stresses and cracks when deforming the two-phase steel. The selection of rolling temperatures for austenite-ferritic steels presents difficulties, for the increased resistance to deformation of these steels requires higher rolling temperatures, but the presence of the α-phase which reduces the metal plasticity at high temperatures

compels that the rolling be done within a narrow range of temperatures.

Stainless steels harden during cold rolling and this makes it necessary to decrease the pass reductions and increase the number of passings (up to 3-5) with intermediate heat treatment operations, to improve the metal plasticity. Austenitic steels are heated to 1100-1150°C with subsequent water quenching or air cooling which provides for their uniphasic structure and better plasticity. Low-plasticity ferritic steels are annealed at 740-780°C to obtain a ferrite-carbide structure.

At present stainless sheet is produced either in specialized shops or in production areas specially allotted for this purpose in the cold-rolling shops. At modern works the stainless steel blooms are rolled in bloomings, slabbings or in combination blooming—plate rolling mills. Continuous casting of slabs is also in progress. Turning, planing, milling, grinding and flame scarfing are employed for a complete removal of surface defects. Recently use has been made of the so-called thermomilling machines capable of scalping the ingots heated to a temperature of 1100-1200°C. Hot rolling is carried out with rolling mills of various designs, including reversing and non-reversing plate mills, reversing mills with furnace coilers, multiroll mills, planetary mills and also with semi-continuous and continuous wide-strip mills, if the volume of production is large enough.

The standard technological pattern for cold rolling of stainless sheet less than 3 mm in thickness is as follows: softening heat treatment, pickling, cold rolling, heat treatment, pickling. The cold rolling, heat treatment and pickling operations may be repeated several times, depending upon the final sheet thickness. The first softening heat treatment and pickling operations are often carried out on a continuous line. Pickling is combined with shot blasting. Austenitic chrome-nickel steels permitting rapid heating and cooling are annealed in continuous furnaces, while ferritic chrome steels, in hood-type annealing furnaces.

More often than not cold rolling is done in four-high reversing mills or in multiroll mills, or else in continuous mills, provided the volume of production is high enough.

The intermediate heat treatment operations are performed on continuous combination heat treatment—pickling lines. In use are also continuous quench and pickle lines which provide for a strip speed up to 45 m/min and a maximum output of up to 40 t/h. After preliminary shot blasting the strip is pickled in a sodium hydride solution and in acids, passivated, rinsed and then coiled. Continuous muffle furnaces of the horizontal and vertical types are used for bright annealing of stainless steel.

Expensive alloying materials (especially nickel) can be saved by making the so-called clad steel.

Clad steel may be produced in various ways: by pouring metal of different chemical composition into moulds with partitions; by pouring metal onto one or several plates, by rolling welded packs made up of different steels, by welding the cladding metal onto base plates by the method of electroslag welding.

12.7. Specific Features of the Equipment Used in Cold Rolling

Working Stand Equipment

As concerns their design the working stands of sheet mills differ but little from those used in mills of other types. However, considering high pressures involved in sheet rolling, their housings, chocks and pressure screws are made with increased cross-sectional areas.

Apart from the high strength and stiffness of the working stands—roll housings (usually of the close-top type)—as well as of the rolls with chocks and bearings and of the mill screws the dimensional accuracy of the rolled sheet is also achieved by employing a special design of screwdowns and adjusting devices which meet high demands on the accuracy of the roll gap setting. Stringent requirements are also imposed upon the finish of the working surfaces of mating components and assemblies. The work rolls of four-high stands are supported by roller bearings. Film lubrication bearings have recently found application for the backup rolls.

Accurate setting of the rolls is provided by selsyns. The screwdowns in most sheet mills are designed for setting the

rolls under the full rolling pressure. As a rule, sheet mills use hydraulic balance for the top backup and work rolls. The hydraulic cylinders are installed either on top of the stand or between the chocks of the work and backup rolls.

Lately in use has come the strip gauge control by hydraulic devices operating on the principle of thrusting apart

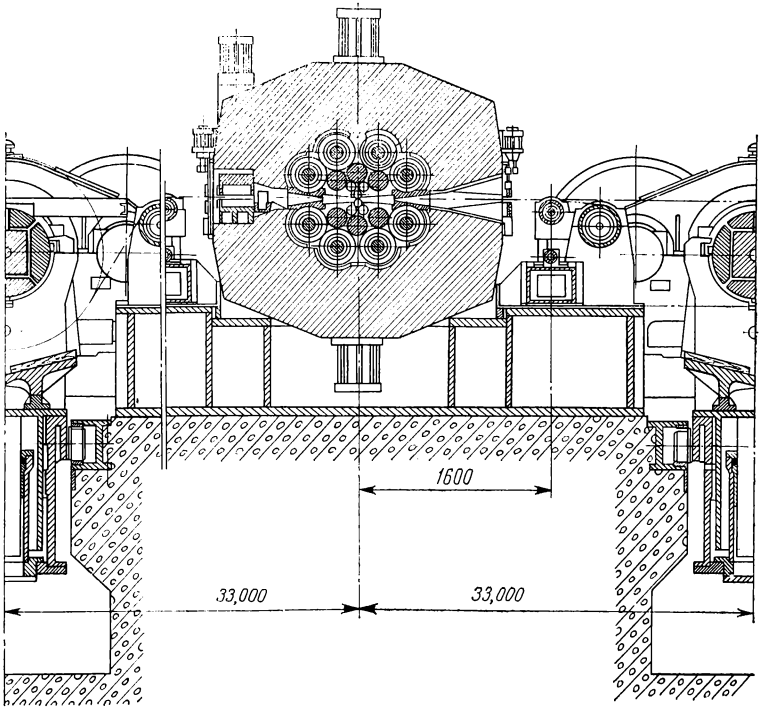


Fig. 12.17. Twenty-high cold-rolling mill

with variable force of 100-500 tf the chocks of the backup rolls in the finishing stands of wide-strip mills.

Multiroll and planetary mills are used for rolling ultra-thin sheet and band with the minimum dimensional tolerances and, in some cases, also for rolling difficult-to-deform special steel grades. With multiroll mills the rolling is done

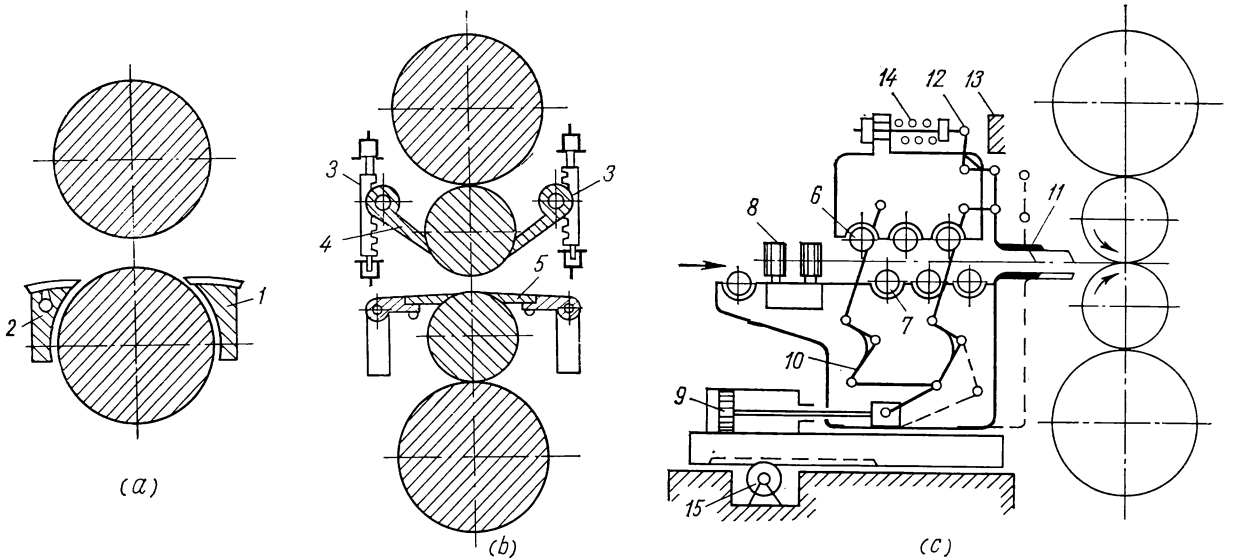


Fig. 12.18. Guides for plate and sheet mills

(*a*) slide bars for two-high roughing stand of plate mill; (*b*) guides for four-high finishing stand of plate mill; (*c*) guide roller table for 1700 mm continuous cold-rolling mill; *1*—front bar; *2*—back bar; *3*—toothed rack of hydraulic drive; *4*—top guide; *5*—bottom hinge-type guide; *6*—top idle rollers; *7*—bottom idle rollers; *8*—side guide rollers; *9*—cylinder; *10*—leverage for lowering top roller; *11*—block; *12*—pressure control arm; *13*—stop; *14*—spring; *15*—rack-and-pinion drive of shifting table

in rolls of a minimum diameter (6.3-9.0 mm) and with a high strip tension.

Figure 12.17 shows a twenty-high cold-rolling mill whose roll housing is made in the form of an all-in-one-piece casting with a shaped base for the set of rolls. The work rolls are made of tungsten carbide which has high wear resistance and elastic modulus. In multiroll Sendzimir mills the work

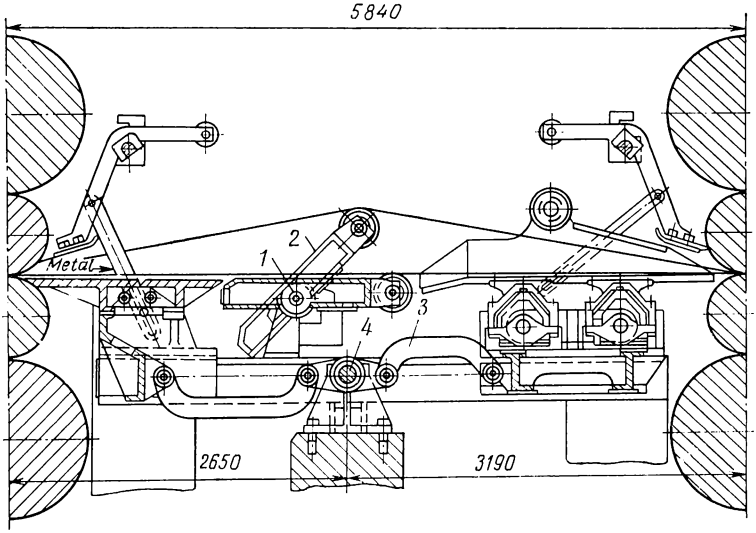


Fig. 12.19. Looper of 1700 mm sheet mill
1—arm- and-drive shaft; *2* – arm; *3*—link rod; *4*—roker arm

rolls practically do not bend, thanks to the high stiffness of the stand. The diameter of the rolls in this mill is 6.3 mm and the mill is capable of rolling high-alloy steel sheet 0.00254 mm in thickness. The ultra-thin tin plate can be rolled through a single passing only.

To properly feed the workpiece into the rolls and to keep it in the required position when it is leaving the rolls sheet mills have guides in the finishing stands and heavy slide bars in the roughing stands. The slide bars frequently carry on their back side nozzles of the hydraulic jet descaler. In cold-rolling mills the workpiece is fed into the rolls by means of special guards or adjustable roller tables (Fig. 12.18).

Three types of loopers are employed in sheet mills, *viz.* pneumatic, hydraulic and electrical. The looper, guides and guards installed between the stands of a 1700 mm continuous mill are shown in Fig. 12.19. The headers and nozzles of the hydraulic jet descaler are either installed separately

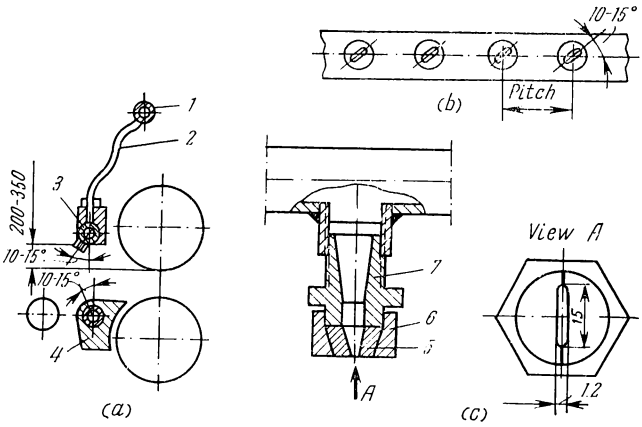


Fig. 12.20. Hydraulic jet descaler

(*a*) vertical section; (*b*) nozzle arrangement; (*c*) nozzle cross section; *1*—pipeline *2*—flexible hose; *3*—top header; *4*—bottom header; *5*—orifice plate; *6*—captive nut; *7*—nozzle body

or built into the slide bars of the stands (Fig. 12.20). Under a pressure of 100-120 at water jets are sprayed at an angle of 75-80° to the strip plate in a direction opposite to its movement through the nozzles with rectangular orifices. The nozzles should be set up above and below the strip plate at a distance of 200-300 mm from its surface. The hydraulic jet descaler includes a pump station, air-water bottle accumulator, scale settling tank and pipelines with valves. The hydraulic jet descaler is controlled automatically.

Lubricants used in cold-rolling mills are either emulsions (which serve at the same time to cool the rolls) or vegetable and animal oils. The lubrication is implemented through the use of circulating oil systems provided with filters and temperature controllers. In small-sized multiroll mills use is made of an integrated lubrication-and-rolling oil system with good filtration and cooling.

Auxiliary Equipment

Sheet mills are equipped with main, intermediate and carry-over tables. The table rolls may be fluted or tapered, the roll pitch is held to the minimum. Turning of the strip plate in the horizontal plane is realized by using roll tables consisting of individually driven sections. The rolls of the adjacent sections are alternately tapered and when rotating in the opposite directions they turn the strip. But more often than not such a turning is effected by means of lift-and-turn devices or with special fingers which project above the roll table level. Horizontally moving manipulator heads serve to properly feed the workpiece into the rolls. Three-high stands are equipped with lifting tables. In coil rolling use find various coilers and decoilers which service almost all types of the rolling equipment (rolling mills, shear lines, etc.).

Up and down coilers of the three-roll- or multiroll-type are used for hot strip. Narrow band and skelp are handled with the three-roll- or multiroll-type up coilers. Thin sheet can be tightly coiled by means of roll-drum-type down coilers (Fig. 12.21). The drum of the coiler made up of spring-loaded arc segments can be collapsed and then pushed aside by a special arm to remove the coil.

Furnace coilers are equipped with solid drums made of heat resistant steel. They have no coil removing arrangements because after the last passing the strip is not coiled.

Tension reels are employed in cold-rolling mills and finishing lines.

To uncoil the strip and feed its tail into the rolls in the course of rolling or during various finishing operations use is made of decoilers equipped with magnet- (Fig. 12.22) or stripper-type tail forming devices. Coil *6* rolls from racks down onto arm *1* and is held in place by rollers *2* and *3*. Driven roller *2* turns the coil until its tail faces electromagnet *4*. Then a crank gear lowers rollers *7* and *8*, pinion *10* fixed on shaft *12* turns toothed sector *11* which makes arm *13* go down to allow electromagnet *4* attract the coil tail. Thereupon arm *13* is lifted, thus stripping off the coil tail, rollers *7* and *8* go up and press the tail against driven roller *9* which feeds it into a mill stand or levelling rolls. A special device which actuates rod *5* helps maintain the axis of the

coil at one and the same level while the coil diameter is changing. Tail strippers of a very simple design are used for uncoiling non-magnetic steels.

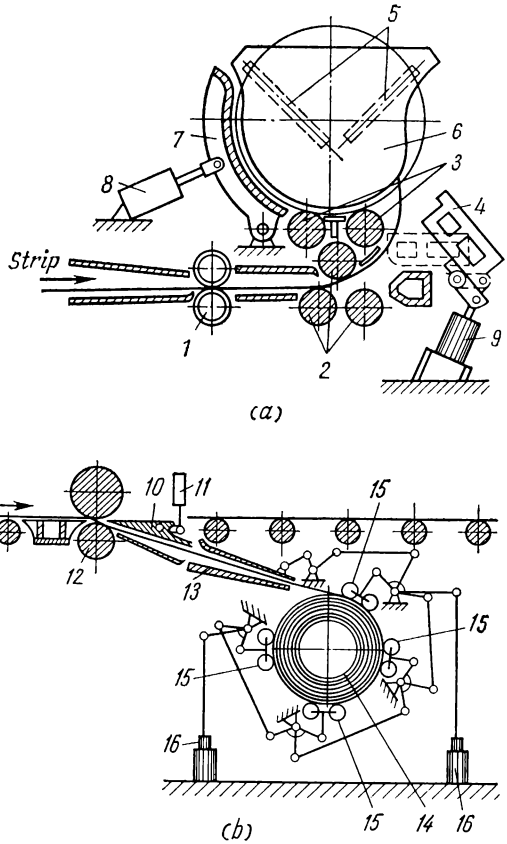


Fig. 12.21. Coilers

(*a*) three-roll up coiler; (*b*) roll-drum-type down coiler; *1*—guide rolls; *2*—forming rolls; *3*—supporting rolls for coil; *4*—plate; *5*—side guide rolls; *6*—expanding side plates; *7*—coil pusher; *8*—pneumatic coil pusher drive; *9*—pneumatic plate turning drive; *10*—hinge-type gate; *11*—pneumatic gate drive; *12*—guide rolls (bottom roll is driven); *13*—guide; *14*—drum; *15*—forming rolls; *16*—pneumatic cylinders of forming roll linkage drive

Guillotine, rotary disc, flying and scrap chopper shears are employed in the manufacture of plate and sheet steel.

Guillotine shears serve the purpose of hot and cold shearing of stationary sheet. Inclined throat (α = 3-5°) downcut guillotine shears are most common.

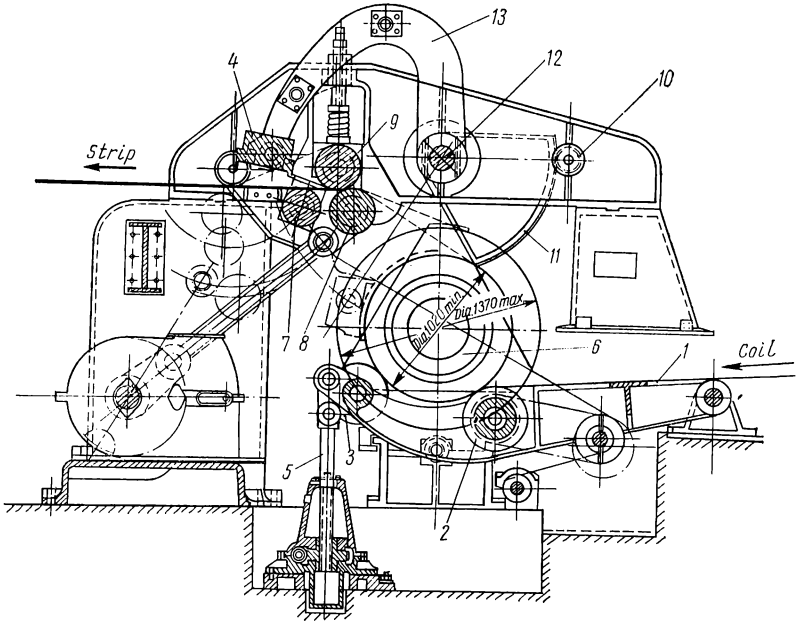


Fig. 12.22. Decoiler with magnet-type tail forming device for 1700 mm cold-rolling mill

Shearing force P depends upon thickness h of the metal to be cut, angle of the blade inclination α, ultimate strength σ_B of the metal (or resistance to deformation of the hot metal) and upon the ratio of the depth of shearing before fracture to the initial thickness of the metal (ε_s) which is equal to 0.3-0.4 for low-carbon and 0.1-0.2 for high-carbon steels (for hot metal $\varepsilon_s = 1$):

$$P = \varepsilon_s \frac{h^2}{\tan \alpha} \sigma_B$$

The permissible side clearance between the shear blades is 0.2-1.2 mm. The upper knife slide is balanced. The shears

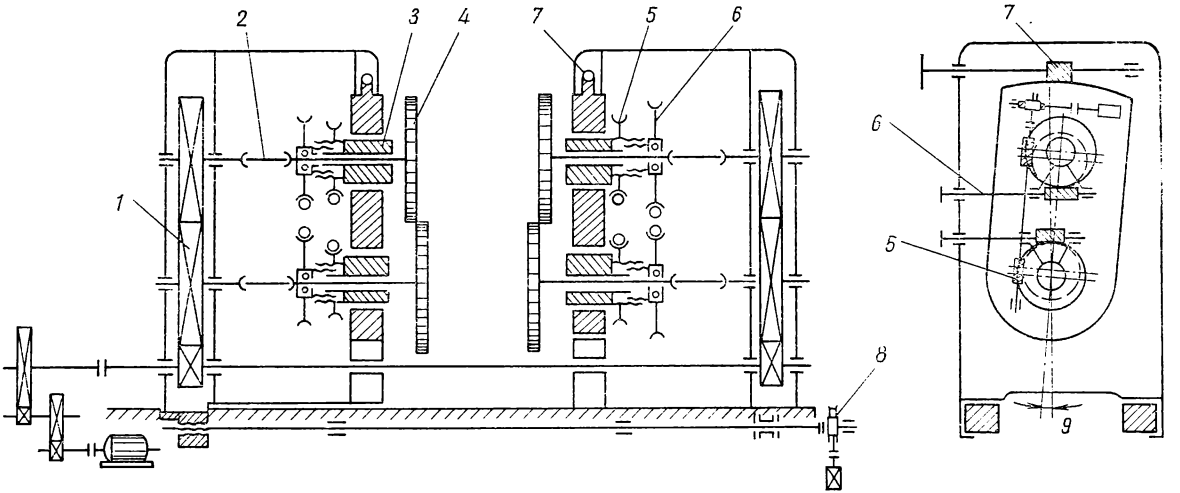


Fig. 12.23. Kinematic diagram of rotary disc shears

1—reducing gear; *2*—universal spindles; *3*—eccentric sleeves; *4*—blades; *5*—mechanism for adjusting vertical clearance between blades; *6*—mechanism for adjusting side clearance between blades; *7*—mechanism for adjusting cutting gear inclination; *8*—housing drive mechanism; *9*—angle of top blade inclination

are equipped with hold-downs for the sheet being sheared and also with guides for the scrap edge. Separately installed shears are fitted out with ball or roller supports to facilitate handling of the sheet.

The rotary disc shears (Fig. 12.23) serve to trim the side edges of sheets and plates up to 25 mm in thickness and also

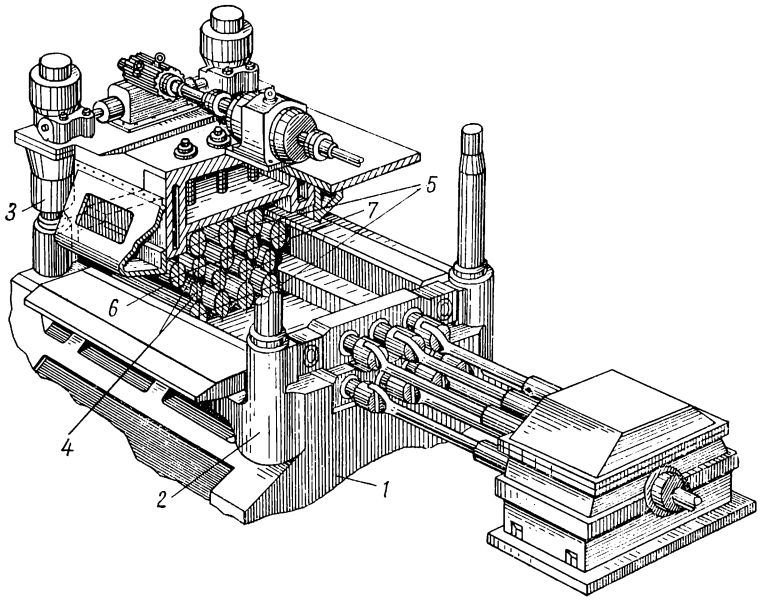


Fig. 12.24. Hot roller leveller

to slit coiled steel into narrow strips and bands. In the latter case use is made of gang slitting shears with three or more pairs of discs. Trimming shears have two pairs of discs installed in separate housings (one of them is fixed while the other can be traversed to suit the required sheet width). With a correct trimming the scrap edge bends down and then goes to the scrap chopper. To achieve this the top disc is displaced with respect to the bottom one or the diameter of the top disc is made smaller than that of the bottom one.

Flying shears are extensively used in modern hot-rolling mills and cold shear lines operating within a wide range of

speeds. These include drum-, crank- and crank-lever-type shears. The design of all these shears is similar to that of the previously described flying shears for cutting section steel.

Multiroll levellers are employed for flattening cold and hot sheet steel; occasionally use is also made of stretcher levellers wherein the sheet is flattened by stretching it with a stress slightly exceeding the yield strength of the metal it is made of.

The hot sheet leveller (Fig. 12.24) consists of housing *1* with cross members *2* and *3*, work and backup rolls *4-7*. The work rolls (with internal water cooling) are driven ones. The gap between the rolls is set by means of a screw-down.

12.8. Automation of Plate and Sheet Steel Production

The characteristic trends in the automation of the plate and sheet steel manufacture are as follows.

Partial automation provides for automatic control and interlocking of individual mechanisms or groups of production equipment. Such an automation system cannot correct even the slightest changes in the production process parameters.

Automation of continuous gauging of technological parameters (temperature, pressure and tension in rolling, dimensions of semi-finished and finished products) makes provisions for manual and automatic control of the production process.

Integrated automation envisages the use of computers and in most advanced systems, of control computers. On the basis of the main criteria (sometimes initial and final data) stored in the machine memory and of continuous information on the varying process parameters the machine selects optimum processing conditions and sends control pulses to all the actuating mechanisms, conformable to the control programme formulated in the machine language (algorithm).

Modern plate and sheet steel production places special demands on the continuous non-contact gauging of the sheet

thickness and width. Such a control is accomplished by using X-ray and radiation thickness gauges and electronic width gauges.

For a long period of time work is in progress aimed at automation of the strip gauge control in hot and cold rolling. With continually growing demands on the quality and dimensional accuracy of sheet steel the increase of its output and of the rolling speed can be achieved only through introduction of the automatic control systems. The introduction of an automatic programmed sheet gauge control in continuous hot-rolling mills makes it possible to decrease in a number of cases the longitudinal strip gauge variations from 0.3-0.4 down to 0.1-0.15 mm. In continuous cold-rolling mills this system provides for the production of sheet with a tolerance on the thickness within the range of ± 6 microns and helps to substantially reduce the strip cambering and buckling.

12.9. Technical and Economical Characteristics of Sheet Steel Production and Accident Prevention

Procedures involved in estimating the productive capacity of plate and sheet mills differ in no way from those mentioned earlier for section mills.

The annual output of the latest continuous hot-rolling mills with a roll body length of 2000 mm amounts to 6 million tons, that of continuous four-stand cold-rolling mills, to 2 million tons and of five-stand cold-rolling mills, to 800 thousand tons.

The metal consumption factors are the key economical characteristics, since the cost of the metal amounts to 80-95% of the total cost of plate and sheet products.

Depending upon the method of production, the overall ingot consumption factors for hot-rolled sheet come to 1.2-1.26 for rimming carbon steel and 1.28-1.34 for killed steel; the slab consumption factors range from 1.025 to 1.070. The overall ingot consumption factors for cold-rolled sheet and tin plate come to 1.34-1.37 for rimming steel and 1.4-1.5 for killed steel.

The per ton fuel consumption in heating slabs is $4 \cdot 10^5$ kcal and that in heat treatment, $250 \cdot 10^3$ kcal.

The power consumption per ton of metal rolled in continuous hot-rolling mills amounts to 50-60 kW·h and that in continuous cold-rolling mills, 90-120 kW·h.

The water consumption per ton of steel rolled in continuous hot-rolling mills is 12-15 m^3, that in cold-rolling mills, 20-25 m^3.

The roll consumption per ton of rolled metal is 1.7-1.8 kg for continuous hot-rolling mills and 1.4-2.0 kg for continuous cold-rolling mills.

Mechanization and automation of modern plate and sheet mills envisages the organization of safe work. Provisions are made for safe access to the equipment including the arrangement of safe passages, platforms, stairs, protective panels and other types of guards.

The operating personnel is supplied with means of individual protection, *viz.* metal nets, leather breast-plates, asbestos covers, etc.

Pits for equipment (coilers, conveyors) should be guarded by handrails. Safe access to the down coilers is secured by arranging special passages in the cellar which are fenced off from mechanisms by nettings. The accident prevention regulations require that the passages be provided with doors interlocked with the coiler drive.

All the tunnels should be well ventilated, and their overall dimensions should meet the requirements stipulated by the regulations (the width of the passages for men in tunnels intended to handle hot coils should be not less than 1 m). Conveyors and transfers are equipped with barriers not less than 400 mm in height.

The workplaces at the hot-rolling mills are forced-ventilated, the ventilating air being cooled in the summer time.

The inspection and marking out of steel plates are done at a temperature of not higher than 60-70°C. Safe passageways with handrails should be arranged along the inspection tables for the movement of men, gangways being provided for crossovers. Inspection, marking out and identification are, as a rule, carried out from safe stagings and foot bridges.

The marking paint is stored and prepared for use in special isolated and well ventilated rooms with water or steam central heating and explosion-proof lighting.

Threading of coilers should be mechanized; manual threading is permissible only when the coiler is stopped.

Control apparatus are installed separately from the stands. The stands are equipped with deflectors to prevent splashing of the emulsion and also with exhaust hoods to collect vapours and emulsion mist. Scrubbing machines are covered and provided with exhaust ventilation.

Annealing rooms should be isolated from other production areas of the rolling shop and aerated. Gas-fired annealing furnaces must comply with safety regulations. Hood-type annealing furnaces should be equipped with exhaust ventilation and outdoor scavenge bleeders.

Electrical annealing furnaces should also comply with the regulations. The leadins should be furnished with heat-resistant insulation and the stands should be reliably earthed. To prevent the metal contacting the heating elements use is made of special insulating spacers (alundum, etc.).

Cleaning lines should be reliably fenced. The looping towers are guarded by nettings and equipped with exhaust ventilation, as well as with sound and light signalling devices.

Threading of all the mechanisms of continuous furnaces is done when they are cold. To thread the hot furnace should the strip break, use is made of special plugged manholes which can be opened only after the gas (or power) supply has been cut off.

It is of great importance that in degreasing, pickling and plating safe conditions be ensured. In addition to overall sealing and ventilation, cleaning divisions are to be equipped with facilities for dust removal at the site of its generation. Continuous pickling vats have to be hermetically sealed and the collection of acid fumes, which should be cleaned from harmful constituents prior to their exhaustion into the atmosphere, is to be provided for.

No manual labour whatsoever should be employed in handling pickling facilities; reliable signalling as to changes occurring in the normal course of the process and an automatic remote control must be available. Should the strip break all operations involved in joining the broken ends are to be carried out from special foot bridges with reliable protective guards.

In electrolytic pickling use has to be made of acid-proof electrical insulation and the voltage must not exceed 30 V.

The control and regulation equipment should be set up beyond the range of the action of noxious fumes. Local exhaust devices (hoods, shafts) made of acid-proof materials are employed to remove the evolving gases. Acid storage tanks are of an underground type, the tanks covers being kept under lock and key.

The acid is poured into the pickling vats after these have been filled with water. Waste liquors are let out through a special sewerage system. The tanks are cleaned following their ventilation. This work must be carried out by at least three men with the foreman among them.

The pickling division operators are provided with protective goggles.

All the continuous lines are equipped with on and off light and sound signalling systems and with emergency switches.

CHAPTER XIII. MANUFACTURE OF TUBULAR PRODUCTS

The manufacture of pipes and tubes is a progressive, rapidly developing branch of the metal rolling industry. The process of making tubular products and its technological principles possess a number of specific features which make it different from other metal rolling processes.

In accordance with the method of their manufacture steel pipes may be seamless and welded and also hot-rolled, cold-rolled and cold-drawn. As to the method of joining there are butt-welded, screwed (sleeve, nipple, etc.) and flanged pipes.

The industry puts out round, oval, rectangular, ribbed, and tapered pipes, as well as those of variable wall thickness. Pipes and tubes may also be of general and special purposes.

13.1. Hot Rolling of Pipes

Pipes are hot rolled from cast ingots, solid or pierced billets produced by rolling or forging and also from centrifugally cast shells. In their turn hot-rolled pipes may represent semi-finished products, intended for further cold rolling and drawing. Pierced forged billets and cast shells are used in making pipes of high-alloy and difficult-to-deform steels, whereas cast ingots with a diameter of more than 270-300 mm are employed for hot rolling of large-diameter pipes.

More often than not use is made of round billets, but when making pipes in presses and units with the rack-type draw-benches and small-sized pilger mills, square and polyhedral billets are employed.

There is a certain specificity in the pretreatment of the metal for rolling. Prior to piercing the metal to obtain shells the billet is divided into comparatively short lengths by various methods (cutting, breaking in a press). Considering the specific shape of the billet, in addition to conventional continuous furnaces use is also made of roll-down furnaces with a hearth slope of 12° along with rotary-hearth and barrel-type furnaces. Walking-beam type furnaces are employed in some metal conversion operations and in the heat treatment of pipes. Application also finds electric heating, including that of induction type. Heating and heat treatment are done in a protective atmosphere or in vacuum.

The general pattern of hot rolling of seamless pipes is as follows.

1. Piercing of a solid, usually round billet (ingot) to obtain a hollow, thick-walled shell. This operation is preceded by the pretreatment (inspection, cleaning, cutting to length, centring) and heating of the billets.

2. Rolling of the shell into a finished pipe or into a semi-finished product, intended for further rolling in finishing mills. According to the method of rolling the shell and the type of the mill used the following units for making hot-rolled seamless pipes are distinguished: plug-rolling, continuous, three-roll and pilgrim. These units are equipped with piercing and finishing (reeling, sizing, reducing) mills

of various designs. Rolling is sometimes done with intermediate heating operations.

3. Finishing of pipes. In addition to cutting, straightening and chamfering the pipe finishing operations include threading, galvanizing, varnishing, cladding, etc.

Piercing Mills

The main operation in the hot rolling of pipes is piercing of the billet, i.e. making of the shell.

Save for comparatively seldom used drilling out and centrifugal casting the main methods of piercing are helical

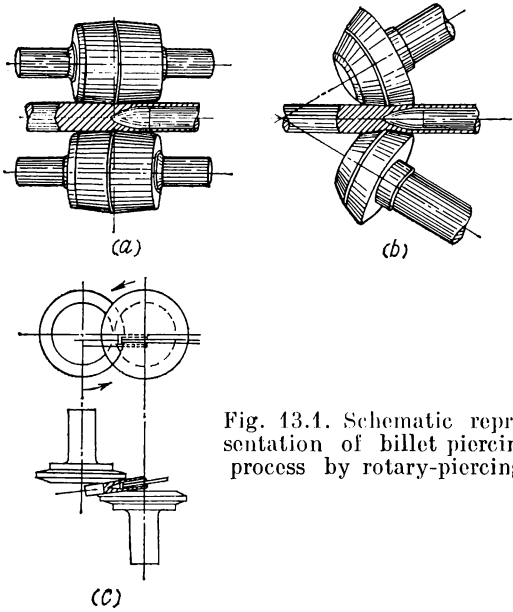


Fig. 13.1. Schematic representation of billet piercing process by rotary-piercing

rolling and broaching in presses. The helical rolling, which combines the elements of conventional longitudinal and cross rolling operations, is done in mills equipped with barrel-, cone- and disc-shaped rolls (Fig. 13.1). In barrel-type roll piercing mills the angle between the rolls in the vertical plane ranges from 5 to 12°; in cone-roll piercing

mills the rolls are inclined with respect to the axis of rolling at an angle of 30° in the horizontal plane and of 7.5° in the vertical plane, and in disc mills the rolls are set parallel to each other, while the axis or rolling is displaced by approximately 50 mm below the horizontal plane passing through the roll axes. With the helical rolling the rolls rotate in the same direction and the billet being pierced is thus set in rotary and translatory motion.

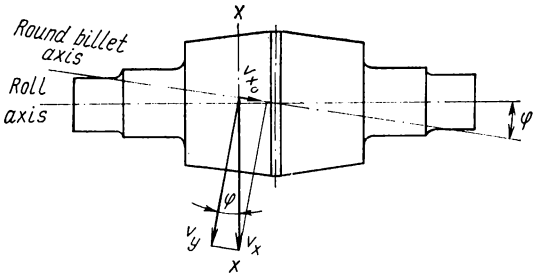


Fig. 13.2. Resolving of peripheral speed vector

Below under consideration is the diagram of forces acting in the zone of deformation during helical rolling (Fig. 13.2). In the zone of piercing the peripheral speed of a point on the surface of the roll barrel in section x—x is

$$v_x = \frac{\pi D_x n}{60}$$

where D_x = diameter of the roll barrel in section x—x
n = roll revolutions per minute

Resolving speed v_x into two components, *viz.* feed speed v_{x_0} and shell rotation speed v_y, in accordance with the parallelogram law of velocities, we have

$$v_{x_0} = \frac{\pi D_x n}{60} \sin \varphi$$

$$v_y = \frac{\pi D_x n}{60} \cos \varphi$$

where φ = angel of the roll inclination relative to the axis of piercing.

The billet revolutions per minute n_x may be found proceeding from the equality of the peripheral speeds of a

point on the roll barrel and the corresponding point on the billet, account being taken of the metal slippage on the rolls:

$$\frac{\pi D_x n}{60} \cos \varphi = \frac{\pi d_x n_x}{60} \eta \xi_x$$

whence

$$n_x = \frac{D_x}{\xi_x d_x \eta} n \cos \varphi$$

where d_x = diameter of the billet in section $x - x$
η = coefficient of slip (η = 0.60-0.98)
ξ_x = coefficient of the billet ovality in the given section

Coefficient of ovality ξ_x ranges from 1.014 to 1.023, depending on the length of the piercing cone.

The axial travel of the given section of the pierced billet per one revolution, i.e. the lead of the feed helix (S_x), may be determined proceeding from the equality of the second volumes of the metal in different sections of the body being deformed:

$$F_0 v_0 = F_x v_x \ldots = F_{sh} v_{sh}$$

where F_0, F_x = cross-sectional areas of the billet
F_{sh} = cross-sectional area of the shell
v_0, v_x = speed of the billet travel in different sections of the zone of deformation
v_{sh} = speed of the shell travel at the point of delivery

The travel speed in any section of the zone of deformation is

$$v_x = \frac{F_{sh}}{F_x} v_{sh}$$

By taking account of the metal slippage relative to the rolls and designating through v_r the axial travel speed of a point on the roll surface at the shell exit and through η_0 the coefficient of slip in the axial direction, we get

$$v_{sh} = v_r \eta_0$$

and

$$v_x = \frac{F_{sh}}{F_x} v_r \eta_0$$

Substituting the value of the axial travel speed of a point on the roll surface yields

$$v_x = \frac{\pi D_r n}{60} \sin \varphi \frac{F_{sh}}{F_x} \eta_0$$

Designating the duration of one revolution of the billet by τ_x we obtain

$$S_x = v_x \tau_x$$

and

$$\tau_x = \frac{1}{n_x} = \frac{60 d_x \xi_x \eta}{D_x n \cos \varphi}$$

Based on the condition that the second volumes of the metal are constant, the coefficient of draught in piercing is

$$\frac{F_0}{F_{sh}} = \frac{v_{sh}}{v_0} = \mu$$

The actual coefficient of draught in piercing ranges from 1.5 to 4.5, while the axial travel speed, which is proportional to the increase of the roll diameter along the tapered roll barrel, augments only by 8%. Therefore the metal slippage on the surface of contact between the metal and the rolls is inevitable.

Many investigators explain the mechanism of the cavity formation in the metal being pierced by the action of tangential and radial tensile stresses in the axial zone of the billet which cause its caving before it actually touches the mandrel. The main deformations in piercing occur between the surfaces of the rolls and mandrel.

The barrel-type roll piercing mills have found wide application. In these mills the roll barrel comprises two truncated cones with their large bases joined together. The angle of the inlet (entry) cone (cone of piercing) is $\varphi_1 = 3\text{-}4°$, while that of the outlet (delivery) cone (cone of cross rolling off) is $\varphi_2 = 3°30' = 5°$. Larger values of φ_2 are taken when the difference between the diameters of the billet and shell is great. Both cones are frequently made of the same length. The rolls of the piercing mills for making large-diameter shells (intended for further rolling into large-diameter pipes in pilger mills) have a special fullering and elongated delivery cone.

According to the data obtained by Yu.Matveev and L.Vatkin the diameter of the piercing mill rolls is

$$D_0 = 2D_{p.max} + (400\text{-}600)$$

where $D_{p.max}$ = maximum diameter of pipes rolled in the given mill, mm

Lately successful experiments have been carried out with piercing in three-roll mills. Here the barrel-shaped rolls are arranged at an angle of 120° to one another and their axes are inclined to the horizon at an angle of up to 15°.

When piercing in a three-roll mill the tensile stresses in the zone of deformation (in front of the mandrel) are sharply reduced, this being contributive to an improved quality of the metal.

Rolling Pipes in Units Equipped with Plug Mills

Figure 13.3 shows the process chart for rolling pipes in a unit equipped with a plug mill. The plug mills are employed for rolling pipes within a wide range of sizes: heavy mills (type 400) roll pipes 127-426 mm in diameter, medium mills (type 250), 76-250 mm dia pipes and light mills (type 140), 70-140 mm dia pipes.

When rolling large-diameter pipes in a plug mill the shell is rolled in two piercing stands with intermediate heating prior to final rolling in the finishing stand. Small-diameter pipes are rolled with intermediate heating and reducing.

The plug mill as such is a two-high non-reversing stand. The mandrel comprising a bar with a plug on its end is inserted into the roll pass. A movable table with a trough for the shell and with a pusher is installed right before the stand. With the plug mill most steels are rolled in two passings, only some difficult-to-deform steels being rolled in three and more passings.

To roll the pipe for the second time stripper rolls arranged after the stand return the pipe to the entry side of the mill through the same roll pass which is made idle by slightly lifting the top roll. Prior to the second passing the pipe is turned through 90°.

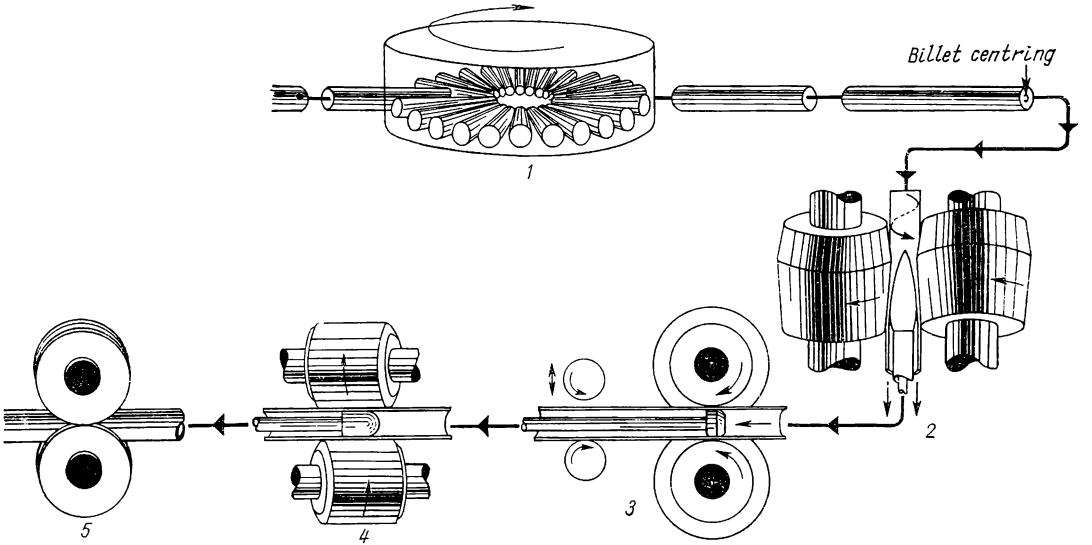


Fig. 13.3. Process chart for rolling pipes in unit equipped with plug mill

1—billet reheating in rotary-hearth furnace; *2*—billet piercing in piercing mill; *3*—rolling shell into pipe in plug mill; *4*—reeling pipe in reeling machine; *5*—sizing pipe in sizing mill

The overall coefficient of draught (per two passings) obtained in the plug mill amounts to 1.5-2.0, the maximum draught being adopted for the first passing. The round roll pass is made with large fillet radii (taper) at the roll parting line and the pass width is 1-3 mm greater than the shell diameter. The diameter of the mandrel should be 2-5 mm smaller than the inside diameter of the shell.

The pipe delivered by the plug mill is then rolled in a reeling mill to improve its surface finish and reduce variations in its diameter and wall thickness. The design of the reeling mill is similar to that of the plug mill, the work rolls of the reeling mill being permanently inclined at an angle of 6-7° 30′ with respect to the axis of rolling. In the reeling mill the pipe diameter is somewhat increased (by 3-9%) and the length, reduced (by 1-6%). The diameter of the reeling mandrel is usually 1-6 mm greater than the inside diameter of the pipe delivered by the plug mill.

The reeling mills are equipped with work rolls which are just slightly tapered (the angle of the entry cone is up to 2° and that of the delivery cone, up to 1.5°). The dimensions of individual elements of the rolls, mandrels, guides, etc. are determined by using empirical formulas.

After reeling, the pipe is rolled in a continuous sizing mill where it is made to assume the exact final dimensions. The sizing mill usually has 5-7 stands equipped with 400-800 mm dia rolls. The stands are alternately inclined to opposite sides at 45° to the horizon so that any two adjacent stands are at right angles. The round pass in the last stand has a true circular shape and its diameter corresponds exactly to that of the finished pipe, whereas in the remaining stands the passes are made with a small taper. The roll pass design of such mills provides only for exact sizing of the pipe, the pipe diameter being reduced by 3-7 mm only.

The pipe rolling units equipped with plug mills may also include reducing mills capable of reducing the pipe diameter by as much as 70%. Basically, these mills operate like the sizing ones, but have a much greater number of stands (9-26). When rolling in the reducing mill which, as a rule, is done without using the mandrel (like in sizing) the thickness of the pipe wall is increased, but can be kept constant

or even reduced by tensioning the pipe during the process (so-called stretch-reducing).

There are various designs of the reducing mill stands which are usually equipped with two, three or four rolls. In old mills the stands were group-driven; lately application have found individual adjustable-speed drives which make it possible to control the wall thickness of the pipe by varying its tension between individual stands. The adopted pass reduction for these mills ranges from 3 to 5%, depending on the wall thickness of the pipe. From the first to the last stand the passes are made with a varying ovality.

The minimum diameter of the rolls for reducing mills is selected conformable to the given size of the pipe and the required roll strength. The rolling speed in these mills reaches 11-12 m/s.

The three- and four-roll reducing mills are more compact and ensure a better surface finish as compared with the two-roll mills, but at the same time they are more complex and less convenient in adjustment.

In stretch-reducing, like in continuous rolling, the pipe is produced with somewhat swollen head and tail ends. To reduce the metal losses with the swollen pipe ends no efforts are spared to make the stretch-rolling unit as compact as possible and thus reduce the length of the pipe ends rolled without tension. For the same reason the length of the rolled pipe is increased, for with a large overall length of the pipe the relative length and weight of the swollen ends are small.

Rolling Pipes in Units Equipped with Continuous Mills

These units (Fig. 13.4) roll pipes with an outside diameter ranging from 50 to 114 mm; pipes of a smaller diameter (down to 17 mm) are also produced in these units by subsequent reducing. Round billets 140 mm in diameter are heated either in continuous barrel-type or rotary hearth furnaces, centred with a pneumatic device and then pierced in a barrel-type roll piercing mill.

The shell obtained after piercing the billet (billets of one and the same diameter of 140 mm are used for the entire range of pipes rolled in the unit) is then rolled on a cylindrical rod (mandrel) in a nine-stand continuous mill. The

lubricated mandrel is inserted into the shell by means of a special pusher. After rolling in the continuous mill the mandrel is withdrawn from the pipe with a stripper. If necessary, the pipe delivered by the mill may be reheated and then sized or reduced. After rolling the finished pipes are cut in half, cooled on chain skid banks and straightened.

The continuous seamless-tube rolling mill consists of two-high stands, the rolls in the adjacent stands being

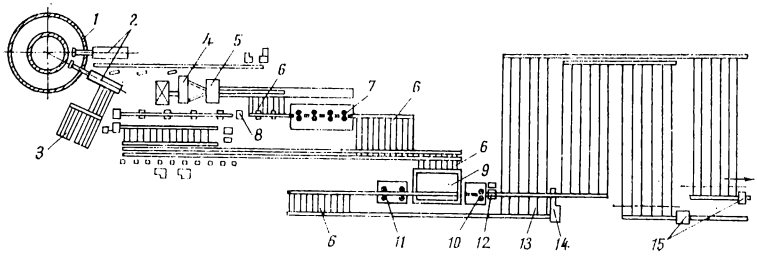


Fig. 13.4. Layout of pipe-rolling unit equipped with continuous mill
1—rotary-hearth furnace; *2*—machines for loading and unloading billets; *3*—charging platform; *4*—pinion stand; *5*—piercing mill; *6*—sloping racks; *7*—nine-stand continuous mill; *8*—mandrel stripper; *9*—reheating furnace; *10*—twelve-stand reducing mill; *11*—twelve-stand sizing mill; *12*—rotary shears; *13*—cooling bank; *14*—hot saw; *15*—straightening machines

arranged at right angles. In most mills of this type the roll axes are inclined at 45° to the horizontal plane. The round passes used ensure high dimensional accuracy of the pipes.

The continuous seamless-tube mills are much more efficient than the plug mills. In modern pipe rolling units the pipes are butt-welded prior to reducing, i.e. these units realise the principle of endless rolling.

Rolling of Pipes in Units Equipped with Three-Roll Mills

The three-roll mills are used for rolling thick-walled carbon and alloy steel pipes 34-200 mm in diameter. Among the advantages of the three-roll mills are a higher dimensional accuracy of the manufactured pipes and rapid change-overs from one size of the pipe to another.

In these units the operations preparatory to rolling are similar to the ones described earlier for the other pipe-rolling

units, namely the round billets are heated in rotary hearth or continuous barrel-type furnaces, then centred by punching a dent in the head end face and pierced to produce shells in a piercing mill with adjustable roll inclination angle. Thereafter the shells are rolled in a three-roll mill.

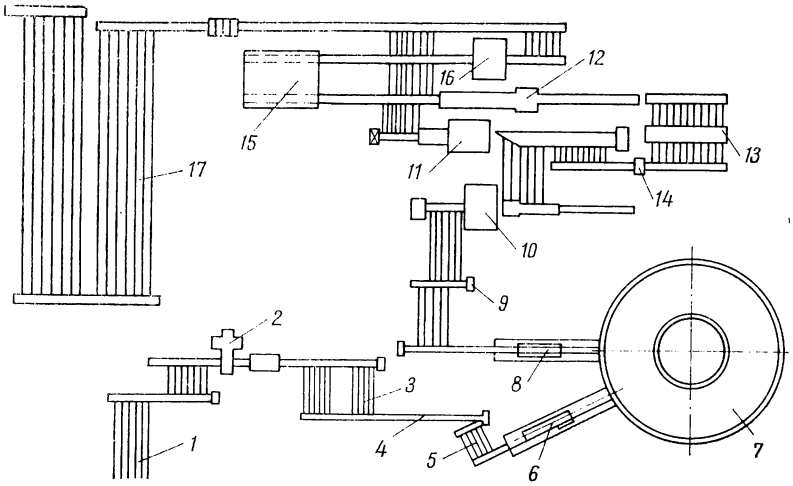


Fig. 13.5. Layout of pipe-rolling unit equipped with three-roll mill

1—charging rack; *2*—billet breaking press; *3*—skid transfer; *4*—transfer table; *5*—charging rack; *6*—charging machine; *7*—rotary-hearth furnace; *8*—discharging machine; *9*—billet centring device; *10*—three-roll type piercing mill; *11*—three-roll type reeling mill; *12*—mandrel stripper; *13*—mandrel cooling vat; *14*—mandrel lubricating machine; *15*—reheating furnace; *16*—three-roll type sizing mill with barrel-shaped rolls; *17*—cooling table

The final sizing of pipes is done in a sizing mill which is also of the three-roll type. The layout of a pipe-rolling unit equipped with a three-roll mill is shown in Fig. 13.5.

In the three-roll mills the rolls are arranged at 120° with respect to one another and rotate in one and the same direction. The roll axes are inclined with respect to the axis of rolling in both the horizontal and vertical planes (Fig. 13.6). The roll inclination angle relative to the vertical plane passing through the axis of rolling is called the feed angle (α) and that relative to the horizontal plane bears the name of reeling angle (φ).

Such a roll arrangement makes it possible to set the shell in rotary and translatory motion. On the delivery side of the mill the pipe is prevented from wobbling by means of a guide and a centring device. The necessary reduction of the outside diameter and of the wall thickness of the shell is accomplished by employing a special roll profile and by varying within a wide range (from 4° 30′ to 10°) the feed

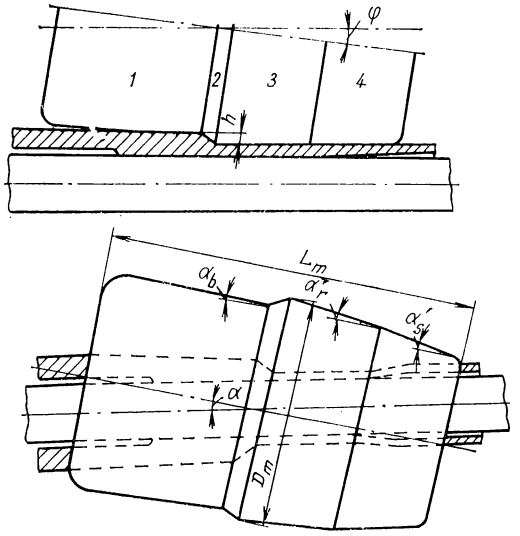


Fig. 13.6. Schematic representation of rolling process in three-roll type mill

1—entry cone; *2*—collar; *3*—reeling cone; *4*—sizing cone

angle α (the reeling angle is only slightly changed; $\varphi \approx 7°$). The coefficient of draught obtained in the three-roll mill amounts to 1.5-3.5. The deformation of the shell is effected mainly by the collar. The entry cone ensures biting of the shell and slightly reduces its diameter and wall thickness. The reeling cone smoothes out irregularities and sizes the pipe wall. The sizing cone secures the exact diameter of the pipe, reduces its ovality and makes a clearance between the pipe and the mandrel which facilitates the withdrawal of the latter.

Rolling Pipes in Units Equipped with Pilger Mills

The rolling of shells in pilgrim (pilger) mills, where the shell being rolled on a mandrel is set in reciprocal motion by means of a special device—feeder, is one of the oldest methods of making seamless pipes with a diameter of up to 680 mm. Most pipe-rolling units equipped with heavy pilger mills roll ingots into pipes 325-700 mm in diameter, 6-13 mm in wall thickness and 12-36 m in length. Medium pilger mills roll ingots into pipes 140-325 mm in diameter, 4.5-8 mm in wall thickness and 12-38 m in length. Light pilger mills are employed to produce pipes 22-140 mm in diameter, 2.25-4.0 mm in wall thickness and 8-50 m in length from rolled billets, with subsequent reducing. In addition to round pipes pilger mills also produce square, hexagonal, ribbed, stepped and tapered pipes and tubes, as well as the ones with swaged ends. Pilger mills are employed for rolling carbon and high-alloy steels.

The pilger rolls which form a pass of varying cross section (both the pass height and width are variable) constantly rotate in opposite directions so that the shell being rolled is made to move in a direction opposite to that of its feed. During each revolution the rolls work a shell section whose length corresponds to that of feed. The pipe is fed between the rolls at the moment when the varying cross section of the pass formed by the revolving rolls reaches its maximum and the metal ceases contacting the rolls. After rolling the mandrel is withdrawn and the pipe is sawn, reheated in a batch-type furnace, sized in a sizing mill and then hot straightened in a straightener with cross concave rolls. The most popular layout of a pipe-rolling unit equipped with a pilger mill is shown in Fig. 13.7.

Figure 13.8 is illustrative of a pilger roll profile. The deformation of the shell is effected mainly by the crest of the roll (so-called 'striker') which corresponds to central angle 0_c (section *1* of variable radius).

Section *2* of constant radius which corresponds to central angle 0_p is called the polishing section. This section of the roll profile sizes the pipe diameter and wall thickness.

Section *3* corresponding to central angle 0_o bears the name of outlet (delivery) section. It makes for a smooth exit of the

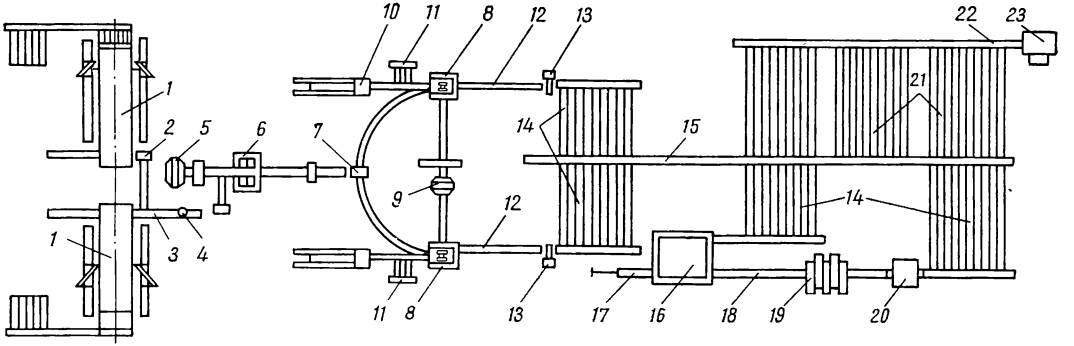


Fig. 13.7. Layout of pipe-rolling unit equipped with pilger mill

1—reheating furnace; *2*—ingot car; *3* and *18*—feed tables; *4*—turn-table; *5*—piercing mill drive; *6*—piercing mill; *7*—shell car; *8*—pilger stands; *9*—pilger mill drive; *10*—feeding mechanism; *11*—mandrel cooling vats; *12*—discharge trough; *13*—shuttle saws; *14*—benches; *15* and *22*—delivery tables; *16*—reheating furnace; *17*—pusher; *19*—five-stand sizing mill; *20*—hot straightener; *21*—inspection benches; *23*—gag press

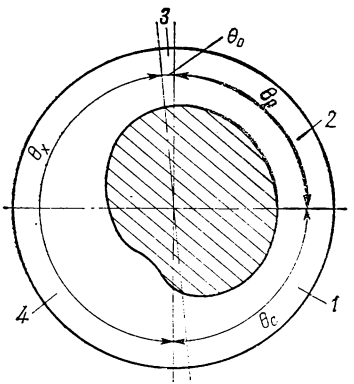


Fig. 13.8. Pilger roll profile

pipe from the zone of deformation and smoothes out irregularities on the pipe surface.

Section *4* —the remaining portion of the roll profile— which corresponds to central angle 0_x [$0_x = 360° - (0_c + 0_p + 0_o) \geqslant 150\text{-}160°$] is of the minimum variable radius and serves to smoothly join the other portions of the roll surface, makes for free passage of the non-deformed shell and provides optimum time necessary to feed the shell for the next rolling cycle.

The crest is the most critical section of the roll profile, for it determines the mill efficiency, power parameters of the process and wall thickness.

The process of rolling in the pilger mill combines elements of forging (in the beginning of the cycle) and rolling with rolls of variable radius.

The process is illustrated by the diagram shown in Fig. 13.9. In the beginning of rolling of the shell portion fed into the rolls (position *I*) the shell is reduced and forced together with the mandrel backwards into position *II*, the length of the reduced portion of the shell being equal to that of feed. While the shell moves from position *II* into position *III* the surface of its rolled portion is smoothed out and gradually leaves the zone of deformation. In position *IV* the pass becomes idle and the feeder advances the shell, simultaneously turning it through 90°.

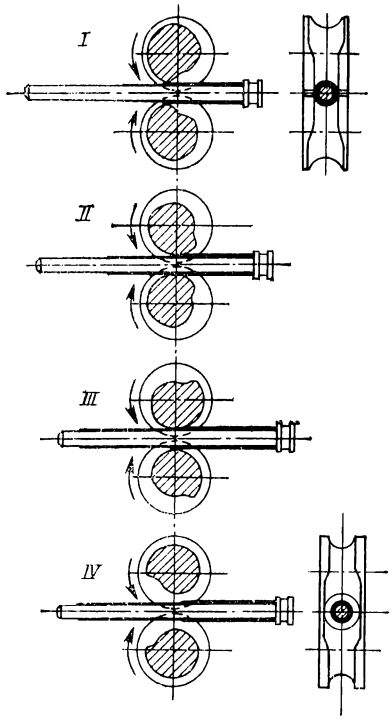


Fig. 13.9. Pilger rolling principle

Prior to rolling the shell end is engaged and the normal process then commences. Somewhat underrolled head and tail ends of the pipe are cut off with a hot saw.

To produce shaped tubes (square, polygonal, ribbed) use is made of roll passes of very complex shapes.

Nowadays a new process pattern is adopted for rolling pipes in units equipped with pilger mills, namely, the ingots are pierced in a hydraulic press (more often of the horizontal type) to produce cup-shaped shells and thereafter the shells are rolled first in elongators with cross rolls and then in pilger mills. The ingots are heated in rotary hearth furnaces and then descaled and broken into desired lengths prior to piercing.

Apart from the described above methods of manufacturing seamless pipes and tubes two other procedures of an entirely different nature have found application for this purpose, namely, extrusion and drawing in the rack-type drawbenches.

Making Pipes by Extrusion

The process of making pipes and shapes from lead by pressing (extrusion) was known as far back as in the 18th century. The method of extrusion is as follows. Hot metal is placed in a heavy container, pierced with a piercing mandrel in a press to make a thick-walled shell and then extruded by the press ram through an annular opening formed by a die and the mandrel.

To implement such a process it is necessary to provide for such a state of stress of the metal (nonuniform triaxial compression) which will secure the required plasticity of the metal without breaking its continuity. The trouble is that to make this, a very high pressure must be created within the container and this causes the development of great friction forces. Besides, during this process the temperature of the metal rapidly drops and its surface becomes considerably oxidized.

The following conditions are required for the successful conduct of the process: thorough preparation of the metal, correct setting of the die and mandrel, adequate descaling and prevention of further scale formation, good lubrication

(glass and graphite), high strength of the tools at high temperatures and availability of presses capable of developing very high pressures during a short pressing cycle (hydraulic presses of the horizontal type have a capacity of up to 12 thousand tons).

Figure 13.10 is illustrative of the operations involved in the extrusion of pipes in a horizontal press. Application also find presses of the vertical type.

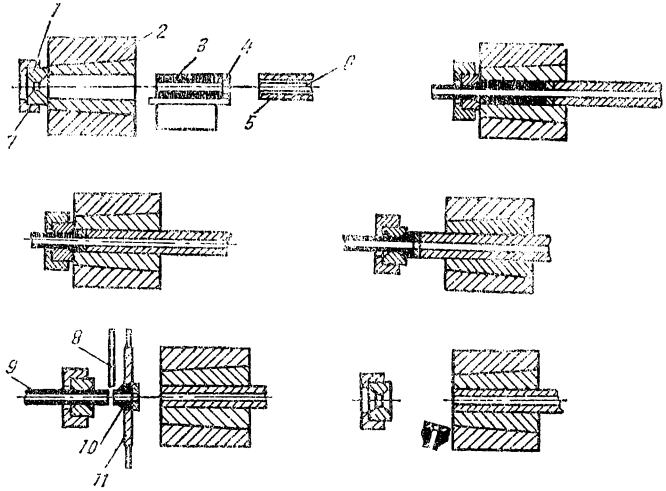


Fig. 13.10. Sequence of operations in extruding of pipes
1—die; *2*—container; *3*—billet; *4*—dummy block; *5*—ram; *6*—piercing mandrel; *7*—die holder; *8*—rotary saw; *9*—pipe; *10*—crop end; *11*—hydraulic clamp

The speed of the extruded pipe delivery relative to the speed of the ram increases in proportion to elongation which may be as great as 25-45 or even 60-70.

When extruding alloy steel pipes a round billet is coated with a protective layer of enamel and heated in a rotary hearth furnace. Thereafter the billet is pierced, reheated in a molten salt bath and then extruded, glass being used as a lubricant. Induction heating in a protective atmosphere is widely applied.

Extrusion enables pipes to be produced from low-plasticity high-alloy steels and this makes it a promising method even though the efficiency of the process is rather low. The

method also holds a good promise for making pipes and tubes of intricate shapes.

Making Pipes in Units Equipped with Rack-Type Drawbenches

In these units a square billet is pierced in a press to produce a round cup-shaped shell which is then drawn into a pipe in a rack-type drawbench. The rack-type drawbench comprises a series of solid or roller-type draw rings through

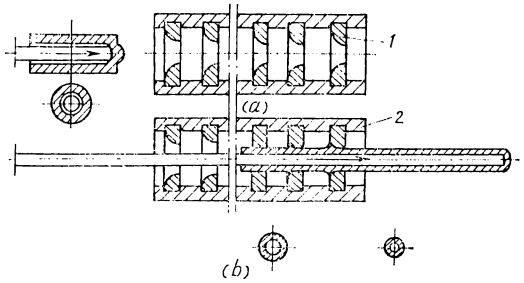


Fig. 13.11. Principle of drawing pipe in rack-type drawbench (*a*) feeding cup with mandrel to drawbench; (*b*) pushing cup through draw rings; *1*—draw ring; *2*—draw-ring stand

which the shell is forced by means of a rack-actuated mandrel inserted into it. The drawbench is shown schematically in Fig. 13.11. The units equipped with the rack-type drawbenches are employed to produce carbon and alloy steel pipes 57-219 mm in diameter, 2.5-8 mm in wall thickness and up to 12 m in length.

The adopted processing pattern provides for the manufacture of high-quality pipes (especially from the standpoint of surface finish and wall thickness variations) and is advantageous in that it enables square billets to be used. This pattern is as follows: (1) heating of square billets in rotary hearth furnaces; (2) descaling (by throwing the billets onto a rack); (3) piercing of the billets in a press to produce cup-shaped shells; (4) reheating; (5) rolling of the shells on a mandrel in an elongator; (6) drawing of the shells in a drawbench fitted out with roller-type draw rings. There-

after the process proceeds as usually: rolling in a reeling mill, sizing or reducing and finishing of pipes.

The rack-type drawbench consists of a guide bed, heavy rack with a speed reducer and motor, mandrel pusher, working and draw-ring stands. The draw-ring stands have slots into which the draw rings with thrust washers are inserted. The roller-type draw rings (used in place of solid ones) include from three to four calibrated rollers. The rollers in the adjacent draw rings are arranged so that the gaps between the rollers in one ring are aligned with the rollers in the next ring. Square-round or square-polygon series are used for the passes formed by the rollers of the draw rings.

Modern rack-type drawbenches usually include six groups of stands with roller-type draw rings. The degree of deformation of the shell cross section in the draw rings is gradually reduced from 20% in the first to 5% in the last ring. To slow down cooling of the shell during drawing the mandrel is preheated to 350-400 °C. After drawing the pipe together with the mandrel is rolled in a reeling mill where a clearance is formed between them. Thereupon the mandrel is withdrawn by means of a chain stripper.

13.2. Cold Rolling and Drawing of Pipes and Tubes

Cold rolling and drawing are used to manufacture highly accurate thin-walled pipes and tubes. Whereas the hot-working methods including stretch reducing make it possible to manufacture pipes with a diameter of not less than 13 mm and with a wall thickness of not less than 2 mm only, cold rolling and drawing enable pipes and tubes to be produced with a diameter of 1-150 mm and with a wall thickness as small as 0.1 mm. The cold-working methods are employed for making various special-purpose pipes and tubes of alloy and high-alloy steels. Hot-rolled seamless pipes are usually used as the starting material for cold working. Auxiliary processing operations (pickling, heat treatment, etc.) in cold working are similar to those involved in cold rolling of sheet steel. Careful inspection of the pipes and cropping of the pipe ends are common to both the cold rolling and cold drawing operations.

Pipes made of alloy steel are preliminarily annealed and then pickled. The pipes are also subjected to intermediate and final annealing, straightening, hydrostatic tests and oiling.

The cold rolling and cold drawing operations are frequently combined.

Cold Rolling of Pipes

This operation is done mainly in two-roll and roller-type cold-rolling mills of intermittent action. In these mills the metal is deformed during reciprocal motion of the working stand itself. Complex planetary and floating-roll type mills are employed less frequently.

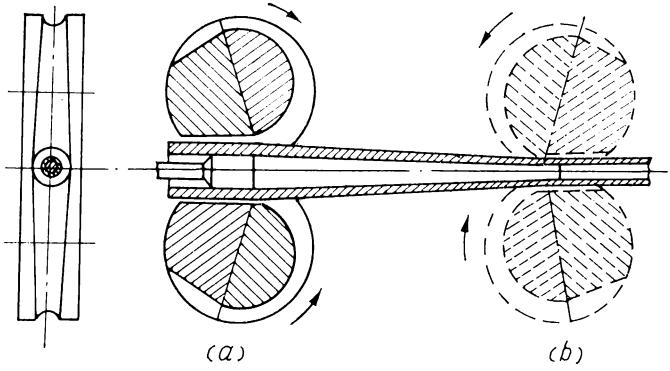


Fig. 13.12. Principle of pipe cold rolling

The two-roll type cold rolling mills make it possible to produce pipes with a wall thickness of 0.65-1.25 mm, the roll diameter ranging from 300 to 550 mm respectively.

In the two-roll type cold rolling mill the pipe is rolled on a stationary tapered mandrel between a pair of rolls (Fig. 13.12). The pipe, which is periodically advanced (fed) by a special cam mechanism (end position *a*), is rolled piecemeal while the working stand is moved forward (forward stroke from position *a* into position *b*) by a crank mechanism. When the rolls cease to contact the metal (end

position *b*) the pipe is turned through 90 or 60° and then the stand is moved backward (backward stroke from position *b* into position *a*) and the rolls roll off the swollen portions of the pipe surface which have been previously formed by the chamfers of the roll pass.

The roll grooves, like in pilger mills, are of varying cross section along the roll circumference. The grooves are formed by segments which are fixed on the roll shafts. The wall thickness of the pipe is determined by the diameter of the tapered mandrel within the zone of deformation and is adjusted by shifting the mandrel along its axis.

The working stand (called saddle), usually of the close-top type, runs back and forth on wheels along guides. It is actuated by a crank mechanism driven by an adjustable-speed d.c. motor via a speed reducer. To synchronize rotation of the rolls so that their grooves are always in a definite position with respect to each other the rolls are geared to fixed toothed racks.

To reduce contact friction in the zone of deformation during cold rolling use is made of billets—pipes—with various coatings (copper, phosphate). Various lubricants and cooling emulsions are to be used without fail.

Cold-rolling mills can be adapted for making pipes of variable cross section. This is usually done by shifting the mandrel within the zone of deformation while rolling (when the saddle is in the extreme position the mandrel can be moved). In this case special mechanisms are used to shift the mandrel as required and withdraw the underrolled portion of the pipe from the rolls.

Some difficult-to-deform special-purpose steels are successfully rolled by using the so-called 'warm-rolling' method which consists in that the pipe portion being deformed is heated to 350-400 °C, this improving the plasticity of the metal, reducing its resistance to deformation and thus reducing the mill load. More often than not use is made of local induction heating.

The roller-type cold-rolling mills differ from the two-roll mills in that instead of rolls with grooves of varying cross section they are equipped with three or five rollers which are movably held in a cage and can smoothly move together with it inside a special socket sleeve, thus changing the

cross section of the pass formed by them. The schematic diagram of such a mill is shown in Fig. 13.13. When the sleeve (carriage) is moving forward, the rollers, bearing against profiled metal strips, are pushed inward and reduce the pipe on a mandrel. After the backward stroke of the carriage the pipe is turned through a certain angle, depending on the number of the rollers (in case of three rollers the pipe is turned through 60°). In this way, when the carriage is

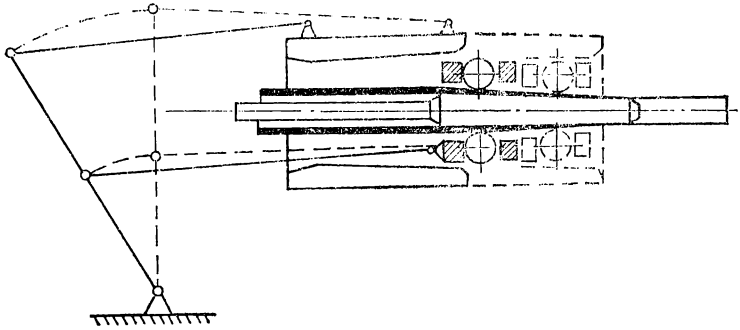


Fig. 13.13. Principle of operation of roller-type cold-rolling mill

reciprocating the rollers alternately move inward and outward and the pipe is thus periodically reduced and then released so as to enable its being freely fed. With a small diameter of the rollers (70 mm) the rolling pressure is reduced and this makes it possible to produce thin-walled pipes even from high-alloy steels.

The rolls and rollers of cold-rolling mills are made of heat-treated ball-bearing steel.

Cold Drawing of Pipes

In the cold drawing of pipes the pipe diameter and wall thickness are subject to reduction, but sometimes only the wall thickness is reduced. The following three cold drawing methods are employed.

1. Without mandrel (sink drawing). The method is used for reducing the pipe diameter only; the pipe is drawn thro-

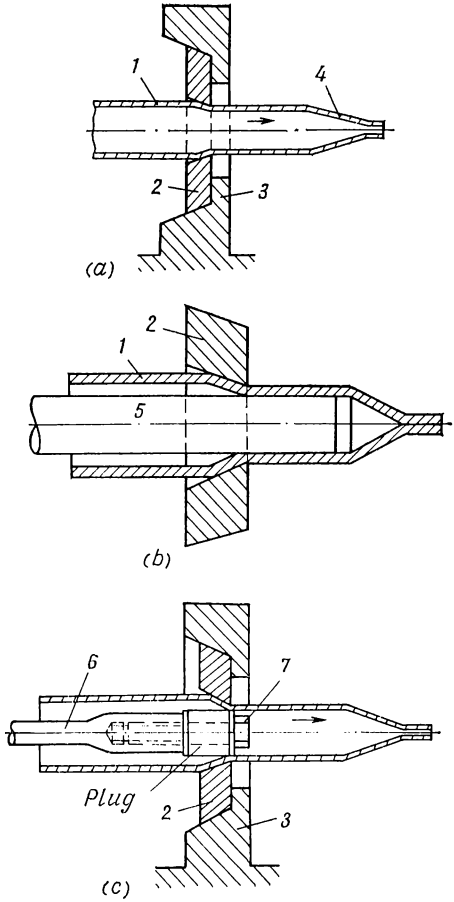


Fig. 13.14. Methods of pipe cold drawing

1—pipe; *2*—ring die; *3*—die stand; *4*—pointed pipe end; *5*—mandrel; *6*—rod; *7*—bolt

ugh a ring die installed in the die stand of the drawbench (Fig. 13.14*a*).

2. With a long mandrel. The method is used to reduce mainly the wall thickness of the pipe; the mandrel moves together with the pipe (Fig. 13.14*b*). With this method there is a possibility of obtaining great deformations in a single passing and for this reason it is employed in the first roughing passes.

3. With a short mandrel. The method is applied to reduce both the pipe diameter and wall thickness. Here a cylindrical mandrel is held in place by means of a long rod it is bolted to (Fig. 13.14*c*).

In modern drawbenches the drawing rate is as high as 100 m/min.

Cold drawing allows pipes of square, rectangular, polygonal and other sections to be made. Such pipes are made from round tubes, whose circumference is somewhat greater than the perimeter of the shaped pipes and the wall thickness, the same as in the latter, by drawing the round tubes through shaped dies without mandrel.

Stainless steel capillary tubes of a diameter less than 1.0 mm (down to several hundredths of a millimetre) are drawn on a long mandrel with reeling and subsequent drawing without mandrel through hard metal and diamond dies.

Pipes and tubes made by any method are mostly in need of an additional finishing conformable to their purpose and standard specifications.

Finishing covers either the entire surcace of pipes (straightening, coating) or their ends alone (cropping, chamfering, sizing, swaging, welding, cutting male and female threads, screwing sleeves and nipples).

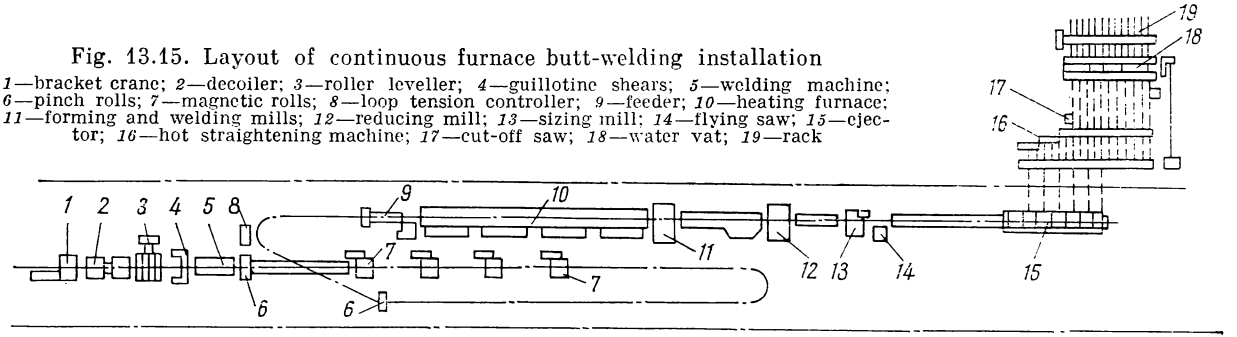
13.3. Welded Pipe Production

Improved technology of making welded pipes, their better quality (at a comparatively low cost) were factors which promoted a rapid increase in the production of such pipes and their extensive application.

Old methods of furnace welding of pipes have given place to highly efficient continuous furnace and electric welding operations.

Fig. 13.15. Layout of continuous furnace butt-welding installation

1—bracket crane; *2*—decoiler; *3*—roller leveller; *4*—guillotine shears; *5*—welding machine; *6*—pinch rolls; *7*—magnetic rolls; *8*—loop tension controller; *9*—feeder; *10*—heating furnace; *11*—forming and welding mills; *12*—reducing mill; *13*—sizing mill; *14*—flying saw; *15*—ejector; *16*—hot straightening machine; *17*—cut-off saw; *18*—water vat; *19*—rack



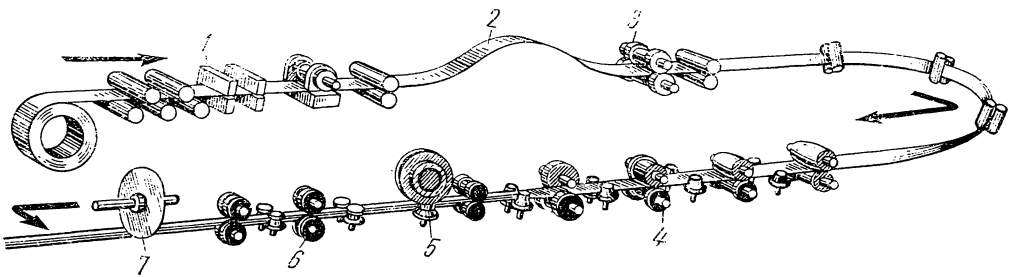


Fig. 13.16. Layout of welding mill for small-diameter pipes

Figure 13.15 represents the equipment layout of a continuous furnace butt-welding installation. Coiled skelp up to 4 t in weight is delivered to a decoiler by a bracket crane and the coil tail is made to enter a straightening machine (if necessary, it is preliminarily sheared). To make the process continuous the ends of successive skelps are machine welded together. Provision is made for a loop whose tension is controlled by a special device to enable the skelp ends to be welded without discontinuing the process. Pinch and magnetic rolls along with a feeder serve to move the skelp. A second welder is usually set up right before the furnace to weld the skelp end to a special needle required for threading the furnace when changing over from one pipe size to another or when changing the mill rolls.

The skelp moving at a considerable speed is heated in a long (40-55 m) continuous multiple-zone fired furnace with a preheating and heating chambers, the latter having a large number of burners (200 and more). These furnaces are of a recuperative type, fired either with gas (coke-oven, natural) or with fuel oil residue. The skelp edges are heated to a temperature higher than that of its centre part. When leaving the furnace the edges are blown off with compressed air and as a result of oxidation their temperature rises and at the same time slag and scale are blown away.

A continuous welding-and-forming mill made up of 6-12 vertical and horizontal roll stands is set up next to the furnace. In new mills the stands, as a rule, have an individual adjustable-speed drive. Forming of the pipe, i.e. its rolling up, starts in the first stand, while in the second and following ones the edges of the rolled up skelp are first tightly brought together and then compressed and welded.

In the welding-and-forming mill all the roll passes except for the forming (first) and finishing ones are oval-shaped. The reduction in the diameter is 3-10%. Upon leaving the mill at a speed of 3-8 m/s the endless pipe is right away cut to length with a flying saw or, as is shown on the attached diagram (Fig. 13.15), it is delivered to reducing and sizing mills. On leaving the sizing mill the pipes are sawn to length and then cooled, straightened and undergo finishing.

The first units for continuous furnace welding of pipes operated with skelp produced in specialized skelp mills;

lately the skelp has been obtained by slitting wide sheet steel.

Modern continuous furnace welding units put out mainly pipes with an outside diameter of 21-114 mm and wall thickness of 2.75-6.0 mm; the skelp used is up to 406 mm wide. The output of these units reaches 25-50 t/h; the pipe delivery speed of the reducing mill exceeds 500 m/min.

The production of electric-welded pipes is on a steady rise. The following three methods of electric welding are in use: resistance, induction and arc welding.

The resistance welding is employed for making pipes 6-660 mm in diameter and 0.3-20 mm in thickness.

The starting material are strips, skelps and sheets, most often in coils.

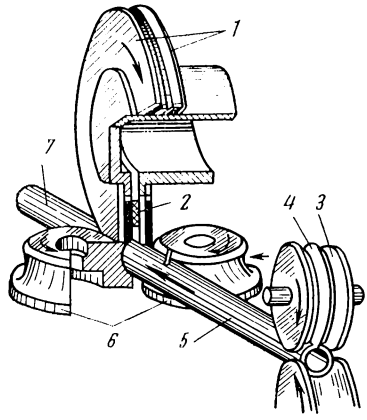


Fig. 13.17. Principle of resistance welding of pipes

1—electrode wheels; *2*—insulating spacer; *3*—guide rolls; *4*—guide fin; *5*—semi-finished pipe; *6*—support welding rolls; *7*—welded pipe

Figure 13.16 is illustrative of the layout of a mill for welding small-diameter pipes. From a decoiler via a straightener the strip goes to shears *1* and then to a welding machine with a weld trimmer. Looper *2* serves to store the strip needed to provide for the continuous operation. The strip edges are trimmed with shears *3* and then the strip goes to forming mill *4* with shaped horizontal and vertical rolls which roll it up to form a semi-finished pipe with a gap between the edges. The semi-finished pipe is welded in welding mill *5*, sized in sizing mill *6* and then cut to length with flying saw *7*.

The resistance welding of pipes is carried out by means of a device which is shown schematically in Fig. 13.17. An alternating current is fed to the edges of the semi-finished pipe to be welded through wheel electrodes separated by an insulating spacer. The pipe is moved by feeding rolls, and the edges are compressed during welding by support wel-

ding rolls. The edges become heated because their resistance is much higher than that of the rest of the pipe. Successive use finds welding current of an elevated frequency (200-300 Hz).

The welding current is supplied to the electrodes through sliding contacts where large power losses occur and for this reason welding units equipped with rotary transformers are employed. Provisions are made for water-cooling of the transformer and electrodes.

The design of the support welding rollers limits the possibilities to control the welding pressure. Of prime importance are the welding speed and the current frequency. A high-quality weld is obtainable only when the welding speed does not exceed a certain critical value. The maximum welding temperature is taken at 1540°C and the minimum, about 1200°C. Higher temperatures bring about undesirable heat effects which disturb the structure of the weld metal (coarse-grain structure, burning). The frequency of the welding current greatly influences the stability of welding. The weld quality has been found to improve when elevated frequencies are used, i.e. 200-400 Hz and even more. But even this does not completely rule out non-uniform heating along the weld.

The use of alternating current in welding yields a non-uniform internal flash. The external flash can be comparatively easily removed with a cutter, but the internal flash is very difficult to take off, particularly in pipes of a small diameter.

Recently there have been developed d.c. pipe-welding installations equipped with rotary solid-state rectifiers. There also exist resistance-welding units equipped with mercury contacts to supply the welding current.

An essentially new method of pipe welding with *radio-frequency currents* of up to 450 thousand Hz and more coming from valve oscillators and supplied to the welded pipe by the non-contact (induction) method has helped resolve a number of outstanding important problems. It has enabled the strength of the weld to be increased and made it possible to achieve higher welding speeds with reduced and better distributed internal flash and also with lesser power consumption.

Figure 13.18 represents a diagram of the induction welding process. The semi-finished pipe is produced, as usual, in a continuous forming mill and then it goes to the radio-frequency welding mill. There exist two methods of feeding power to the pipe edges to be welded, *viz.* by using either a flat contact inductor or coil (non-contact) inductor.

When the pipe is reduced by the rolls of the welding mill the molten metal of the edges is squeezed from the weld, forming internal and external flash and pure plastic metal of

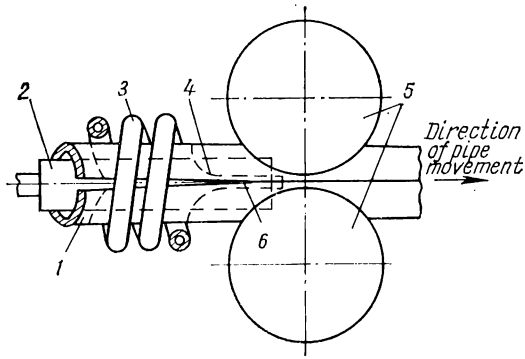


Fig. 13.18. Induction welding principle

1—semi-finished pipe; *2*—ferrite core; *3*—inductor; *4*—lines of forces; *5*—rolls of support-welding unit; *6*—welding zone

the transitional zone forms the final seam. To concentrat the welding current on the pipe edges a ferrite core is inser ted into the pipe.

Arc welding is used for making large-sized pipes (400-1500 mm and more). As to the configuration of the seam there are distinguished helical-seam, single-straight-seam and double-straight-seam pipes (pipes of large diameters are welded from two sheets).

Helical-seam pipes are formed from coils. Straight-seam pipes with a diameter of more than 700 mm are formed of hot-rolled sheet of ordinary length (6-12.5 m). Of great importance in flash and submerged arc welding of helical- and straight-seam pipes is the pretreatment of the sheet and, chiefly, of its edges, i.e. shot blasting to remove oxides and bevelling.

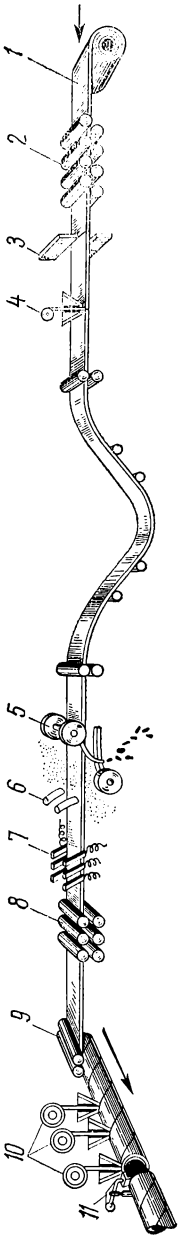


Fig. 13.19. Layout of mill for welding helical-seam pipes

The welding of helical-seam pipes is a continuous process (Fig. 13.19). Upon performing operations usual for the pretreatment of coiled metal, namely, straightening *2*, cutting *3*, welding of coil ends *4*, edge preparation *5*, *6* and *7*, and straightening *8*, strip *1* goes to combination forming-and-welding mill *9* and *10* and after welding is cut by gas-cutting torch *11*. The weld consists of the edge metal (up to 85%) and of the electrode wire metal (15-20%). The structure of the weld metal is largely dependent upon the welding speed and the weld cooling rate.

The submerged arc welding of straight-seam pipes is conducted in accord with the following standard pattern: pretreatment of the sheet (straightening, bevelling, sometimes shot blasting), forming, which for pipes of a small diameter is done in continuous forming mills and for those of a larger diameter, in forming presses (of preliminary and final forming). Then the external and internal seams are welded by welding heads on scaffolds. The welded pipes go to a sizing press (expander) which simultaneously performs the following operations: shaping in dies (the pipe is made to assume true round shape by delivering water under a high pressure inside it), expanding (the pipe diameter is increased by as much as up to 3%), strengthening of the metal (by stressing it above the elastic limit) and, finally, hydraulic testing. The conclusive operation involves chamfering of the pipe edges and X-ray inspection of the welds.

Thin-walled high-alloy pipes (made of austenitic steels) are argon- and helium-arc welded. Welding of stainless steels is done with titania-type electrodes. Radio-frequency welding is also used in dealing with high-alloy pipes.

The high quality of the pipe seam obtained by any known welding method can be achieved only with an adequate pipe

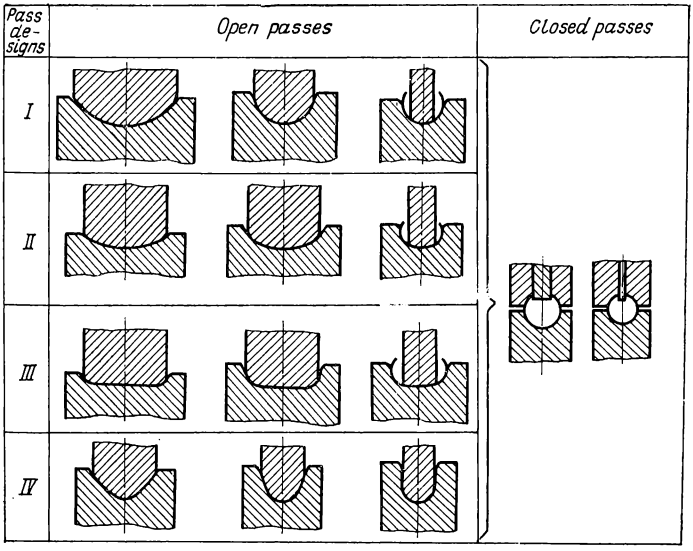


Fig. 13.20. Four types of roll pass design for forming mill

forming, which is determined by the roll pass design of the forming mills. During welding it is already impossible to correct the defects and irregularities in the edges of the semi-finished pipe.

The degree of deformation adopted in designing roll passes for pipe forming mills is based on experimental data.

Figure 13.20 illustrates four types of the roll pass design for the forming mill with open and closed passes, of which, considering the convenience of adjustment and a number of other factors, preference is given to the passes of the fourth type, the third and the second types coming next.

13.4. Manufacture of Special Products and Shapes

Production of Wheels and Tyres

The wheels of the rolling stock and locomotives 950-1810 mm in diameter are made either solid or tired. Wheels and tyres are produced from ingots or rolled billets. For this purpose steel containing 0.5-0.8% C, 0.6-0.8% Mn and

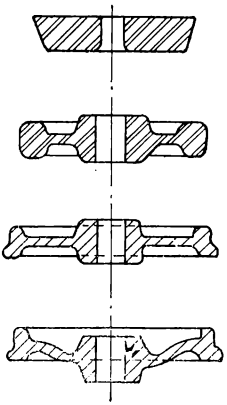


Fig. 13.21. Diagram of wheel forming

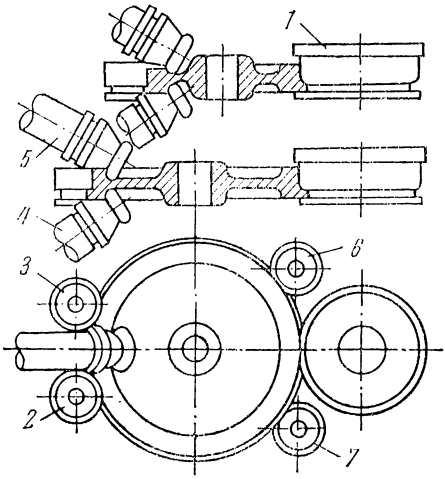


Fig. 13.22. Wheel-rolling mill

0.15-0.35% Si, with an ultimate strength of 70-90 kgf/mm^2 and elongation per unit length of 8-13% is used.

Figure 13.21 shows the wheel forming diagram.

The wheel blanks are heated in rotary-hearth furnaces, forged and pierced in a press and then formed in another press. Upon forming the wheel goes to a wheel-rolling mill equipped with seven rolls: one vertical roll *1* (driven), four vertical idle rolls *2*, *3*, *6* and *7*, two inclined rolls—lower roll *4* and upper one *5* (Fig. 13.22).

After rolling the wheel web is dished in a press. Thereafter the wheels are placed into furnaces for isothermal soaking

at 600 °C for 3 hours, to prevent the formation of flakes. Following assortment the webs and the rolling surfaces are machined, this being followed by the final heat treatment, i.e. heating up to 800-850°C, water-spray quenching of the rolling surface and tempering by heating up to 500-520°C in pits with subsequent air cooling.

The technology of the tyre production is somewhat different. The rolling is preceded by only one pressing operation The rolling of tyres is done in two mills—roughing and finishing—this being followed by stretch-sizing in a special press. The following is the heat treatment pattern for the tyres: slow cooling in pits or under special hoods, followed by oil quenching and tempering in furnaces. The tyres at temperature of 150-200 °C are loaded into the furnaces.

Rolling of Balls

A rather laborious process of making balls by pressing and forging has been successfully replaced by rolling in special mills (Fig. 13.23). In these mills, like in the piercing ones, the rolls are arranged at an angle of 4-7° to the axis of rolling. The rolls have helical grooves cut into them. The

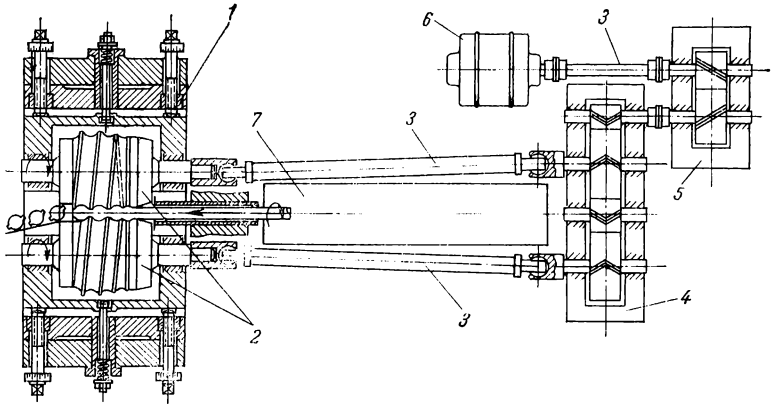


Fig. 13.23. Layout of ball-rolling mill

1—working stand; *2*—rolls; *3*—spindles; *4*—pinion stand; *5*—speed reducer; *6*—electric motor; *7*—inlet trough

rolls rotate in one direction. A heated round billet is fed along a trough from the spindle side of the mill and is held in a guide. First the rolls bite the metal which is then gradually reduced, shaped into a ball and the latter is separated from the workpiece.

The number of balls yielded per each revolution of the rolls depends upon the number of starts of the helical passes. When rolling large balls use is made of single-start passes which yield one ball per each revolution, while in rolling balls 40 mm across a four-start helical pass is employed, yielding 4 balls per each revolution of the rolls.

Depending upon the size and the purpose of balls, billets of different sizes, as well as various heating methods, are used (balls less than 4 mm across are cold rolled).

Bearing balls are made of accurately sized ball bearing steel bars up to 4.5 m long and with a diameter 1.5-2.0 mm less than that of the finished ball. The billets are induction heated to 850-900°C.

Large-sized carbon steel balls are used in ball mills.

The diameter of the rolls of ball-rolling mills is approximately 5-6 times that of the ball produced.

Production of Die-Rolled Sections by Cross Rolling

Solid and hollow (tubular) die-rolled sections for making axles, gas cylinders and other items can be obtained by employing the cross and helical rolling methods *.

Figure 13.24 represents a diagram of a three-roll mill for manufacture of die-rolled sections. A heated round billet is rolled by three rolls rotating in one and the same direction. The rolls can be shifted along the axis of rolling over an equal distance by a hydraulic device. Moreover, the rolls can move along their own axes, the clearance between them being changed by following a cam.

Three-roll mills may operate with either disc or tapered rolls arranged at an angle of 120° to one another and at 4-8° to the axis of rolling, this providing for helical rolling with a round workpiece being set in reciprocal rotary motion.

* The process of longitudinal die rolling of reinforcement bars was described earlier.

The billet heated to 1100-1150°C is clamped in the chuck and the cam follower roller is engaged. The chuck moves

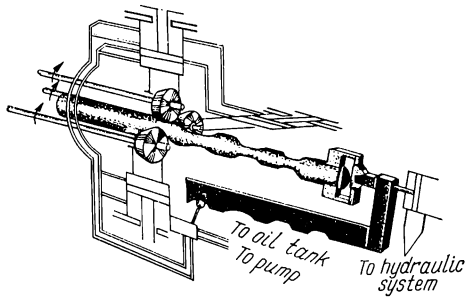


Fig. 13.24. Three-roll mill for making die-rolled sections

the billet whereas the rolls move by following the cam profile, increasing or decreasing the reduction of the metal, and this yields the required section.

Production of Roll-Formed Sections

Thin-walled, light-weight, intricately-shaped sections find application in automobile, aircraft, construction industries, etc. The replacement of rolled metal by roll-formed sections is economically advantageous, since the production of the required shaped sections by hot rolling frequently presents certain technological difficulties (non-uniform deformation, rapid cooling, wear and tear of rolls, etc.). For instance, it is practically impossile to obtain a hot-rolled section less than 2 mm thick. Moreover, rolled sections have a disadvantage of irregular distribution of the metal.

The merit of the roll-formed sections consists chiefly in a considerable saving of metal thanks to the fact that greater moment of inertia and, consequently, higher strength can be achieved with equal cross-sectional areas, but with different shapes. Furthermore, roll-formed sections can be comparatively easily made to assume structural shapes which help materially improve the service of the machine components, members of buildings and edifices, means of transportation, etc. (Fig. 13.25).

In addition to intricately-shaped open sections the method also enables closed sections to be obtained with a comparative ease. In spite of essential differences (no reduction in cross section, no forward and backward slip) the continuous roll-forming has also much in common with the rolling process, such as design features of the equipment, continuous

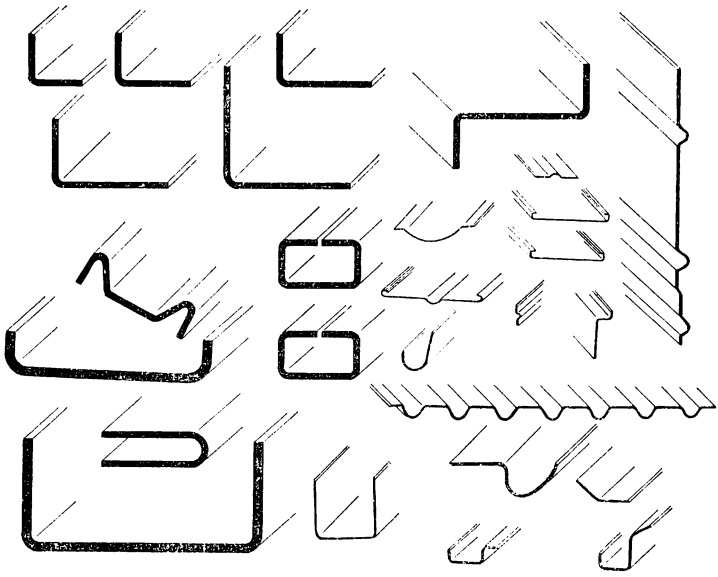


Fig. 13.25. Roll-formed sections

operation, endless workpiece, the effect of stiff ends on the metal deformation, the presence of a zone of non-contact deformation, etc.

The roll passes for the cold-roll-forming machines are designed proceeding from the following basic principles.

The number of passes (stands, roll pairs) is chosen so as to provide for the required bending and reliable metal biting by the rolls. In the first passes the bending angles are small, then they are increased and thereafter, when nearing the finishing passes, decreased again. The number of passes is as great as 15-20.

During gradual forming there is no draught along the longitudinal axis, while in the lateral direction there occurs plastic deformation in the pass itself and all along the non-contact zone. The passes are arranged so as to avoid twisting of the strip and its longitudinal deformation. The main axis of the roll-formed section is made to pass through the points of its cross section which do not change their position in the course of forming.

By gradually increasing the roll diameters in succesive stands a slight interstand tension is built up (0.3-0.5%).

In designing the roll passes account is taken of elastic lag of the strip.

With lesser single deformations, i.e. with a greater number of passes, the degree of the work hardening of the metal decreases.

Vertical rolls, filling mandrels, side rollers and guides are used to obtain closed roll-formed sections.

The design of the cold-roll-forming machines is similar to that of forming mills for welded pipe production.

The delivery speed of the machine is adjusted, depending upon the strip thickness and steel grade and may be as high as 3 m/s. Its output is about 120 thousand tons per year.

13.5. Features Specific for Rolling High-Grade Steels, Alloys and Non-Ferrous Metals

All high-grade steels and alloys undergo a thorough surface conditioning and rather frequently are subjected to an all-round planing or turning.

Considering poor plasticity of such metals, they are rolled at temperatures varying within a comparatively narrow range. The finishing temperature is also strictly fixed in conformity with the specific features of the given steel grade and the purpose of the rolled product. For example, low finishing temperatures are required for tool and stainless chrome steels of the ferritic class to obtain a fine-grained structure. Austenitic stainless chrome-nickel steels require finishing temperature of not below 950-1000°C because they harden at comparatively high temperatures due to slow recrystallization rates (resulting in an increased resistance to deformation).

In cold working of difficult-to-deform brittle steels and alloys a resort is taken to heating them to a temperature of 150-250°C, which corresponds to that the metal would have reached after being passed once through the rolls, this providing for plasticity sufficient to work the metal without any breakages.

The roll pass design adopted for rolling high-grade steels provides a somewhat greater spread than this is the case with carbon steels, sharp corners and changes in the shape of the passes are avoided. Efforts should be made to work the metal as uniformly as possible over its entire cross section. Box, diamond, hexagonal and oval-round series are used. Edging passes are employed for descaling in the first passings.

Most grades of alloy steels require slow cooling after rolling.

Specific Features of Rolling Non-Ferrous Metals and Alloys

From among non-ferrous metals used for making rolled products (sheets, strips, bars, wire and pipes) most prevalent are copper, nickel, zinc, lead, tin, aluminium, magnesium and their alloys, such as copper-tin (tin bronze), copper-tin-zinc (tin-zinc bronze), copper-aluminium (aluminium bronze), copper-nickel (melchoir, nickeline), copper-nickel-zinc (German silver), nickel-chrome (nichrome), aluminium-copper-magnesium-manganese (duralumin), aluminium-magnesium (electron alloy), etc. Non-ferrous rolled-metal products are widely used in a number of industries.

These products are made of ingots of different shape and weight. When remelting and casting the ingots special attention is paid to the purity of starting materials, and to temperature conditions in melting, pouring and cooling.

In rolling, most non-ferrous metals and alloys may break and crack even with small deformations due to improper temperature conditions in pouring of the ingots and because of impurities. The pretreatment for rolling of the ingots involves cold topping and surface conditioning, i.e. planing, turning or milling.

Various rolling schemes are used, depending upon the properties of the metal or alloy: hot rolling alone, hot and cold rolling, and for some alloys, cold rolling only.

Prior to hot working of copper its temperature should be within a range of 800-930°C; of brass and bronze, 420-480°C; of aluminium alloys: for duralumin, 400-450°C, for electron alloy, 280-320°C; of zinc, 110-150°C and of nickel, 1220°C. Most non-ferrous metals are heated in an oxidizing atmosphere.

Non-ferrous metals and alloys have different temperatures and rates of recrystallization and are subject to different degrees of work hardening. They are made to undergo intermediate annealing in cold rolling and also subjected to final annealing. To make the composition more uniform and to improve rollability use is made of homogenizing (for duralumin). In hot rolling of some alloys the roll surfaces are lubricated.

The equipment used for rolling non-ferrous metals is similar in design to that employed in the iron and steel industry and there are many common features in the roll pass designs and technology.

The rolling of non-ferrous metals proceeds with due consideration for the nature of polymorphic transformations in accordance with phase diagrams of binary and ternary alloy systems, as well as for the influence exerted by alloying additions and impurities. For instance, in rolling copper-zinc alloys (tompac, brass) their strength improves with an increase in the zinc content, but the cold plastic properties of such alloys containing more than 30% Zn, worsen.

According to the copper-zinc phase diagram, an alloy containing up to 36% of zinc is called alpha-brass and that with a zinc content of 36-45% is termed beta-brass. The hot plastic properties of alpha- and beta-brass are found to be influenced by the presence of lead. In the alpha-brass lead is concentrated at the grain boundaries and therefore, when heated, it melts and causes red shortness of the alloy. In the beta-brass lead is dispersed undissolved inside the grains and, for this reason, hot rolling of the alloy is also possible. Bismuth, antimony and arsenic are also deleterious impurities in the alpha-brass.

Pure nickel heated to 1230 °C can be rolled with great reductions (up to 80%). Impurities (lead, arsenic, antimony) deteriorate the rollability of nickel. Popular nickel alloy called monel metal (containing up to 70% Ni, 2-3% Fe, up to 1.2% Mn and the balance—copper) hardens after hot rolling (750-800°C) and, for this reason, is subjected to annealing.

Zinc is hot rolled at 150-180°C. Higher temperatures result in a sharply reduced plasticity of zinc.

In its pure form aluminium is very plastic, is little liable to work hardening and admits high single and overall reductions when cold. Iron and silicon impurities (up to 1%) greatly worsen its cold plastic properties. Emulsions are employed in hot rolling of duralumin, to avoid its sticking to the roll surfaces. Cold rolling of duralumin with intermediate annealing presents no difficulties.

Refractory metals (tungsten, molybdenum, rhenium, etc.) are rolled in conventional hot- and cold-rolling mills from billets obtained by powder metallurgy methods.

Rare metals (titanium, zirconium, uranium, thorium) and their alloys are rolled by following special schedules, consideration being given to polymorphic transformations, the nature of crystal structure and the impurities content. Most of these metals and alloys have a tendency towards sticking to the working tools and, therefore, in dealing with them special lubricants are used and measures are also taken to prevent oxidation and gas saturation of the superficial layer. All these specific properties are particularly characteristic of titanium and of alloys with it as a base.

Titanium and its alloys possess high strength characteristics with a low unit weight. They are heated to 1000°C, when heated above 700°C rapid heating is practised. Optimum rollability is achieved at 900-980°C, and, therefore, hot working is effected with intermediate heating operations.

Uranium and thorium are hot and cold rolled. They possess fairly good rollability, but special precautions should be taken because of their radioactivity. Pure zirkonium is hot and cold rolled. Special measures are taken against oxidation, up to and including heating in molten salt bath. Zirkonium-uranium-thorium alloys in round ingots obtained by vacuum smelting are machine scalped, enclosed in steel shells and then forged and rolled.

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TO THE READER

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